Tim Trice

Planned Controlled Document

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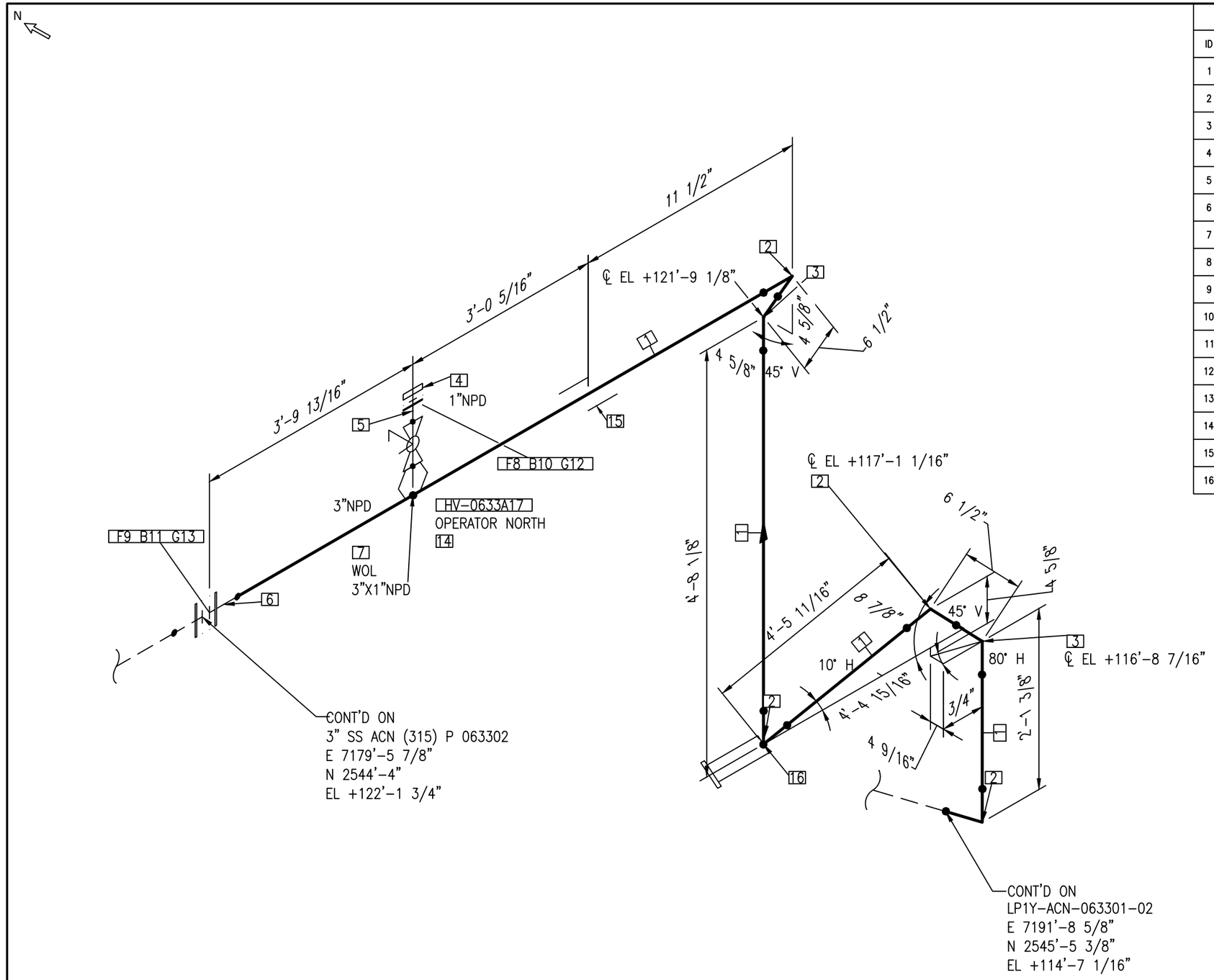

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	16'-8"	3"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	4	3"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
3	2	3"	ELL 45 SCH 10S 316/316L SS A403-W	5450712
4	1	1"	FLG BLIND 150# RF 316/316L SS	5616520
5	1	1"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
6	1	3"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
7	1	3"X1"	PIPET SCH 10S 316/316L SS	5510132
8	1	1"	FLG LAP JOINT 150# 316 SS	5599578L
9	1	3"	FLG LAP JOINT 150# 316 SS	5599578L
10	4	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
11	4	5/8"X2 3/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
12	1	1"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
13	1	3"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
14	1	1"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
15	1	3"	5CI (ISOLATION CRADLE)	5CI-03
16	1	3"	5DS1 (DUMMY SUPPORT) L=1'-1"	5DS1-03

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

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7. PIPE SUPPORTS INDICATED ON THE DRAWING MAY BE PART OF AN ASSEMBLY ON SEVERAL ISOMETRICS. REFER TO THE 3D MODEL FOR THE COMPLETE PIPE SUPPORT ASSEMBLY.
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9. THE BILL OF MATERIAL IS FOR REFERENCE ONLY.
10. SLOPE EQUIVALENTS: 1:27 = 7/16" PER FT, 1:96 = 1/8" PER FT, 1:120 = 1/10" PER FT, 1:192 = 1/16" PER FT, 1:480 = 1/40" PER FT.

NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	ACN
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	P
HYGIENIC: NO	IMPACT: YES

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	09-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3" SS ACN (315) P 063301

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000063	LP1Y-ACN-063301-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 09-MAR-2026

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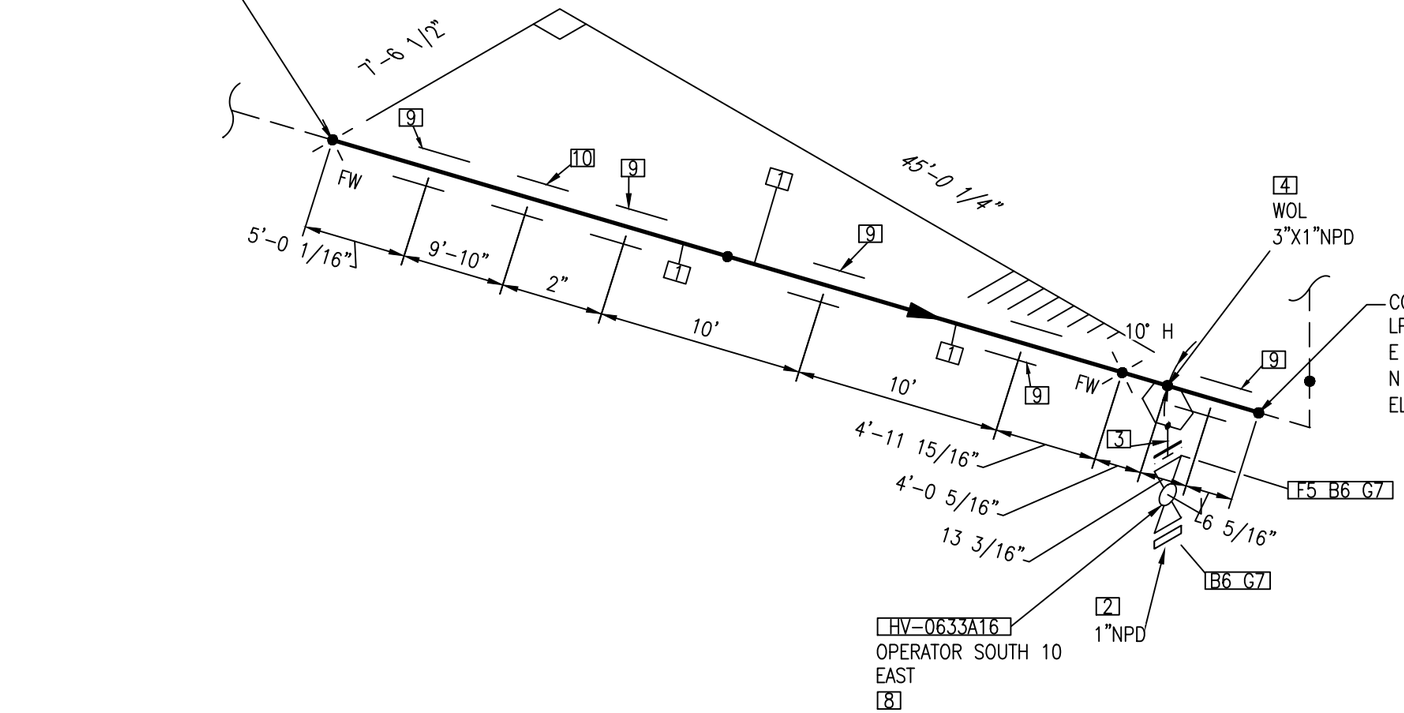

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	45'-8"	3"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	1	1"	FLG BLIND 150# RF 316/316L SS	5616520
3	1	1"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
4	1	3"X1"	PIPET SCH 10S 316/316L SS	5510132
5	1	1"	FLG LAP JOINT 150# 316 SS	5599578L
6	8	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
7	2	1"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
8	1	1"	BALL 150# RF 316SS TFE FP HNDL OP	5352193L1
9	5	3"	5CI (ISOLATION CRADLE)	5CI-03
10	1	3"	5UG (U-BOLT GUIDE)	5UG-03

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	ACN
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	P
HYGIENIC: NO	IMPACT: YES

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	09-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3" SS ACN (315) P 063301

REFERENCE P&ID NUMBER
LP1Y-DI-00000063

ISOMETRIC DRAWING NUMBER
LP1Y-ACN-063301-02

REV
0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 09-MAR-2026

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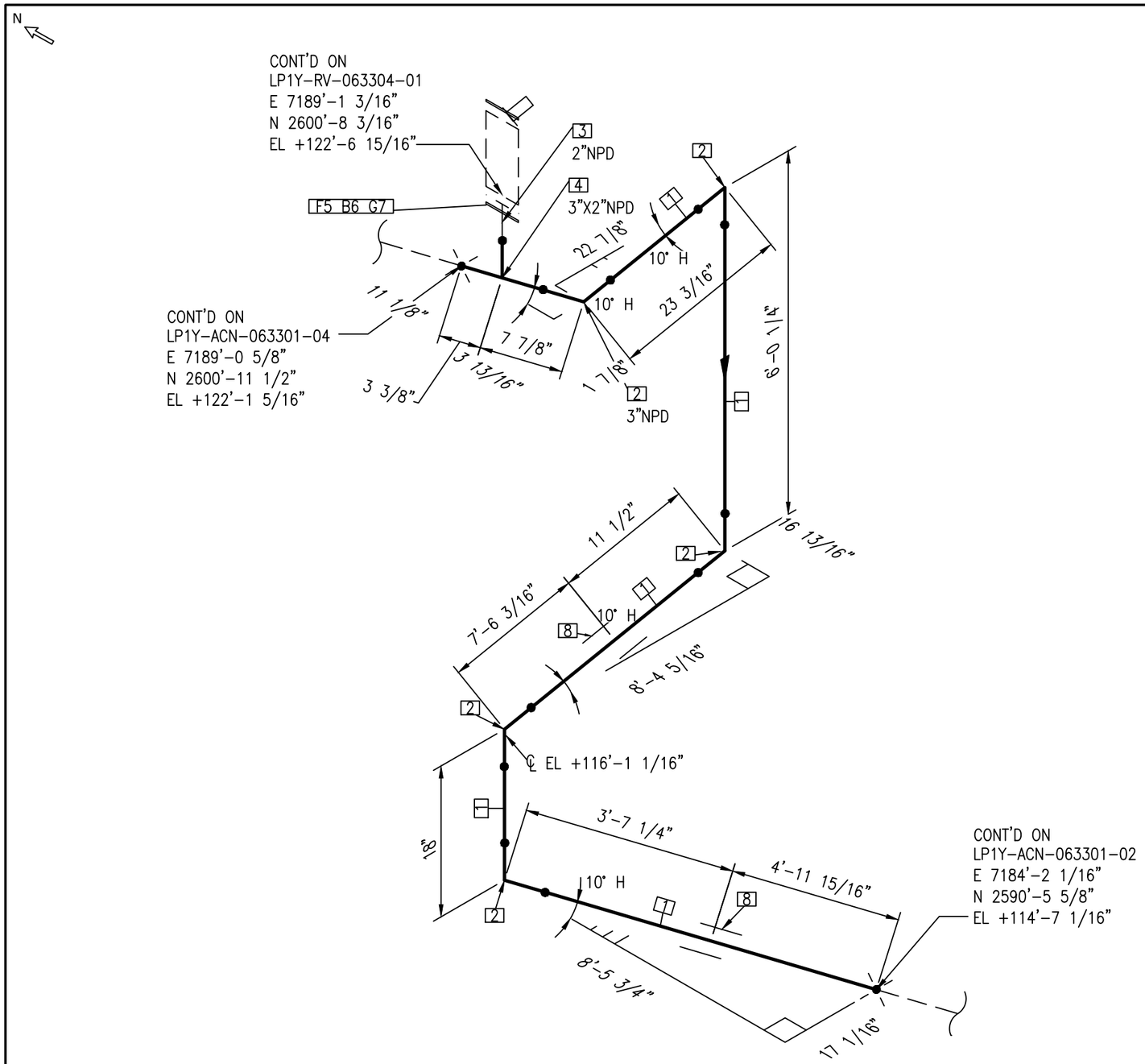

BILL OF MATERIALS

ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	23'-2"	3"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	5	3"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
3	1	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
4	1	3"X2"	TEE RED SCH 10S 316/316L SS A403-W	5450714
5	1	2"	FLG LAP JOINT 150# 316 SS	5599578L
6	4	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
7	1	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
8	2	3"	5CI (ISOLATION CRADLE)	5CI-03

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	ACN
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	P
HYGIENIC: NO	IMPACT: YES

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	09-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3" SS ACN (315) P 063301

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000063	LP1Y-ACN-063301-03	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 09-MAR-2026

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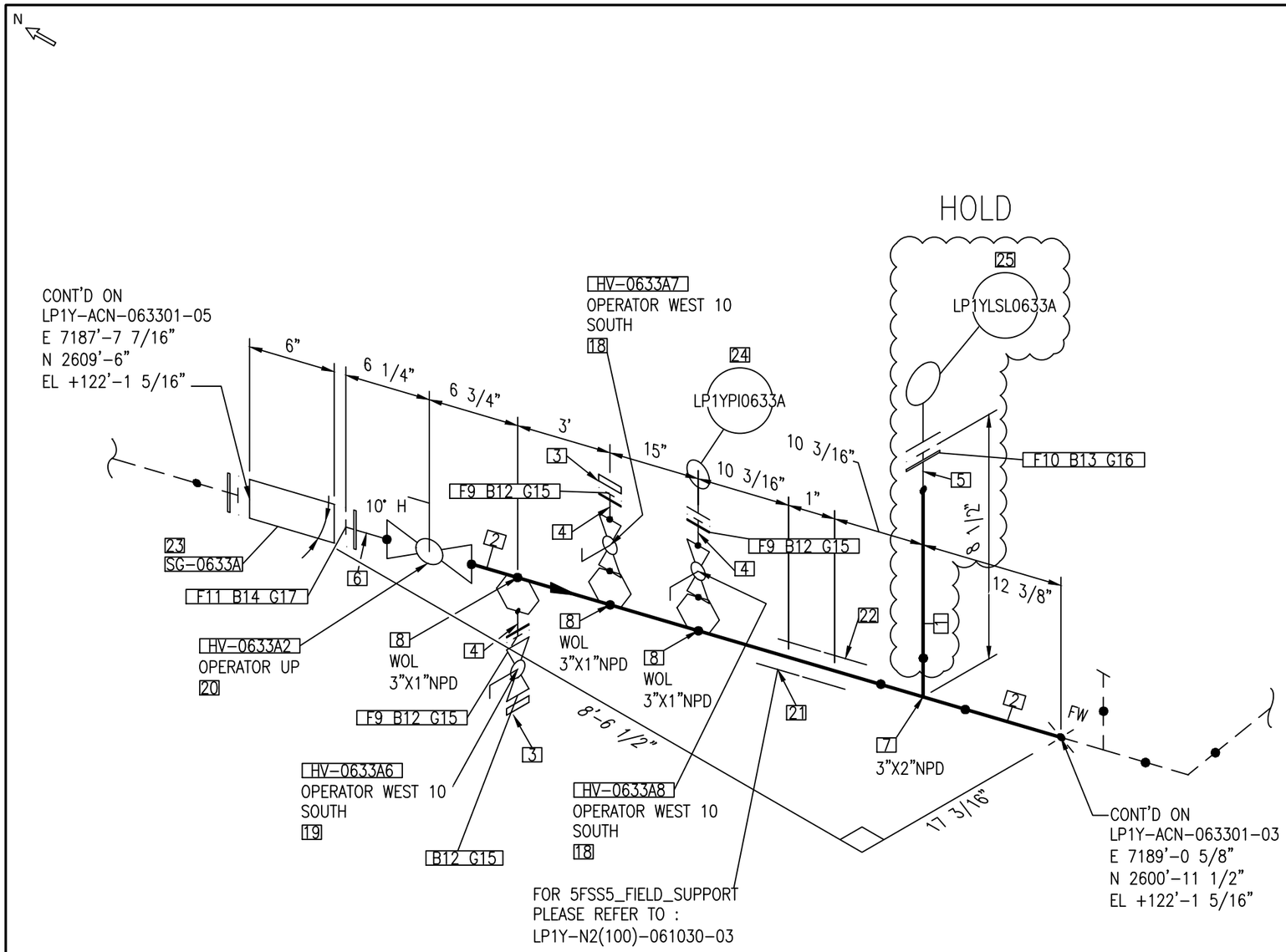

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	3"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	6'-9"	3"	PIPE SCH 10S ERW 316/316L SS A312	5368751
3	2	1"	FLG BLIND 150# RF 316/316L SS	5616520
4	3	1"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
5	1	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
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7	1	3"X2"	TEE RED SCH 10S 316/316L SS A403-W	5450714
8	3	3"X1"	PIPET SCH 10S 316/316L SS	5510132
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12	16	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
13	4	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
14	4	5/8"X2 3/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
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18	2	1"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
19	1	1"	BALL 150# RF 316SS TFE FP HNDL OP	5352193L1
20	1	3"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
21	1	3"	5CI (ISOLATION CRADLE)	5CI-03
22	1	3"	5UG (U-BOLT GUIDE)	5UG-03
23	1	3"	JACOBY TARBOX (F-830HP-PFA) SIGHT GLASS	LP1YSG0633A
24	1	1"	PRESSURE GAUGE	LP1YPI0633A
25	1	2"	LSL	LP1YLSL0633A

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	ACN
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	P
HYGIENIC: NO	IMPACT: YES

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	16-MAR-2026	DRW	SBR	N/A	JHN / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3" SS ACN (315) P 063301

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000063	LP1Y-ACN-063301-04	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 16-MAR-2026

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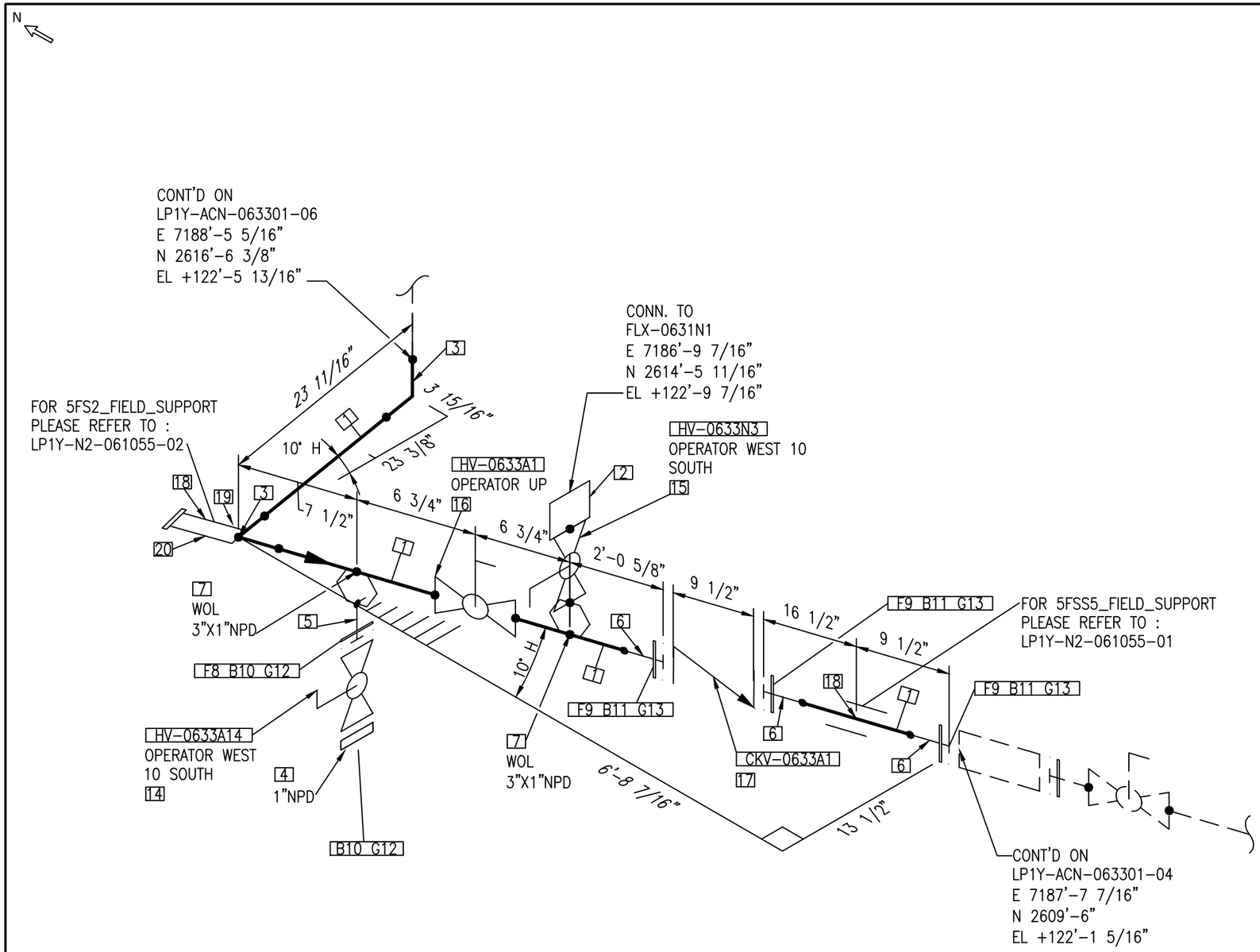

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	5'-7"	3"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	1	1"	BUTTWELD CAMLOCK CONNECTION UNIT	LP1YCCU0633N2
3	2	3"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
4	1	1"	FLG BLIND 150# RF 316/316L SS	5616520
5	1	1"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
6	3	3"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
7	2	3"X1"	PIPET SCH 10S 316/316L SS	5510132
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16	1	3"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
17	1	3"	CHECK 150# RF 316 SS SWING	5257711L
18	2	3"	5CI (ISOLATION CRADLE)	5CI-03
20	1	3"	5UG (U-BOLT GUIDE)	5DS1-03
19	1	3"	5DS1 (DUMMY SUPPORT) L=1'-6"	5DS1-03

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	ACN
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	P
HYGIENIC: NO	IMPACT: YES

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3" SS ACN (315) P 063301

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000063	LP1Y-ACN-063301-05	0

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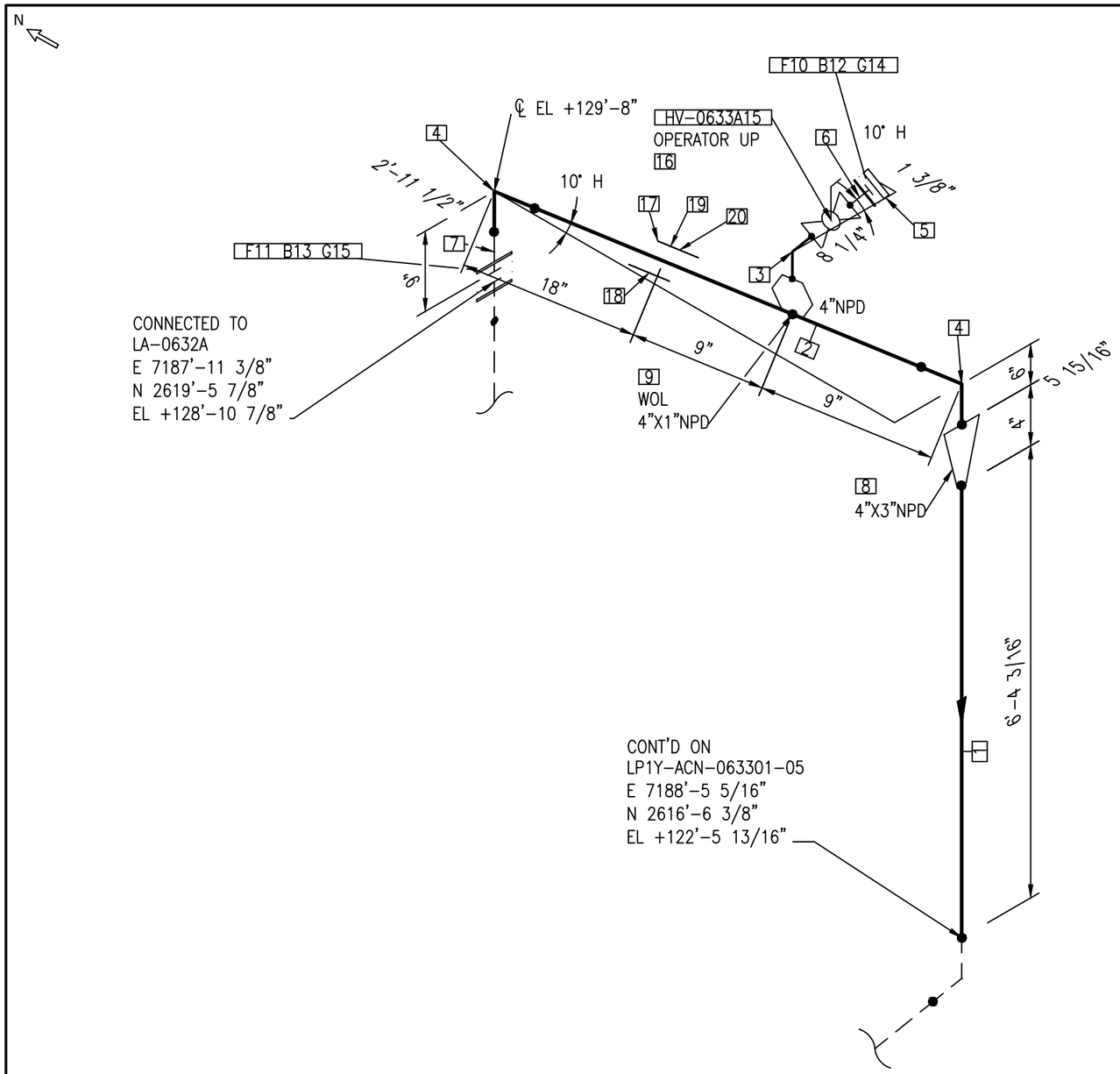

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	6'-5"	3"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	2'-1"	4"	PIPE SCH 10S ERW 316/316L SS A312	5368751
3	1	1"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
4	2	4"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
5	1	1"	FLG BLIND 150# RF 316/316L SS	5616520
6	1	1"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
7	1	4"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
8	1	4"X3"	REDUCER CONC SCH 10S 316/316L SS A403-W	5450717
9	1	4"X1"	PIPET SCH 10S 316/316L SS	5510132
10	1	1"	FLG LAP JOINT 150# 316 SS	5599578L
11	1	4"	FLG LAP JOINT 150# 316 SS	5599578L
12	4	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
13	8	5/8"X2 3/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
14	1	1"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
15	1	4"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
16	1	1"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
17	1	4"	5CI (ISOLATION CRADLE)	5CI-04
18	1	4"	5FS1 (FIELD SUPPORT)	5FS1S2
19	1	4"	5UG (U-BOLT GUIDE)	5UG-04
20	1	4"	5DA1 (DIRECTIONAL ANCHOR)	5DA1-M

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	17-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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7. PIPE SUPPORTS INDICATED ON THE DRAWING MAY BE PART OF AN ASSEMBLY ON SEVERAL ISOMETRICS. REFER TO THE 3D MODEL FOR THE COMPLETE PIPE SUPPORT ASSEMBLY.
8. MECHANICAL CONTRACTOR SHALL NOTE ALL DESIGN CHANGES ON THIS DRAWING AND RETURN IT TO THE CONSTRUCTION WPO FOR AS-BUILT RECORDS.
9. THE BILL OF MATERIAL IS FOR REFERENCE ONLY.
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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	ACN
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	P
HYGIENIC: NO	IMPACT: YES

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3" SS ACN (315) P 063301

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000063	LP1Y-ACN-063301-06	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 17-MAR-2026

Tim Trice

Planned Controlled Document

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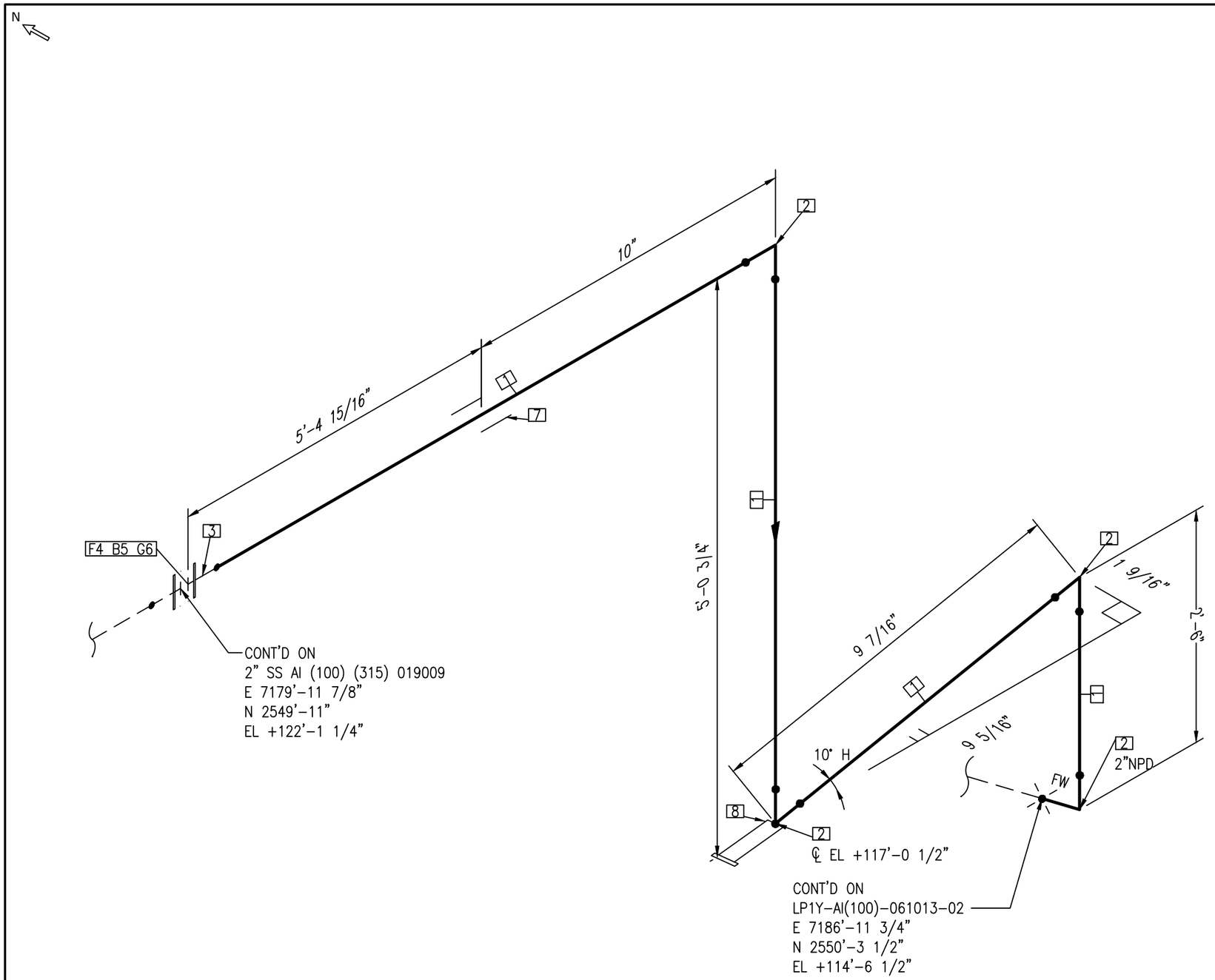

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	12'-8"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	4	2"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
3	1	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
4	1	2"	FLG LAP JOINT 150# 316 SS	5599578L
5	4	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
6	1	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
7	1	2"	5CI (ISOLATION CRADLE)	5CI-02
8	1	2"	5DS1 (DUMMY SUPPORT) L=10"	5DS1-02

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	02-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	AI(100)
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS AI(100) (315) 061013

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-AI(100)-061013-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 02-MAR-2026

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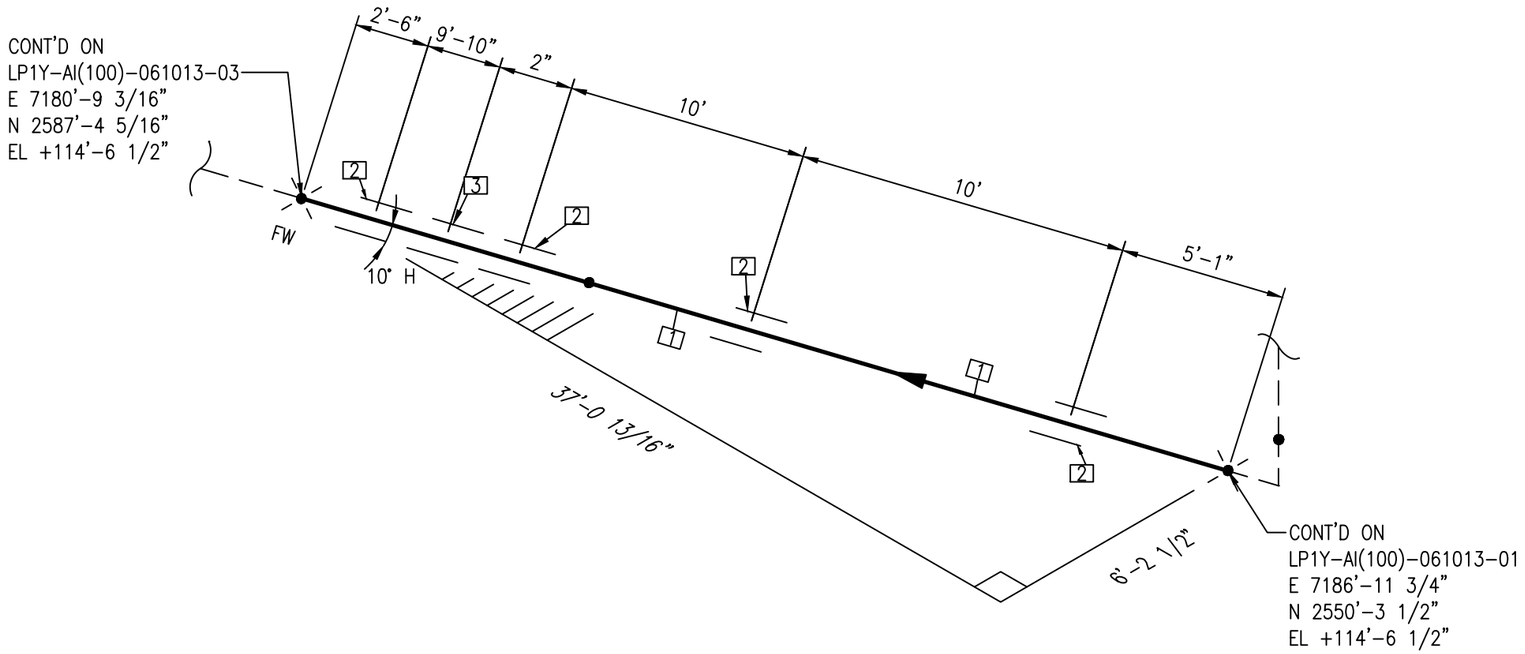
N

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	37'-7"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	4	2"	5CI (ISOLATION CRADLE)	5CI-02
3	1	2"	5UG (U-BOLT GUIDE)	5UG-02



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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	AI(100)
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	02-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS AI(100) (315) 061013

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-AI(100)-061013-02	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 02-MAR-2026

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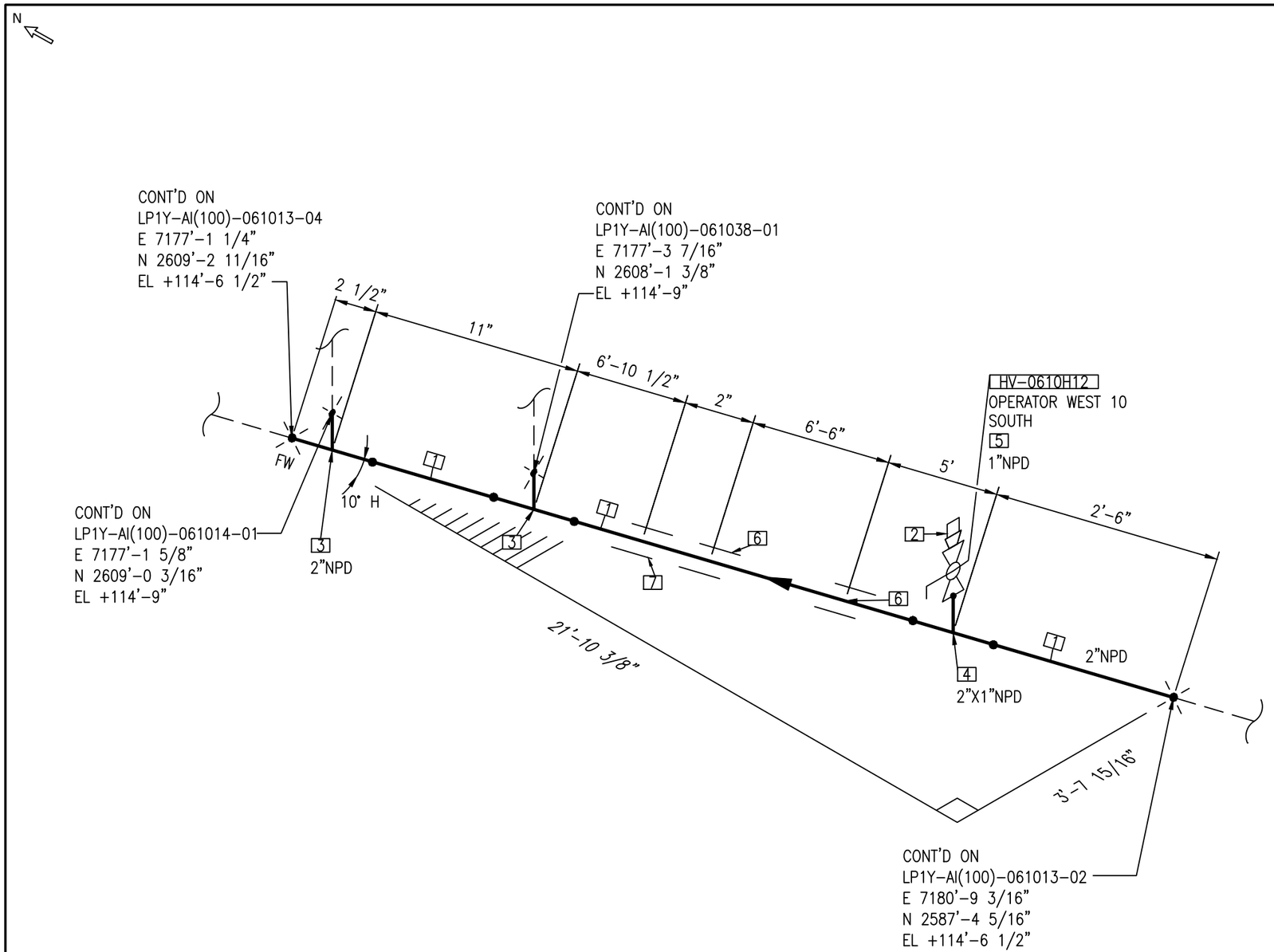

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	21'	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	1	1"	PLUG ROUND HEAD SCRD 316/316L SS	5498383
3	2	2"	TEE SCH 10S 316/316L SS A403-W	5450715
4	1	2"X1"	TEE RED SCH 10S 316/316L SS A403-W	5450714
5	1	1"	BALL 1000# CWP BW X FNPT 316SS TFE FP HNDL OP	5352192L8
6	2	2"	5CI (ISOLATION CRADLE)	5CI-02
7	1	2"	5UG (U-BOLT GUIDE)	5UG-02

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	02-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	AI(100)
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS AI(100) (315) 061013

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-AI(100)-061013-03	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 02-MAR-2026

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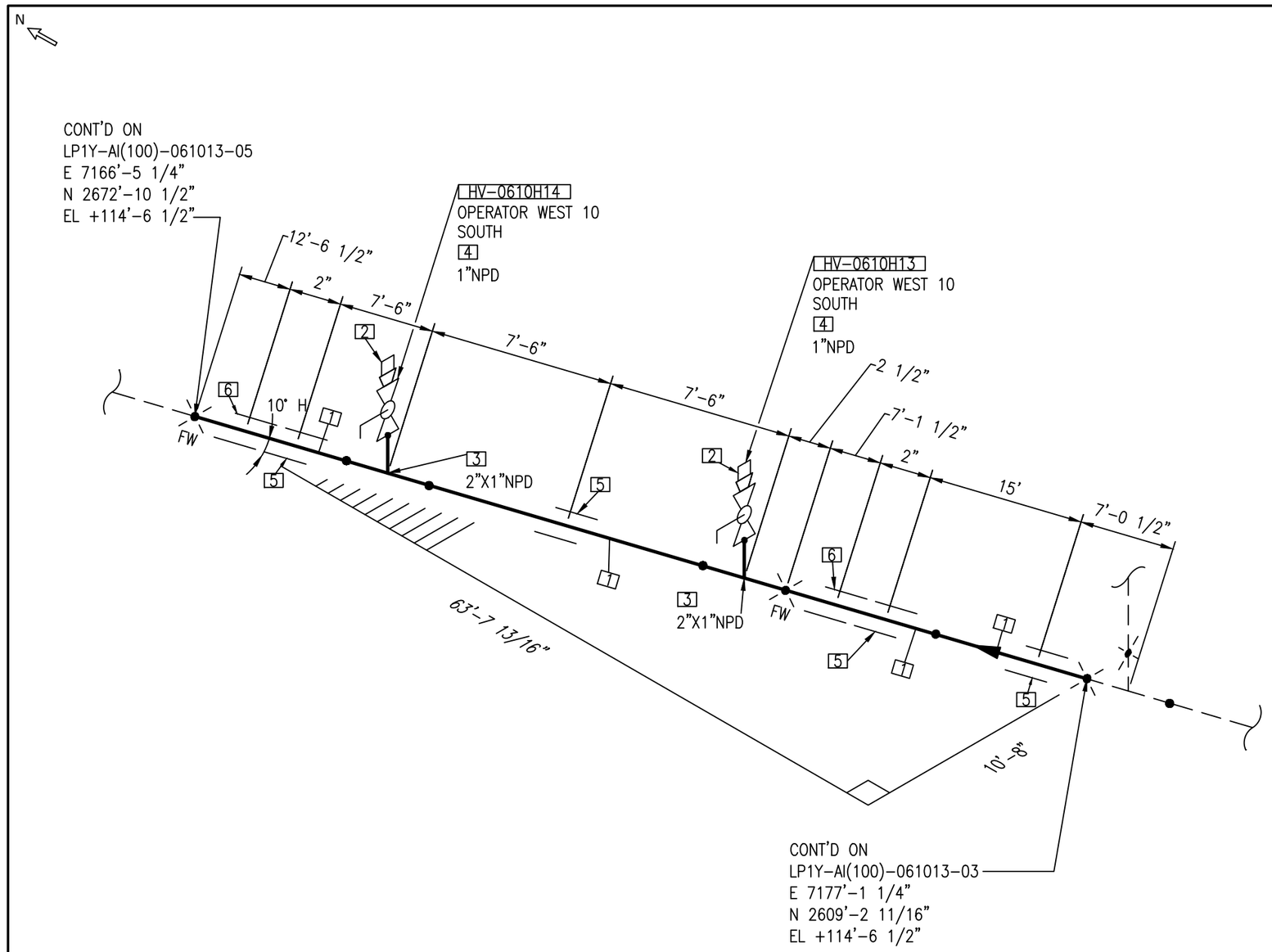

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	63'-9"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	2	1"	PLUG ROUND HEAD SCRD 316/316L SS	5498383
3	2	2"X1"	TEE RED SCH 10S 316/316L SS A403-W	5450714
4	2	1"	BALL 1000# CWP BW X FNPT 316SS TFE FP HNDL OP	5352192L8
5	4	2"	5CI (ISOLATION CRADLE)	5CI-02
6	2	2"	5UG (U-BOLT GUIDE)	5UG-02

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	AI(100)
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	02-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS AI(100) (315) 061013

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-AI(100)-061013-04	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 02-MAR-2026

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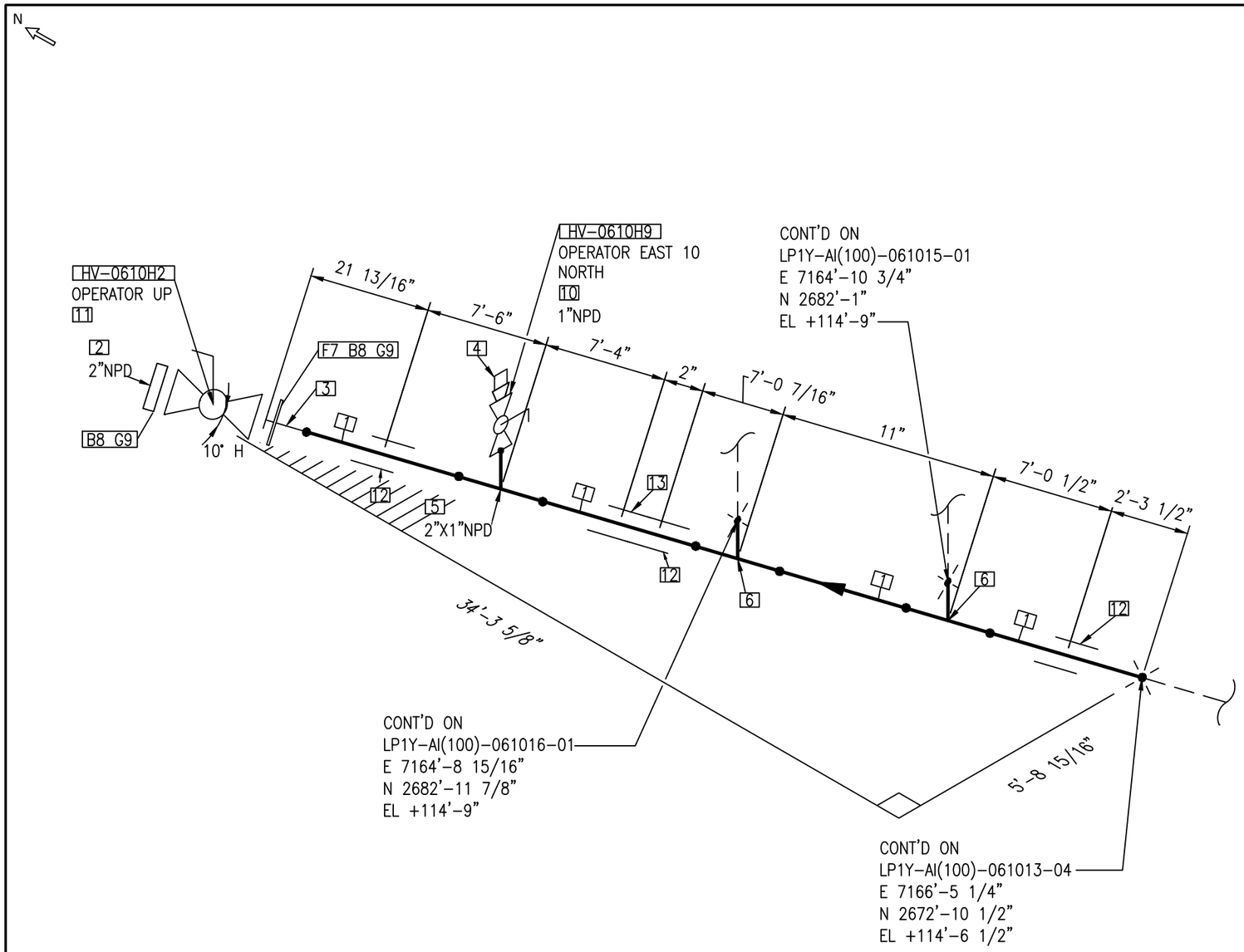

BILL OF MATERIALS

ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	32'-8"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	1	2"	FLG BLIND 150# RF 316/316L SS	5616520
3	1	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
4	1	1"	PLUG ROUND HEAD SCRD 316/316L SS	5498383
5	1	2"X1"	TEE RED SCH 10S 316/316L SS A403-W	5450714
6	2	2"	TEE SCH 10S 316/316L SS A403-W	5450715
7	1	2"	FLG LAP JOINT 150# 316 SS	5599578L
8	8	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
9	2	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
10	1	1"	BALL 1000# CWP BW X FNPT 316SS TFE FP HNDL OP	5352192L8
11	1	2"	BALL 150# RF 316SS TFE FP HNDL OP	5352194L1
12	3	2"	5CI (ISOLATION CRADLE)	5CI-02
13	1	2"	5UG (U-BOLT GUIDE)	5UG-02

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	AI(100)
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	02-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS AI(100) (315) 061013

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-AI(100)-061013-05	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 02-MAR-2026

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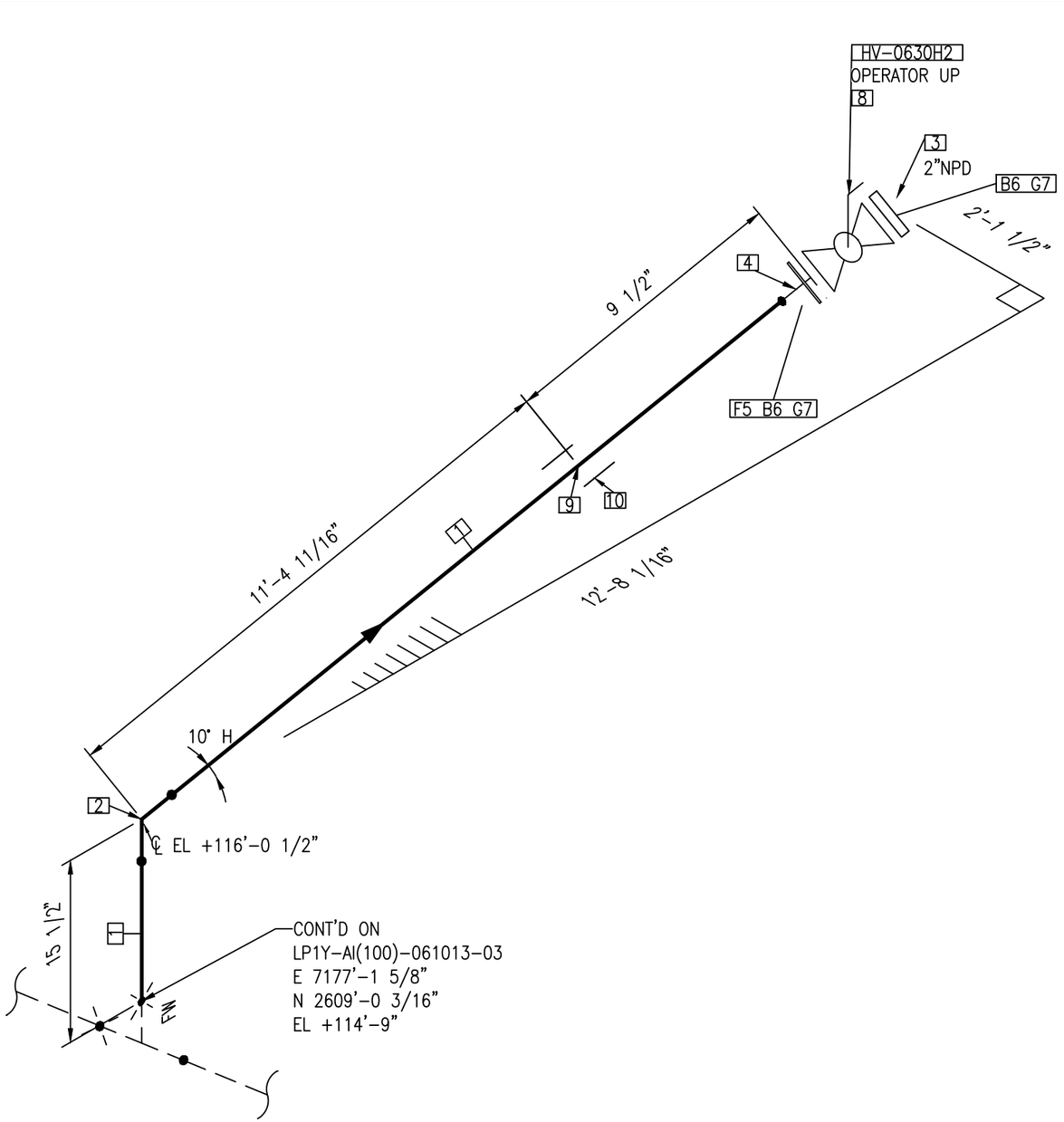

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	12'-10"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	1	2"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
3	1	2"	FLG BLIND 150# RF 316/316L SS	5616520
4	1	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
5	1	2"	FLG LAP JOINT 150# 316 SS	5599578L
6	8	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
7	2	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
8	1	2"	BALL 150# RF 316SS TFE FP HNDL OP	5352194L1
9	1	2"	5CI (ISOLATION CRADLE)	5CI-02
10	1	2"	5UG (U-BOLT GUIDE)	5UG-02

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	02-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	AI(100)
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS AI(100) (315) 061014

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-AI(100)-061014-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 02-MAR-2026

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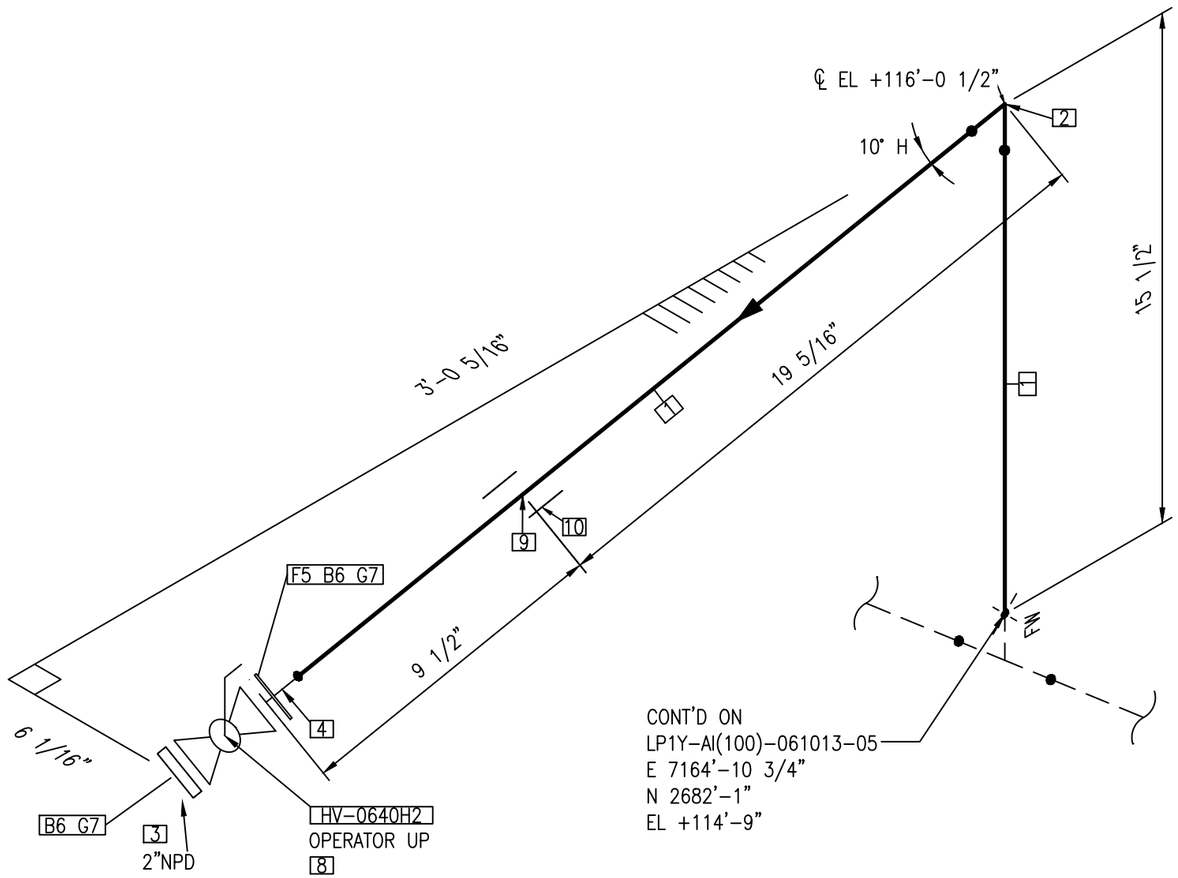

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	3'	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	1	2"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
3	1	2"	FLG BLIND 150# RF 316/316L SS	5616520
4	1	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
5	1	2"	FLG LAP JOINT 150# 316 SS	5599578L
6	8	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
7	2	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
8	1	2"	BALL 150# RF 316SS TFE FP HNDL OP	5352194L1
9	1	2"	5CI (ISOLATION CRADLE)	5CI-02
10	1	2"	5UG (U-BOLT GUIDE)	5UG-02

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	02-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	AI(100)
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS AI(100) (315) 061015

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-AI(100)-061015-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 02-MAR-2026

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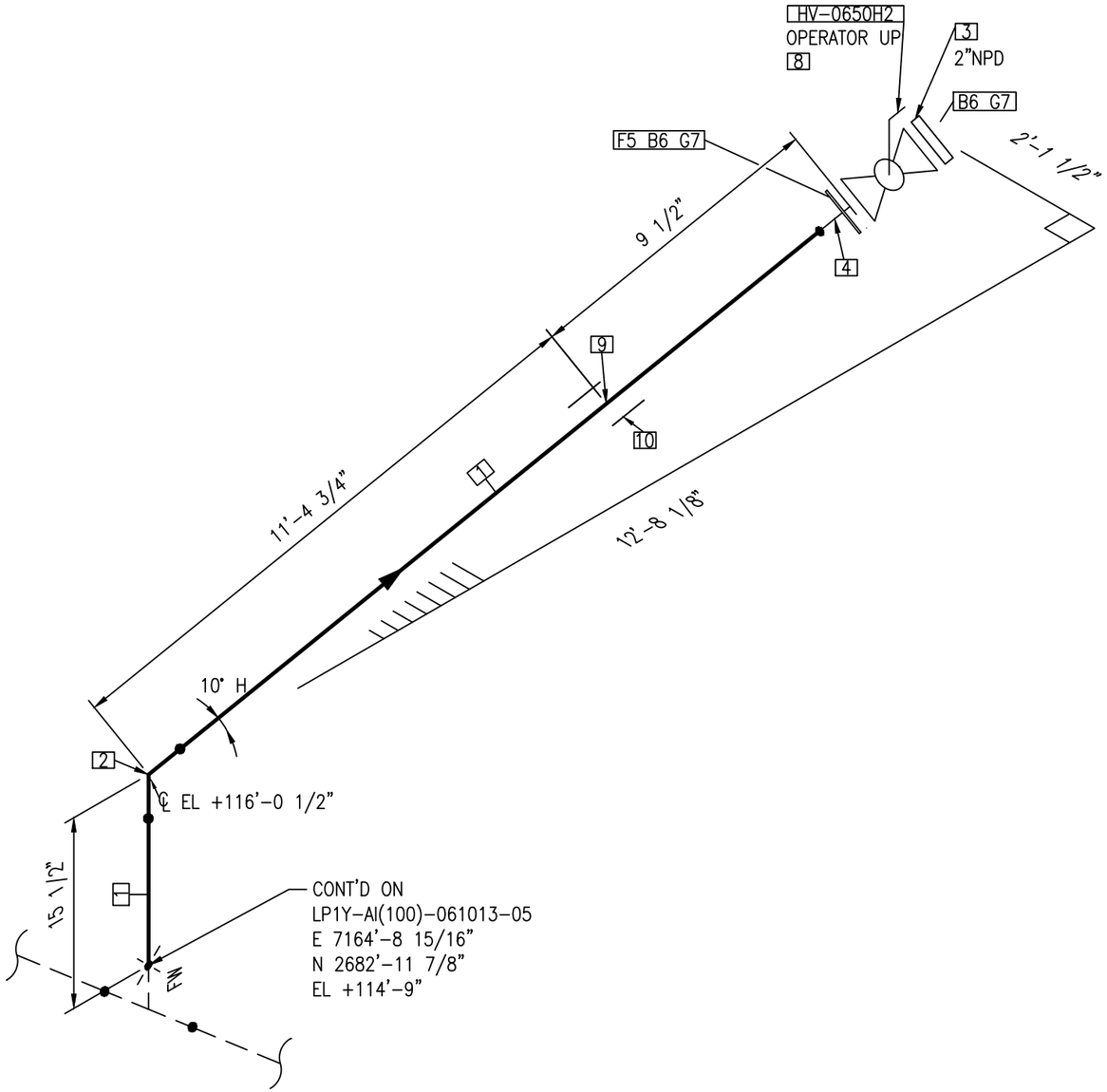

BILL OF MATERIALS				
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3	1	2"	FLG BLIND 150# RF 316/316L SS	5616520
4	1	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
5	1	2"	FLG LAP JOINT 150# 316 SS	5599578L
6	8	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
7	2	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
8	1	2"	BALL 150# RF 316SS TFE FP HNDL OP	5352194L1
9	1	2"	5CI (ISOLATION CRADLE)	5CI-02
10	1	2"	5UG (U-BOLT GUIDE)	5UG-02

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	02-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	AI(100)
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS AI(100) (315) 061016

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-AI(100)-061016-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 02-MAR-2026

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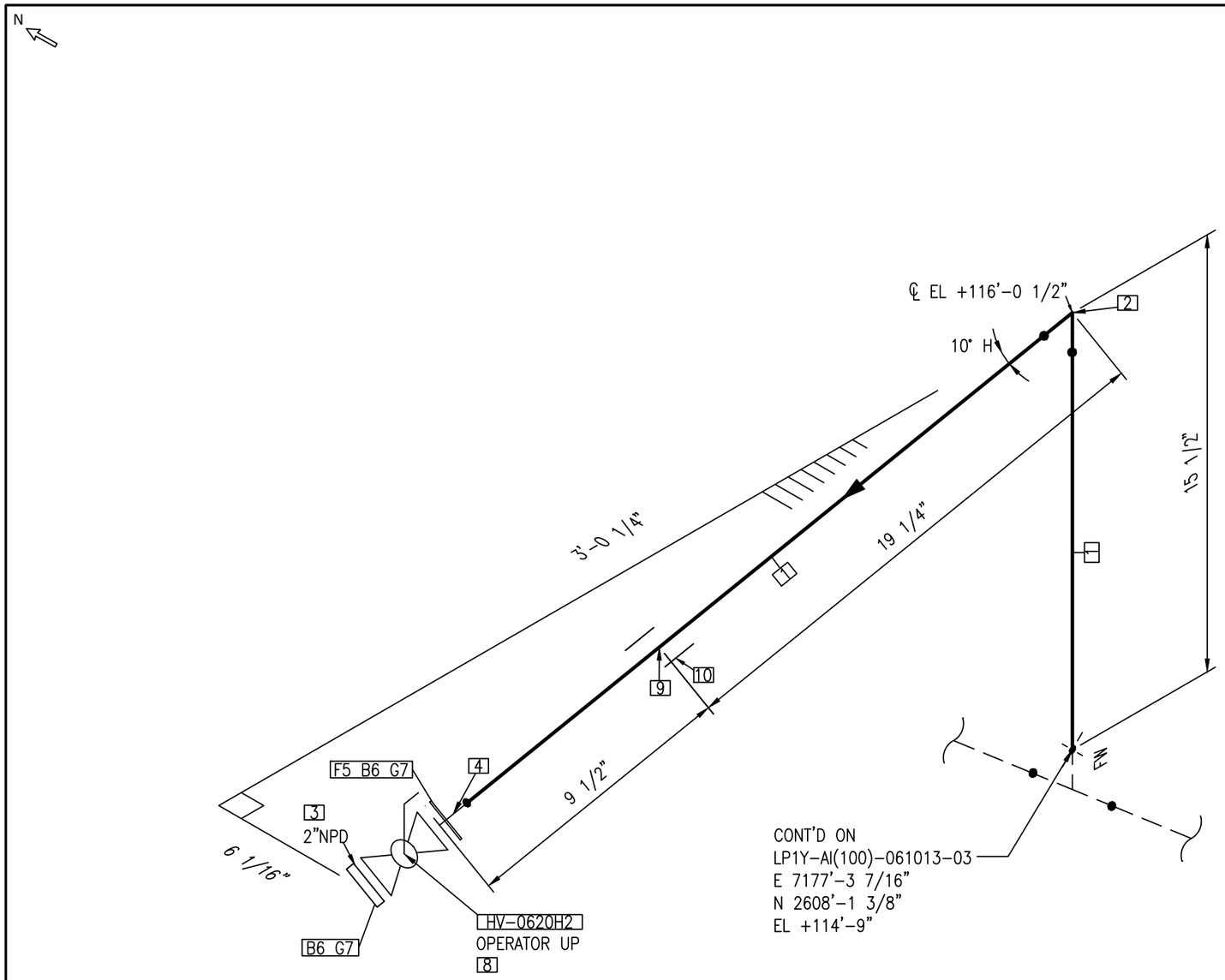

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5	1	2"	FLG LAP JOINT 150# 316 SS	5599578L
6	8	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
7	2	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
8	1	2"	BALL 150# RF 316SS TFE FP HNDL OP	5352194L1
9	1	2"	5CI (ISOLATION CRADLE)	5CI-02
10	1	2"	5UG (U-BOLT GUIDE)	5UG-02

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	02-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	AI(100)
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS AI(100) (315) 061038

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-AI(100)-061038-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 02-MAR-2026

Tim Trice

Planned Controlled Document

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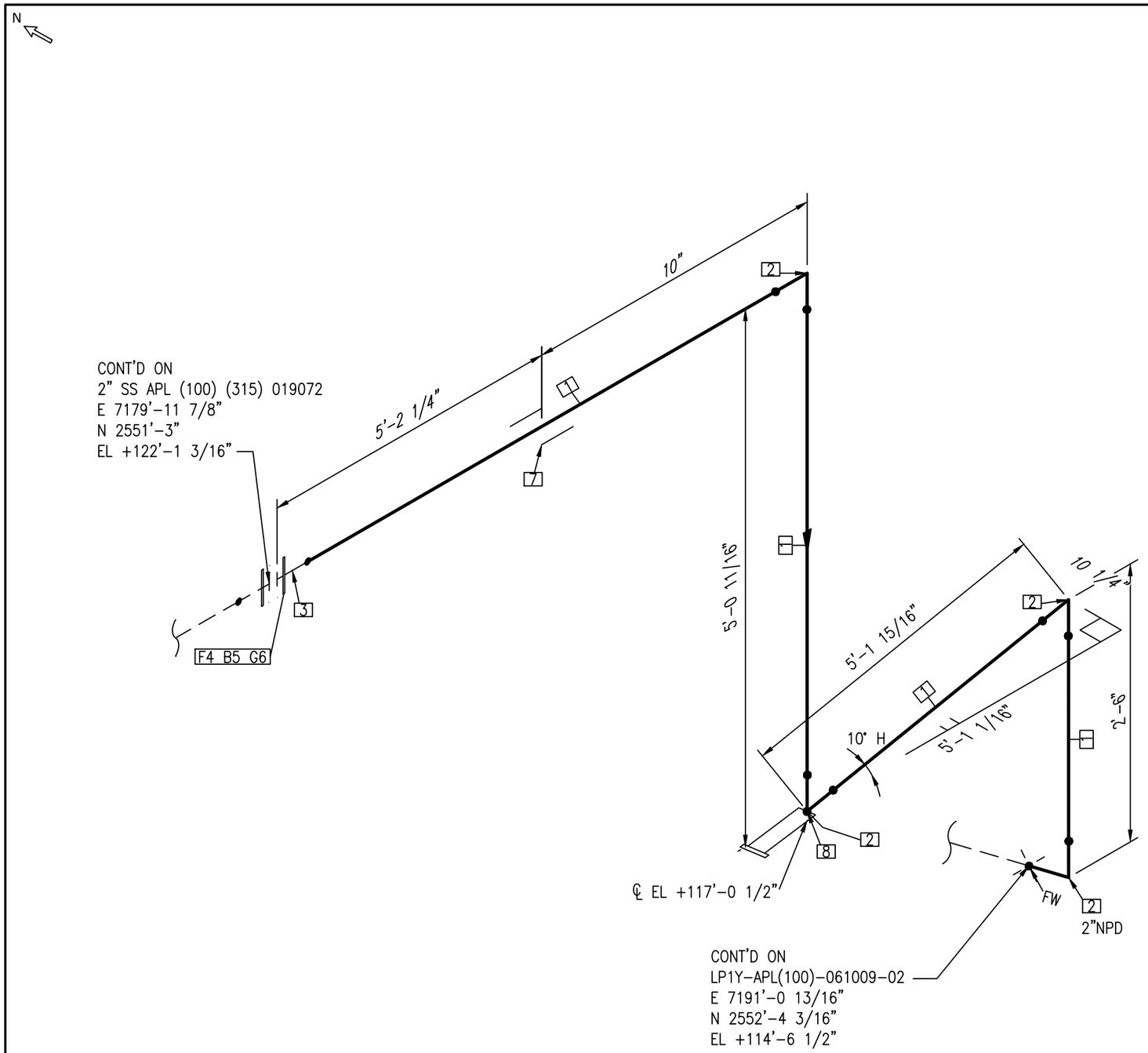

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	16'-10"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	4	2"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
3	1	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
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5	4	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
6	1	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
7	1	2"	5CI (ISOLATION CRADLE)	5CI-02
8	1	2"	5DS1 (DUMMY SUPPORT) L=10"	5DS1-02

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By dac54131 at Apr 10, 2026

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8. MECHANICAL CONTRACTOR SHALL NOTE ALL DESIGN CHANGES ON THIS DRAWING AND RETURN IT TO THE CONSTRUCTION WPO FOR AS-BUILT RECORDS.
9. THE BILL OF MATERIAL IS FOR REFERENCE ONLY.
10. SLOPE EQUIVALENTS: 1:27 = 7/16" PER FT, 1:96 = 1/8" PER FT, 1:120 = 1/10" PER FT, 1:192 = 1/16" PER FT, 1:480 = 1/40" PER FT.

NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	APL(100)
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	02-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS APL(100) (315) 061009

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-APL(100)-061009-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 02-MAR-2026

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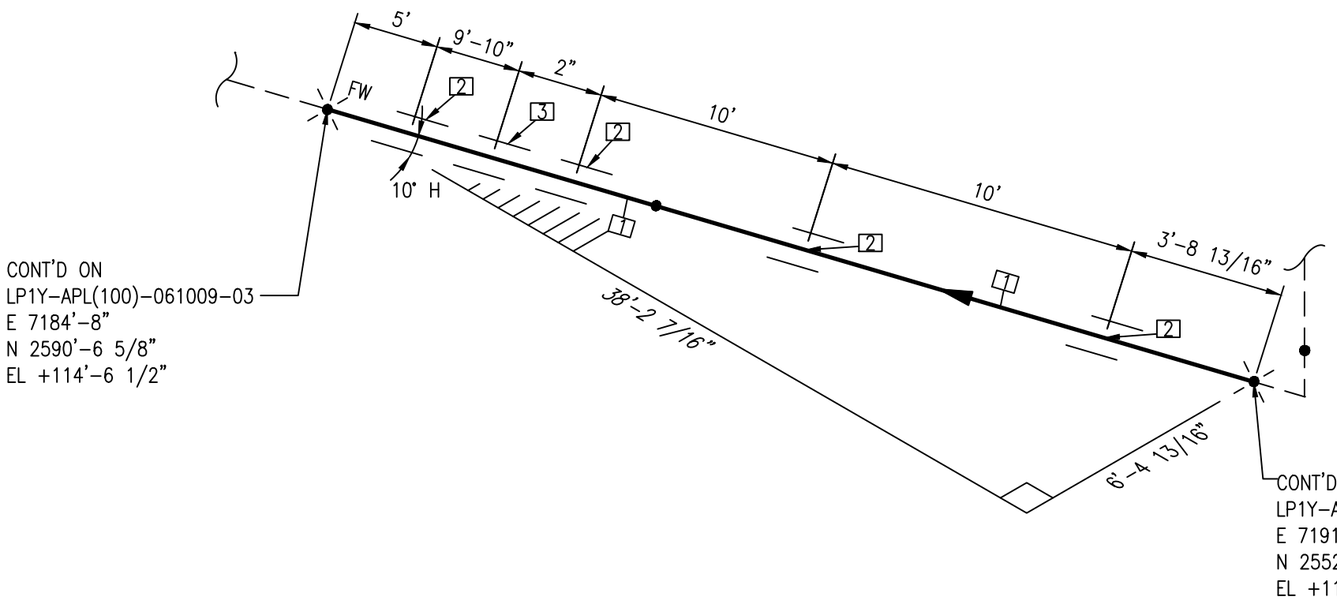
N



BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	38'-9"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	4	2"	5CI (ISOLATION CRADLE)	5CI-02
3	1	2"	5UG (U-BOLT GUIDE)	5UG-02

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	02-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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5. THE MECHANICAL CONTRACTOR IS TO VERIFY ALL DIMENSIONAL DATA WITH ACTUAL FIELD CONDITIONS PRIOR TO INSTALLATION OF THE SYSTEM.
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7. PIPE SUPPORTS INDICATED ON THE DRAWING MAY BE PART OF AN ASSEMBLY ON SEVERAL ISOMETRICS. REFER TO THE 3D MODEL FOR THE COMPLETE PIPE SUPPORT ASSEMBLY.
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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	APL(100)
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS APL(100) (315) 061009

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-APL(100)-061009-02	0

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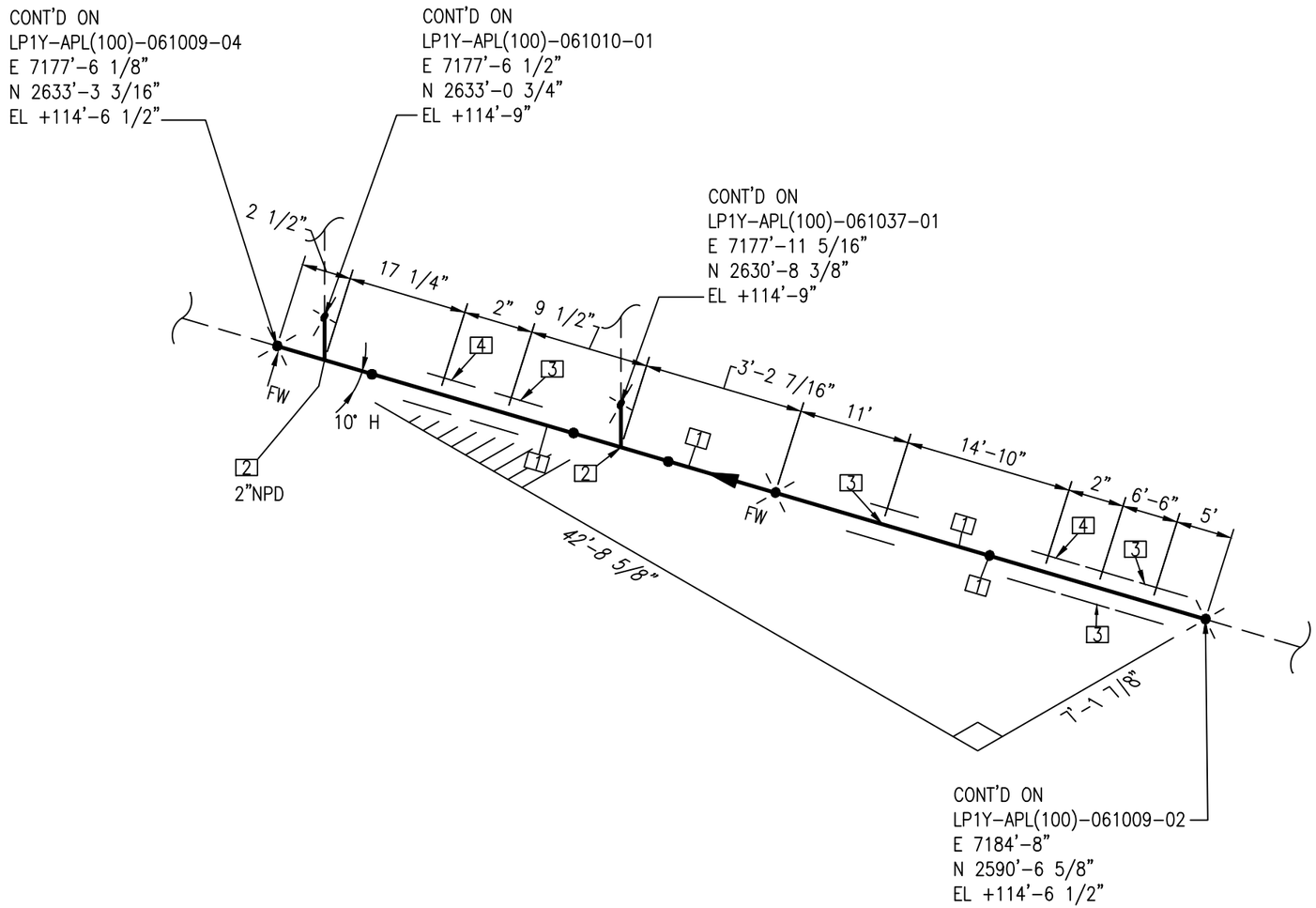

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	42'-6"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	2	2"	TEE SCH 10S 316/316L SS A403-W	5450715
3	4	2"	5CI (ISOLATION CRADLE)	5CI-02
4	2	2"	5UG (U-BOLT GUIDE)	5UG-02

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0	ISSUE FOR CONSTRUCTION	02-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	APL(100)
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS APL(100) (315) 061009

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-APL(100)-061009-03	0

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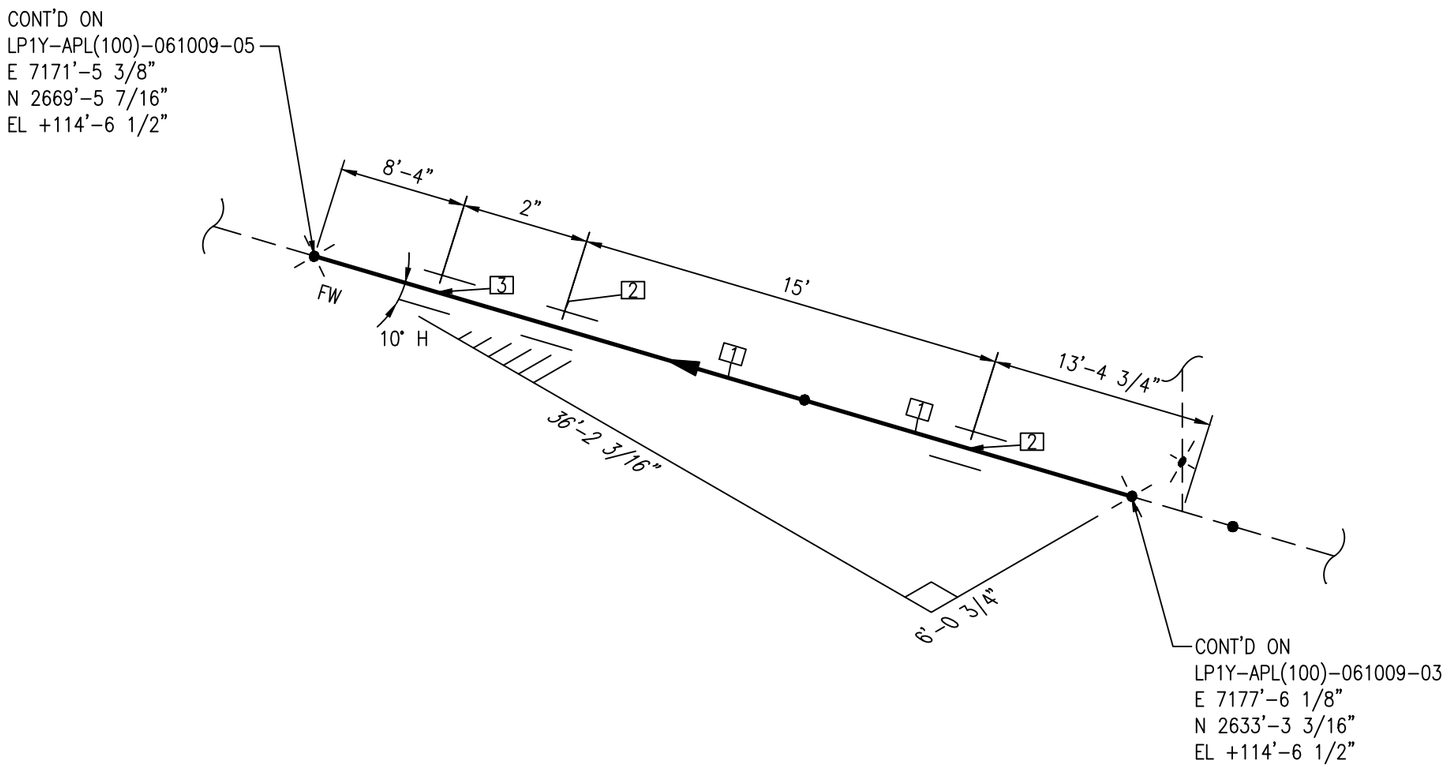
N

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	36'-9"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	2	2"	5CI (ISOLATION CRADLE)	5CI-02
3	1	2"	5UG (U-BOLT GUIDE)	5UG-02



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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	02-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	APL(100)
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS APL(100) (315) 061009

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-APL(100)-061009-04	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 02-MAR-2026

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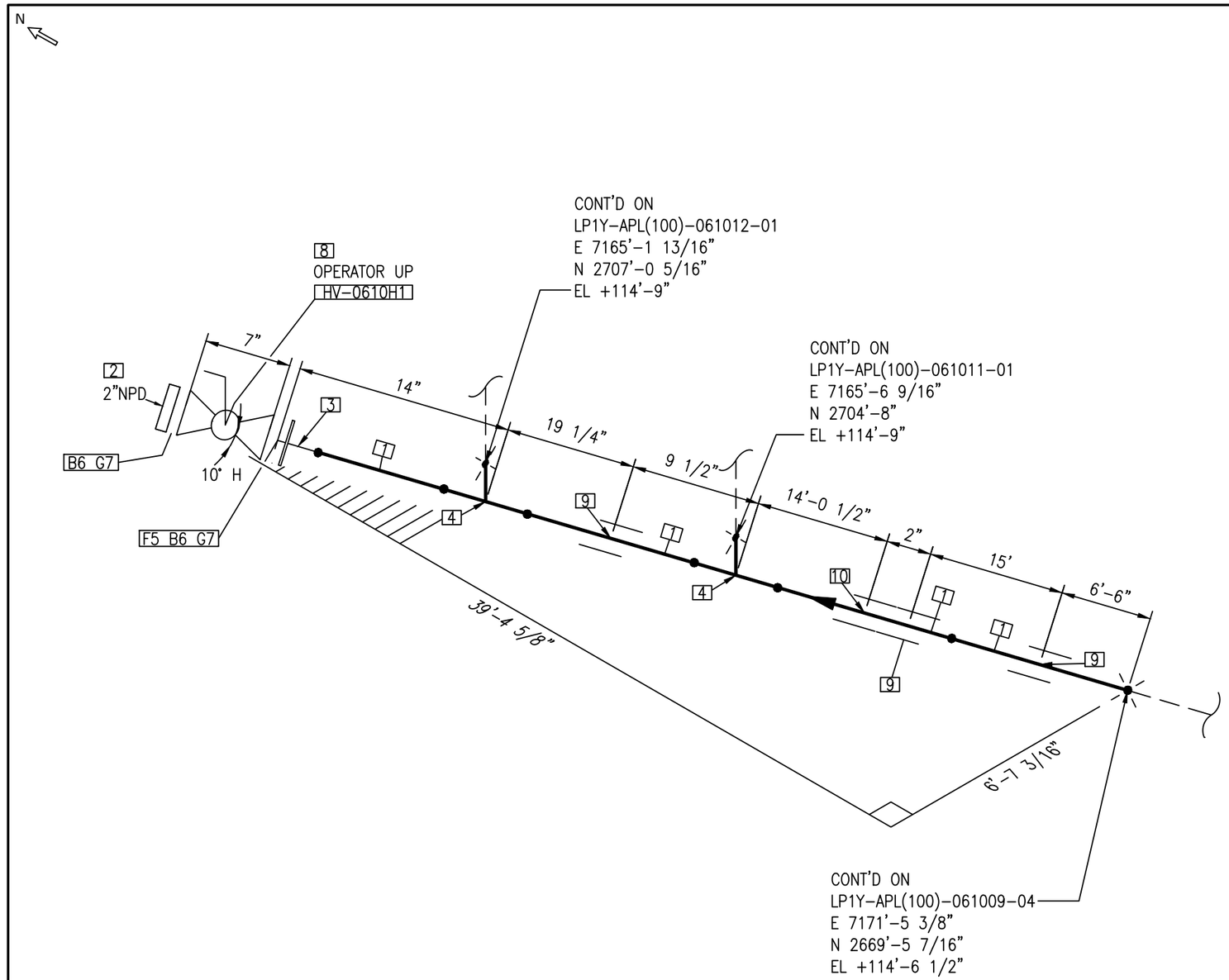

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	38'-3"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	1	2"	FLG BLIND 150# RF 316/316L SS	5616520
3	1	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
4	2	2"	TEE SCH 10S 316/316L SS A403-W	5450715
5	1	2"	FLG LAP JOINT 150# 316 SS	5599578L
6	8	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
7	2	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
8	1	2"	BALL 150# RF 316SS TFE FP HNDL OP	5352194L1
9	3	2"	5CI (ISOLATION CRADLE)	5CI-02
10	1	2"	5UG (U-BOLT GUIDE)	5UG-02

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	APL(100)
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	02-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS APL(100) (315) 061009

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-APL(100)-061009-05	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 02-MAR-2026

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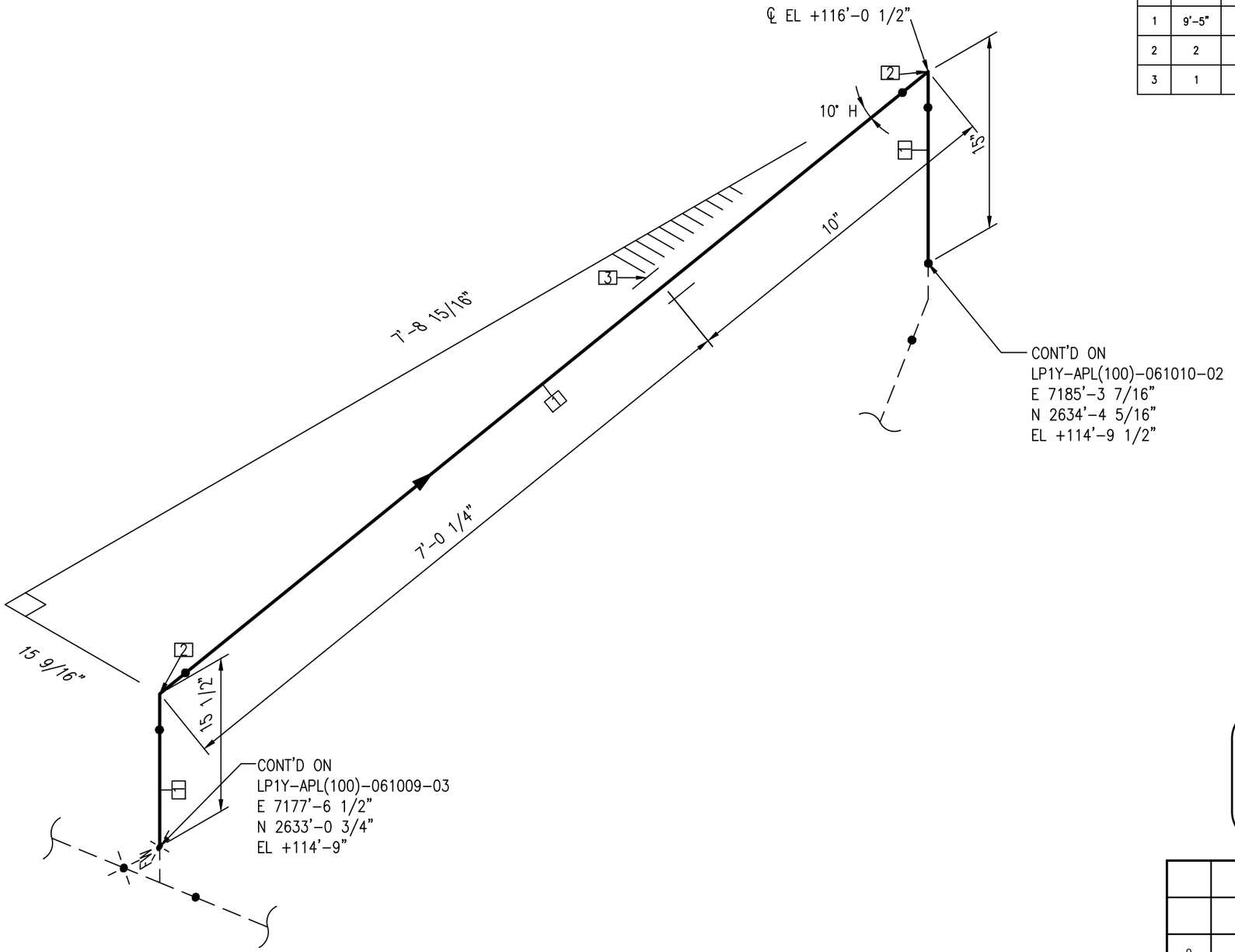

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	9'-5"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	2	2"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
3	1	2"	5CI (ISOLATION CRADLE)	5CI-02

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	02-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	APL(100)
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS APL(100) (315) 061010

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-APL(100)-061010-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 02-MAR-2026

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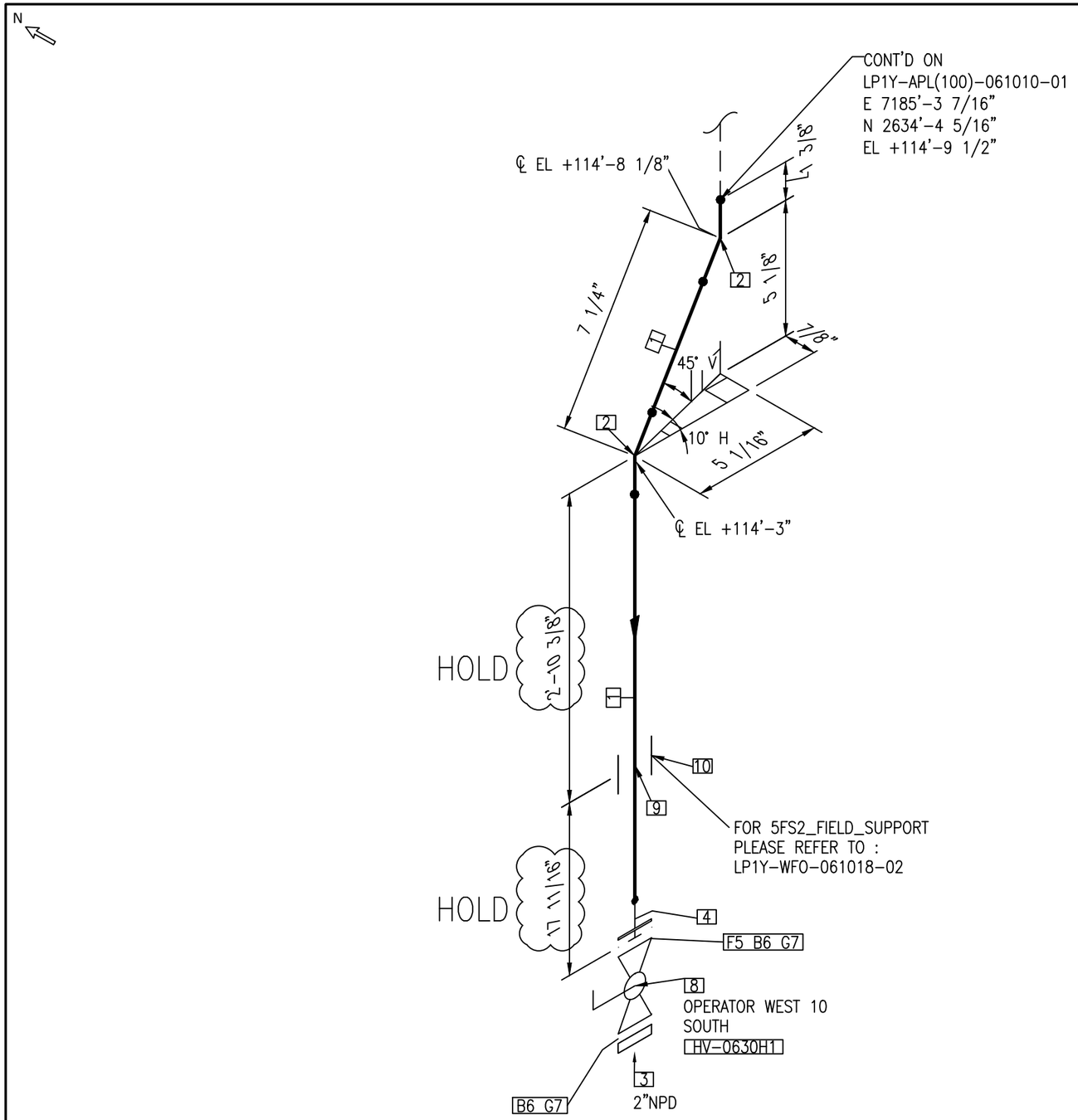

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
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2	2	2"	ELL 45 SCH 10S 316/316L SS A403-W	5450712
3	1	2"	FLG BLIND 150# RF 316/316L SS	5616520
4	1	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
5	1	2"	FLG LAP JOINT 150# 316 SS	5599578L
6	8	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
7	2	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
8	1	2"	BALL 150# RF 316SS TFE FP HNDL OP	5352194L1
9	1	2"	5CI (ISOLATION CRADLE)	5CI-02
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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	APL(100)
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	02-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS APL(100) (315) 061010

REFERENCE P&ID NUMBER
LP1Y-DI-00000061

ISOMETRIC DRAWING NUMBER
LP1Y-APL(100)-061010-02

REV
0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 02-MAR-2026

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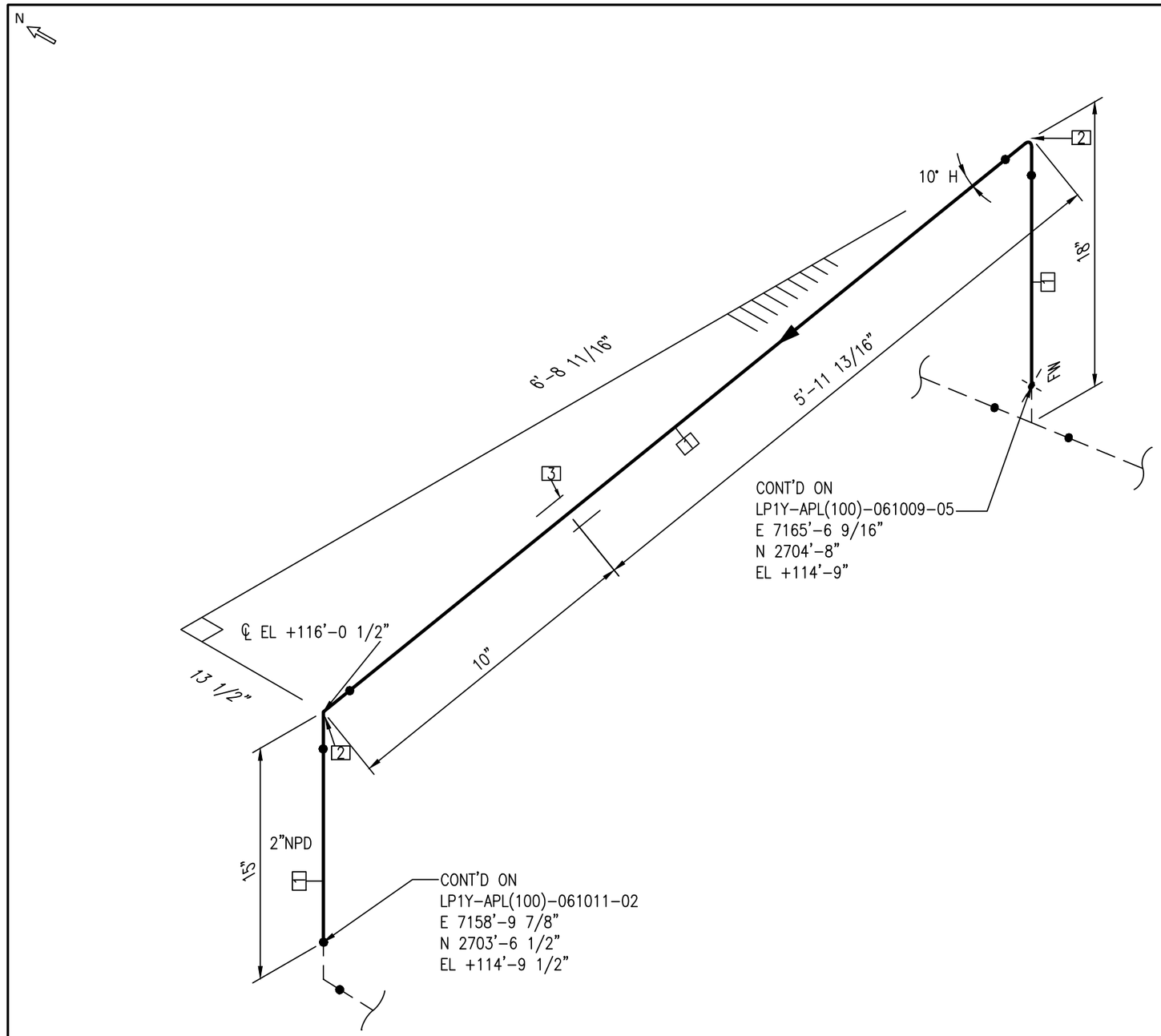

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	8'-5"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	2	2"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
3	1	2"	5CI (ISOLATION CRADLE)	5CI-02

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	APL(100)
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	02-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS APL(100) (315) 061011

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-APL(100)-061011-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 02-MAR-2026

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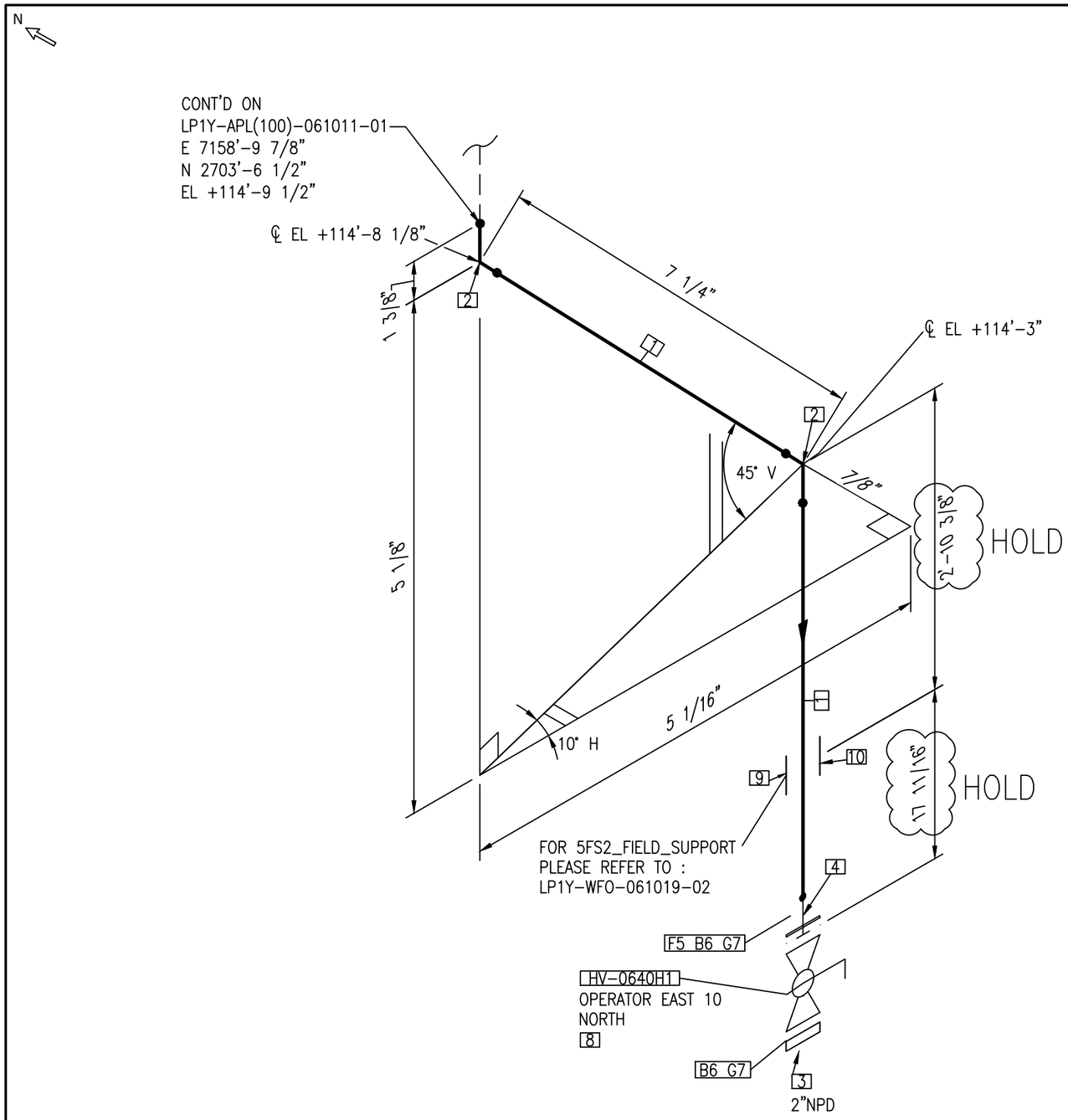

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4	1	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
5	1	2"	FLG LAP JOINT 150# 316 SS	5599578L
6	8	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
7	2	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
8	1	2"	BALL 150# RF 316SS TFE FP HNDL OP	5352194L1
9	1	2"	5CI (ISOLATION CRADLE)	5CI-02
10	1	2"	5UG (U-BOLT GUIDE)	5UG-02

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	APL(100)
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS APL(100) (315) 061011

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-APL(100)-061011-02	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 02-MAR-2026

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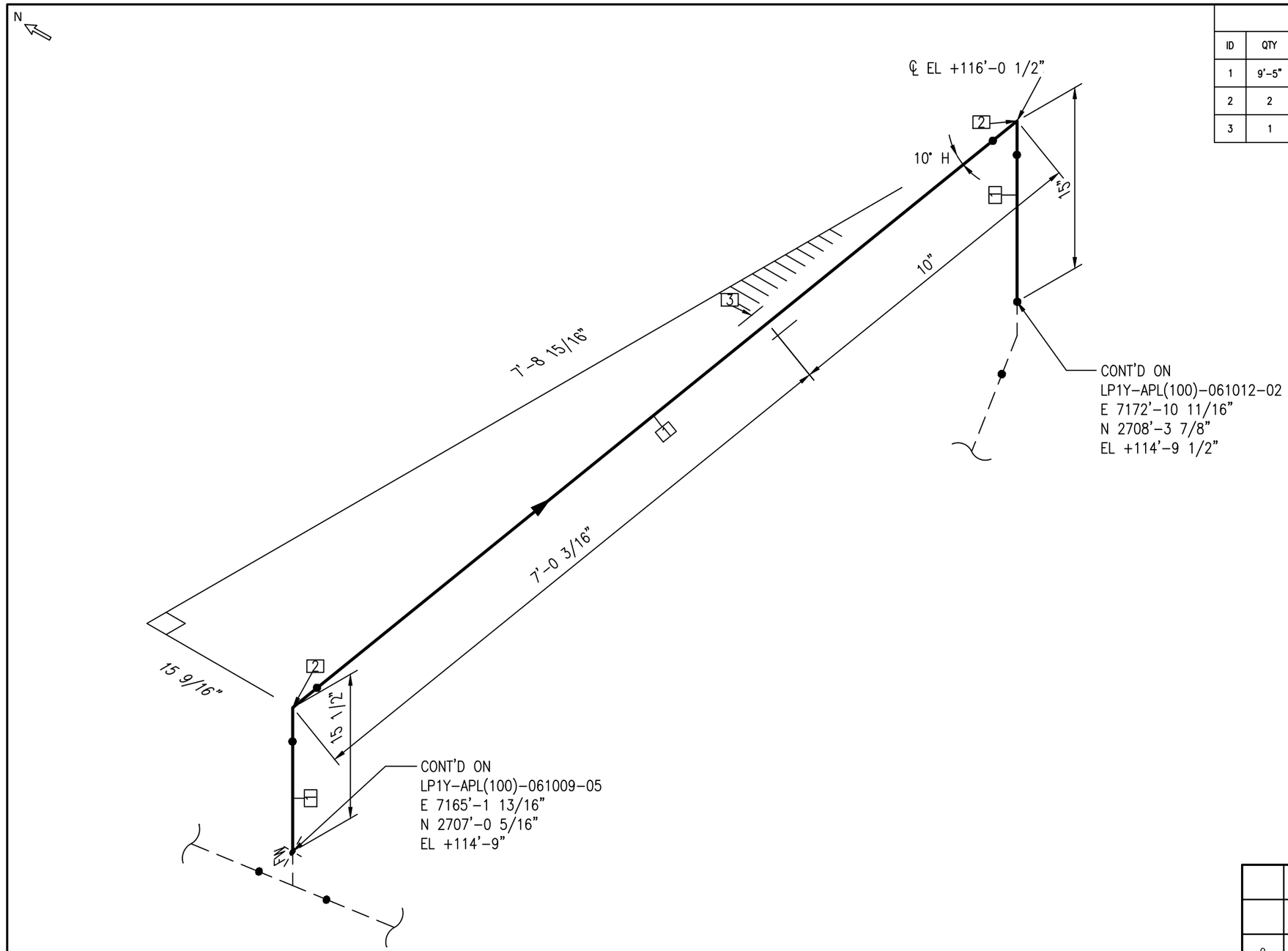

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	APL(100)
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS APL(100) (315) 061012

REFERENCE P&ID NUMBER
LP1Y-DI-00000061

ISOMETRIC DRAWING NUMBER
LP1Y-APL(100)-061012-01

REV
0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 02-MAR-2026

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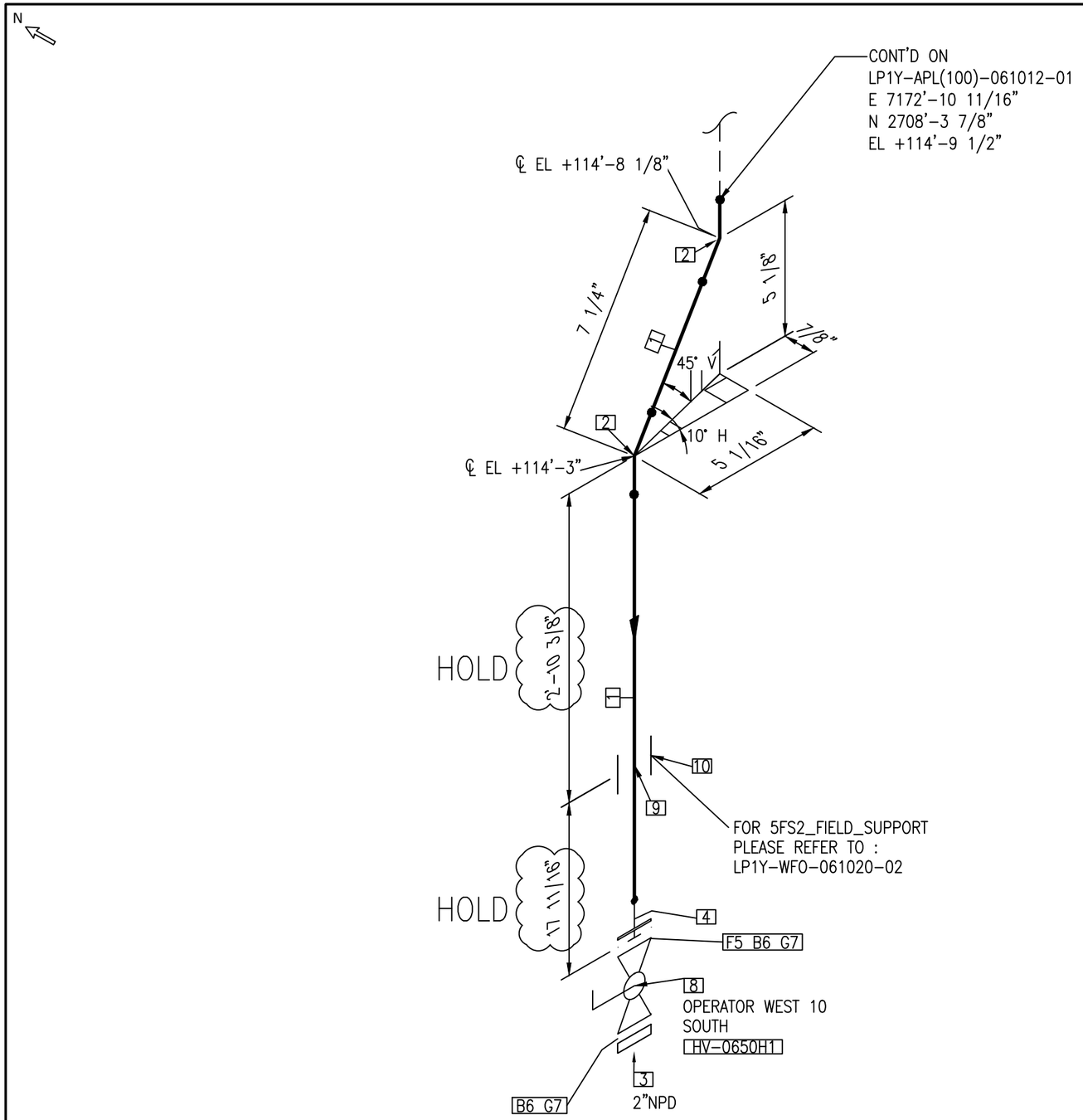

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	APL(100)
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	02-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS APL(100) (315) 061012

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-APL(100)-061012-02	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 02-MAR-2026

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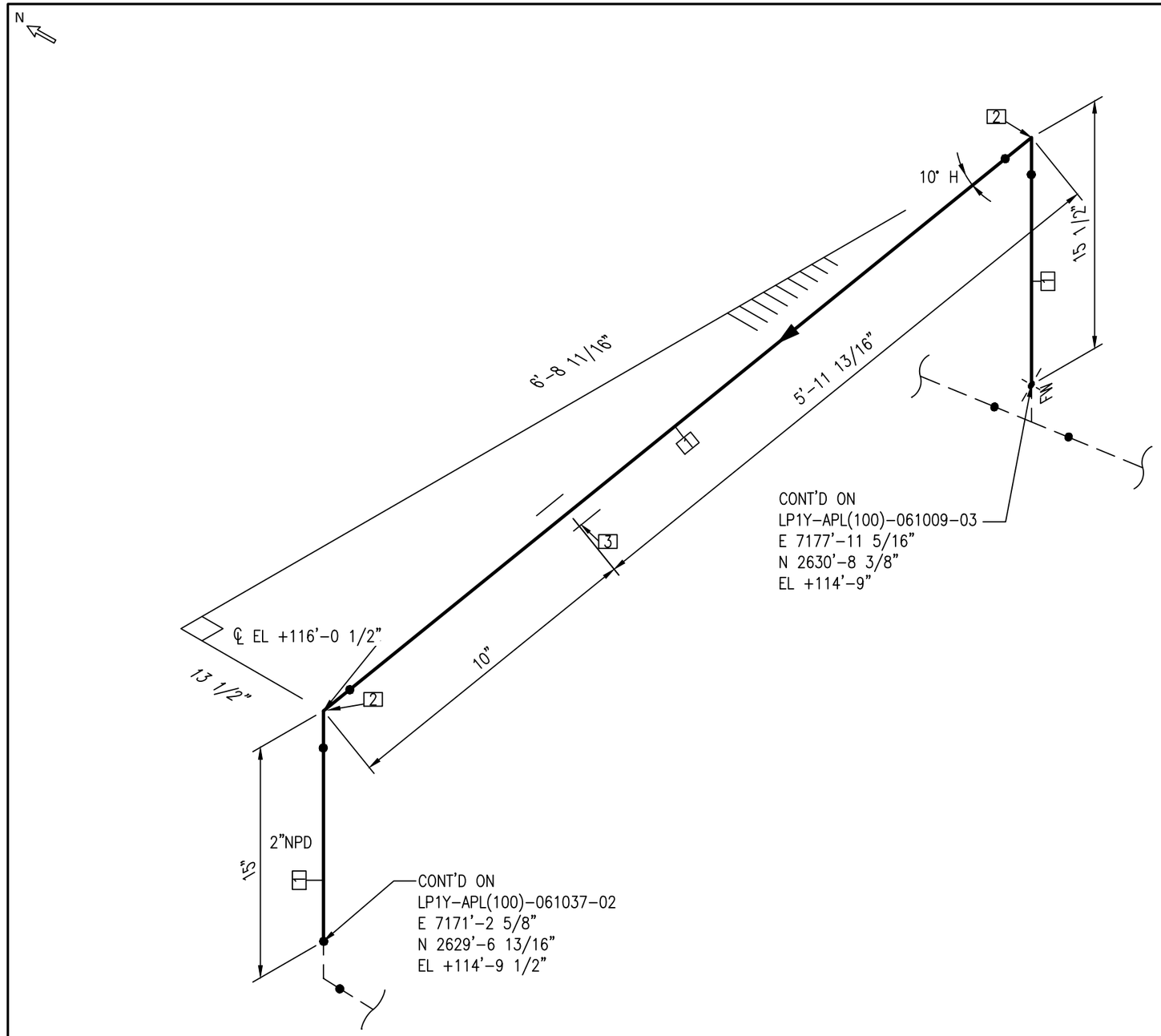

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BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	APL(100)
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS APL(100) (315) 061037

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-APL(100)-061037-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 02-MAR-2026

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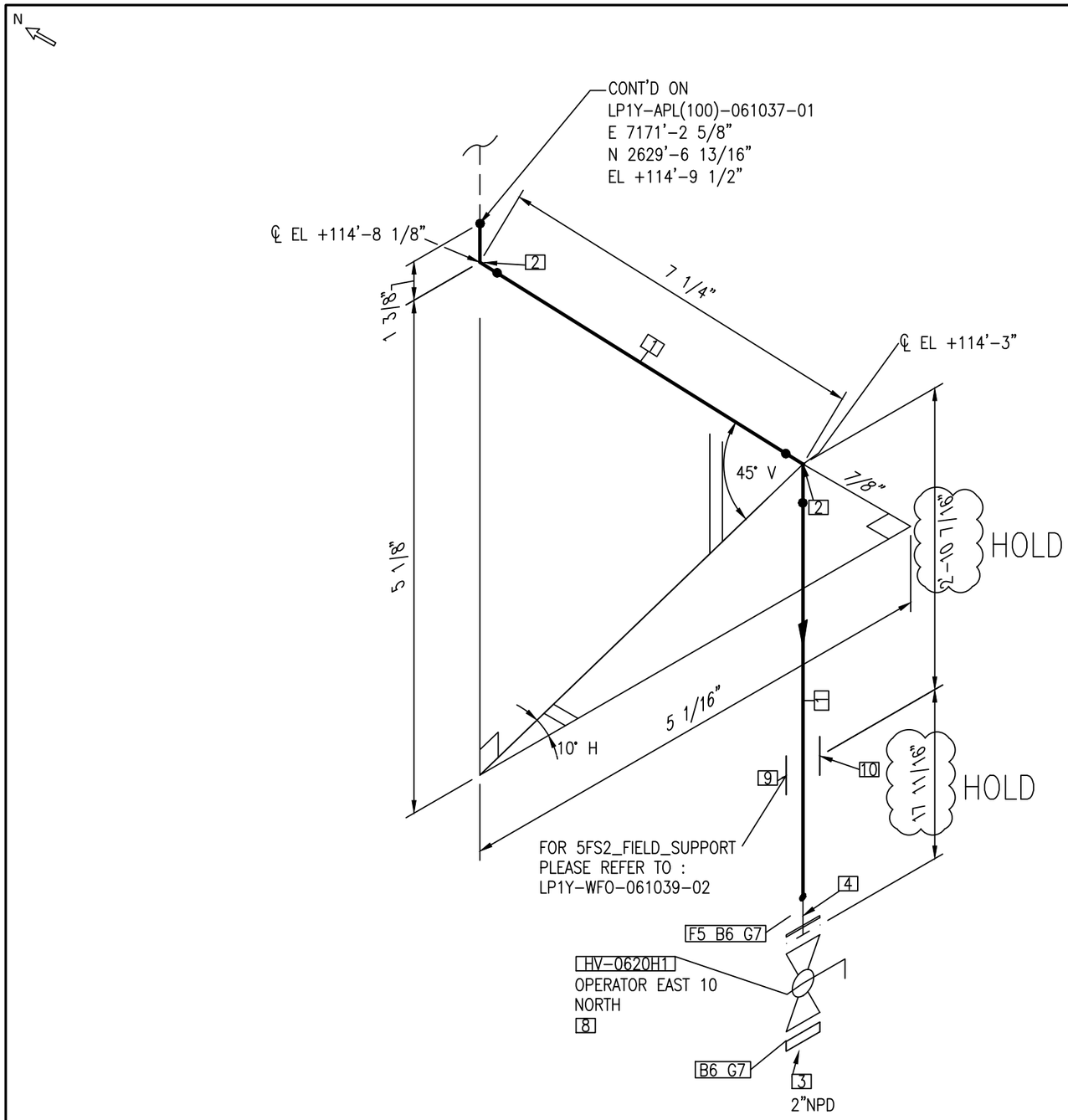

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7	2	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
8	1	2"	BALL 150# RF 316SS TFE FP HNDL OP	5352194L1
9	1	2"	5CI (ISOLATION CRADLE)	5CI-02
10	1	2"	5UG (U-BOLT GUIDE)	5UG-02

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	02-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

GENERAL NOTE:
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5. THE MECHANICAL CONTRACTOR IS TO VERIFY ALL DIMENSIONAL DATA WITH ACTUAL FIELD CONDITIONS PRIOR TO INSTALLATION OF THE SYSTEM.
6. PIPE SUPPORTS SHALL BE FURNISHED AND INSTALLED IN ACCORDANCE WITH THIS DRAWING AND SPECIFICATION LP100-250-EMS-15110. MECHANICAL CONTRACTOR SHALL PROVIDE TEMPORARY SUPPORTS AS REQUIRED TO AID PRESSURE TESTING, INSTALLATION ETC.
7. PIPE SUPPORTS INDICATED ON THE DRAWING MAY BE PART OF AN ASSEMBLY ON SEVERAL ISOMETRICS. REFER TO THE 3D MODEL FOR THE COMPLETE PIPE SUPPORT ASSEMBLY.
8. MECHANICAL CONTRACTOR SHALL NOTE ALL DESIGN CHANGES ON THIS DRAWING AND RETURN IT TO THE CONSTRUCTION WPO FOR AS-BUILT RECORDS.
9. THE BILL OF MATERIAL IS FOR REFERENCE ONLY.
10. SLOPE EQUIVALENTS: 1:27 = 7/16" PER FT, 1:96 = 1/8" PER FT, 1:120 = 1/10" PER FT, 1:192 = 1/16" PER FT, 1:480 = 1/40" PER FT.

NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	APL(100)
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS APL(100) (315) 061037

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-APL(100)-061037-02	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 02-MAR-2026

Tim Trice

Planned Controlled Document

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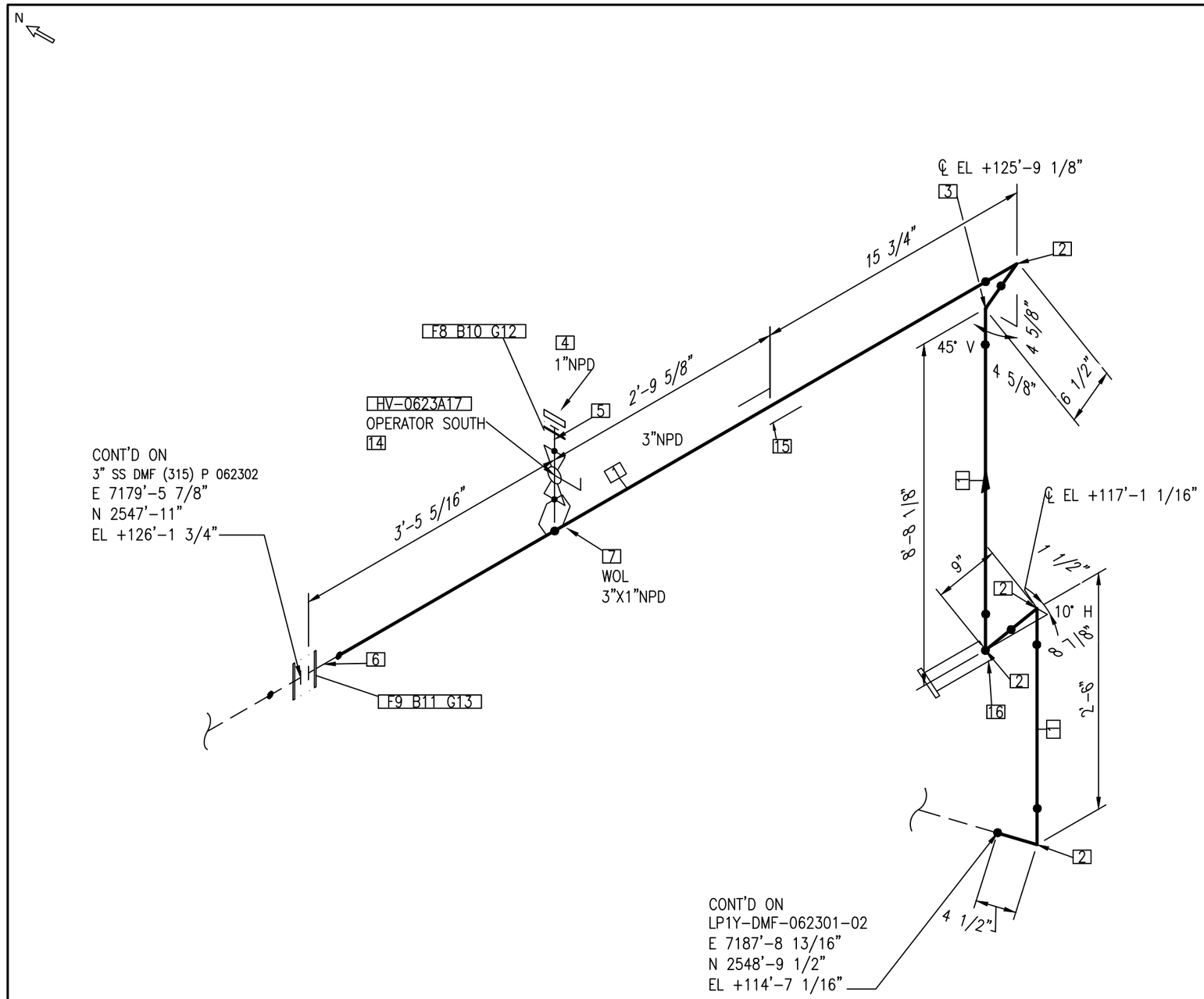

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	16'-11"	3"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	4	3"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
3	1	3"	ELL 45 SCH 10S 316/316L SS A403-W	5450712
4	1	1"	FLG BLIND 150# RF 316/316L SS	5616520
5	1	1"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
6	1	3"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
7	1	3"X1"	PIPET SCH 10S 316/316L SS	5510132
8	1	1"	FLG LAP JOINT 150# 316 SS	5599578L
9	1	3"	FLG LAP JOINT 150# 316 SS	5599578L
10	4	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
11	4	5/8"X2 3/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
12	1	1"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
13	1	3"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
14	1	1"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
15	1	3"	5CI (ISOLATION CRADLE)	5CI-03
16	1	3"	5DS1 (DUMMY SUPPORT) L=1'-5"	5DS1-03

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	09-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	DMF
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	P
HYGIENIC: NO	IMPACT: YES

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

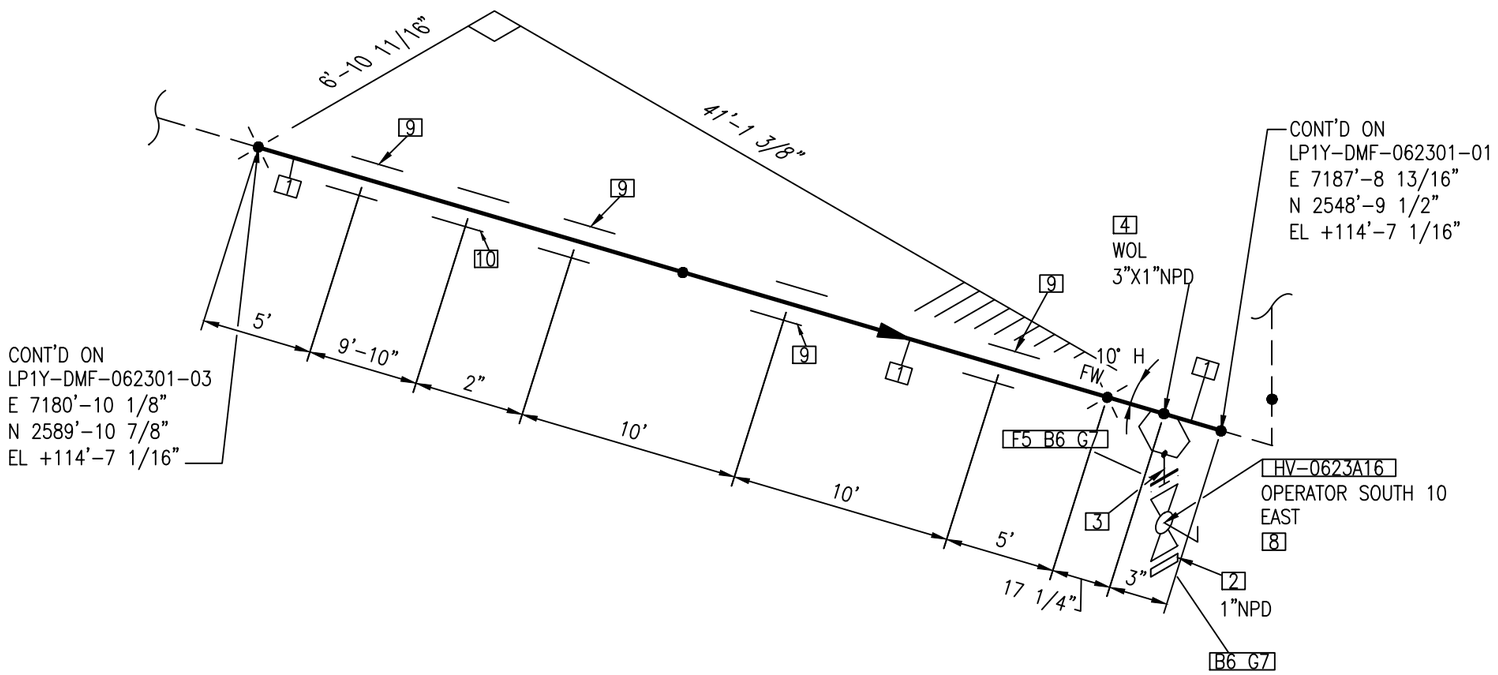
LINE NUMBER
LP1Y 3" SS DMF (315) P 062301

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000062	LP1Y-DMF-062301-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 09-MAR-2026

NOTICE: THIS DRAWING HAS NOT BEEN PUBLISHED AND IS THE SOLE PROPERTY OF FLUOR AND IS LENT TO THE BORROWER FOR THEIR CONFIDENTIAL USE ONLY, AND IN CONSIDERATION OF THE LOAN OF THIS DRAWING, THE BORROWER PROMISES AND AGREES TO RETURN IT UPON REQUEST AND AGREES THAT IT WILL NOT BE REPRODUCED, COPIED, LENT, OR OTHERWISE DISPOSED OF DIRECTLY OR INDIRECTLY, NOR USED FOR ANY PURPOSE OTHER THAN THAT FOR WHICH IT IS FURNISHED.

N



BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	41'-9"	3"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	1	1"	FLG BLIND 150# RF 316/316L SS	5616520
3	1	1"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
4	1	3"X1"	PIPET SCH 10S 316/316L SS	5510132
5	1	1"	FLG LAP JOINT 150# 316 SS	5599578L
6	8	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
7	2	1"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
8	1	1"	BALL 150# RF 316SS TFE FP HNDL OP	5352193L1
9	4	3"	5CI (ISOLATION CRADLE)	5CI-03
10	1	3"	5UG (U-BOLT GUIDE)	5UG-03

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	DMF
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	P
HYGIENIC: NO	IMPACT: YES

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	09-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3" SS DMF (315) P 062301

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000062	LP1Y-DMF-062301-02	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 09-MAR-2026

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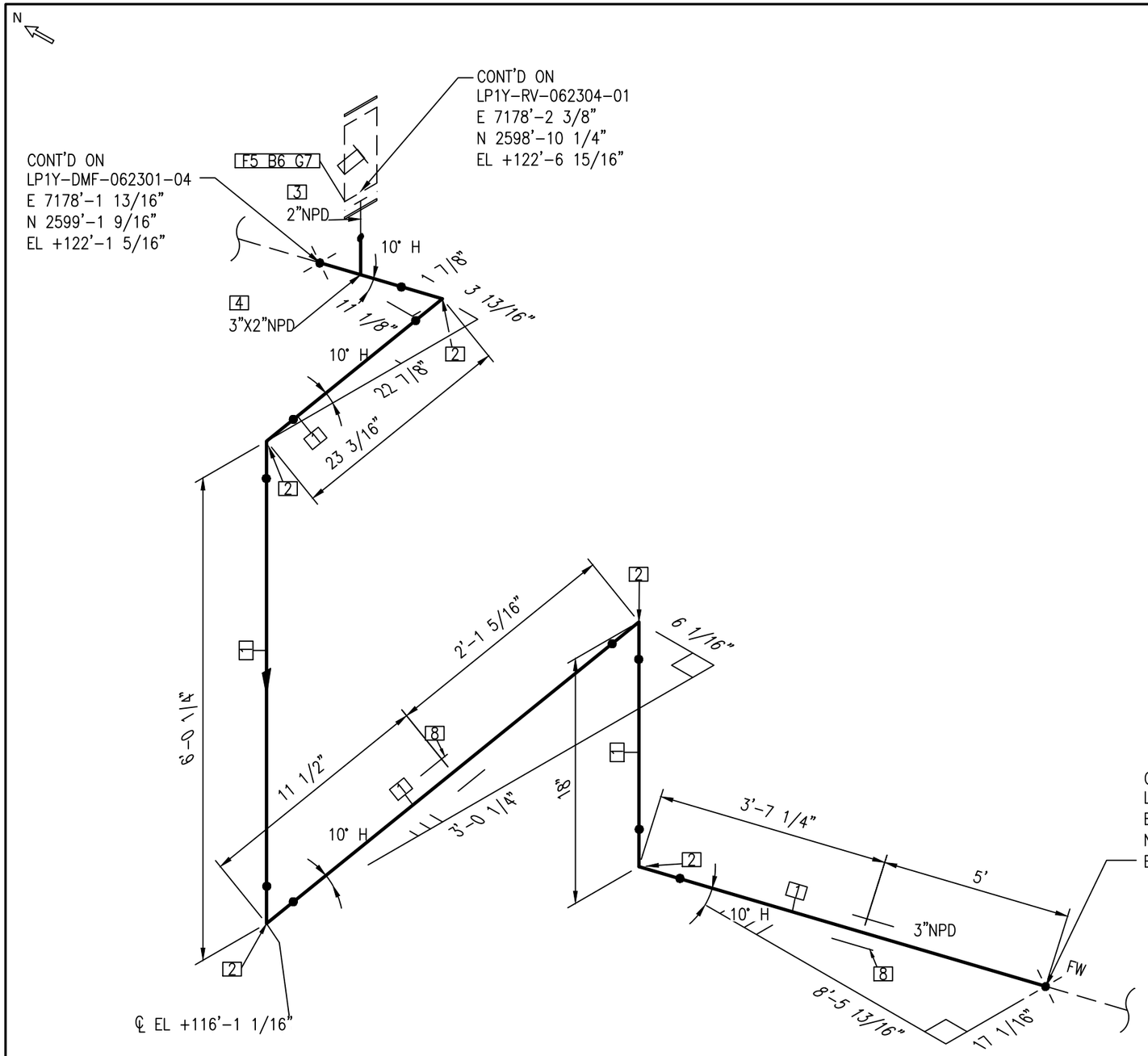

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	17'-9"	3"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	5	3"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
3	1	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
4	1	3"X2"	TEE RED SCH 10S 316/316L SS A403-W	5450714
5	1	2"	FLG LAP JOINT 150# 316 SS	5599578L
6	4	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
7	1	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
8	2	3"	5CI (ISOLATION CRADLE)	5CI-03

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	DMF
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	P
HYGIENIC: NO	IMPACT: YES

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	09-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3" SS DMF (315) P 062301

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000062	LP1Y-DMF-062301-03	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 09-MAR-2026

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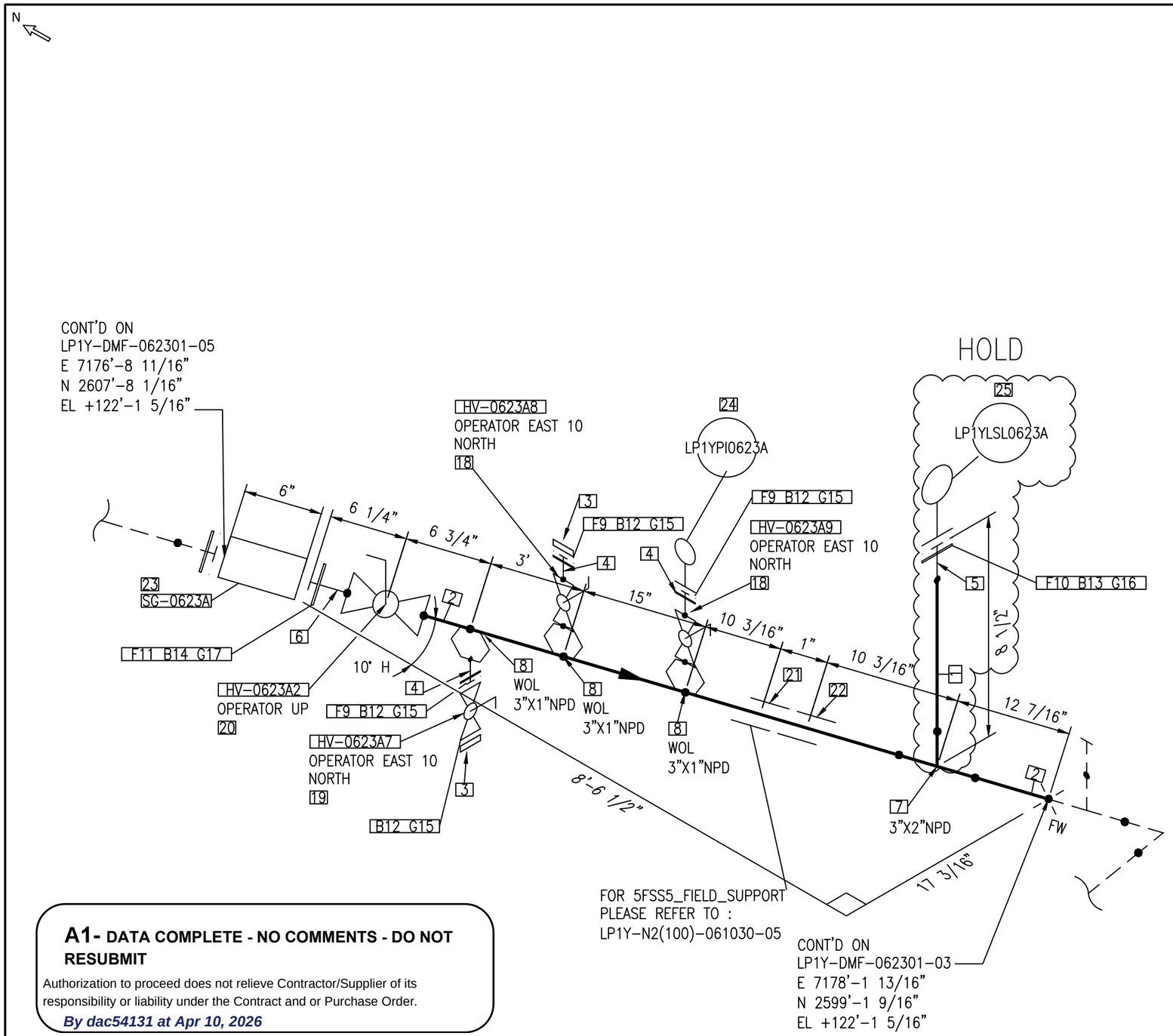

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	3"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	6'-9"	3"	PIPE SCH 10S ERW 316/316L SS A312	5368751
3	2	1"	FLG BLIND 150# RF 316/316L SS	5616520
4	3	1"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
5	1	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
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18	2	1"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
19	1	1"	BALL 150# RF 316SS TFE FP HNDL OP	5352193L1
20	1	3"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
21	1	3"	5CI (ISOLATION CRADLE)	5CI-03
22	1	3"	5UG (U-BOLT GUIDE)	5UG-03
23	1	3"	JACOBY TARBOX (F-830HP-PFA) SIGHT GLASS	LP1YSG0623A
24	1	1"	PRESSURE GAUGE	LP1YPI0623A
25	1	2"	LSL	LP1YLSL0623A

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	DMF
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	P
HYGIENIC: NO	IMPACT: YES

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	16-MAR-2026	DRW	SBR	N/A	JHN / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3" SS DMF (315) P 062301

REFERENCE P&ID NUMBER
LP1Y-DI-00000062

ISOMETRIC DRAWING NUMBER
LP1Y-DMF-062301-04

REV
0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 16-MAR-2026

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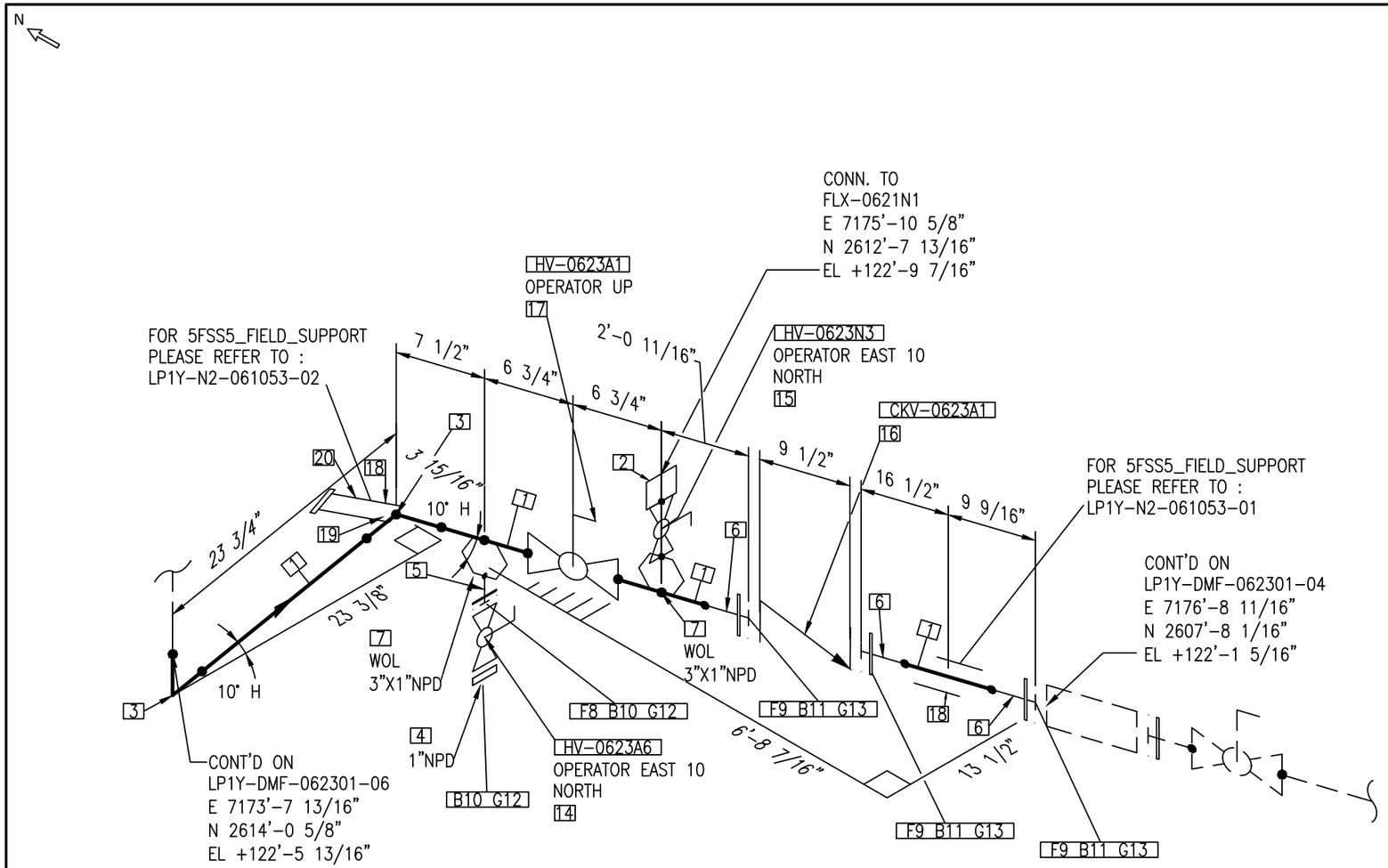

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	5'-7"	3"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	1	1"	BUTTWELD CAMLOCK CONNECTION UNIT	LP1YCCU0623N1
3	2	3"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
4	1	1"	FLG BLIND 150# RF 316/316L SS	5616520
5	1	1"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
6	3	3"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
7	2	3"X1"	PIPET SCH 10S 316/316L SS	5510132
8	1	1"	FLG LAP JOINT 150# 316 SS	5599578L
9	3	3"	FLG LAP JOINT 150# 316 SS	5599578L
10	8	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
11	12	5/8"X2 3/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
12	2	1"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
13	3	3"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
14	1	1"	BALL 150# RF 316SS TFE FP HNDL OP	5352193L1
15	1	1"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
16	1	3"	CHECK 150# RF 316 SS SWING	5257711L
17	1	3"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
18	2	3"	5CI (ISOLATION CRADLE)	5CI-03
19	1	3"	5UG (U-BOLT GUIDE)	5UG-03
20	1	3"	5DS1 (DUMMY SUPPORT) L=1'-6"	5DS1-03

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

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3. WELDS AND/OR COUPLINGS DEFINING RANDOM PIPE SPOOL LENGTHS ARE FOR REFERENCE ONLY. FINAL WELD OR COUPLING LOCATION IS TO BE DETERMINED BY THE CONTRACTOR.
4. THE MECHANICAL CONTRACTOR TO FIELD VERIFY INTERFACE POINTS BETWEEN EQUIPMENT & PLUMBING FIXTURES IN 3rd FIX (3F SERIES CWP) AND YS YARD STICK BUILT (YS SERIES CWP EQUIPMENT CONNECTIONS) ISOMETRICS PRIOR TO FABRICATION.
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6. PIPE SUPPORTS SHALL BE FURNISHED AND INSTALLED IN ACCORDANCE WITH THIS DRAWING AND SPECIFICATION LP100-250-EMS-15110. MECHANICAL CONTRACTOR SHALL PROVIDE TEMPORARY SUPPORTS AS REQUIRED TO AID PRESSURE TESTING, INSTALLATION ETC.
7. PIPE SUPPORTS INDICATED ON THE DRAWING MAY BE PART OF AN ASSEMBLY ON SEVERAL ISOMETRICS. REFER TO THE 3D MODEL FOR THE COMPLETE PIPE SUPPORT ASSEMBLY.
8. MECHANICAL CONTRACTOR SHALL NOTE ALL DESIGN CHANGES ON THIS DRAWING AND RETURN IT TO THE CONSTRUCTION WPO FOR AS-BUILT RECORDS.
9. THE BILL OF MATERIAL IS FOR REFERENCE ONLY.
10. SLOPE EQUIVALENTS: 1:27 = 7/16" PER FT, 1:96 = 1/8" PER FT, 1:120 = 1/10" PER FT, 1:192 = 1/16" PER FT, 1:480 = 1/40" PER FT.

NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	DMF
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	P
HYGIENIC: NO	IMPACT: YES

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	09-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

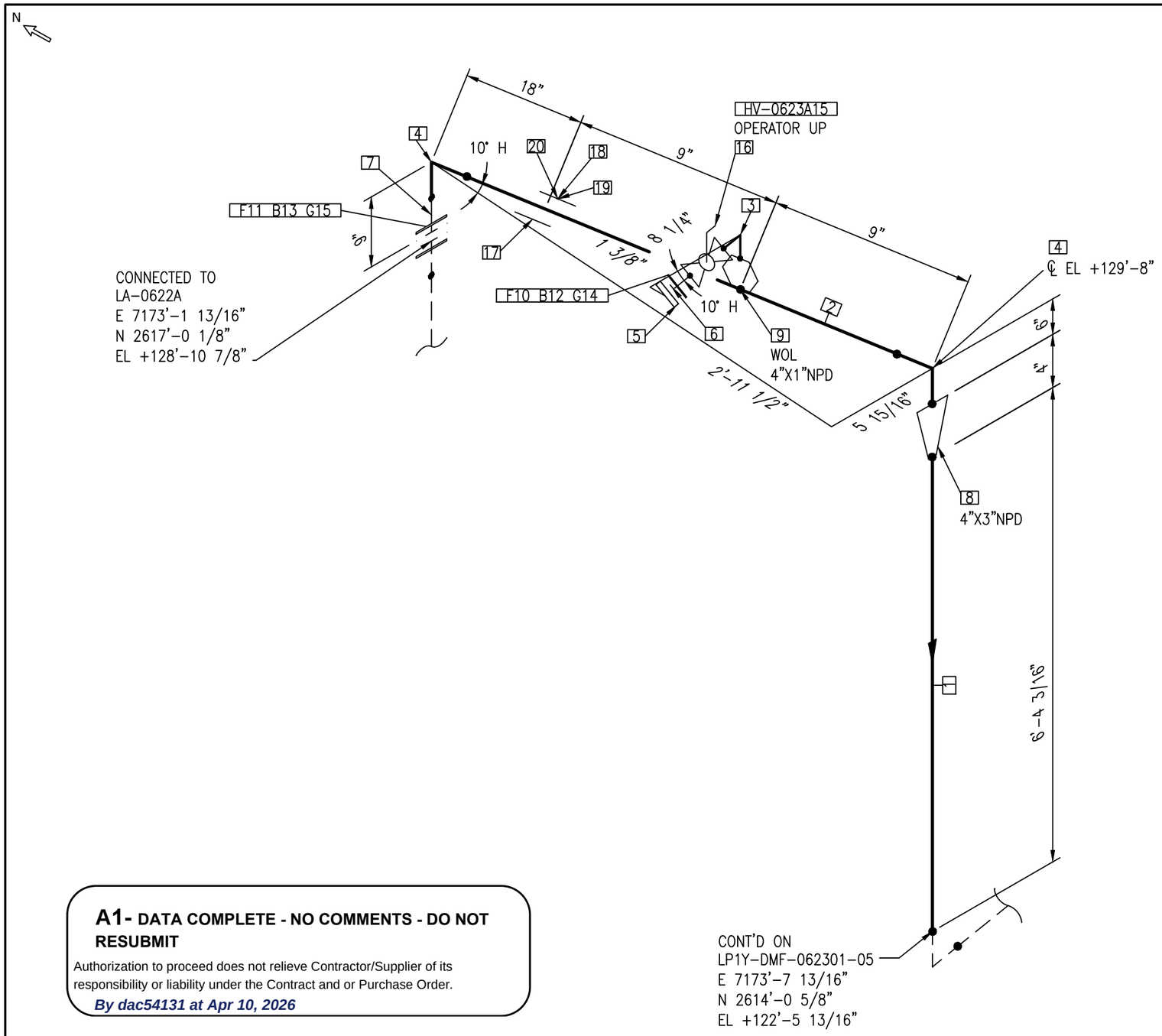
HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3" SS DMF (315) P 062301

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000062	LP1Y-DMF-062301-05	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 09-MAR-2026

NOTICE: THIS DRAWING HAS NOT BEEN PUBLISHED AND IS THE SOLE PROPERTY OF FLUOR. AND IS LENT TO THE BORROWER FOR THEIR CONFIDENTIAL USE ONLY, AND IN CONSIDERATION OF THE LOAN OF THIS DRAWING, THE BORROWER PROMISES AND AGREES TO RETURN IT UPON REQUEST AND AGREES THAT IT WILL NOT BE REPRODUCED, COPIED, LENT, OR OTHERWISE DISPOSED OF DIRECTLY OR INDIRECTLY, NOR USED FOR ANY PURPOSE OTHER THAN THAT FOR WHICH IT IS FURNISHED.



BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	6'-5"	3"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	2'-1"	4"	PIPE SCH 10S ERW 316/316L SS A312	5368751
3	1	1"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
4	2	4"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
5	1	1"	FLG BLIND 150# RF 316/316L SS	5616520
6	1	1"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
7	1	4"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
8	1	4"X3"	REDUCER CONC SCH 10S 316/316L SS A403-W	5450717
9	1	4"X1"	PIPET SCH 10S 316/316L SS	5510132
10	1	1"	FLG LAP JOINT 150# 316 SS	5599578L
11	1	4"	FLG LAP JOINT 150# 316 SS	5599578L
12	4	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
13	8	5/8"X2 3/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
14	1	1"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
15	1	4"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
16	1	1"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
17	1	4"	5CI (ISOLATION CRADLE)	5CI-04
18	1	4"	5FS1 (FIELD SUPPORT)	5FS1S2
19	1	4"	5UG (U-BOLT GUIDE)	5UG-04
20	1	4"	5DA1 (DIRECTIONAL ANCHOR)	5DA1-M

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	DMF
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	P
HYGIENIC: NO	IMPACT: YES

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
						/	
						/	
0	ISSUE FOR CONSTRUCTION	17-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3" SS DMF (315) P 062301

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000062	LP1Y-DMF-062301-06	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 17-MAR-2026

Tim Trice

Planned Controlled Document

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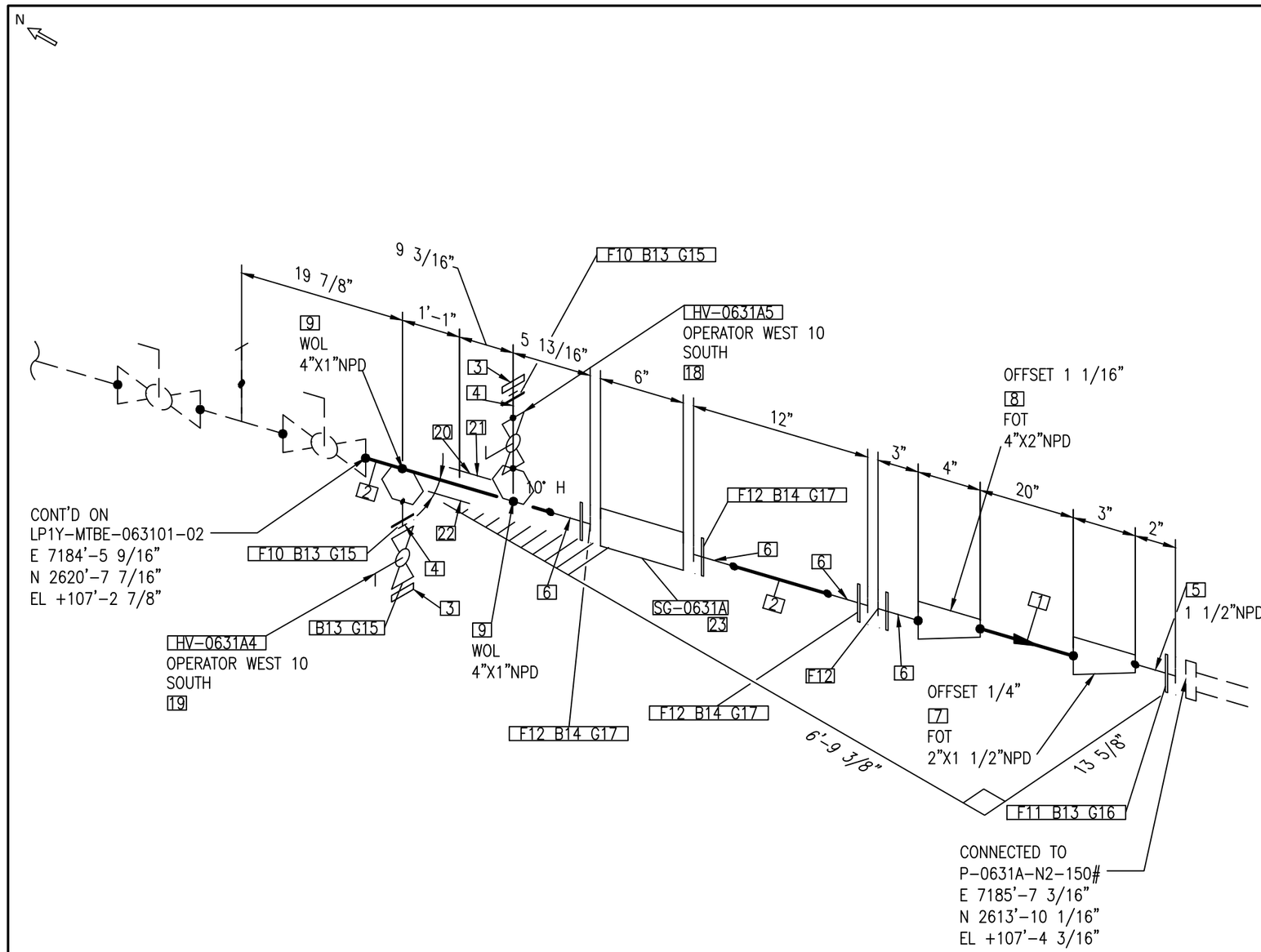

BILL OF MATERIALS

ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	1'-8"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	3'	4"	PIPE SCH 10S ERW 316/316L SS A312	5368751
3	2	1"	FLG BLIND 150# RF 316/316L SS	5616520
4	2	1"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
5	1	1 1/2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
6	4	4"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
7	1	2"X1 1/2"	REDUCER ECC SCH 10S 316/316L SS A403-W	5450718
8	1	4"X2"	REDUCER ECC SCH 10S 316/316L SS A403-W	5450718
9	2	4"X1"	PIPET SCH 10S 316/316L SS	5510132
10	2	1"	FLG LAP JOINT 150# 316 SS	5599578L
11	1	1 1/2"	FLG LAP JOINT 150# 316 SS	5599578L
12	4	4"	FLG LAP JOINT 150# 316 SS	5599578L
13	16	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
14	24	5/8"X2 3/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
15	3	1"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
16	1	1 1/2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
17	3	4"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
18	1	1"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
19	1	1"	BALL 150# RF 316SS TFE FP HNDL OP	5352193L1
20	1	4"	5CI (ISOLATION CRADLE)	5CI-04
21	1	4"	5FSS5 (FIELD BASE SUPPORT)	5FSS5-L-1
22	1	4"	5UG (U-BOLT GUIDE)	5UG-04
23	1	4"	JACOBY TARBOX (F-830HP-PFA) SIGHT GLASS	LP1YSG0631A

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	11-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

GENERAL NOTE:
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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	MTBE
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	P
HYGIENIC: NO	IMPACT: YES

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 4" SS MTBE (315) P 063101

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000063	LP1Y-MTBE-063101-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 11-MAR-2026

NOTICE: THIS DRAWING HAS NOT BEEN PUBLISHED AND IS THE SOLE PROPERTY OF FLUOR AND IS LENT TO THE BORROWER FOR THEIR CONFIDENTIAL USE ONLY, AND IN CONSIDERATION OF THE LOAN OF THIS DRAWING, THE BORROWER PROMISES AND AGREES TO RETURN IT UPON REQUEST AND AGREES THAT IT WILL NOT BE REPRODUCED, COPIED, LENT, OR OTHERWISE DISPOSED OF DIRECTLY OR INDIRECTLY, NOR USED FOR ANY PURPOSE OTHER THAN THAT FOR WHICH IT IS FURNISHED.

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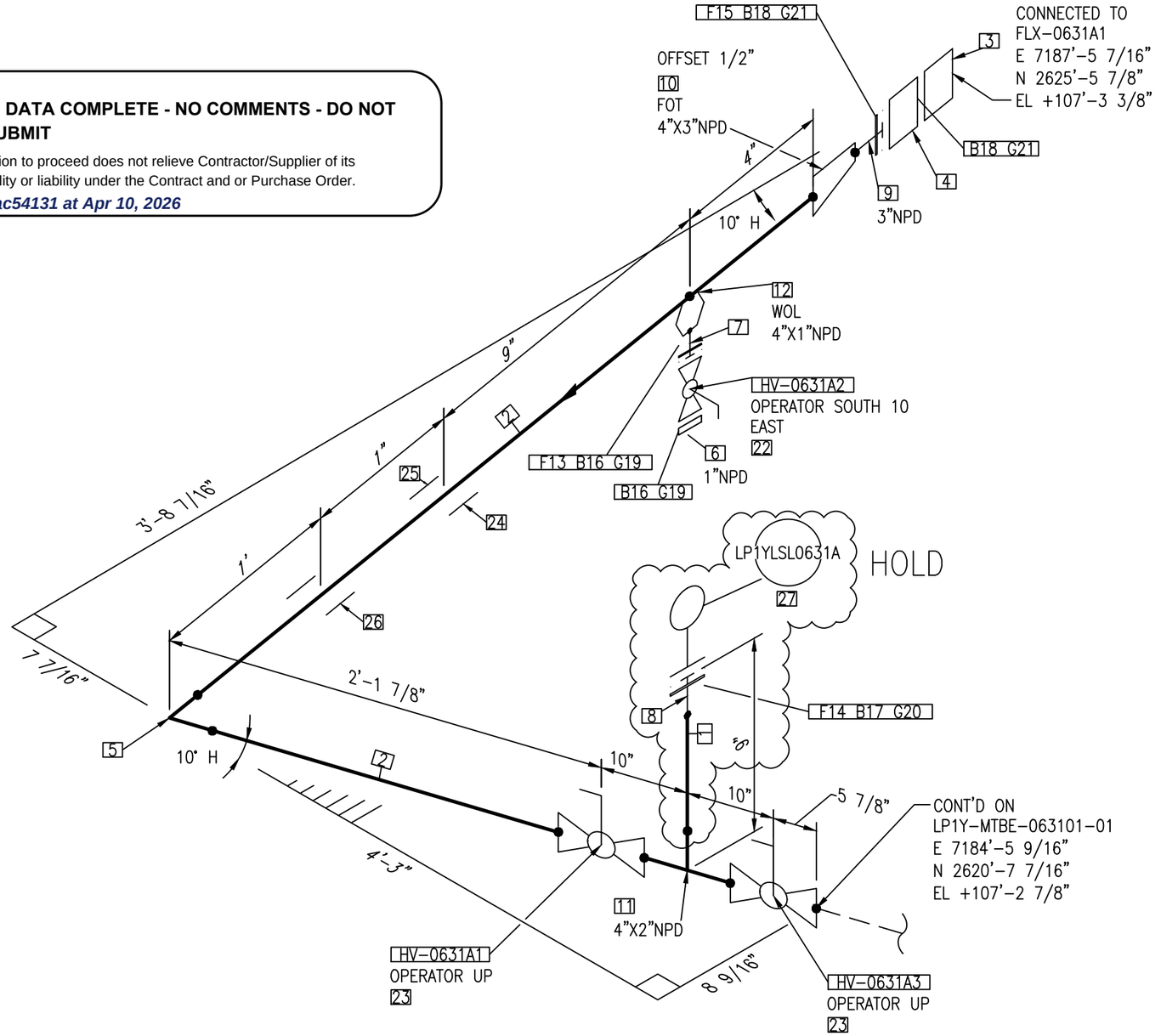
By dac54131 at Apr 10, 2026



BILL OF MATERIALS

ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	3"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	2'-11"	4"	PIPE SCH 10S ERW 316/316L SS A312	5368751
3	1	3"	FLANGED QUICK CONNECTION UNIT	LP1YQCU0631A2
4	1	3"	MANNTEK FLANGED BREAKAWAY COUPLING	LP1YBCU0631A1
5	1	4"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
6	1	1"	FLG BLIND 150# RF 316/316L SS	5616520
7	1	1"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
8	1	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
9	1	3"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
10	1	4"X3"	REDUCER ECC SCH 10S 316/316L SS A403-W	5450718
11	1	4"X2"	TEE RED SCH 10S 316/316L SS A403-W	5450714
12	1	4"X1"	PIPET SCH 10S 316/316L SS	5510132
13	1	1"	FLG LAP JOINT 150# 316 SS	5599578L
14	1	2"	FLG LAP JOINT 150# 316 SS	5599578L
15	1	3"	FLG LAP JOINT 150# 316 SS	5599578L
16	8	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
17	4	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
18	8	5/8"X2 3/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
19	2	1"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
20	1	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
21	2	3"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
22	1	1"	BALL 150# RF 316SS TFE FP HNDL OP	5352193L1
23	2	4"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
24	1	4"	5FSS5 (FIELD BASE SUPPORT)	5FSS5-L-1
25	1	4"	5CI (ISOLATION CRADLE)	5CI-04
26	1	4"	5UG (U-BOLT GUIDE)	5UG-04
27	1	2"	LSL	LP1YLSL0631A

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	MTBE
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	P
HYGIENIC: NO	IMPACT: YES

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	16-MAR-2026	DRW	SBR	N/A	JHN / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 4" SS MTBE (315) P 063101

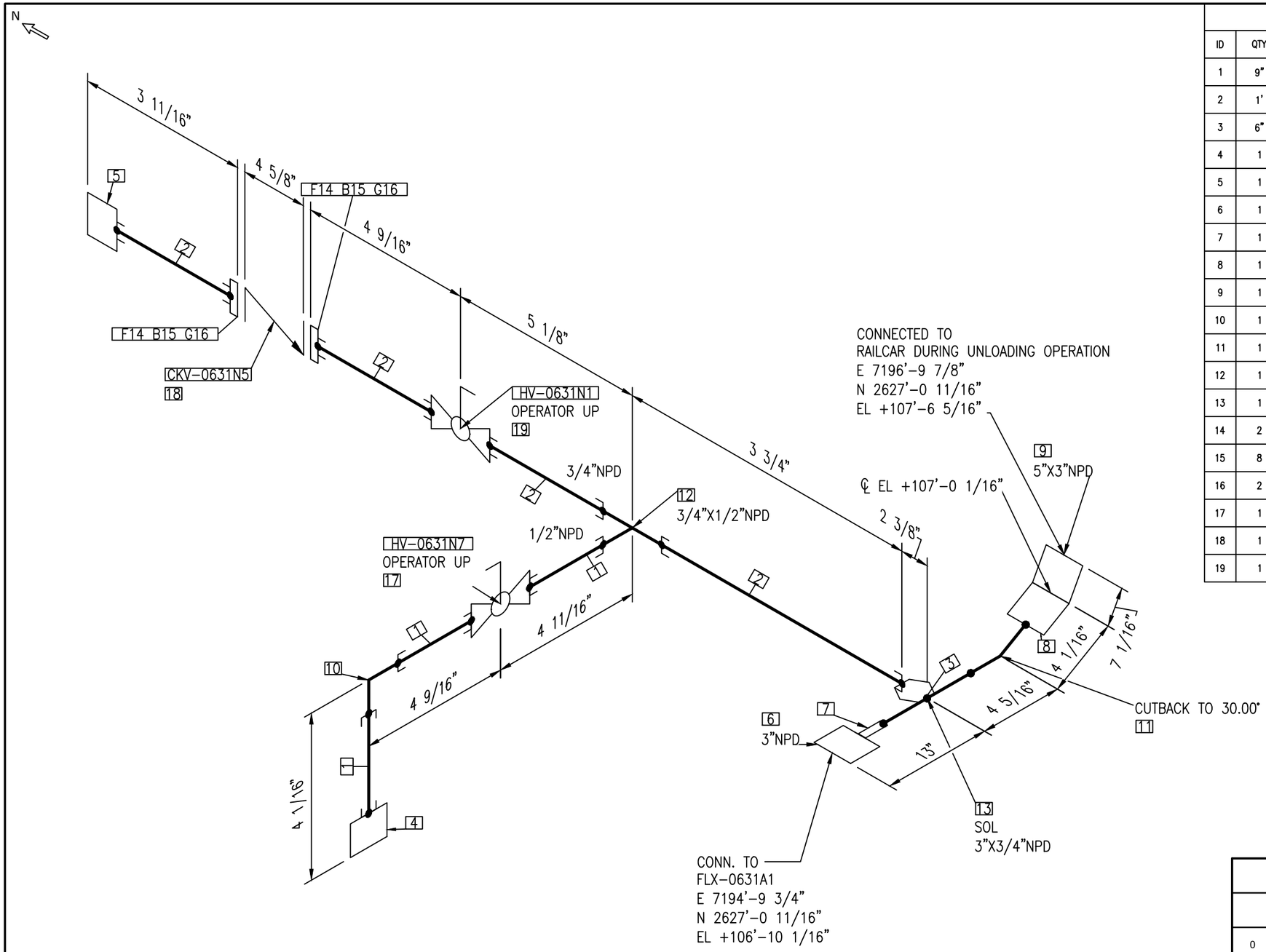
REFERENCE P&ID NUMBER
LP1Y-DI-00000063

ISOMETRIC DRAWING NUMBER
LP1Y-MTBE-063101-02

REV
0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 16-MAR-2026

NOTICE: THIS DRAWING HAS NOT BEEN PUBLISHED AND IS THE SOLE PROPERTY OF FLUOR AND IS LENT TO THE BORROWER FOR THEIR CONFIDENTIAL USE ONLY, AND IN CONSIDERATION OF THE LOAN OF THIS DRAWING, THE BORROWER PROMISES AND AGREES TO RETURN IT UPON REQUEST AND AGREES THAT IT WILL NOT BE REPRODUCED, COPIED, LENT, OR OTHERWISE DISPOSED OF DIRECTLY OR INDIRECTLY, NOR USED FOR ANY PURPOSE OTHER THAN THAT FOR WHICH IT IS FURNISHED.



BILL OF MATERIALS

ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	9"	1/2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	1'	3/4"	PIPE SCH 10S ERW 316/316L SS A312	5368751
3	6"	3"	PIPE SCH 10S ERW 316/316L SS A312	5368751
4	1	1/2"	SOCKETWELD CAMLOCK CONNECTION UNIT	LP1YCCU0631N1
5	1	3/4"	SOCKETWELD CAMLOCK CONNECTION UNIT	LP1YCCU0631N3
6	1	3"	THREADED QUICK CONNECTION UNIT	LP1YQCU0631A1
7	1	3"	NIPPLE SCH 80S 316/316L SS BOETOE 6" LONG	5758143L0
8	1	3"	BUTTWELD CAMLOCK CONNECTION UNIT	LP1YCCU0631A2
9	1	5"X3"	DIXON RAILCAR ADAPTER	LP1YEI0631A1
10	1	1/2"	ELL 90 DEG 3000# SW 316/316L SS	5480138
11	1	3"	ELL 45 SCH 10S 316/316L SS A403-W	5450712
12	1	3/4"X1/2"	TEE RED 3000# SW 316/316L SS	5480152
13	1	3"X3/4"	PIPET 10S X 3000# SW 316/316L SS	5537081
14	2	3/4"	FLG SW 150# RF 316/316L SS 10S BORE	5608148
15	8	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
16	2	3/4"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
17	1	1/2"	BALL 1000# CWP SW 316SS TFE FP HNDL OP	5352192L
18	1	3/4"	CHECK 150# RF 316 SS SWING	5257711L
19	1	3/4"	BALL 1000# CWP SW 316SS TFE FP HNDL OP	5352192L

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	12-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	MTBE
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	P
HYGIENIC: NO	IMPACT: YES

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 4" SS MTBE (315) P 063101

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000063	LP1Y-MTBE-063101-03	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 12-MAR-2026

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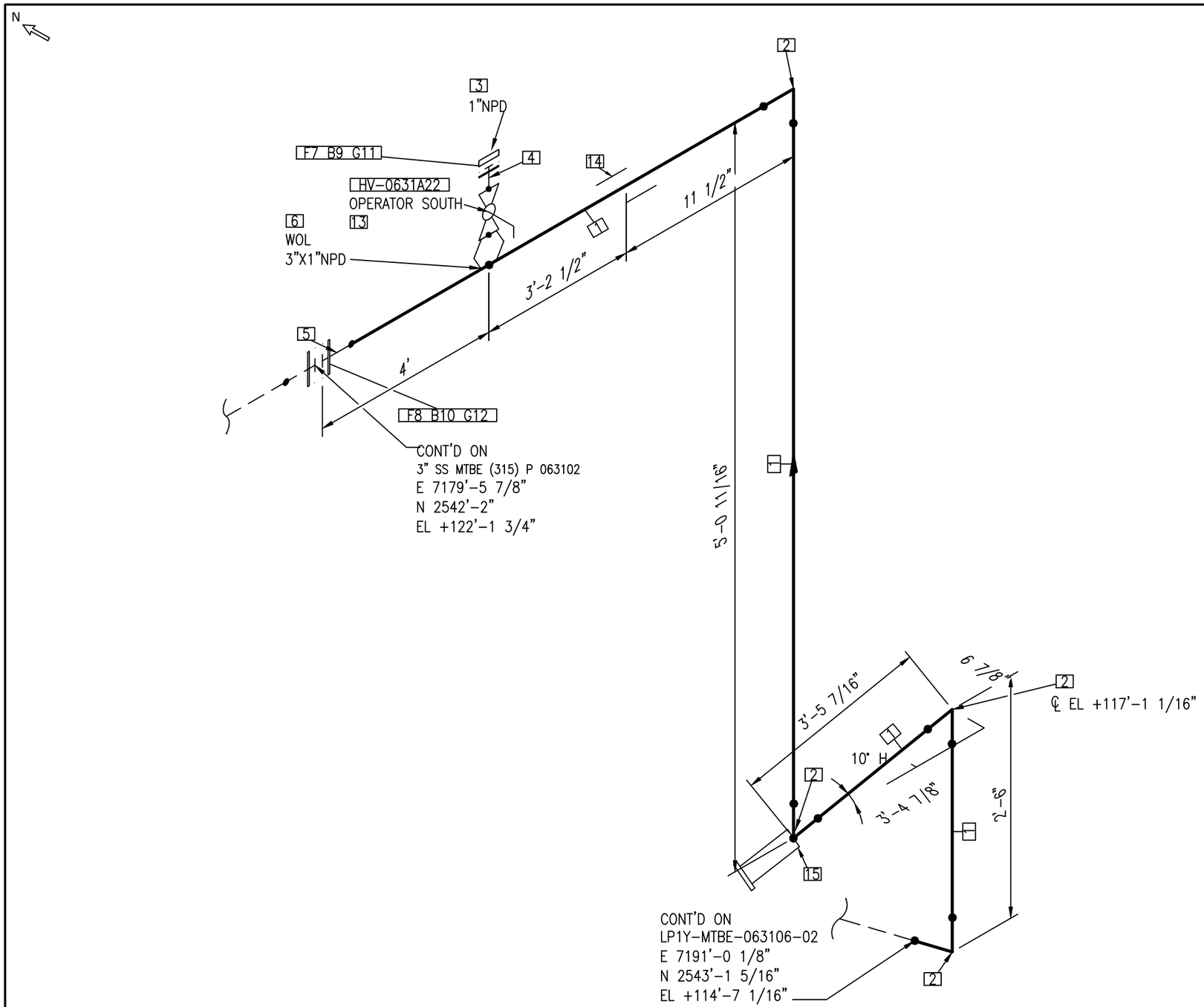

BILL OF MATERIALS

ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	16'-5"	3"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	4	3"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
3	1	1"	FLG BLIND 150# RF 316/316L SS	5616520
4	1	1"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
5	1	3"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
6	1	3"X1"	PIPET SCH 10S 316/316L SS	5510132
7	1	1"	FLG LAP JOINT 150# 316 SS	5599578L
8	1	3"	FLG LAP JOINT 150# 316 SS	5599578L
9	4	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
10	4	5/8"X2 3/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
11	1	1"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
12	1	3"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
13	1	1"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
14	1	3"	5CI (ISOLATION CRADLE)	5CI-03
15	1	3"	5DS1 (DUMMY SUPPORT) L=1'	5DS1-03

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	MTBE
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	P
HYGIENIC: NO	IMPACT: YES

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
						/	
						/	
0	ISSUE FOR CONSTRUCTION	10-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3" SS MTBE (315) P 063106

REFERENCE P&ID NUMBER
LP1Y-DI-00000063

ISOMETRIC DRAWING NUMBER
LP1Y-MTBE-063106-01

REV
0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 10-MAR-2026

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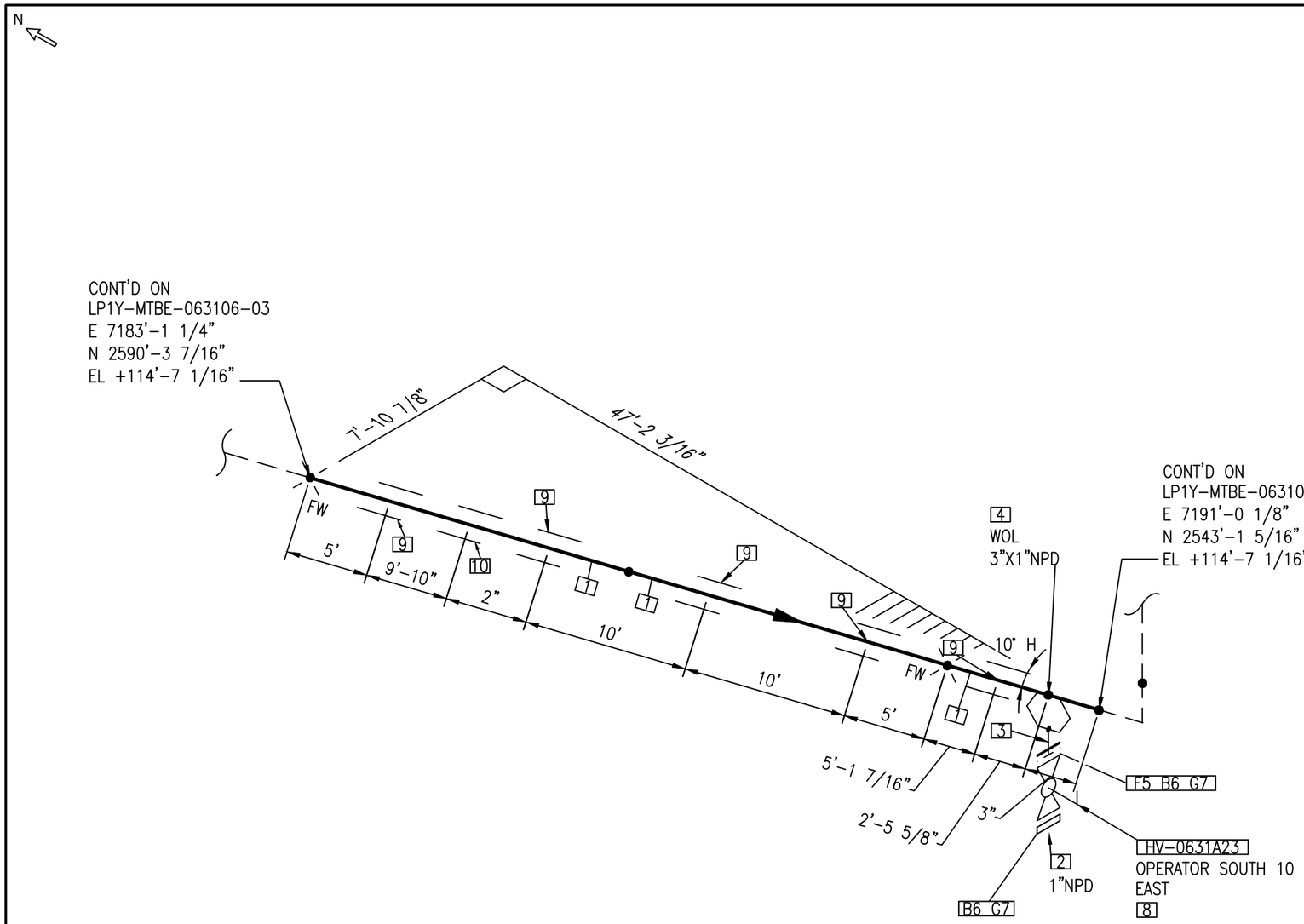

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	47'-11"	3"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	1	1"	FLG BLIND 150# RF 316/316L SS	5616520
3	1	1"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
4	1	3"X1"	PIPET SCH 10S 316/316L SS	5510132
5	1	1"	FLG LAP JOINT 150# 316 SS	5599578L
6	8	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
7	2	1"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
8	1	1"	BALL 150# RF 316SS TFE FP HNDL OP	5352193L1
9	5	3"	5CI (ISOLATION CRADLE)	5CI-03
10	1	3"	5UG (U-BOLT GUIDE)	5UG-03

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	MTBE
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	P
HYGIENIC: NO	IMPACT: YES

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3" SS MTBE (315) P 063106

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000063	LP1Y-MTBE-063106-02	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 10-MAR-2026

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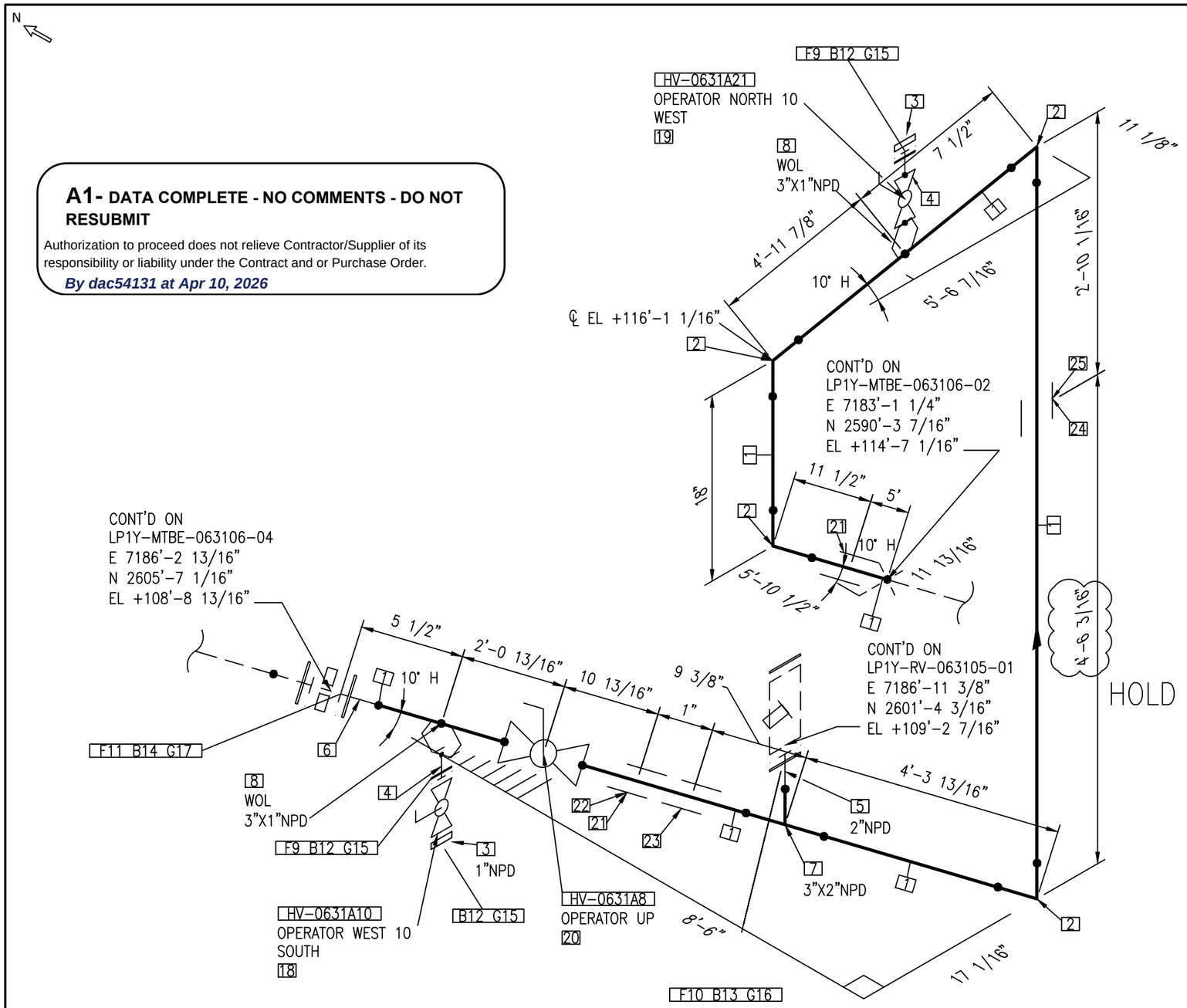

BILL OF MATERIALS

ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	24'-8"	3"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	4	3"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
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7	1	3"X2"	TEE RED SCH 10S 316/316L SS A403-W	5450714
8	2	3"X1"	PIPET SCH 10S 316/316L SS	5510132
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15	3	1"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
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19	1	1"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
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21	2	3"	5CI (ISOLATION CRADLE)	5CI-03
22	1	3"	5FSS5 (FIELD BASE SUPPORT)	5FSS5-L-1
23	1	3"	5UG (U-BOLT GUIDE)	5UG-03
24	1	3"	5US (U-BOLT SUPPORT)	5US-03
25	1	3"	5FS1 (FIELD SUPPORT)	5FS1S2

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	10-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	MTBE
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	P
HYGIENIC: NO	IMPACT: YES

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3" SS MTBE (315) P 063106

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000063	LP1Y-MTBE-063106-03	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 10-MAR-2026

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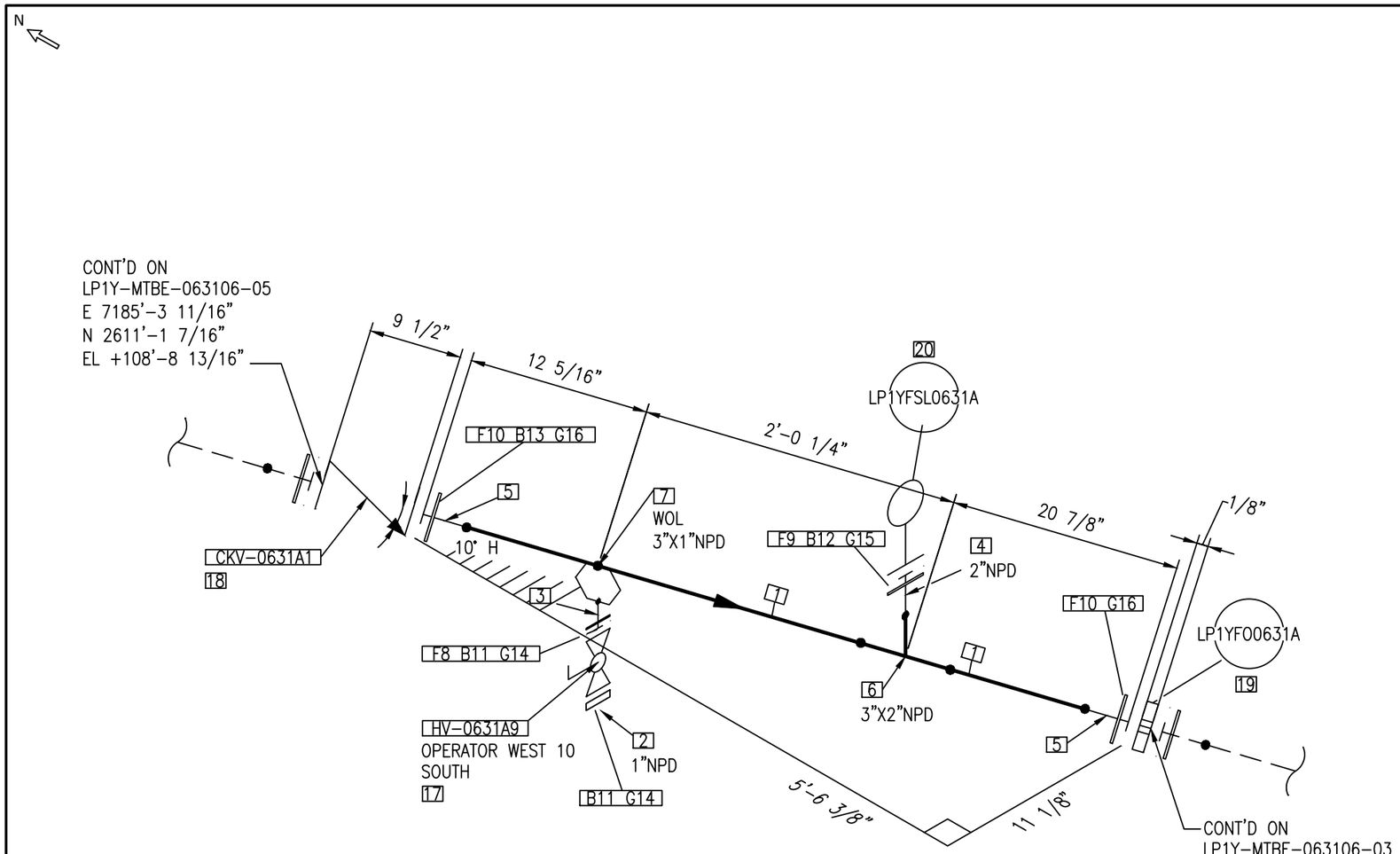

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17	1	1"	BALL 150# RF 316SS TFE FP HNDL OP	5352193L1
18	1	3"	CHECK 150# RF 316 SS SWING	5257711L
19	1	3"	FLOW ORIFICE	LP1YF00631A
20	1	2"	FSL	LP1YFSL0631A

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CWP NO:	N/A
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TRACING TYPE:	NT
ELASTOMER:	P
HYGIENIC: NO	IMPACT: YES

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Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3" SS MTBE (315) P 063106

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
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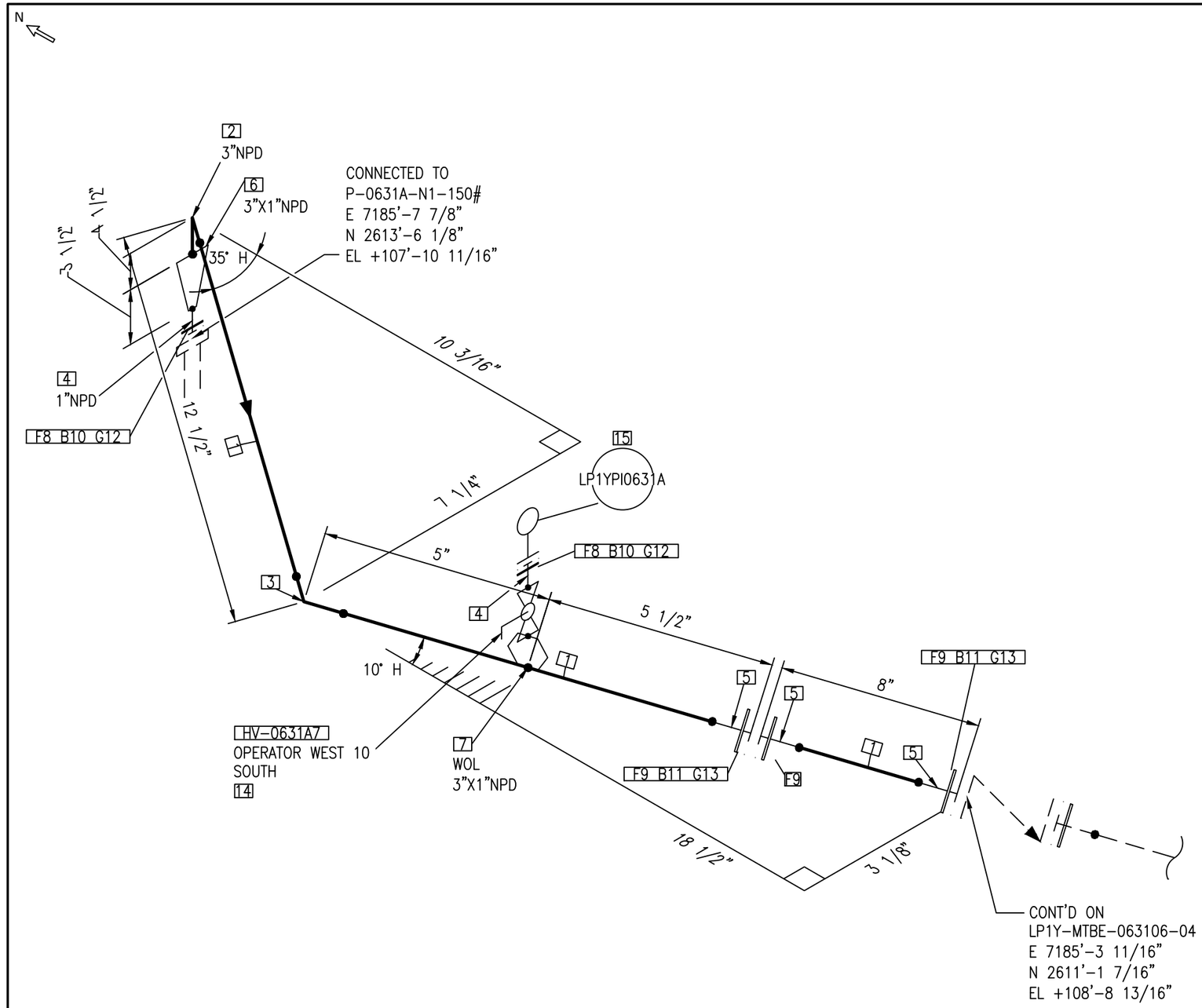

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1	1'-3"	3"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	1	3"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
3	1	3"	ELL 45 SCH 10S 316/316L SS A403-W	5450712
4	2	1"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
5	3	3"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
6	1	3"X1"	REDUCER CONC SCH 10S 316/316L SS A403-W	5450717
7	1	3"X1"	PIPET SCH 10S 316/316L SS	5510132
8	2	1"	FLG LAP JOINT 150# 316 SS	5599578L
9	3	3"	FLG LAP JOINT 150# 316 SS	5599578L
10	8	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
11	8	5/8"X2 3/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
12	2	1"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
13	2	3"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
14	1	1"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
15	1	1"	PRESSURE GAUGE	LP1YPI0631A

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
						/	
						/	
0	ISSUE FOR CONSTRUCTION	10-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

GENERAL NOTE:
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7. PIPE SUPPORTS INDICATED ON THE DRAWING MAY BE PART OF AN ASSEMBLY ON SEVERAL ISOMETRICS. REFER TO THE 3D MODEL FOR THE COMPLETE PIPE SUPPORT ASSEMBLY.
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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	MTBE
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	P
HYGIENIC: NO	IMPACT: YES

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3" SS MTBE (315) P 063106

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000063	LP1Y-MTBE-063106-05	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 10-MAR-2026

Tim Trice

Planned Controlled Document

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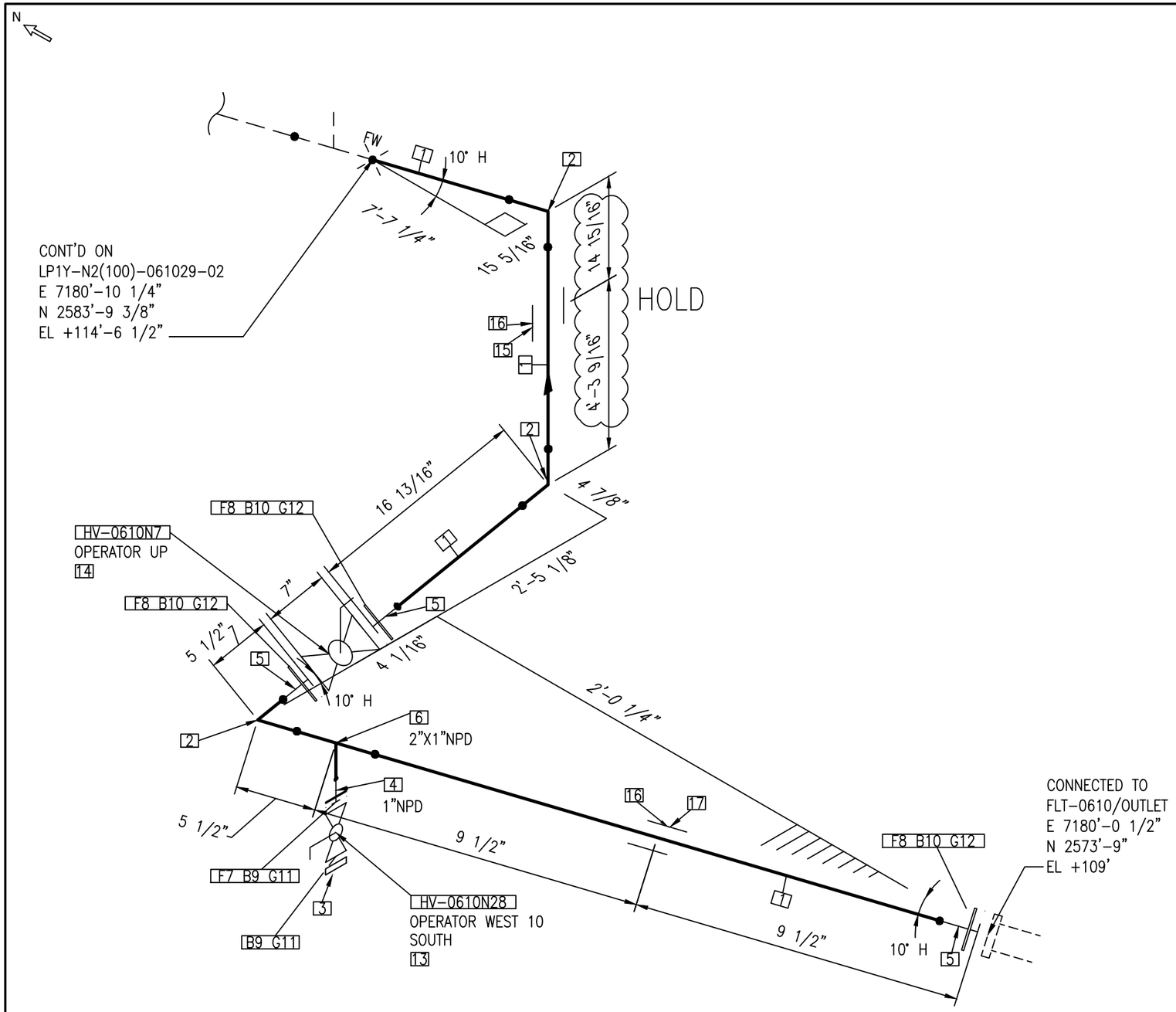

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	14'-8"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	3	2"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
3	1	1"	FLG BLIND 150# RF 316/316L SS	5616520
4	1	1"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
5	3	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
6	1	2"X1"	TEE RED SCH 10S 316/316L SS A403-W	5450714
7	1	1"	FLG LAP JOINT 150# 316 SS	5599578L
8	3	2"	FLG LAP JOINT 150# 316 SS	5599578L
9	8	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
10	12	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
11	2	1"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
12	3	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
13	1	1"	BALL 150# RF 316SS TFE FP HNDL OP	5352193L1
14	1	2"	BALL 150# RF 316SS TFE FP HNDL OP	5352194L1
15	1	2"	5US (U-BOLT SUPPORT)	5US-02
16	2	2"	5FS1 (FIELD SUPPORT)	5FS1S1
17	1	2"	5CI (ISOLATION CRADLE)	5CI-02

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	19-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2(100)
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

Lilly

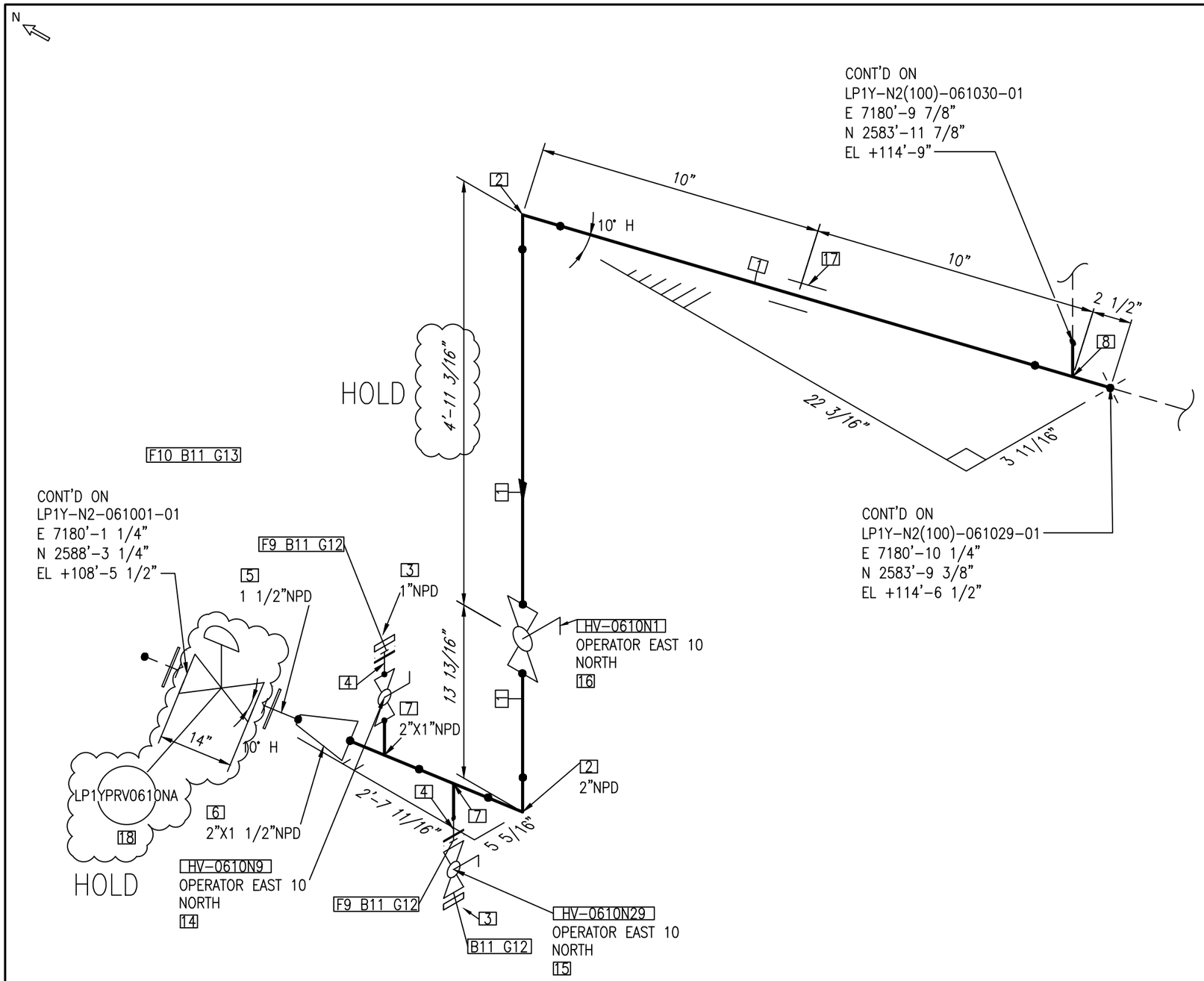
HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2(100) (315) E 061029

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-N2(100)-061029-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 19-MAR-2026

NOTICE: THIS DRAWING HAS NOT BEEN PUBLISHED AND IS THE SOLE PROPERTY OF FLUOR AND IS LENT TO THE BORROWER FOR THEIR CONFIDENTIAL USE ONLY, AND IN CONSIDERATION OF THE LOAN OF THIS DRAWING, THE BORROWER PROMISES AND AGREES TO RETURN IT UPON REQUEST AND AGREES THAT IT WILL NOT BE REPRODUCED, COPIED, LENT, OR OTHERWISE DISPOSED OF DIRECTLY OR INDIRECTLY, NOR USED FOR ANY PURPOSE OTHER THAN THAT FOR WHICH IT IS FURNISHED.



BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	6'-4"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	2	2"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
3	2	1"	FLG BLIND 150# RF 316/316L SS	5616520
4	2	1"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
5	1	1 1/2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
6	1	2"X1 1/2"	REDUCER CONC SCH 10S 316/316L SS A403-W	5450717
7	2	2"X1"	TEE RED SCH 10S 316/316L SS A403-W	5450714
8	1	2"	TEE SCH 10S 316/316L SS A403-W	5450715
9	2	1"	FLG LAP JOINT 150# 316 SS	5599578L
10	1	1 1/2"	FLG LAP JOINT 150# 316 SS	5599578L
11	16	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
12	3	1"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
13	1	1 1/2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
14	1	1"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
15	1	1"	BALL 150# RF 316SS TFE FP HNDL OP	5352193L1
16	1	2"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
17	1	2"	5CI (ISOLATION CRADLE)	5CI-02
18	1	1 1/2"	PRV CL 150# RF	LP1YPRV0610NA

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	13-MAR-2026	DRW	SBR	N/A	JHN / N/A	MDW

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10. SLOPE EQUIVALENTS: 1:27 = 7/16" PER FT, 1:96 = 1/8" PER FT, 1:120 = 1/10" PER FT, 1:192 = 1/16" PER FT, 1:480 = 1/40" PER FT.

NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2(100)
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2(100) (315) E 061029

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-N2(100)-061029-02	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 13-MAR-2026

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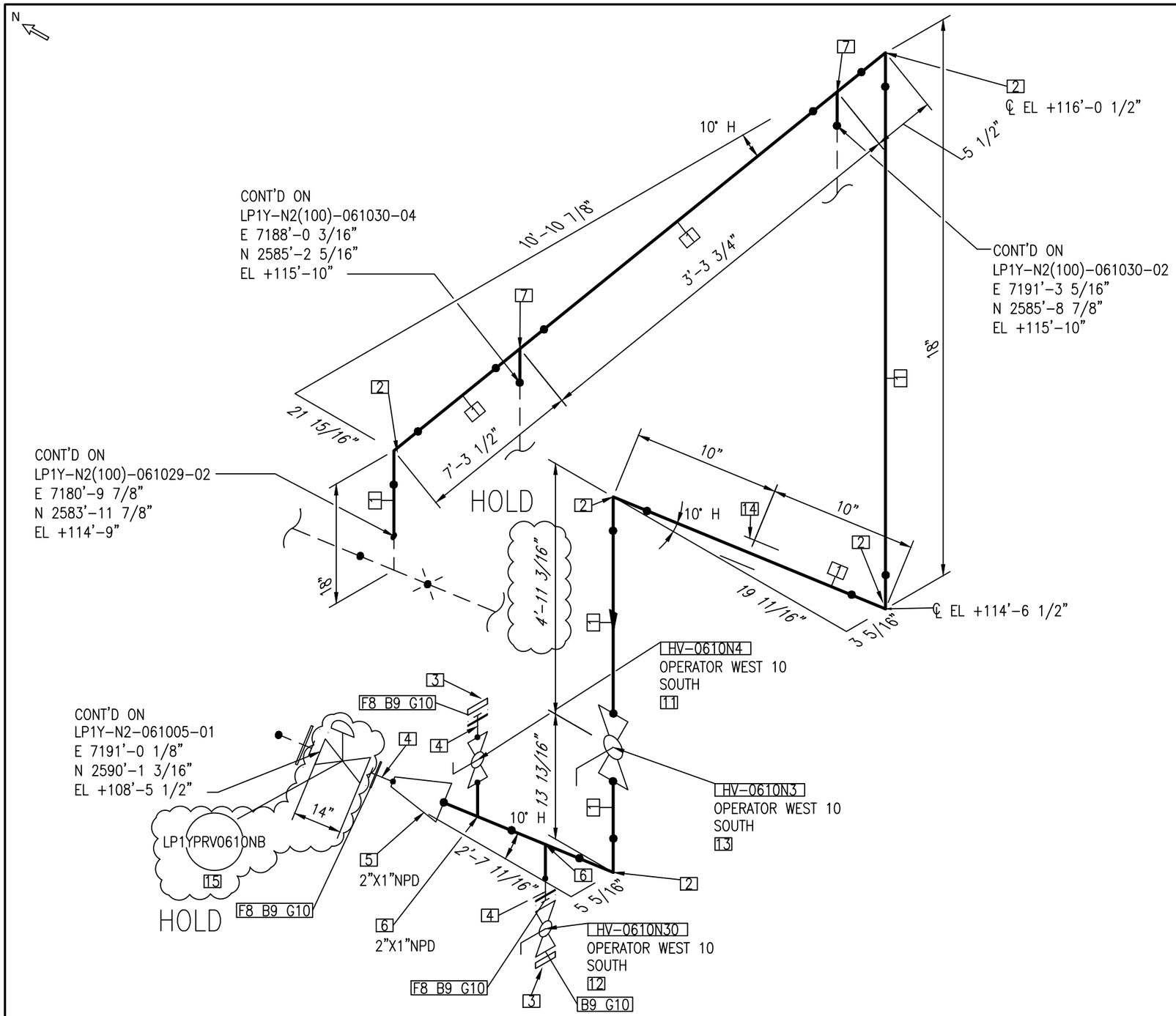

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	18'-1"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	5	2"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
3	2	1"	FLG BLIND 150# RF 316/316L SS	5616520
4	3	1"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
5	1	2"X1"	REDUCER CONC SCH 10S 316/316L SS A403-W	5450717
6	2	2"X1"	TEE RED SCH 10S 316/316L SS A403-W	5450714
7	2	2"	TEE SCH 10S 316/316L SS A403-W	5450715
8	3	1"	FLG LAP JOINT 150# 316 SS	5599578L
9	16	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
10	4	1"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
11	1	1"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
12	1	1"	BALL 150# RF 316SS TFE FP HNDL OP	5352193L1
13	1	2"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
14	1	2"	5CI (ISOLATION CRADLE)	5CI-02
15	1	1"	PRV CL 150# RF	LP1YPRV0610NB

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	10-MAR-2026	DRW	SBR	N/A	JHN / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2(100)
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2(100) (315) E 061030

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-N2(100)-061030-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 10-MAR-2026

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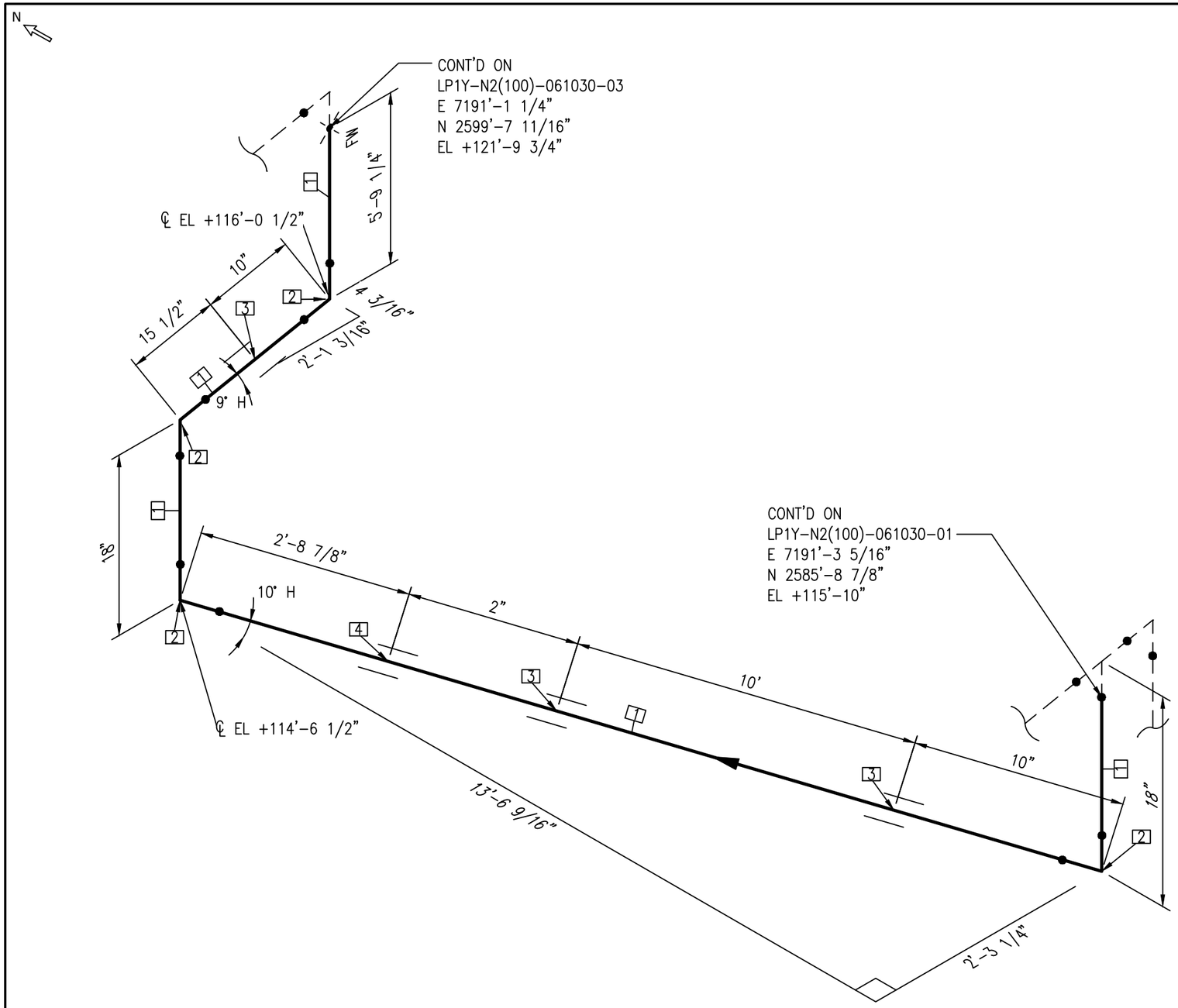

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	22'-6"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	4	2"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
3	3	2"	5CI (ISOLATION CRADLE)	5CI-02
4	1	2"	5UG (U-BOLT GUIDE)	5UG-02

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	10-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2(100)
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2(100) (315) E 061030

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-N2(100)-061030-02	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 10-MAR-2026

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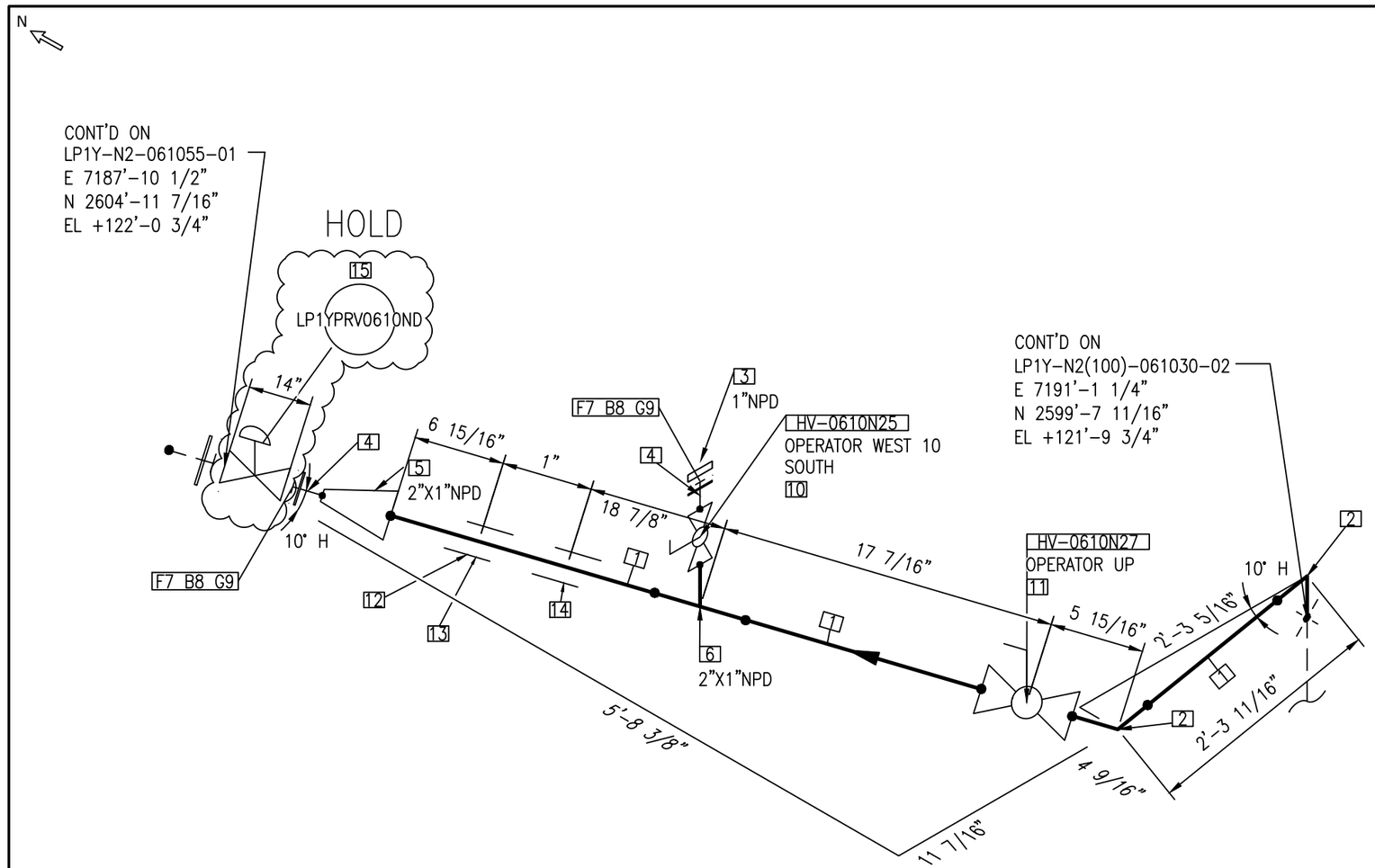

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	4'-11"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	2	2"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
3	1	1"	FLG BLIND 150# RF 316/316L SS	5616520
4	2	1"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
5	1	2"X1"	REDUCER CONC SCH 10S 316/316L SS A403-W	5450717
6	1	2"X1"	TEE RED SCH 10S 316/316L SS A403-W	5450714
7	2	1"	FLG LAP JOINT 150# 316 SS	5599578L
8	8	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
9	2	1"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
10	1	1"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
11	1	2"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
12	1	2"	5CI (ISOLATION CRADLE)	5CI-02
13	1	2"	5FSS5 (FIELD BASE SUPPORT)	5FSS5-L-1
14	1	2"	5UG (U-BOLT GUIDE)	5UG-02
15	1	1"	PRV CL 150# RF	LP1YPRV0610ND

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	10-MAR-2026	DRW	SBR	N/A	JHN / N/A	MDW

GENERAL NOTE:
1. REFER TO THE PIPING LINE LIST FOR PROCESS DATA/CONDITIONS AND OTHER FABRICATION DETAILS SUCH AS TESTING, CLEANING, AND PAINTING REQUIREMENTS.
2. UNLESS OTHERWISE NOTED, ALL WELDS ARE INDICATED AS SHOP WELDS. CONTRACTOR TO DETERMINE FIELD WELDS LOCATIONS AND DOCUMENT ACCORDINGLY ON AS-BUILT RECORDS.
3. WELDS AND/OR COUPLINGS DEFINING RANDOM PIPE SPOOL LENGTHS ARE FOR REFERENCE ONLY. FINAL WELD OR COUPLING LOCATION IS TO BE DETERMINED BY THE CONTRACTOR.
4. THE MECHANICAL CONTRACTOR TO FIELD VERIFY INTERFACE POINTS BETWEEN EQUIPMENT & PLUMBING FIXTURES IN 3rd FIX (3F SERIES CWP) AND YS YARD STICK BUILT (YS SERIES CWP EQUIPMENT CONNECTIONS) ISOMETRICS PRIOR TO FABRICATION.
5. THE MECHANICAL CONTRACTOR IS TO VERIFY ALL DIMENSIONAL DATA WITH ACTUAL FIELD CONDITIONS PRIOR TO INSTALLATION OF THE SYSTEM.
6. PIPE SUPPORTS SHALL BE FURNISHED AND INSTALLED IN ACCORDANCE WITH THIS DRAWING AND SPECIFICATION LP100-250-EMS-15110. MECHANICAL CONTRACTOR SHALL PROVIDE TEMPORARY SUPPORTS AS REQUIRED TO AID PRESSURE TESTING, INSTALLATION ETC.
7. PIPE SUPPORTS INDICATED ON THE DRAWING MAY BE PART OF AN ASSEMBLY ON SEVERAL ISOMETRICS. REFER TO THE 3D MODEL FOR THE COMPLETE PIPE SUPPORT ASSEMBLY.
8. MECHANICAL CONTRACTOR SHALL NOTE ALL DESIGN CHANGES ON THIS DRAWING AND RETURN IT TO THE CONSTRUCTION WPO FOR AS-BUILT RECORDS.
9. THE BILL OF MATERIAL IS FOR REFERENCE ONLY.
10. SLOPE EQUIVALENTS: 1:27 = 7/16" PER FT, 1:96 = 1/8" PER FT, 1:120 = 1/10" PER FT, 1:192 = 1/16" PER FT, 1:480 = 1/40" PER FT.

NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2(100)
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2(100) (315) E 061030

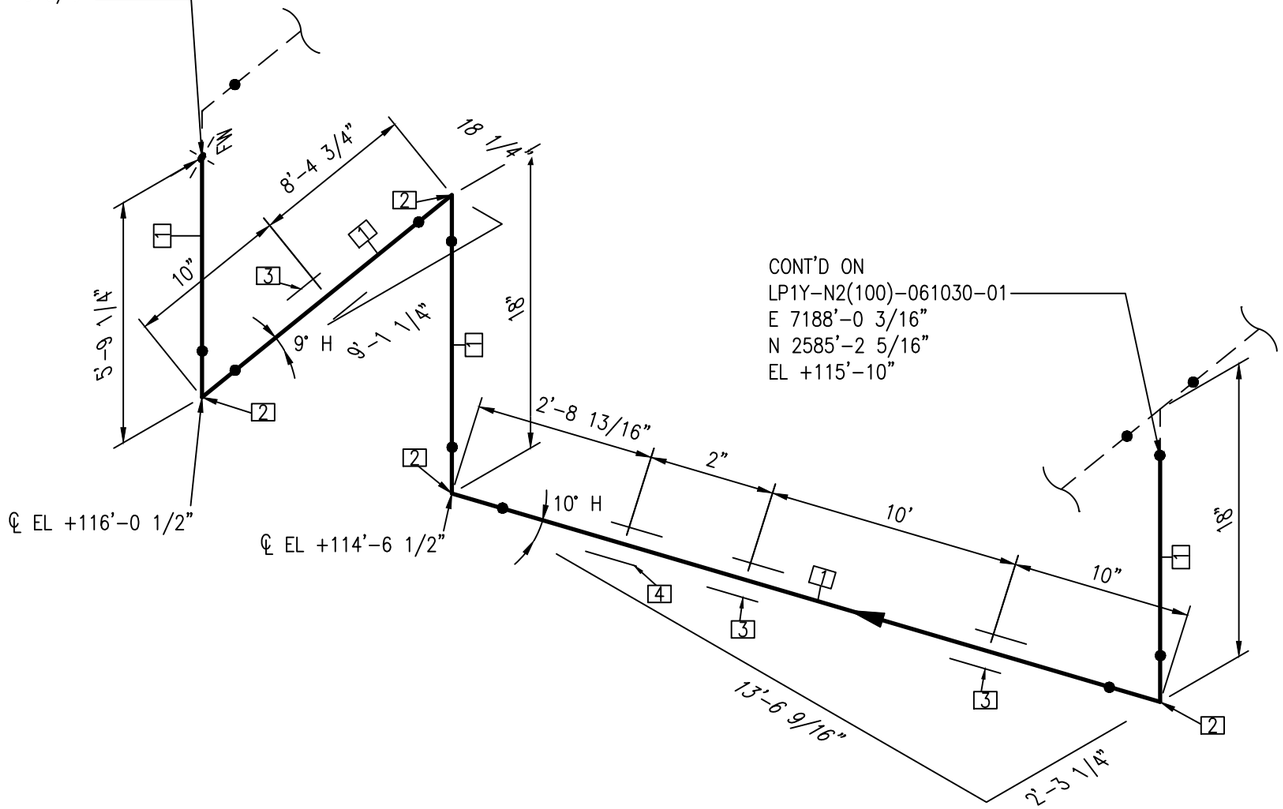
REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-N2(100)-061030-03	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 10-MAR-2026

NOTICE: THIS DRAWING HAS NOT BEEN PUBLISHED AND IS THE SOLE PROPERTY OF FLUOR AND IS LENT TO THE BORROWER FOR THEIR CONFIDENTIAL USE ONLY, AND IN CONSIDERATION OF THE LOAN OF THIS DRAWING, THE BORROWER PROMISES AND AGREES TO RETURN IT UPON REQUEST AND AGREES THAT IT WILL NOT BE REPRODUCED, COPIED, LENT, OR OTHERWISE DISPOSED OF DIRECTLY OR INDIRECTLY, NOR USED FOR ANY PURPOSE OTHER THAN THAT FOR WHICH IT IS FURNISHED.

N

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	29'-7"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	4	2"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
3	3	2"	5CI (ISOLATION CRADLE)	5CI-02
4	1	2"	5UG (U-BOLT GUIDE)	5UG-02



A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	10-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

GENERAL NOTE:
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6. PIPE SUPPORTS SHALL BE FURNISHED AND INSTALLED IN ACCORDANCE WITH THIS DRAWING AND SPECIFICATION LP100-250-EMS-15110. MECHANICAL CONTRACTOR SHALL PROVIDE TEMPORARY SUPPORTS AS REQUIRED TO AID PRESSURE TESTING, INSTALLATION ETC.
7. PIPE SUPPORTS INDICATED ON THE DRAWING MAY BE PART OF AN ASSEMBLY ON SEVERAL ISOMETRICS. REFER TO THE 3D MODEL FOR THE COMPLETE PIPE SUPPORT ASSEMBLY.
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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2(100)
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2(100) (315) E 061030

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-N2(100)-061030-04	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 10-MAR-2026

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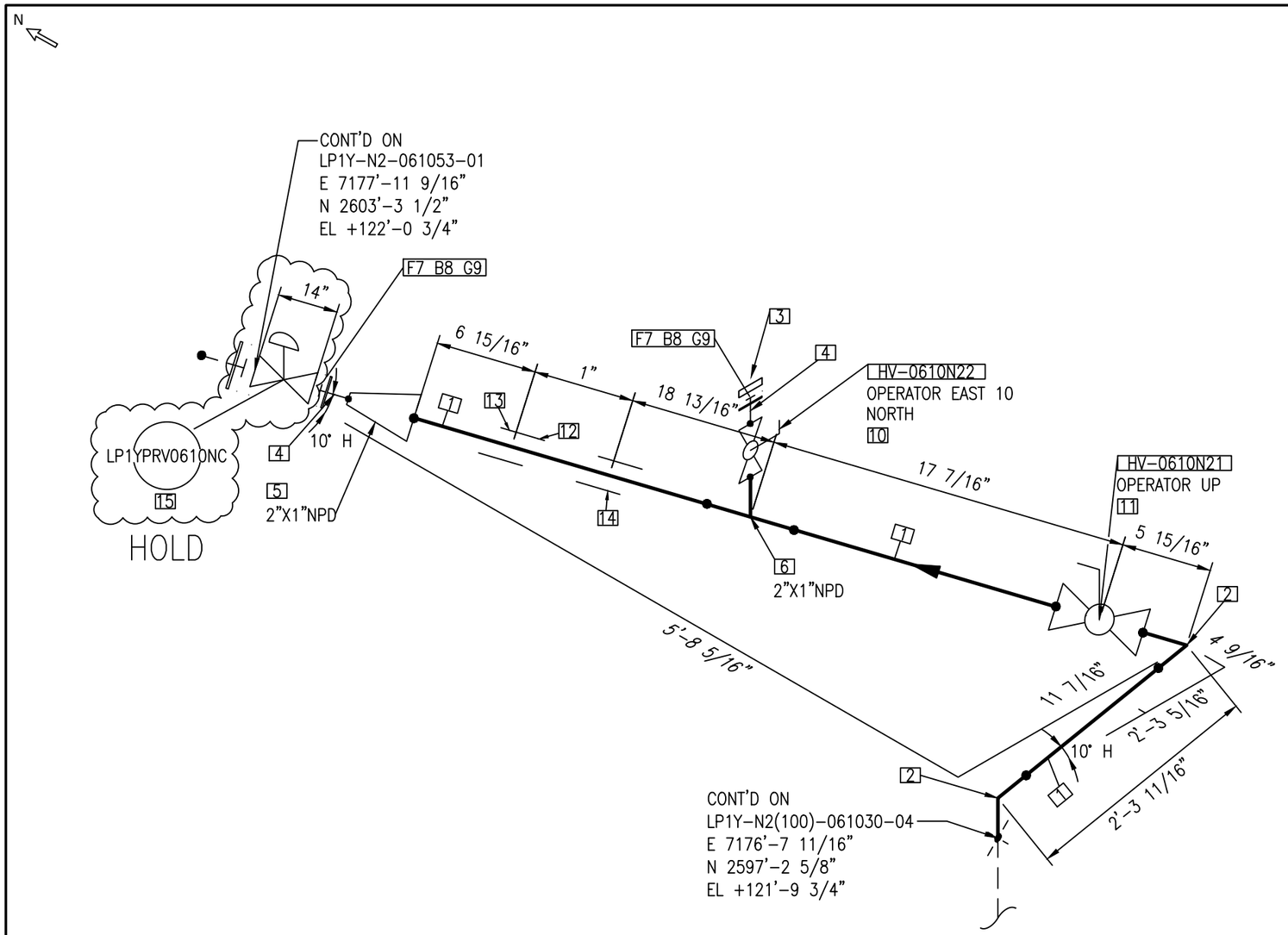

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	4'-10"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	2	2"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
3	1	1"	FLG BLIND 150# RF 316/316L SS	5616520
4	2	1"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
5	1	2"X1"	REDUCER CONC SCH 10S 316/316L SS A403-W	5450717
6	1	2"X1"	TEE RED SCH 10S 316/316L SS A403-W	5450714
7	2	1"	FLG LAP JOINT 150# 316 SS	5599578L
8	8	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
9	2	1"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
10	1	1"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
11	1	2"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
12	1	2"	5CI (ISOLATION CRADLE)	5CI-02
13	1	2"	5FSS5 (FIELD BASE SUPPORT)	5FSS5-L-1
14	1	2"	5UG (U-BOLT GUIDE)	5UG-02
15	1	1"	PRV CL 150# RF	LP1YPRV0610NC

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2(100)
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	10-MAR-2026	DRW	SBR	N/A	JHN / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2(100) (315) E 061030

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-N2(100)-061030-05	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 10-MAR-2026

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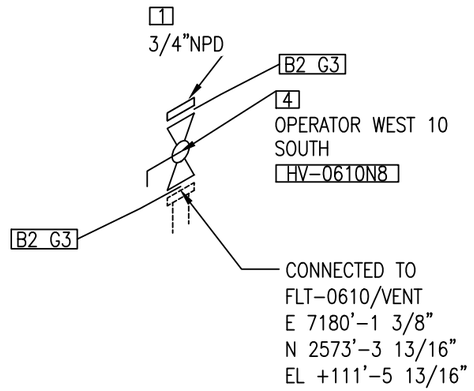
N

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	1	3/4"	FLG BLIND 150# RF 316/316L SS	5616520
2	8	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
3	2	3/4"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
4	1	3/4"	BALL 150# RF 316SS TFE FP HNDL OP	5352193L1



A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	13-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

GENERAL NOTE:
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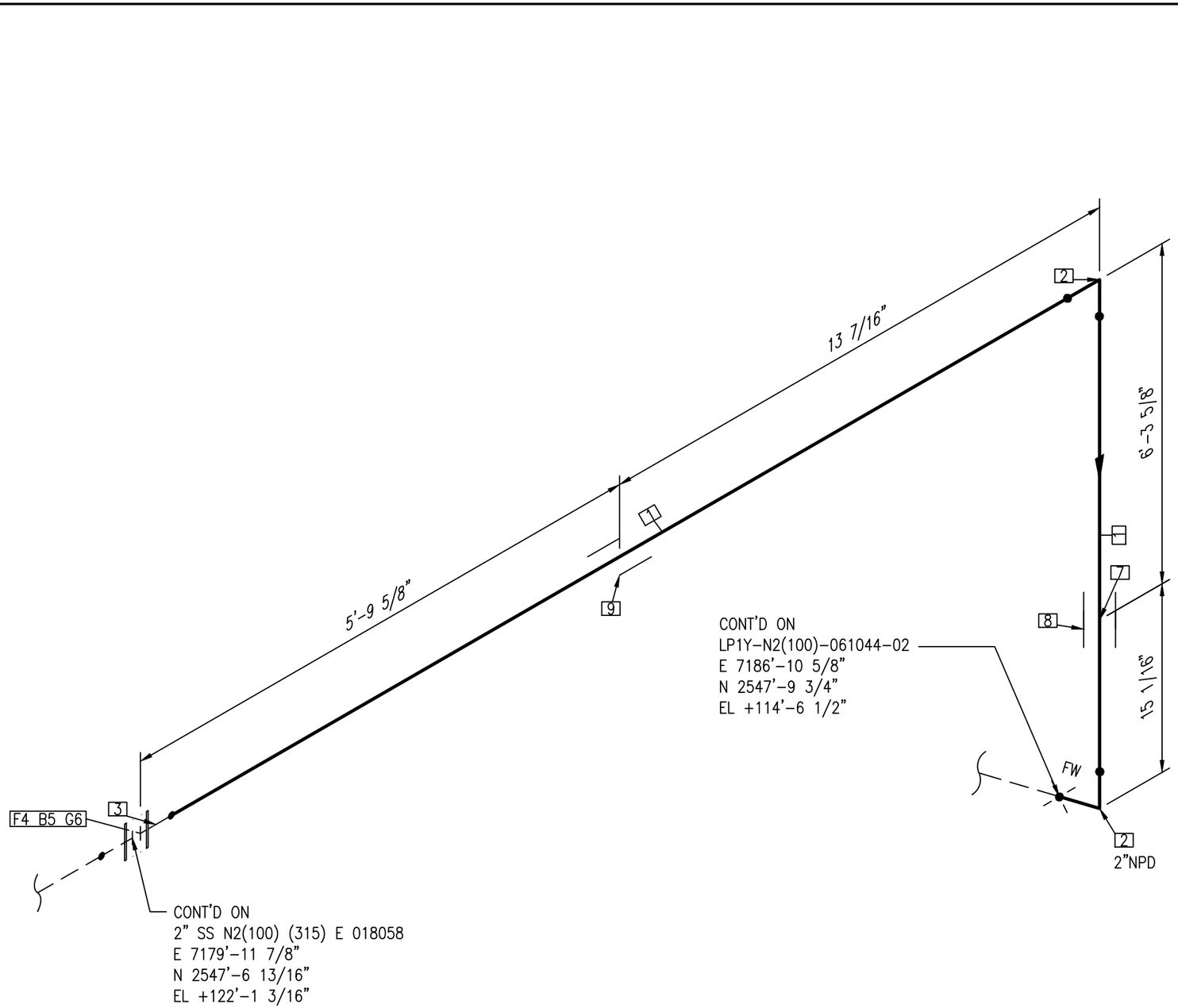
NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2(100)
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

LINE NUMBER
LP1Y 2" SS N2(100) (315) E 061043

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-N2(100)-061043-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 13-MAR-2026

NOTICE: THIS DRAWING HAS NOT BEEN PUBLISHED AND IS THE SOLE PROPERTY OF FLUOR AND IS LENT TO THE BORROWER FOR THEIR CONFIDENTIAL USE ONLY, AND IN CONSIDERATION OF THE LOAN OF THIS DRAWING, THE BORROWER PROMISES AND AGREES TO RETURN IT UPON REQUEST AND AGREES THAT IT WILL NOT BE REPRODUCED, COPIED, LENT, OR OTHERWISE DISPOSED OF DIRECTLY OR INDIRECTLY, NOR USED FOR ANY PURPOSE OTHER THAN THAT FOR WHICH IT IS FURNISHED.



BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	13'-7"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	2	2"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
3	1	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
4	1	2"	FLG LAP JOINT 150# 316 SS	5599578L
5	4	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
6	1	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
7	1	2"	5US (U-BOLT SUPPORT)	5US-02
8	1	2"	5FS1 (FIELD SUPPORT)	5FS1S1
9	1	2"	5CI (ISOLATION CRADLE)	5CI-02

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	13-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2(100)
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

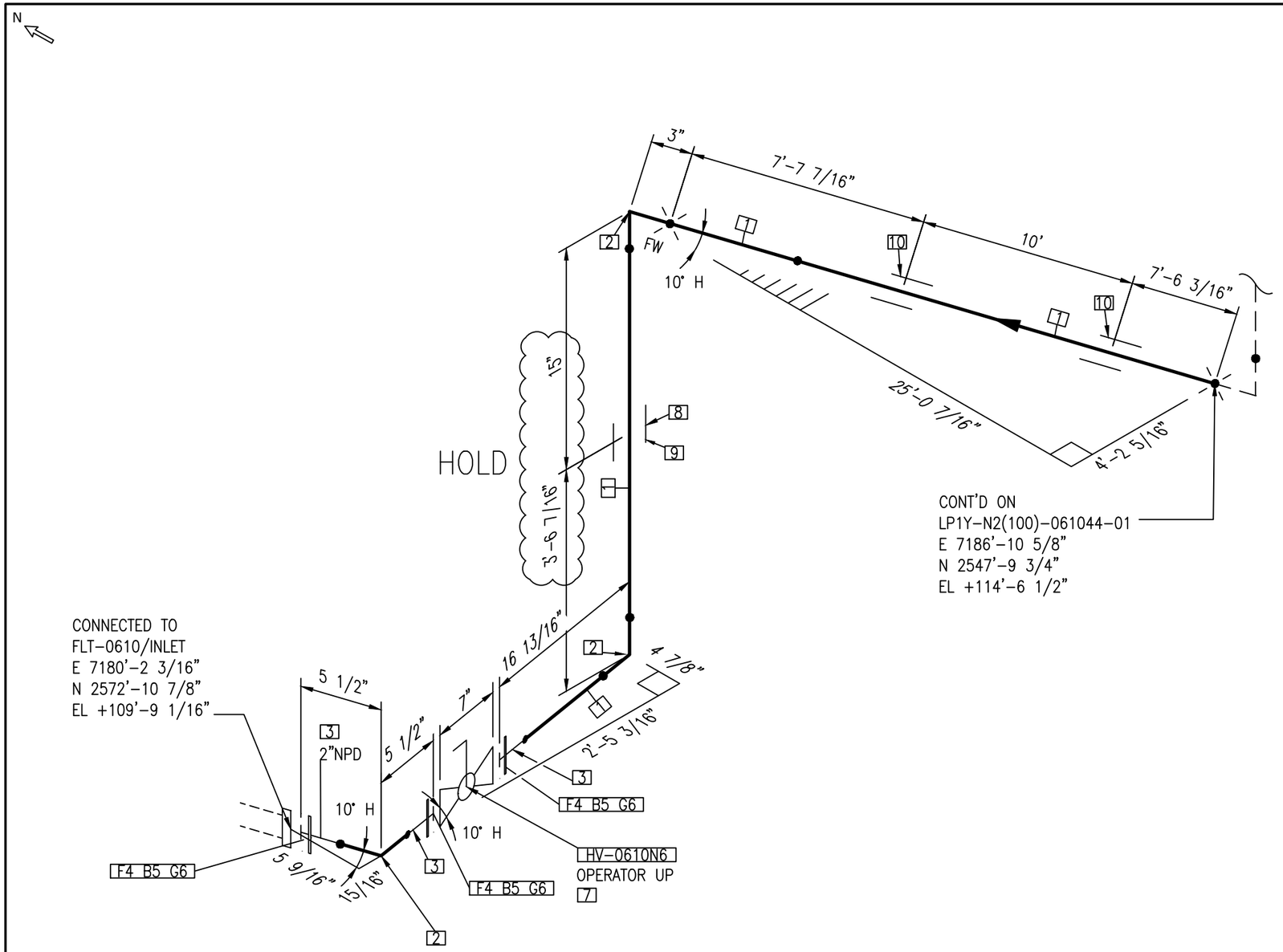
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2(100) (315) E 061044

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-N2(100)-061044-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 13-MAR-2026

NOTICE: THIS DRAWING HAS NOT BEEN PUBLISHED AND IS THE SOLE PROPERTY OF FLUOR AND IS LENT TO THE BORROWER FOR THEIR CONFIDENTIAL USE ONLY, AND IN CONSIDERATION OF THE LOAN OF THIS DRAWING, THE BORROWER PROMISES AND AGREES TO RETURN IT UPON REQUEST AND AGREES THAT IT WILL NOT BE REPRODUCED, COPIED, LENT, OR OTHERWISE DISPOSED OF DIRECTLY OR INDIRECTLY, NOR USED FOR ANY PURPOSE OTHER THAN THAT FOR WHICH IT IS FURNISHED.



BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	30'-5"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	3	2"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
3	3	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
4	3	2"	FLG LAP JOINT 150# 316 SS	5599578L
5	12	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
6	3	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
7	1	2"	BALL 150# RF 316SS TFE FP HNDL OP	5352194L1
8	1	2"	5US (U-BOLT SUPPORT)	5US-02
9	1	2"	5FS1 (FIELD SUPPORT)	5FS1S1
10	2	2"	5CI (ISOLATION CRADLE)	5CI-02

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	19-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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5. THE MECHANICAL CONTRACTOR IS TO VERIFY ALL DIMENSIONAL DATA WITH ACTUAL FIELD CONDITIONS PRIOR TO INSTALLATION OF THE SYSTEM.
6. PIPE SUPPORTS SHALL BE FURNISHED AND INSTALLED IN ACCORDANCE WITH THIS DRAWING AND SPECIFICATION LP100-250-EMS-15110. MECHANICAL CONTRACTOR SHALL PROVIDE TEMPORARY SUPPORTS AS REQUIRED TO AID PRESSURE TESTING, INSTALLATION ETC.
7. PIPE SUPPORTS INDICATED ON THE DRAWING MAY BE PART OF AN ASSEMBLY ON SEVERAL ISOMETRICS. REFER TO THE 3D MODEL FOR THE COMPLETE PIPE SUPPORT ASSEMBLY.
8. MECHANICAL CONTRACTOR SHALL NOTE ALL DESIGN CHANGES ON THIS DRAWING AND RETURN IT TO THE CONSTRUCTION WPO FOR AS-BUILT RECORDS.
9. THE BILL OF MATERIAL IS FOR REFERENCE ONLY.
10. SLOPE EQUIVALENTS: 1:27 = 7/16" PER FT, 1:96 = 1/8" PER FT, 1:120 = 1/10" PER FT, 1:192 = 1/16" PER FT, 1:480 = 1/40" PER FT.

NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2(100)
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

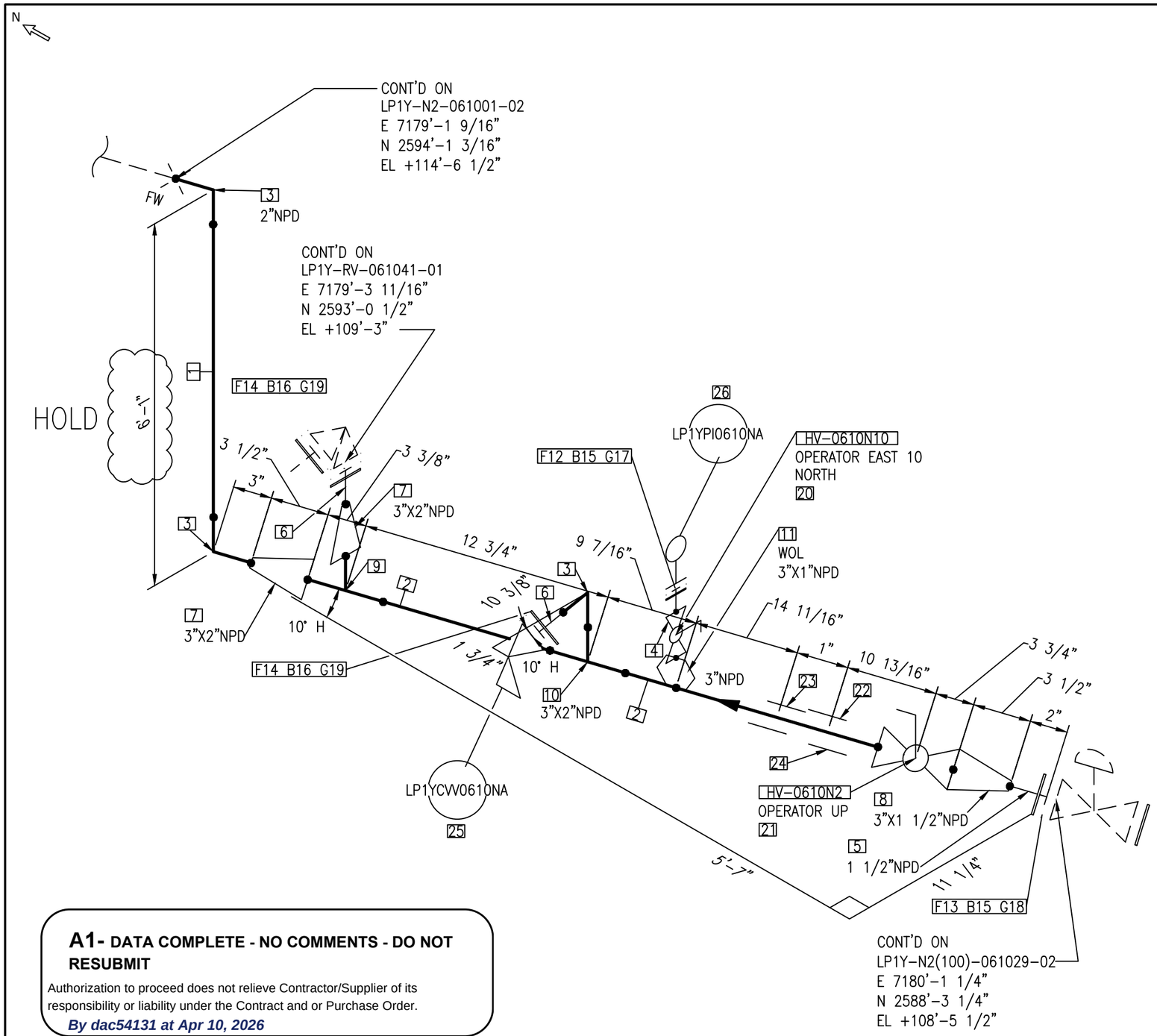
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2(100) (315) E 061044

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-N2(100)-061044-02	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 19-MAR-2026

NOTICE: THIS DRAWING HAS NOT BEEN PUBLISHED AND IS THE SOLE PROPERTY OF FLUOR AND IS LENT TO THE BORROWER FOR THEIR CONFIDENTIAL USE ONLY, AND IN CONSIDERATION OF THE LOAN OF THIS DRAWING, THE BORROWER PROMISES AND AGREES TO RETURN IT UPON REQUEST AND AGREES THAT IT WILL NOT BE REPRODUCED, COPIED, LENT, OR OTHERWISE DISPOSED OF DIRECTLY OR INDIRECTLY, NOR USED FOR ANY PURPOSE OTHER THAN THAT FOR WHICH IT IS FURNISHED.



BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	5'-7"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	2'-11"	3"	PIPE SCH 10S ERW 316/316L SS A312	5368751
3	3	2"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
4	1	1"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
5	1	1 1/2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
6	2	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
7	2	3"X2"	REDUCER CONC SCH 10S 316/316L SS A403-W	5450717
8	1	3"X1 1/2"	REDUCER CONC SCH 10S 316/316L SS A403-W	5450717
9	1	3"	TEE SCH 10S 316/316L SS A403-W	5450715
10	1	3"X2"	TEE RED SCH 10S 316/316L SS A403-W	5450714
11	1	3"X1"	PIPET SCH 10S 316/316L SS	5510132
12	1	1"	FLG LAP JOINT 150# 316 SS	5599578L
13	1	1 1/2"	FLG LAP JOINT 150# 316 SS	5599578L
14	2	2"	FLG LAP JOINT 150# 316 SS	5599578L
15	8	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
16	8	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
17	1	1"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
18	1	1 1/2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
19	2	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
20	1	1"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
21	1	3"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
22	1	3"	5CI (ISOLATION CRADLE)	5CI-03
23	1	3"	5UG (U-BOLT GUIDE)	5UG-03
24	1	3"	5FSS5 (FIELD BASE SUPPORT)	5FSS5-L-1
25	1	2"	CVV (VACUUM RELIEF VALVE)	LP1YCVV0610NA
26	1	1"	PRESSURE GAUGE	LP1YPI0610NA

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	10-MAR-2026	DRW	SBR	N/A	JHN / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2 (315) E 061001

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-N2-061001-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 10-MAR-2026

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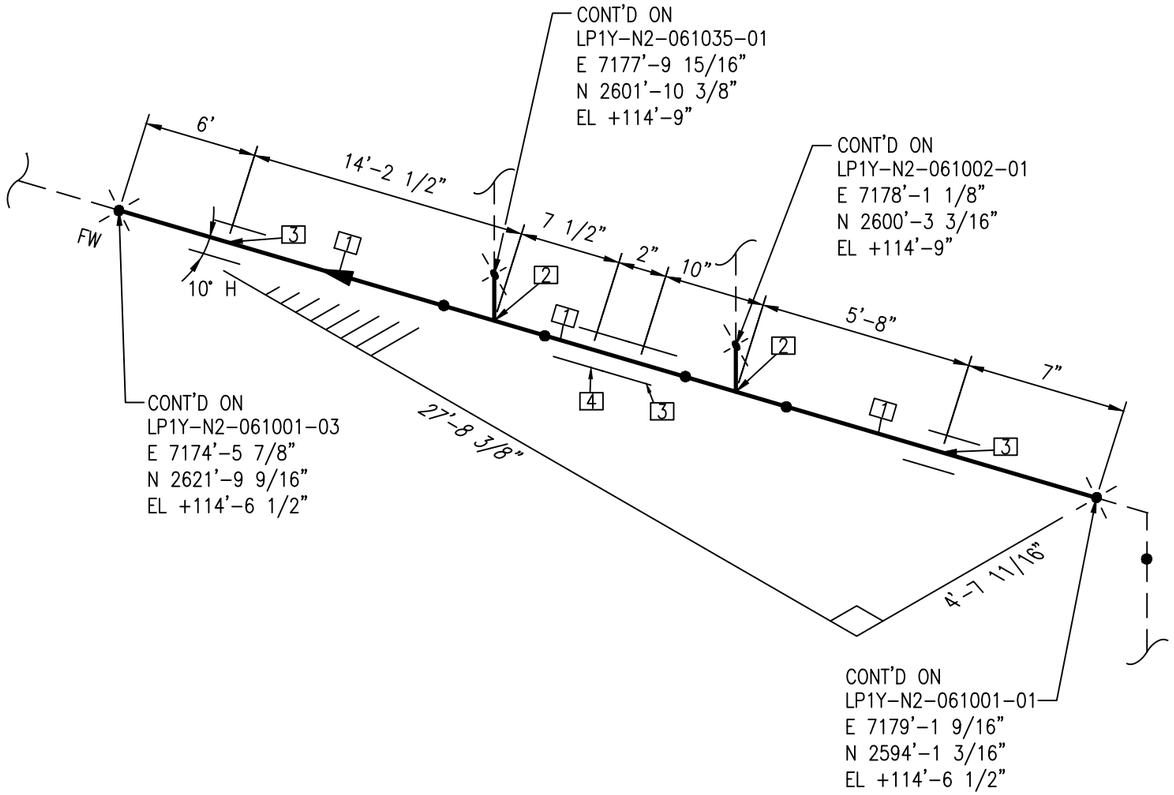
N

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	27'-3"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	2	2"	TEE SCH 10S 316/316L SS A403-W	5450715
3	3	2"	5CI (ISOLATION CRADLE)	5CI-02
4	1	2"	5UG (U-BOLT GUIDE)	5UG-02



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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	10-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2 (315) E 061001

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-N2-061001-02	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 10-MAR-2026

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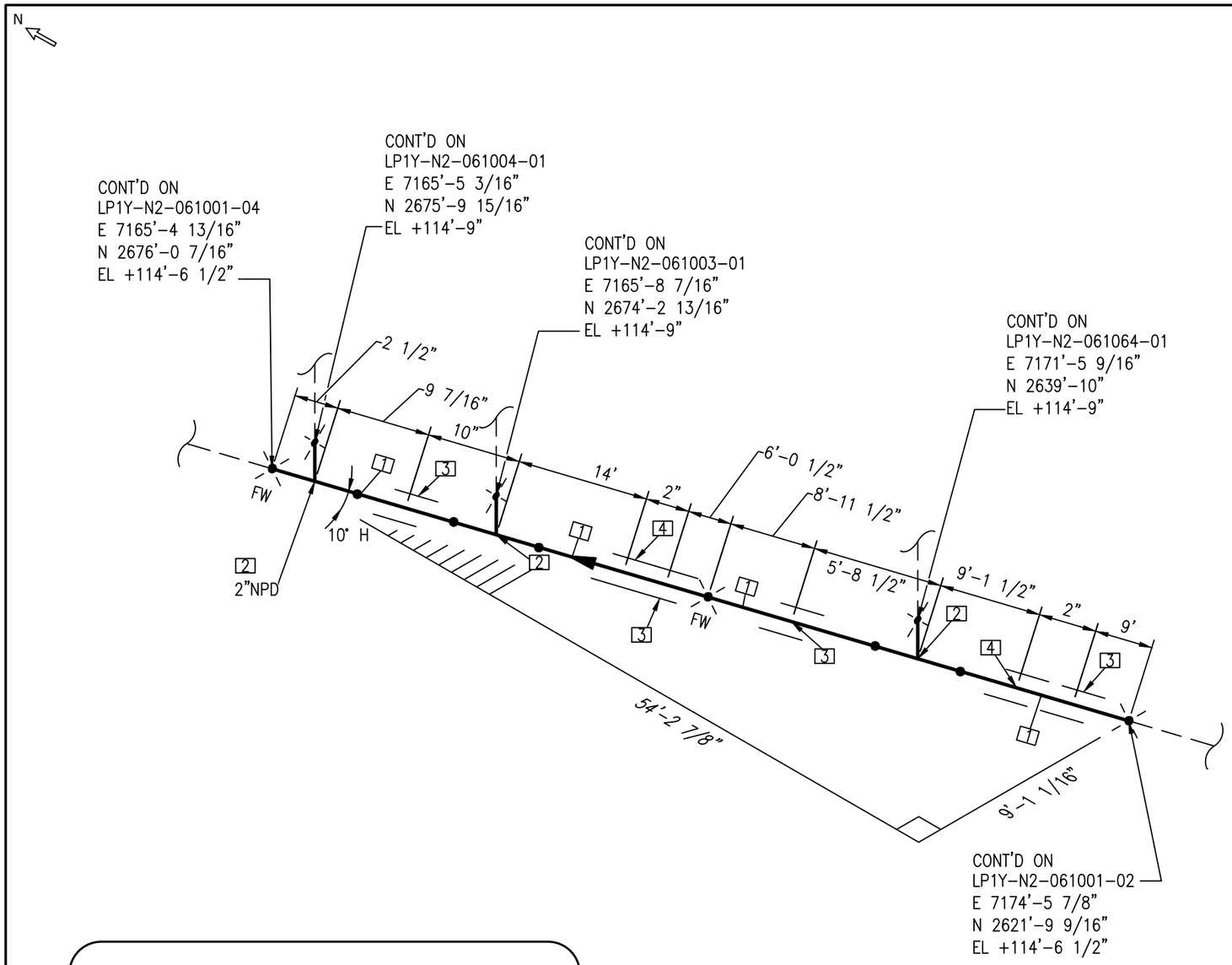

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	53'-9"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	3	2"	TEE SCH 10S 316/316L SS A403-W	5450715
3	4	2"	5CI (ISOLATION CRADLE)	5CI-02
4	2	2"	5UG (U-BOLT GUIDE)	5UG-02

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	10-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2 (315) E 061001

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-N2-061001-03	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 10-MAR-2026

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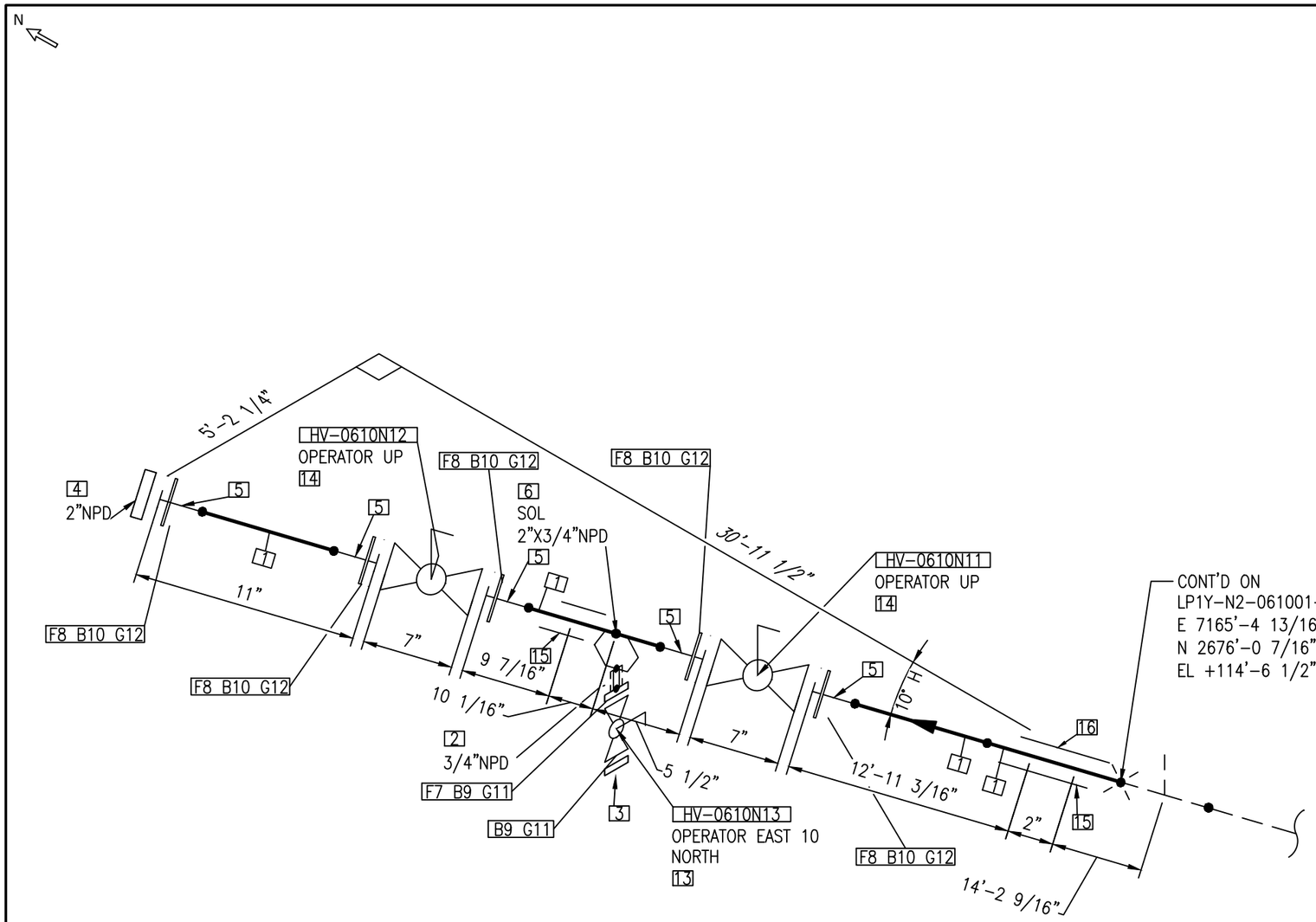

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	29'-1"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	1	3/4"	NIPPLE SCH 10S 316/316L SS PBE 3" LONG	5540491
3	1	3/4"	FLG BLIND 150# RF 316/316L SS	5616520
4	1	2"	FLG BLIND 150# RF 316/316L SS	5616520
5	5	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
6	1	2"X3/4"	PIPET 10S X 3000# SW 316/316L SS	5537081
7	1	3/4"	FLG SW 150# RF 316/316L SS 10S BORE	5608148
8	5	2"	FLG LAP JOINT 150# 316 SS	5599578L
9	8	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
10	20	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
11	2	3/4"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
12	5	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
13	1	3/4"	BALL 150# RF 316SS TFE FP HNDL OP	5352193L1
14	2	2"	BALL 150# RF 316SS TFE FP HNDL OP	5352194L1
15	2	2"	5CI (ISOLATION CRADLE)	5CI-02
16	1	2"	5UG (U-BOLT GUIDE)	5UG-02

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	10-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2 (315) E 061001

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-N2-061001-04	0

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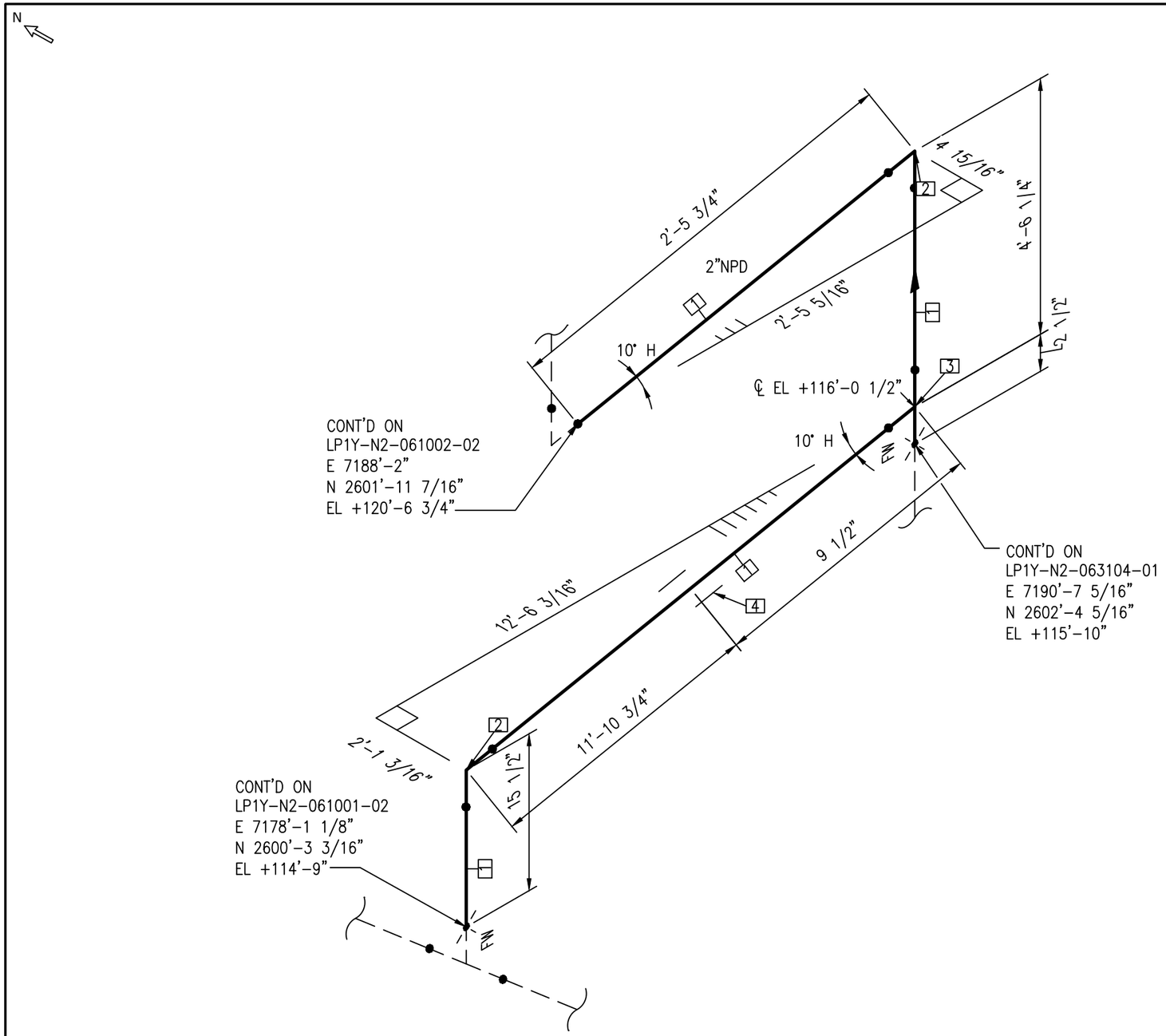

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	19'-7"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	2	2"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
3	1	2"	TEE SCH 10S 316/316L SS A403-W	5450715
4	1	2"	5CI (ISOLATION CRADLE)	5CI-02

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	10-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2 (315) E 061002

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-N2-061002-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 10-MAR-2026

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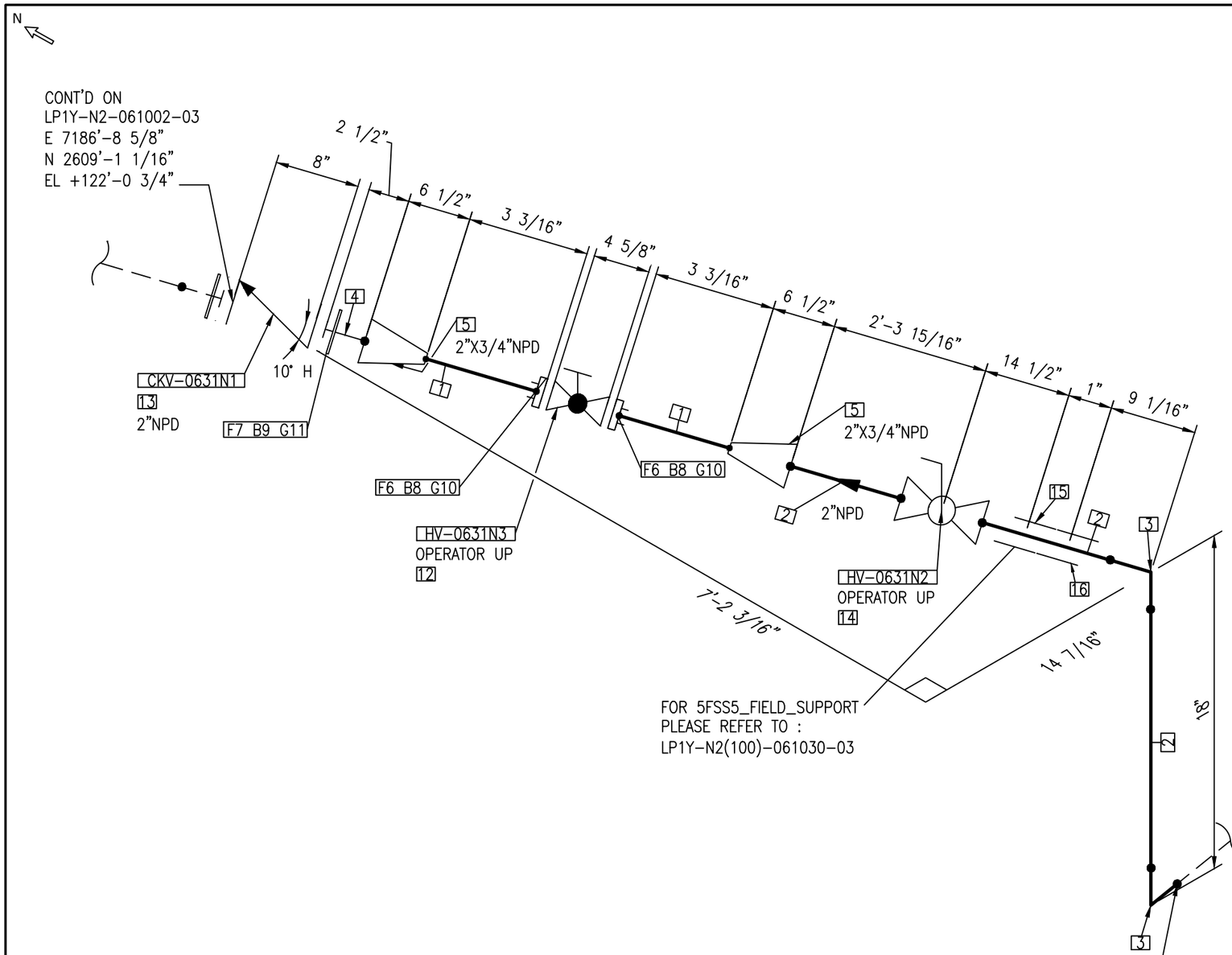

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	7"	3/4"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	4'-8"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
3	2	2"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
4	1	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
5	2	2"X3/4"	SWAGE CONC SCH 10S 316/316L SS BBE	5546561
6	2	3/4"	FLG SW 150# RF 316/316L SS 10S BORE	5608148
7	1	2"	FLG LAP JOINT 150# 316 SS	5599578L
8	8	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
9	4	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
10	2	3/4"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
11	1	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
12	1	3/4"	GLOBE 150# RF 316 SS	5254829
13	1	2"	CHECK 150# RF 316 SS SWING	5257711L
14	1	2"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
15	1	2"	5CI (ISOLATION CRADLE)	5CI-02
16	1	2"	5UG (U-BOLT GUIDE)	5UG-02

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	10-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2 (315) E 061002

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-N2-061002-02	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 10-MAR-2026

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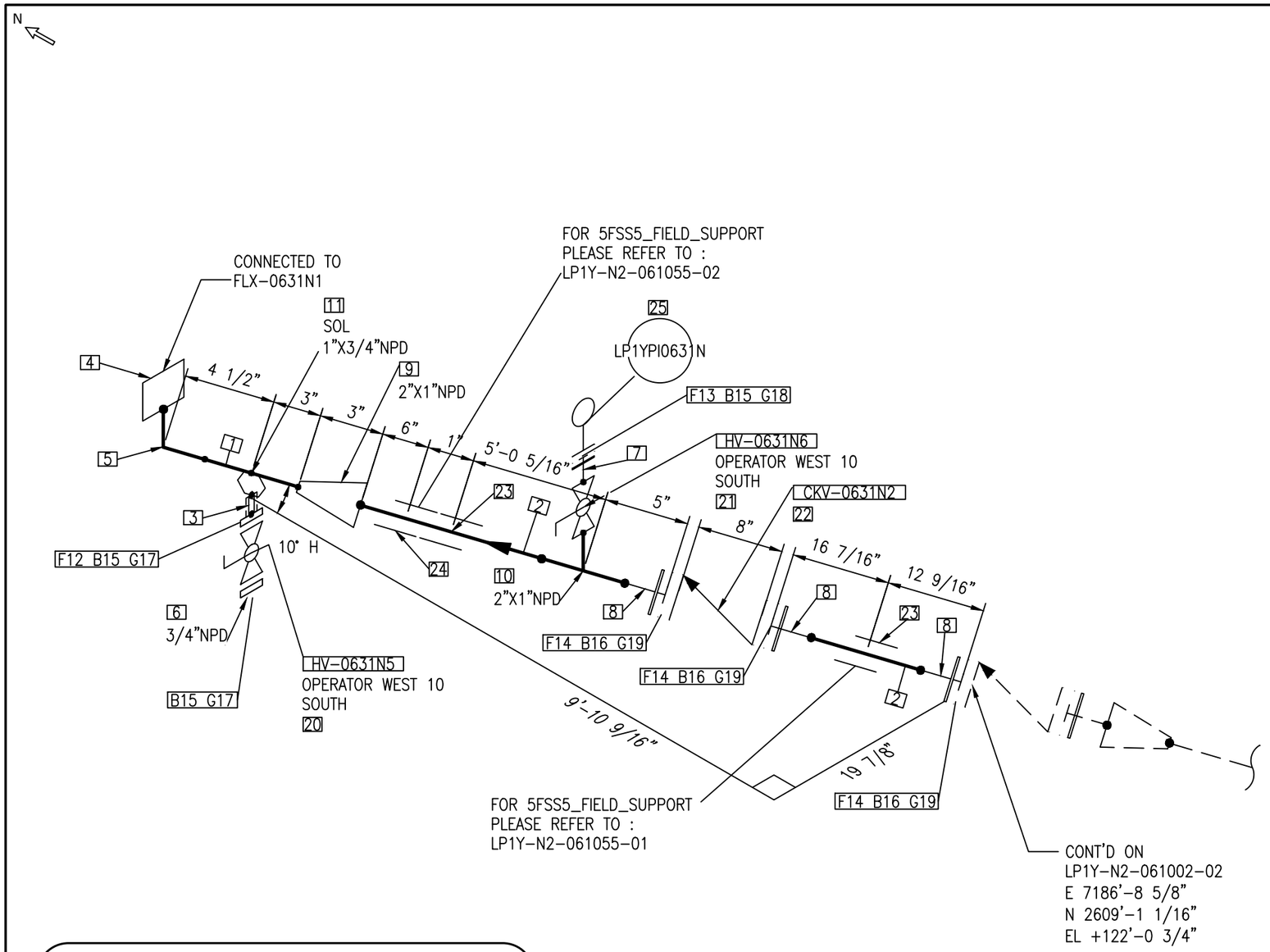

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	6"	1"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	7'-5"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
3	1	3/4"	NIPPLE SCH 10S 316/316L SS PBE 3" LONG	5540491
4	1	1"	BUTTWELD CAMLOCK CONNECTION UNIT	LP1YCCU0631N2
5	1	1"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
6	1	3/4"	FLG BLIND 150# RF 316/316L SS	5616520
7	1	1"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
8	3	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
9	1	2"X1"	REDUCER CONC SCH 10S 316/316L SS A403-W	5450717
10	1	2"X1"	TEE RED SCH 10S 316/316L SS A403-W	5450714
11	1	1"X3/4"	PIPET 10S X 3000# SW 316/316L SS	5537081
12	1	3/4"	FLG SW 150# RF 316/316L SS 10S BORE	5608148
13	1	1"	FLG LAP JOINT 150# 316 SS	5599578L
14	3	2"	FLG LAP JOINT 150# 316 SS	5599578L
15	12	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
16	12	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
17	2	3/4"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
18	1	1"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
19	3	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
20	1	3/4"	BALL 150# RF 316SS TFE FP HNDL OP	5352193L1
21	1	1"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
22	1	2"	CHECK 150# RF 316 SS SWING	5257711L
23	2	2"	5CI (ISOLATION CRADLE)	5CI-02
24	1	2"	5UG (U-BOLT GUIDE)	5UG-02
25	1	1"	PRESSURE GAUGE	LP1YPI0631N

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	10-MAR-2026	DRW	SBR	N/A	JHN / N/A	MDW

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NODE:	TFS		
BUILDING NO:	Y		
CWP NO:	N/A		
MODULE NO:	N/A		
SERVICE CODE:	N2		
PIPE SCHEDULE:	315		
INSULATION CODE:	-		
TRACING TYPE:	NT		
ELASTOMER:	E		
HYGIENIC:	NO	IMPACT:	YES

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2 (315) E 061002

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-N2-061002-03	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 10-MAR-2026

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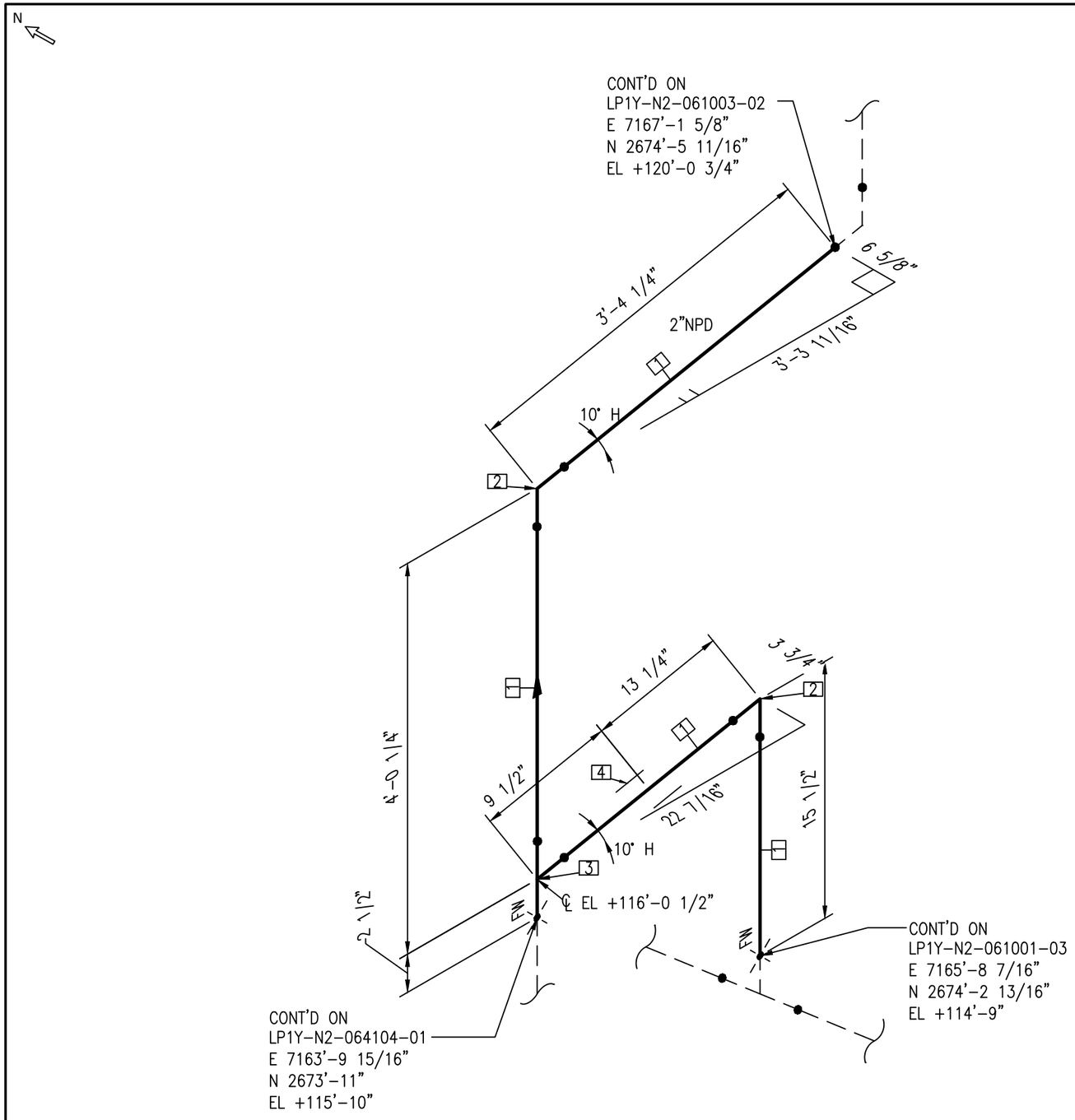

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	9'-2"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	2	2"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
3	1	2"	TEE SCH 10S 316/316L SS A403-W	5450715
4	1	2"	5CI (ISOLATION CRADLE)	5CI-02

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By dac54131 at Apr 10, 2026

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0	ISSUE FOR CONSTRUCTION	10-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2 (315) E 061003

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-N2-061003-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 10-MAR-2026

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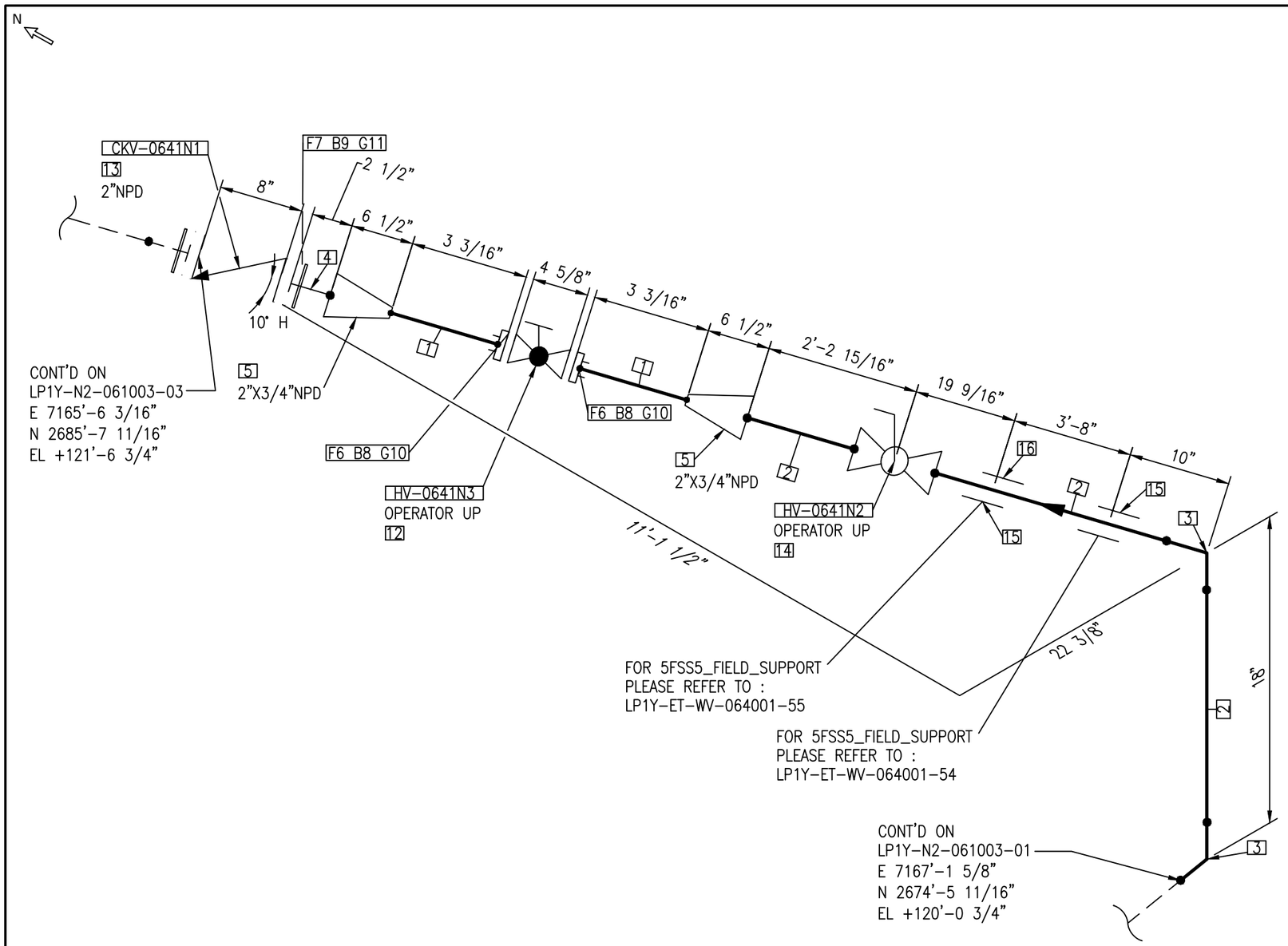

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12	1	3/4"	GLOBE 150# RF 316 SS	5254829
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By dac54131 at Apr 10, 2026

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0	ISSUE FOR CONSTRUCTION	10-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2 (315) E 061003

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-N2-061003-02	0

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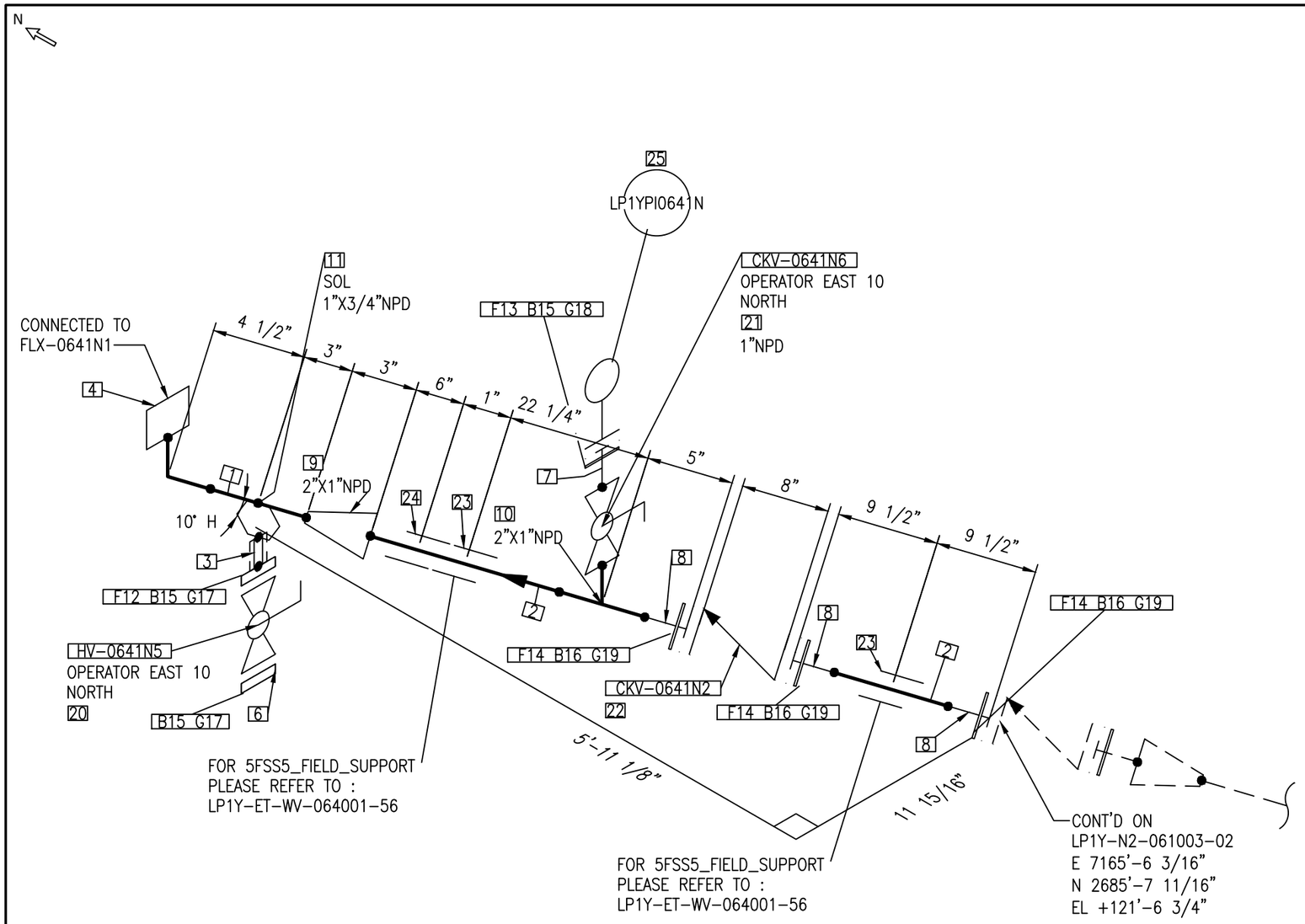

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By dac54131 at Apr 10, 2026

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5. THE MECHANICAL CONTRACTOR IS TO VERIFY ALL DIMENSIONAL DATA WITH ACTUAL FIELD CONDITIONS PRIOR TO INSTALLATION OF THE SYSTEM.
6. PIPE SUPPORTS SHALL BE FURNISHED AND INSTALLED IN ACCORDANCE WITH THIS DRAWING AND SPECIFICATION LP100-250-EMS-15110. MECHANICAL CONTRACTOR SHALL PROVIDE TEMPORARY SUPPORTS AS REQUIRED TO AID PRESSURE TESTING, INSTALLATION ETC.
7. PIPE SUPPORTS INDICATED ON THE DRAWING MAY BE PART OF AN ASSEMBLY ON SEVERAL ISOMETRICS. REFER TO THE 3D MODEL FOR THE COMPLETE PIPE SUPPORT ASSEMBLY.
8. MECHANICAL CONTRACTOR SHALL NOTE ALL DESIGN CHANGES ON THIS DRAWING AND RETURN IT TO THE CONSTRUCTION WPO FOR AS-BUILT RECORDS.
9. THE BILL OF MATERIAL IS FOR REFERENCE ONLY.
10. SLOPE EQUIVALENTS: 1:27 = 7/16" PER FT, 1:96 = 1/8" PER FT, 1:120 = 1/10" PER FT, 1:192 = 1/16" PER FT, 1:480 = 1/40" PER FT.

NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	17-MAR-2026	DRW	SBR	N/A	JHN / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2 (315) E 061003

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-N2-061003-03	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 17-MAR-2026

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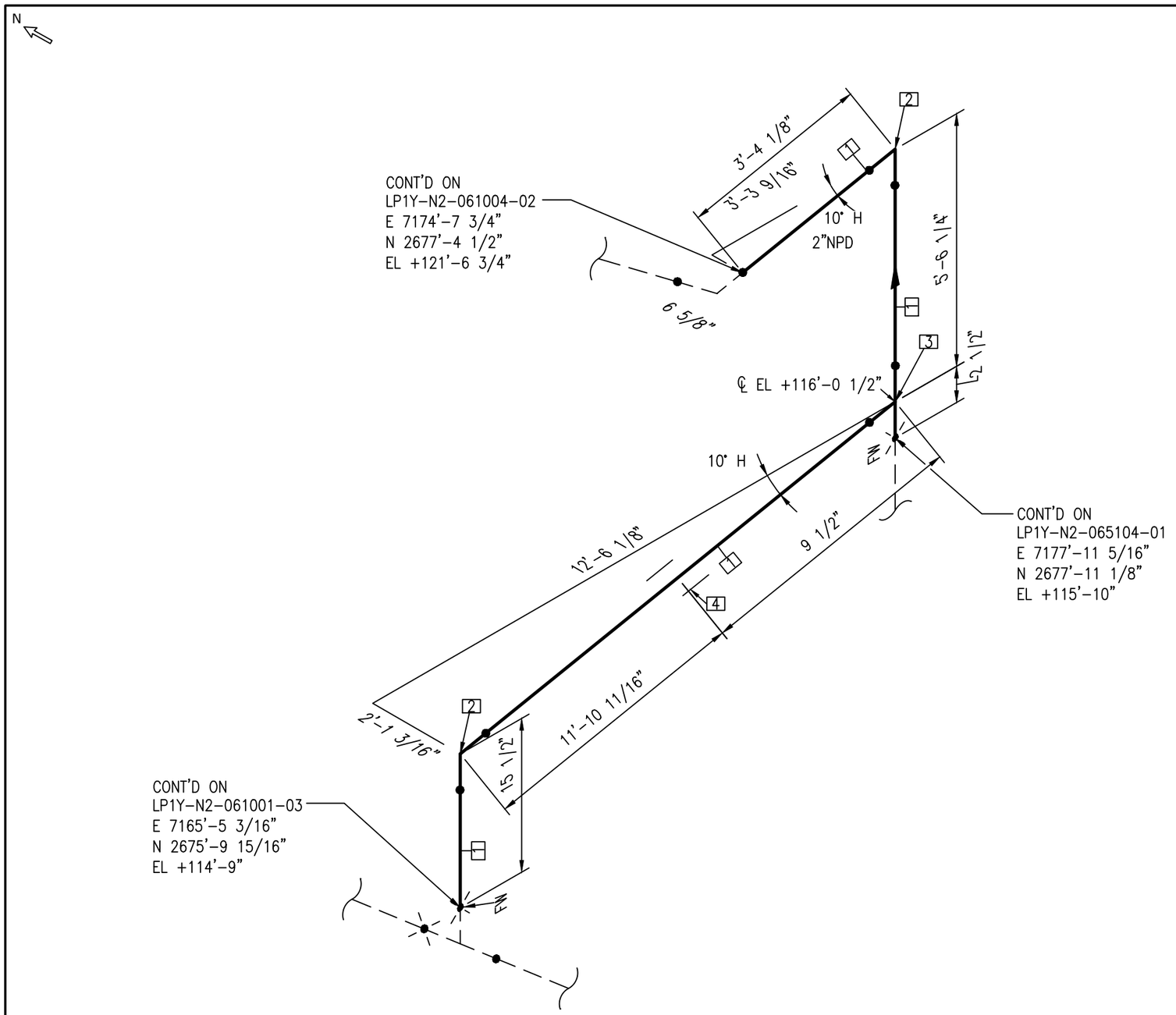

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	21'-6"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	2	2"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
3	1	2"	TEE SCH 10S 316/316L SS A403-W	5450715
4	1	2"	5CI (ISOLATION CRADLE)	5CI-02

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	10-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2 (315) E 061004

REFERENCE P&ID NUMBER
LP1Y-DI-00000061

ISOMETRIC DRAWING NUMBER
LP1Y-N2-061004-01

REV
0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 10-MAR-2026

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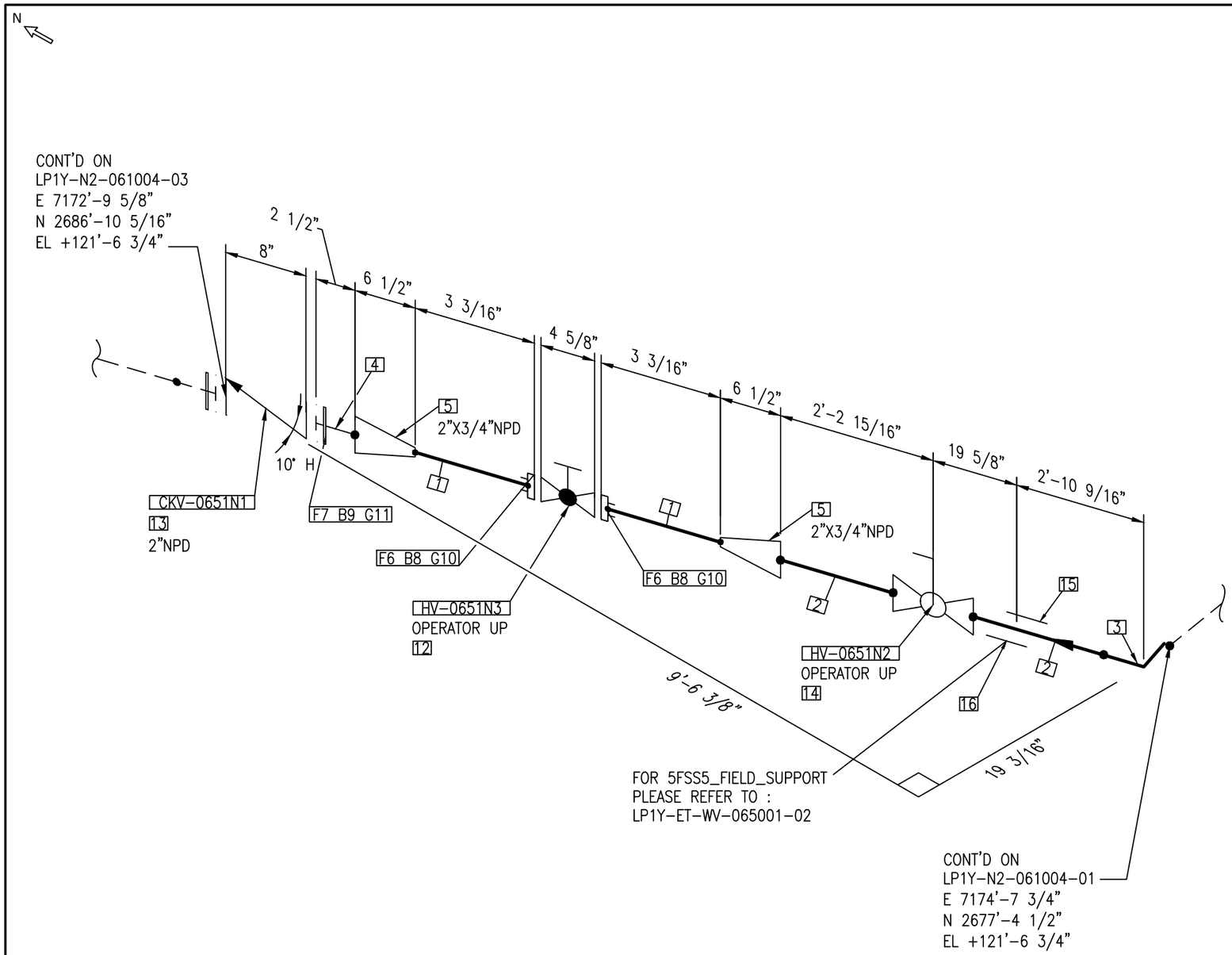

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	7"	3/4"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	6'-1"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
3	1	2"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
4	1	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
5	2	2"X3/4"	SWAGE CONC SCH 10S 316/316L SS BBE	5546561
6	2	3/4"	FLG SW 150# RF 316/316L SS 10S BORE	5608148
7	1	2"	FLG LAP JOINT 150# 316 SS	5599578L
8	8	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
9	4	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
10	2	3/4"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
11	1	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
12	1	3/4"	GLOBE 150# RF 316 SS	5254829
13	1	2"	CHECK 150# RF 316 SS SWING	5257711L
14	1	2"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
15	1	2"	5CI (ISOLATION CRADLE)	5CI-02
16	1	2"	5UG (U-BOLT GUIDE)	5UG-02

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	10-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2 (315) E 061004

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-N2-061004-02	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 10-MAR-2026

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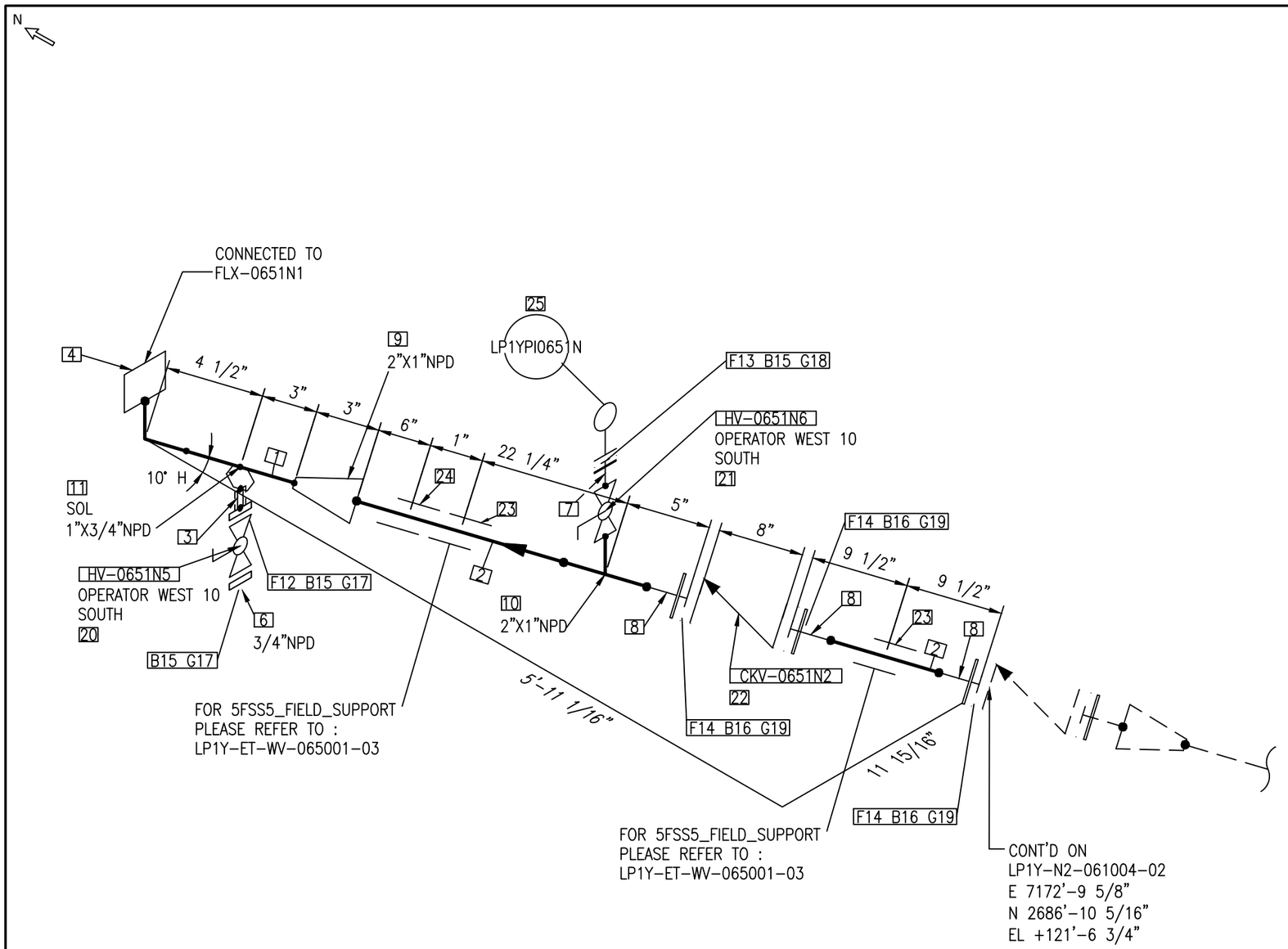

BILL OF MATERIALS

ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	6"	1"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	3'-5"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
3	1	3/4"	NIPPLE SCH 10S 316/316L SS PBE 3" LONG	5540491
4	1	1"	BUTTWELD CAMLOCK CONNECTION UNIT	LP1YCCU0651N2
5	1	1"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
6	1	3/4"	FLG BLIND 150# RF 316/316L SS	5616520
7	1	1"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
8	3	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
9	1	2"X1"	REDUCER CONC SCH 10S 316/316L SS A403-W	5450717
10	1	2"X1"	TEE RED SCH 10S 316/316L SS A403-W	5450714
11	1	1"X3/4"	PIPET 10S X 3000# SW 316/316L SS	5537081
12	1	3/4"	FLG SW 150# RF 316/316L SS 10S BORE	5608148
13	1	1"	FLG LAP JOINT 150# 316 SS	5599578L
14	3	2"	FLG LAP JOINT 150# 316 SS	5599578L
15	12	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
16	12	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
17	2	3/4"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
18	1	1"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
19	3	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
20	1	3/4"	BALL 150# RF 316SS TFE FP HNDL OP	5352193L1
21	1	1"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
22	1	2"	CHECK 150# RF 316 SS SWING	5257711L
23	2	2"	5CI (ISOLATION CRADLE)	5CI-02
24	1	2"	5UG (U-BOLT GUIDE)	5UG-02
25	1	1"	PRESSURE GAUGE	LP1YPI0651N

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	10-MAR-2026	DRW	SBR	N/A	JHN / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2 (315) E 061004

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-N2-061004-03	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 10-MAR-2026

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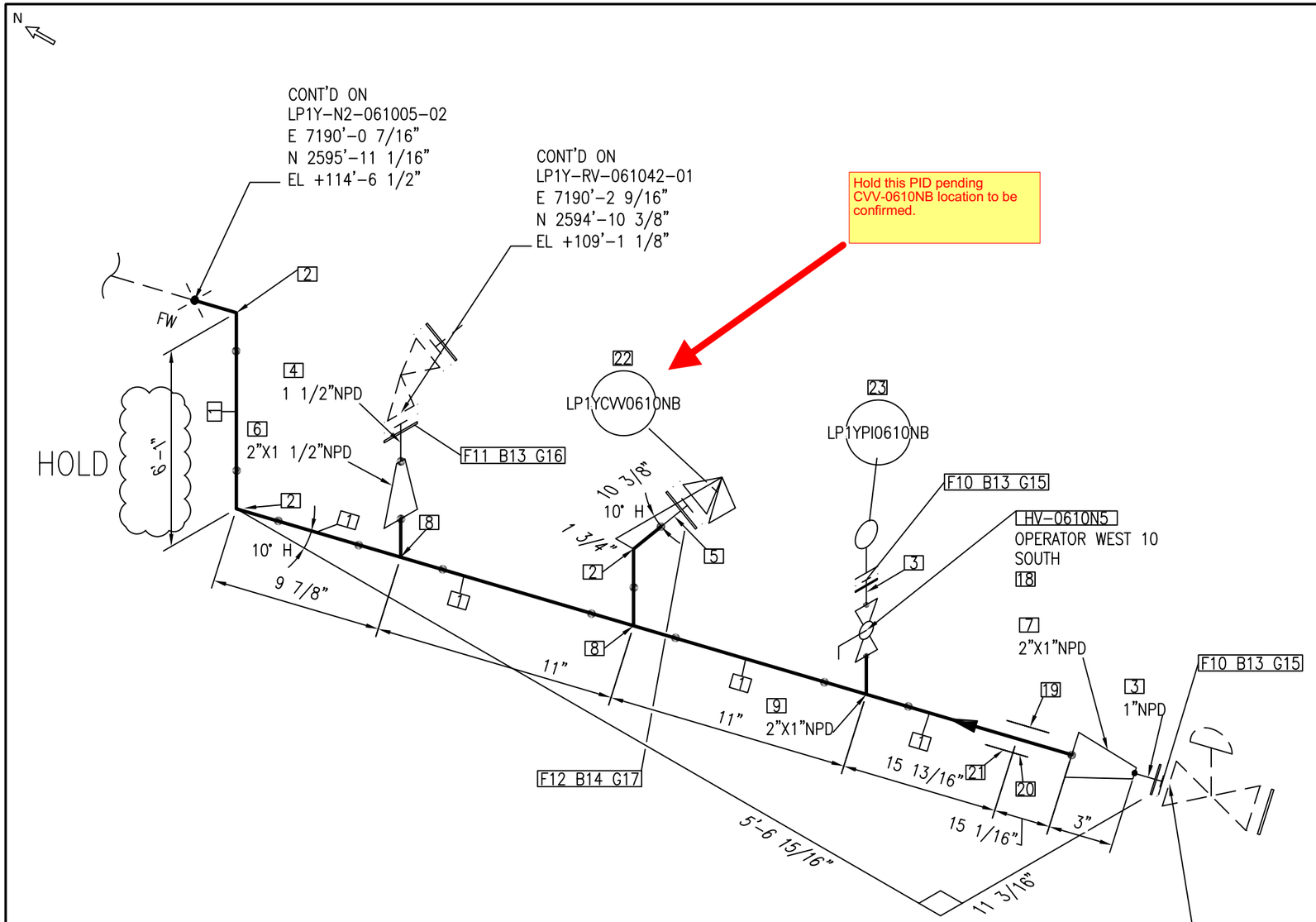

No additional comment

C1 - DATA INCOMPLETE - WITH COMMENTS - HOLD WORK - RESUBMIT

Authorization to proceed does not relieve Contractor/Supplier of its responsibility or liability under the Contract and or Purchase Order.

By: Ricardo DaCosta dac54131 April 22, 2026

BILL OF MATERIALS

ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	9'-4"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	3	2"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
3	2	1"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
4	1	1 1/2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
5	1	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
6	1	2"X1 1/2"	REDUCER CONC SCH 10S 316/316L SS A403-W	5450717
7	1	2"X1"	REDUCER CONC SCH 10S 316/316L SS A403-W	5450717
8	2	2"	TEE SCH 10S 316/316L SS A403-W	5450715
9	1	2"X1"	TEE RED SCH 10S 316/316L SS A403-W	5450714
10	2	1"	FLG LAP JOINT 150# 316 SS	5599578L
11	1	1 1/2"	FLG LAP JOINT 150# 316 SS	5599578L
12	1	2"	FLG LAP JOINT 150# 316 SS	5599578L
13	12	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
14	4	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
15	2	1"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
16	1	1 1/2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
17	1	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
18	1	1"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
19	1	2"	5UG (U-BOLT GUIDE)	5UG-02
20	1	2"	5ABS2 (ANGLE BASE SUPPORT)	5ABS2
21	1	2"	5CI (ISOLATION CRADLE)	5CI-02
22	1	2"	CVV (VACUUM RELIEF VALVE)	LP1YCVV0610NB
23	1	1"	PRESSURE GAUGE	LP1YPI0610NB

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	10-MAR-2026	DRW	SBR	N/A	JHN / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2 (315) E 061005

REFERENCE P&ID NUMBER
LP1Y-DI-00000061

ISOMETRIC DRAWING NUMBER
LP1Y-N2-061005-01

REV
0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 10-MAR-2026

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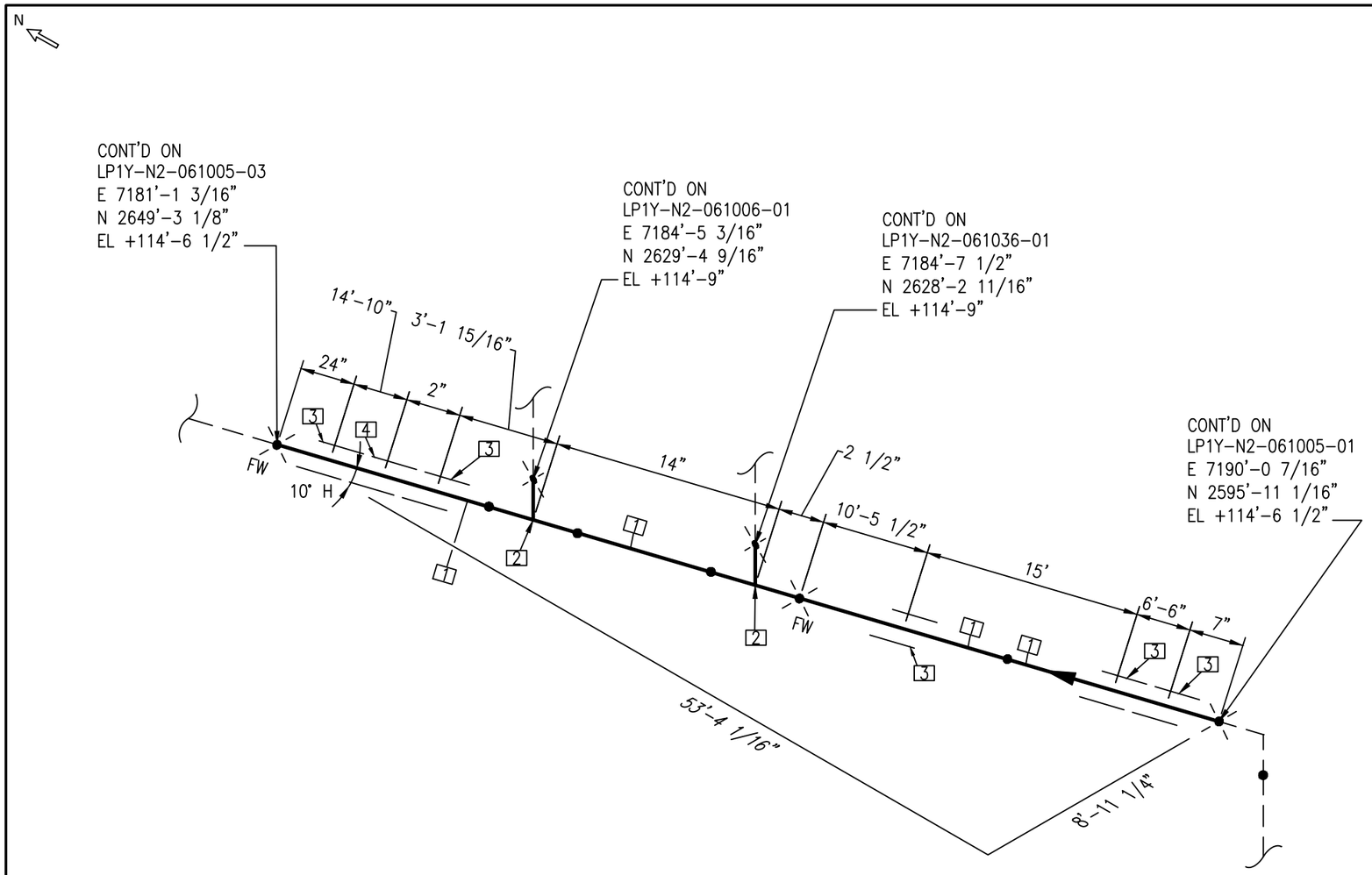

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	53'-3"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	2	2"	TEE SCH 10S 316/316L SS A403-W	5450715
3	5	2"	5CI (ISOLATION CRADLE)	5CI-02
4	1	2"	5UG (U-BOLT GUIDE)	5UG-02

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By dac54131 at Apr 10, 2026

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5. THE MECHANICAL CONTRACTOR IS TO VERIFY ALL DIMENSIONAL DATA WITH ACTUAL FIELD CONDITIONS PRIOR TO INSTALLATION OF THE SYSTEM.
6. PIPE SUPPORTS SHALL BE FURNISHED AND INSTALLED IN ACCORDANCE WITH THIS DRAWING AND SPECIFICATION LP100-250-EMS-15110. MECHANICAL CONTRACTOR SHALL PROVIDE TEMPORARY SUPPORTS AS REQUIRED TO AID PRESSURE TESTING, INSTALLATION ETC.
7. PIPE SUPPORTS INDICATED ON THE DRAWING MAY BE PART OF AN ASSEMBLY ON SEVERAL ISOMETRICS. REFER TO THE 3D MODEL FOR THE COMPLETE PIPE SUPPORT ASSEMBLY.
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10. SLOPE EQUIVALENTS: 1:27 = 7/16" PER FT, 1:96 = 1/8" PER FT, 1:120 = 1/10" PER FT, 1:192 = 1/16" PER FT, 1:480 = 1/40" PER FT.

NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	10-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2 (315) E 061005

REFERENCE P&ID NUMBER
LP1Y-DI-00000061

ISOMETRIC DRAWING NUMBER
LP1Y-N2-061005-02

REV
0

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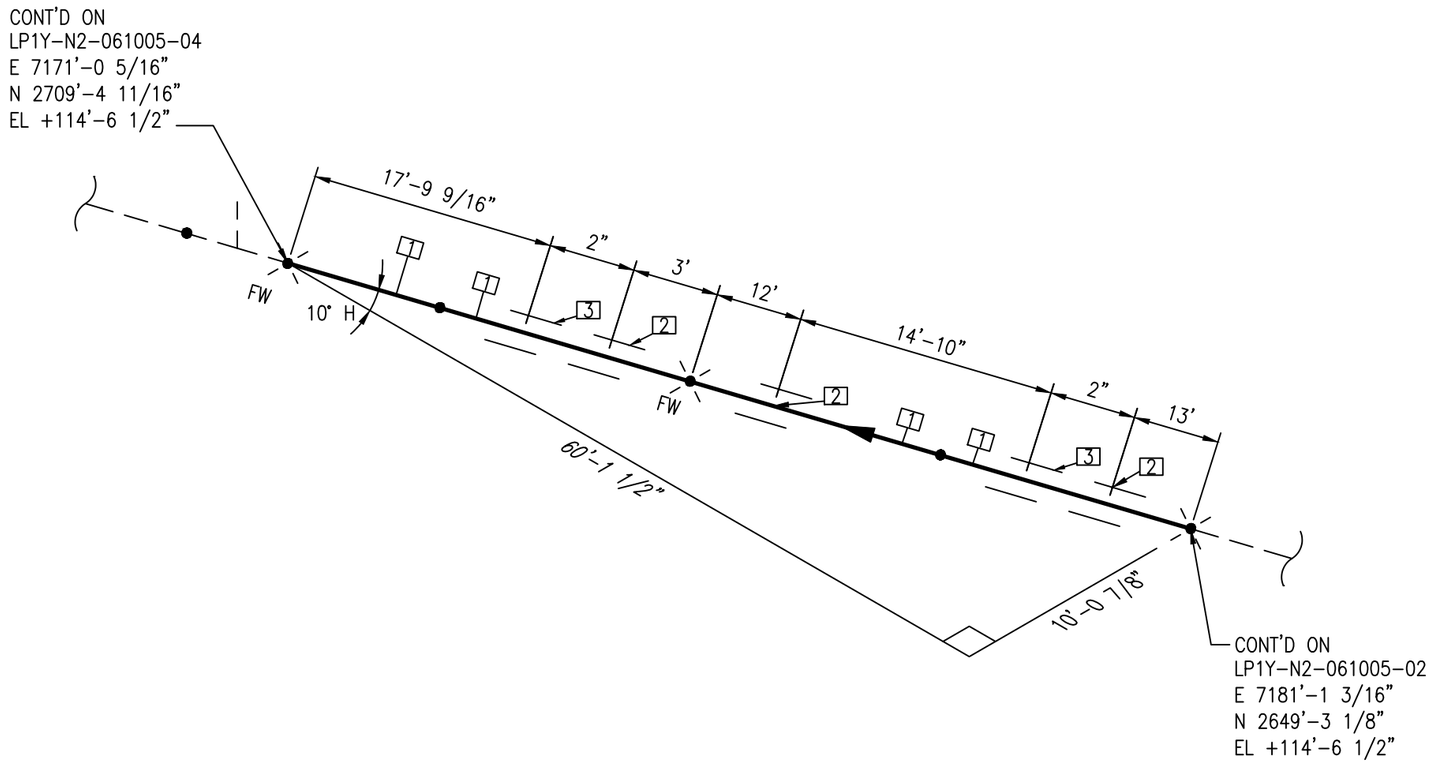
N

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	61'	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	3	2"	5CI (ISOLATION CRADLE)	5CI-02
3	2	2"	5UG (U-BOLT GUIDE)	5UG-02



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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	10-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2 (315) E 061005

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-N2-061005-03	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 10-MAR-2026

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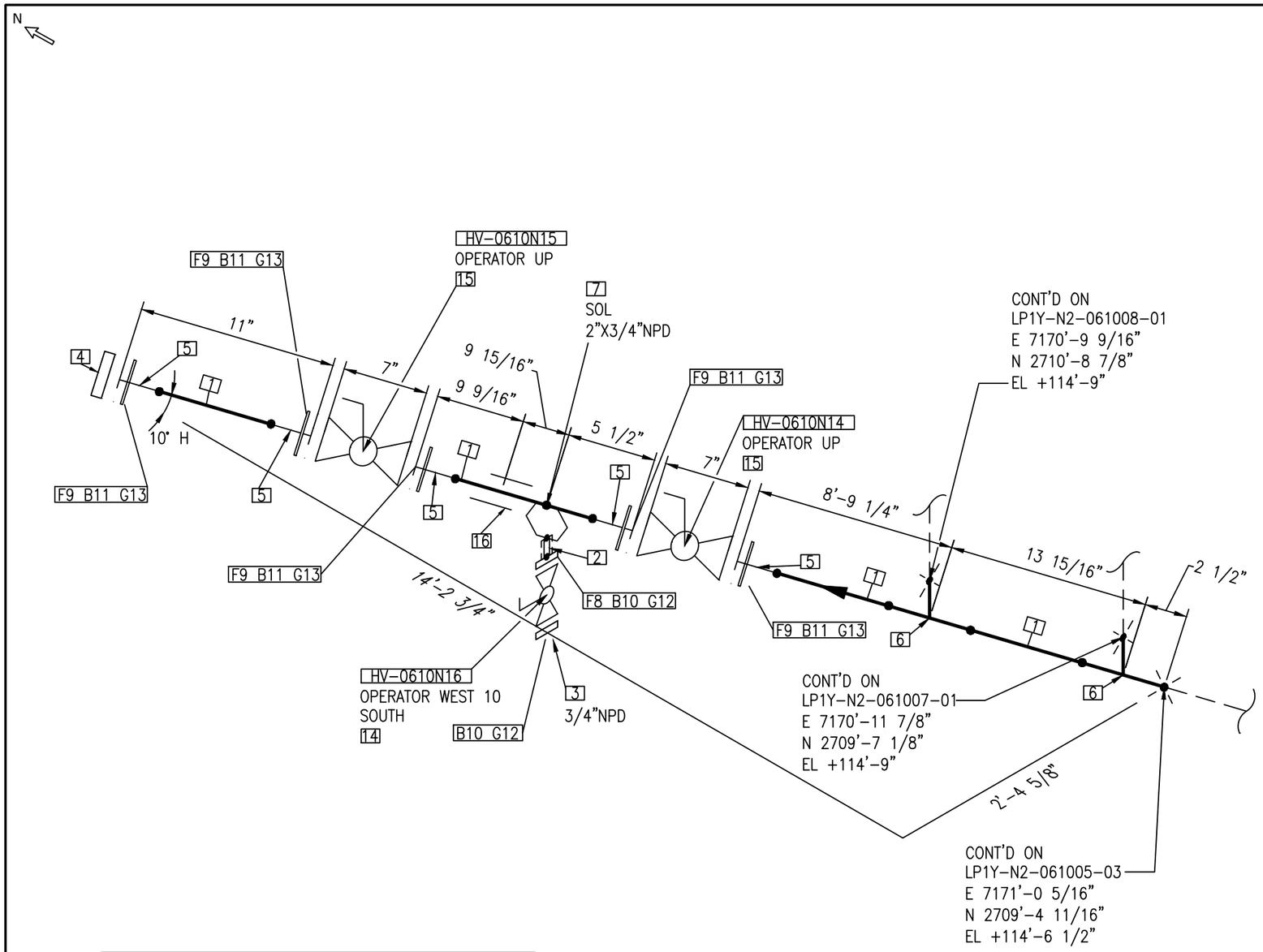

BILL OF MATERIALS

ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	11'-4"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	1	3/4"	NIPPLE SCH 10S 316/316L SS PBE 3" LONG	5540491
3	1	3/4"	FLG BLIND 150# RF 316/316L SS	5616520
4	1	2"	FLG BLIND 150# RF 316/316L SS	5616520
5	5	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
6	2	2"	TEE SCH 10S 316/316L SS A403-W	5450715
7	1	2"X3/4"	PIPET 10S X 3000# SW 316/316L SS	5537081
8	1	3/4"	FLG SW 150# RF 316/316L SS 10S BORE	5608148
9	5	2"	FLG LAP JOINT 150# 316 SS	5599578L
10	8	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
11	20	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
12	2	3/4"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
13	5	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
14	1	3/4"	BALL 150# RF 316SS TFE FP HNDL OP	5352193L1
15	2	2"	BALL 150# RF 316SS TFE FP HNDL OP	5352194L1
16	1	2"	5CI (ISOLATION CRADLE)	5CI-02

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	10-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2 (315) E 061005

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-N2-061005-04	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 10-MAR-2026

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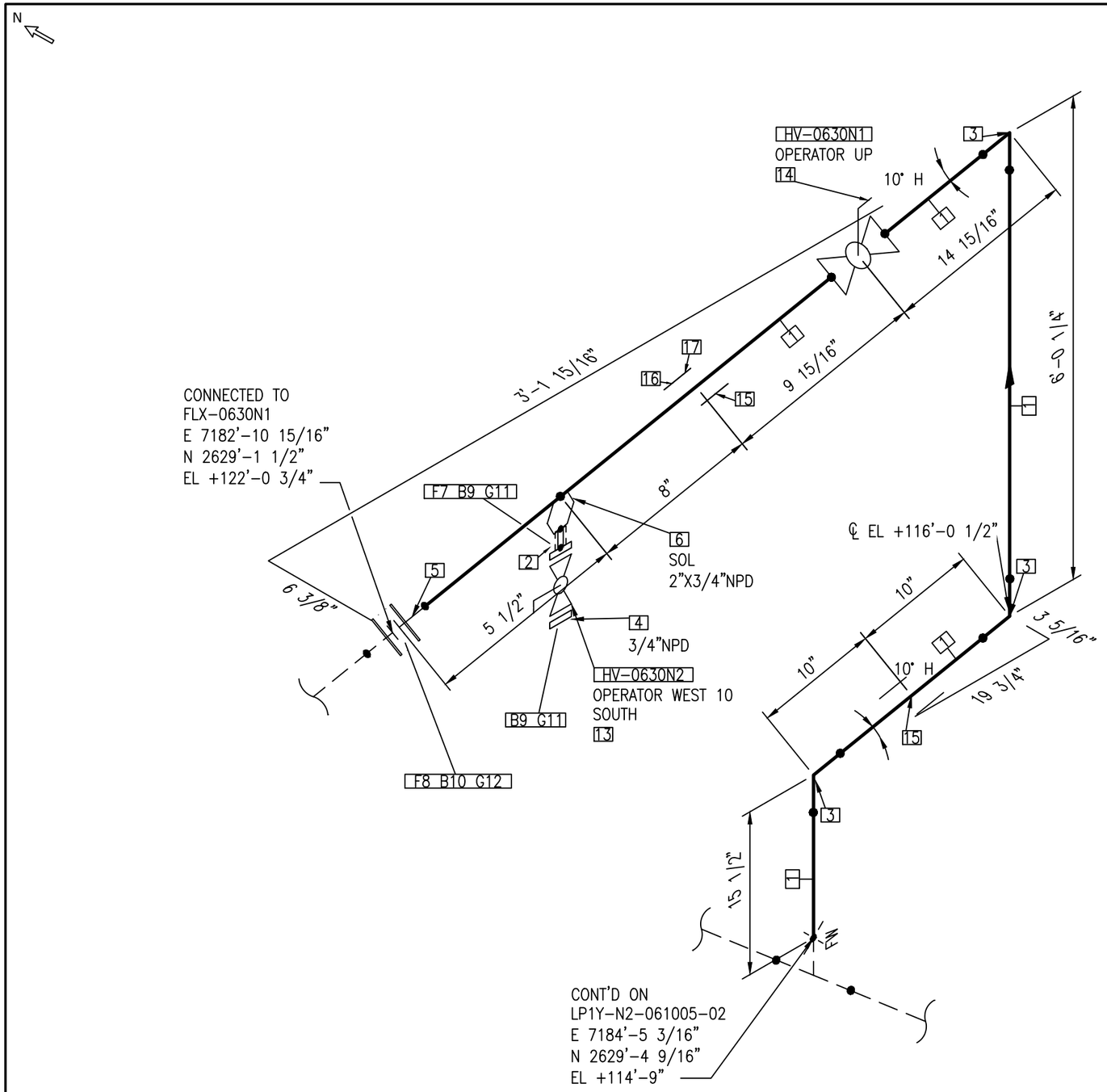

BILL OF MATERIALS

ID	QTY	ND	DESCRIPTION	CMDTY CODE
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2	1	3/4"	NIPPLE SCH 10S 316/316L SS PBE 3" LONG	5540491
3	3	2"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
4	1	3/4"	FLG BLIND 150# RF 316/316L SS	5616520
5	1	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
6	1	2"X3/4"	PIPET 10S X 3000# SW 316/316L SS	5537081
7	1	3/4"	FLG SW 150# RF 316/316L SS 10S BORE	5608148
8	1	2"	FLG LAP JOINT 150# 316 SS	5599578L
9	8	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
10	4	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
11	2	3/4"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
12	1	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
13	1	3/4"	BALL 150# RF 316SS TFE FP HNDL OP	5352193L1
14	1	2"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
15	2	2"	5CI (ISOLATION CRADLE)	5CI-02
16	1	2"	5UG (U-BOLT GUIDE)	5UG-02
17	1	2"	5ABS2 (ANGLE BASE SUPPORT)	5ABS2

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	13-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2 (315) E 061006

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-N2-061006-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 13-MAR-2026

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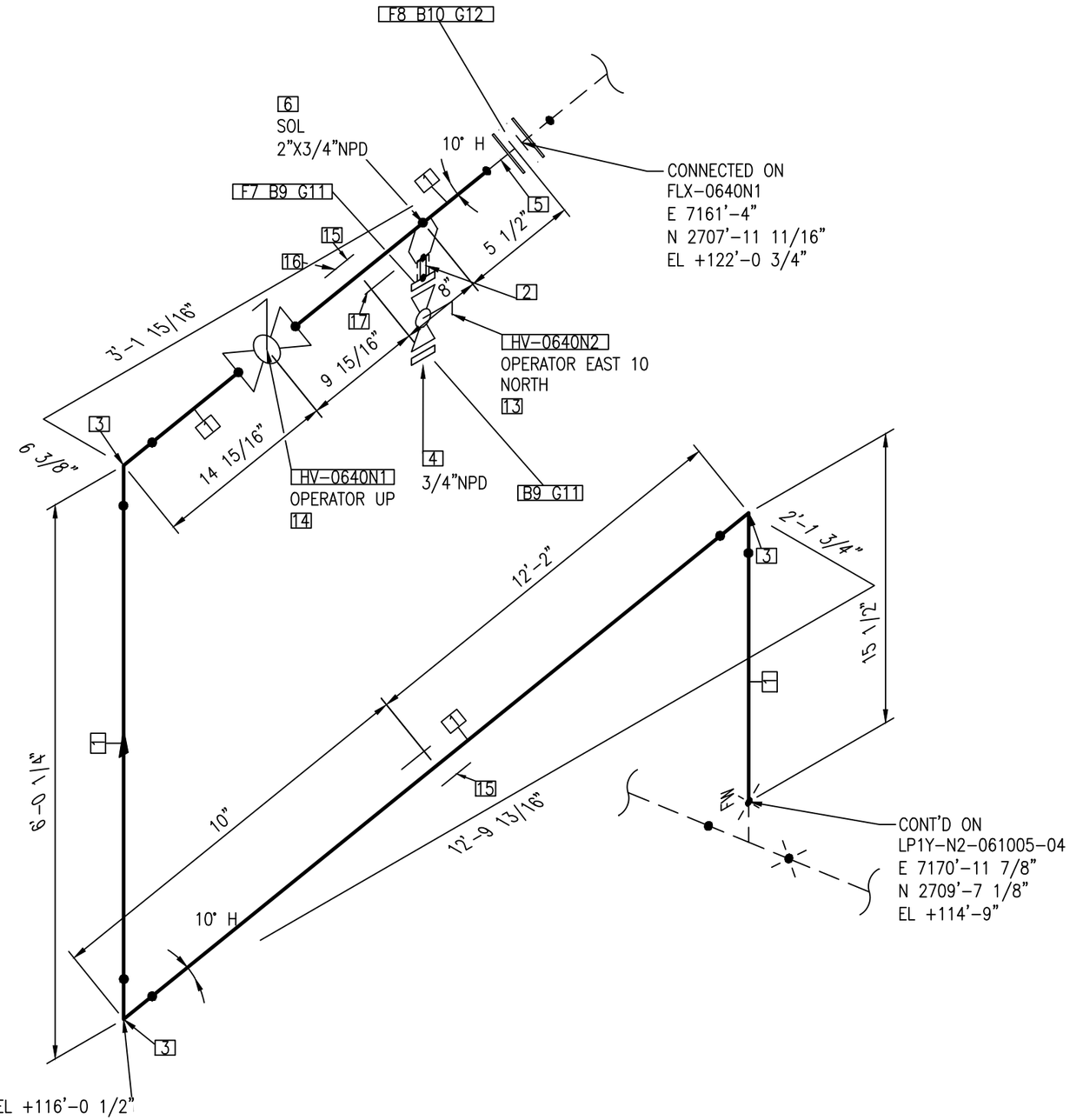

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	21'-4"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	1	3/4"	NIPPLE SCH 10S 316/316L SS PBE 3" LONG	5540491
3	3	2"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
4	1	3/4"	FLG BLIND 150# RF 316/316L SS	5616520
5	1	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
6	1	2"X3/4"	PIPET 10S X 3000# SW 316/316L SS	5537081
7	1	3/4"	FLG SW 150# RF 316/316L SS 10S BORE	5608148
8	1	2"	FLG LAP JOINT 150# 316 SS	5599578L
9	8	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
10	4	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
11	2	3/4"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
12	1	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
13	1	3/4"	BALL 150# RF 316SS TFE FP HNDL OP	5352193L1
14	1	2"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
15	2	2"	5CI (ISOLATION CRADLE)	5CI-02
16	1	2"	5UG (U-BOLT GUIDE)	5UG-02
17	1	2"	5ABS2 (ANGLE BASE SUPPORT)	5ABS2

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
						/	
						/	
0	ISSUE FOR CONSTRUCTION	13-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2 (315) E 061007

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-N2-061007-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 13-MAR-2026

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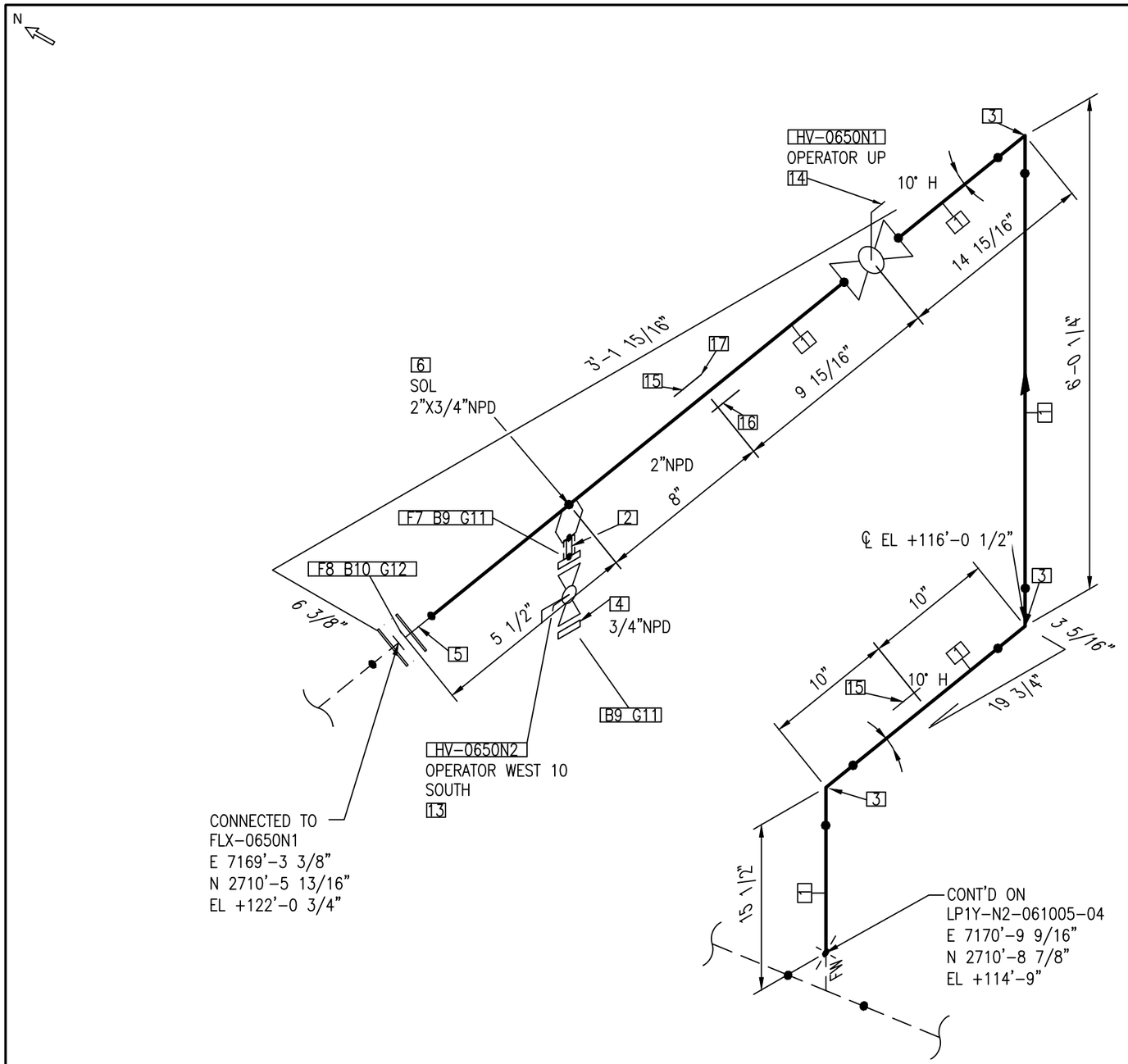

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	10'	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	1	3/4"	NIPPLE SCH 10S 316/316L SS PBE 3" LONG	5540491
3	3	2"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
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15	2	2"	5CI (ISOLATION CRADLE)	5CI-02
16	1	2"	5UG (U-BOLT GUIDE)	5UG-02
17	1	2"	5ABS2 (ANGLE BASE SUPPORT)	5ABS2

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By dac54131 at Apr 10, 2026

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1. REFER TO THE PIPING LINE LIST FOR PROCESS DATA/CONDITIONS AND OTHER FABRICATION DETAILS SUCH AS TESTING, CLEANING, AND PAINTING REQUIREMENTS.
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3. WELDS AND/OR COUPLINGS DEFINING RANDOM PIPE SPOOL LENGTHS ARE FOR REFERENCE ONLY. FINAL WELD OR COUPLING LOCATION IS TO BE DETERMINED BY THE CONTRACTOR.
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7. PIPE SUPPORTS INDICATED ON THE DRAWING MAY BE PART OF AN ASSEMBLY ON SEVERAL ISOMETRICS. REFER TO THE 3D MODEL FOR THE COMPLETE PIPE SUPPORT ASSEMBLY.
8. MECHANICAL CONTRACTOR SHALL NOTE ALL DESIGN CHANGES ON THIS DRAWING AND RETURN IT TO THE CONSTRUCTION WPO FOR AS-BUILT RECORDS.
9. THE BILL OF MATERIAL IS FOR REFERENCE ONLY.
10. SLOPE EQUIVALENTS: 1:27 = 7/16" PER FT, 1:96 = 1/8" PER FT, 1:120 = 1/10" PER FT, 1:192 = 1/16" PER FT, 1:480 = 1/40" PER FT.

NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	13-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2 (315) E 061008

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-N2-061008-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 13-MAR-2026

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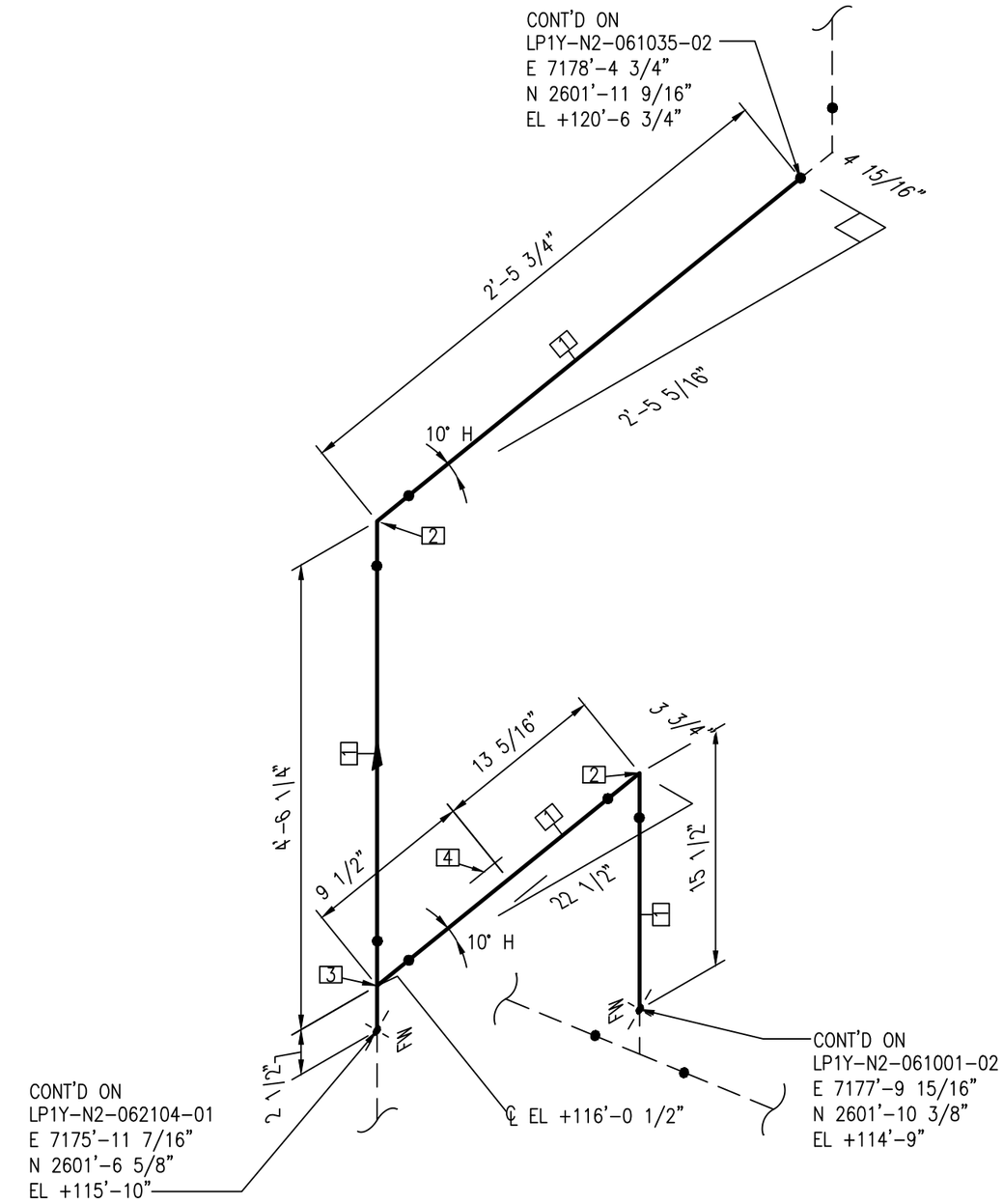
N



BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	8'-10"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	2	2"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
3	1	2"	TEE SCH 10S 316/316L SS A403-W	5450715
4	1	2"	5CI (ISOLATION CRADLE)	5CI-02

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
						/	
						/	
0	ISSUE FOR CONSTRUCTION	10-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2 (315) E 061035

REFERENCE P&ID NUMBER
LP1Y-DI-00000061

ISOMETRIC DRAWING NUMBER
LP1Y-N2-061035-01

REV
0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 10-MAR-2026

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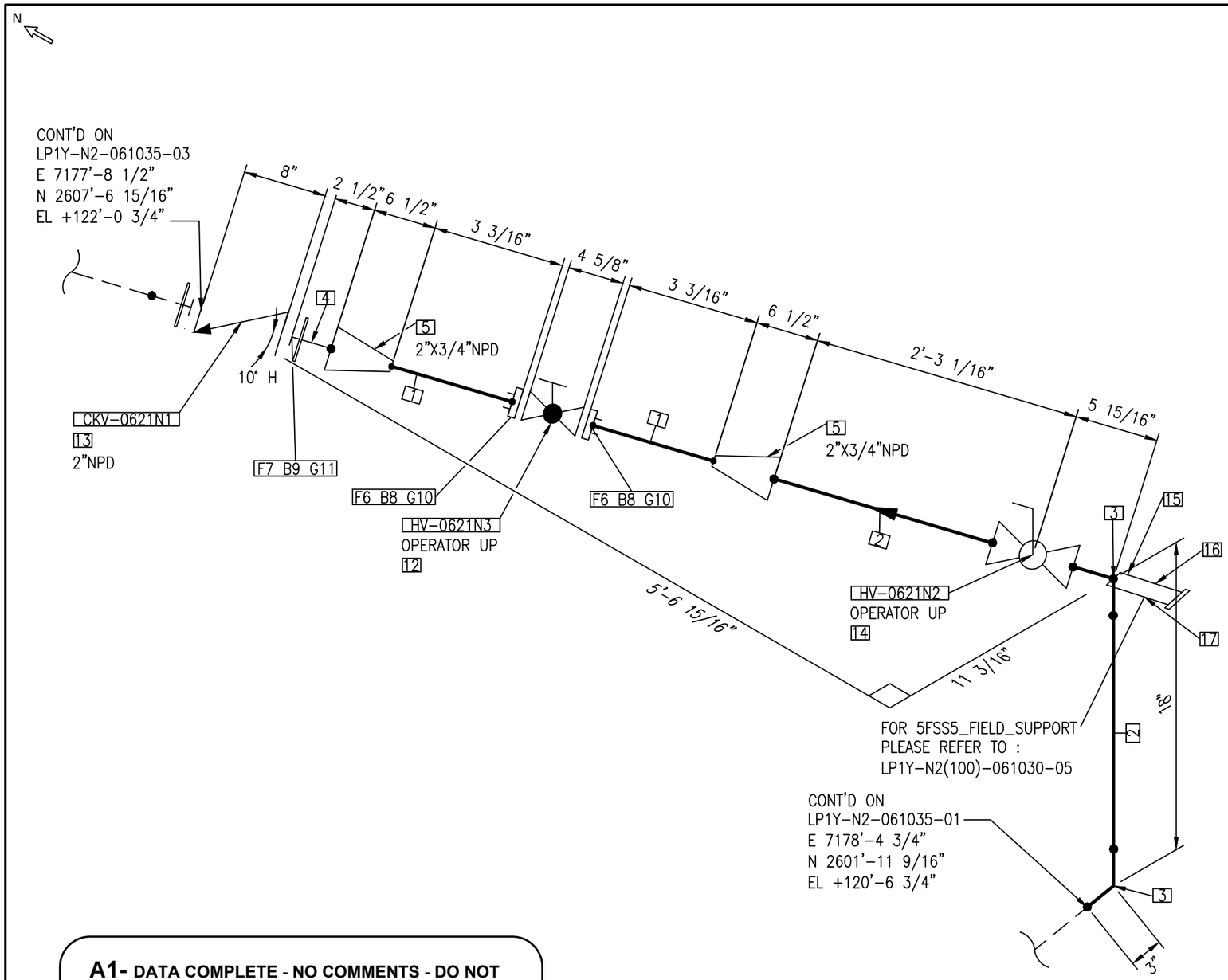

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	6"	3/4"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	3'-1"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
3	2	2"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
4	1	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
5	2	2"X3/4"	SWAGE CONC SCH 10S 316/316L SS BBE	5546561
6	2	3/4"	FLG SW 150# RF 316/316L SS 10S BORE	5608148
7	1	2"	FLG LAP JOINT 150# 316 SS	5599578L
8	8	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
9	4	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
10	2	3/4"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
11	1	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
12	1	3/4"	GLOBE 150# RF 316 SS	5254829
13	1	2"	CHECK 150# RF 316 SS SWING	5257711L
14	1	2"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
15	1	2"	5CI (ISOLATION CRADLE)	5CI-02
16	1	2"	5UG (U-BOLT GUIDE)	5UG-02
17	1	2"	5DS1 (DUMMY SUPPORT) L=10"	5DS1-02

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	10-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

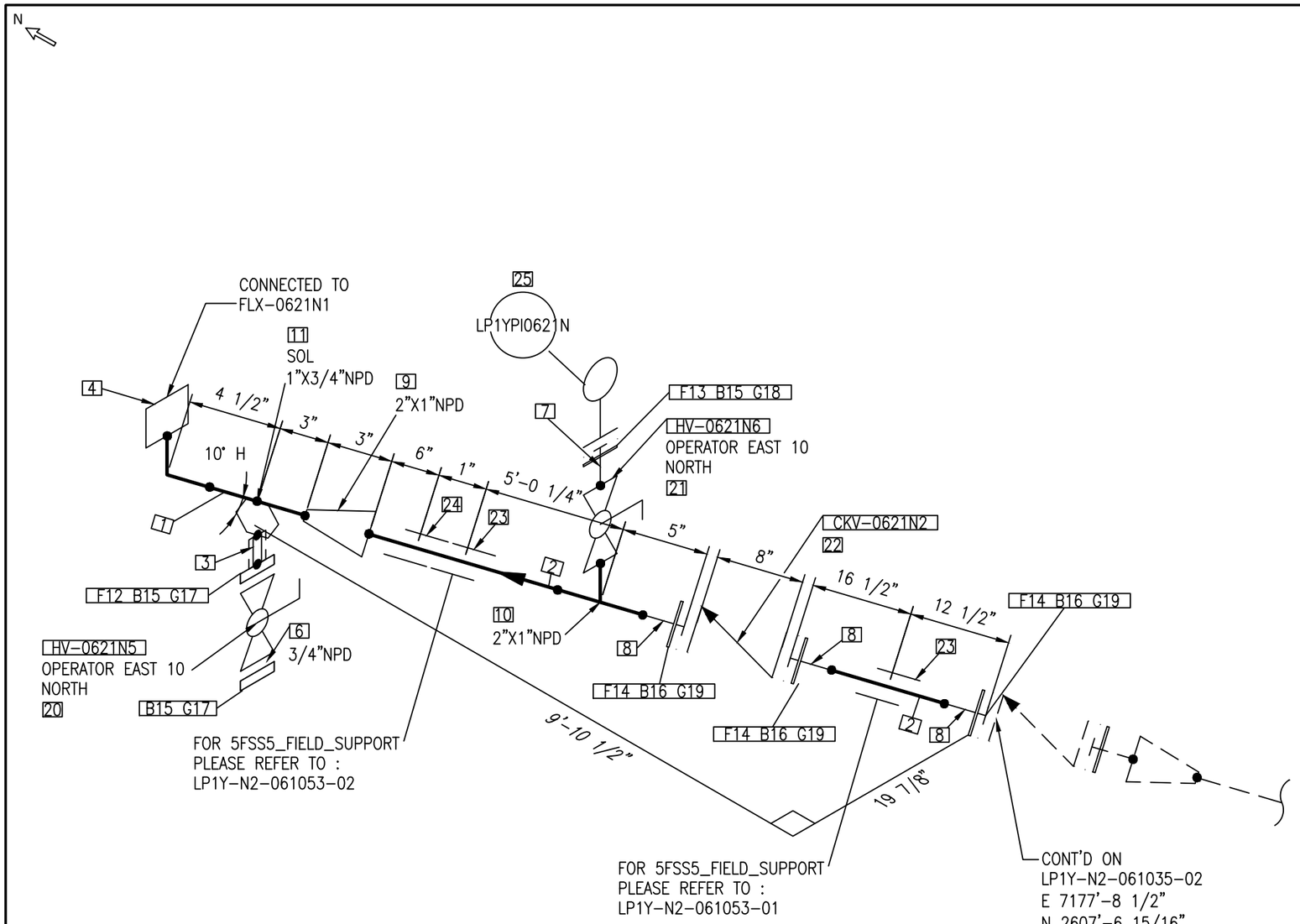
HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2 (315) E 061035

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-N2-061035-02	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 10-MAR-2026

NOTICE: THIS DRAWING HAS NOT BEEN PUBLISHED AND IS THE SOLE PROPERTY OF FLUOR AND IS LENT TO THE BORROWER FOR THEIR CONFIDENTIAL USE ONLY, AND IN CONSIDERATION OF THE LOAN OF THIS DRAWING, THE BORROWER PROMISES AND AGREES TO RETURN IT UPON REQUEST AND AGREES THAT IT WILL NOT BE REPRODUCED, COPIED, LENT, OR OTHERWISE DISPOSED OF DIRECTLY OR INDIRECTLY, NOR USED FOR ANY PURPOSE OTHER THAN THAT FOR WHICH IT IS FURNISHED.



BILL OF MATERIALS

ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	6"	1"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	7'-5"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
3	1	3/4"	NIPPLE SCH 10S 316/316L SS PBE 3" LONG	5540491
4	1	1"	BUTTWELD CAMLOCK CONNECTION UNIT	LP1YCCU0621N2
5	1	1"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
6	1	3/4"	FLG BLIND 150# RF 316/316L SS	5616520
7	1	1"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
8	3	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
9	1	2"X1"	REDUCER CONC SCH 10S 316/316L SS A403-W	5450717
10	1	2"X1"	TEE RED SCH 10S 316/316L SS A403-W	5450714
11	1	1"X3/4"	PIPET 10S X 3000# SW 316/316L SS	5537081
12	1	3/4"	FLG SW 150# RF 316/316L SS 10S BORE	5608148
13	1	1"	FLG LAP JOINT 150# 316 SS	5599578L
14	3	2"	FLG LAP JOINT 150# 316 SS	5599578L
15	12	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
16	12	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
17	2	3/4"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
18	1	1"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
19	3	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
20	1	3/4"	BALL 150# RF 316SS TFE FP HNDL OP	5352193L1
21	1	1"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
22	1	2"	CHECK 150# RF 316 SS SWING	5257711L
23	2	2"	5CI (ISOLATION CRADLE)	5CI-02
24	1	2"	5UG (U-BOLT GUIDE)	5UG-02
25	1	1"	PRESSURE GAUGE	LP1YPI0621N

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	17-MAR-2026	DRW	SBR	N/A	JHN / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

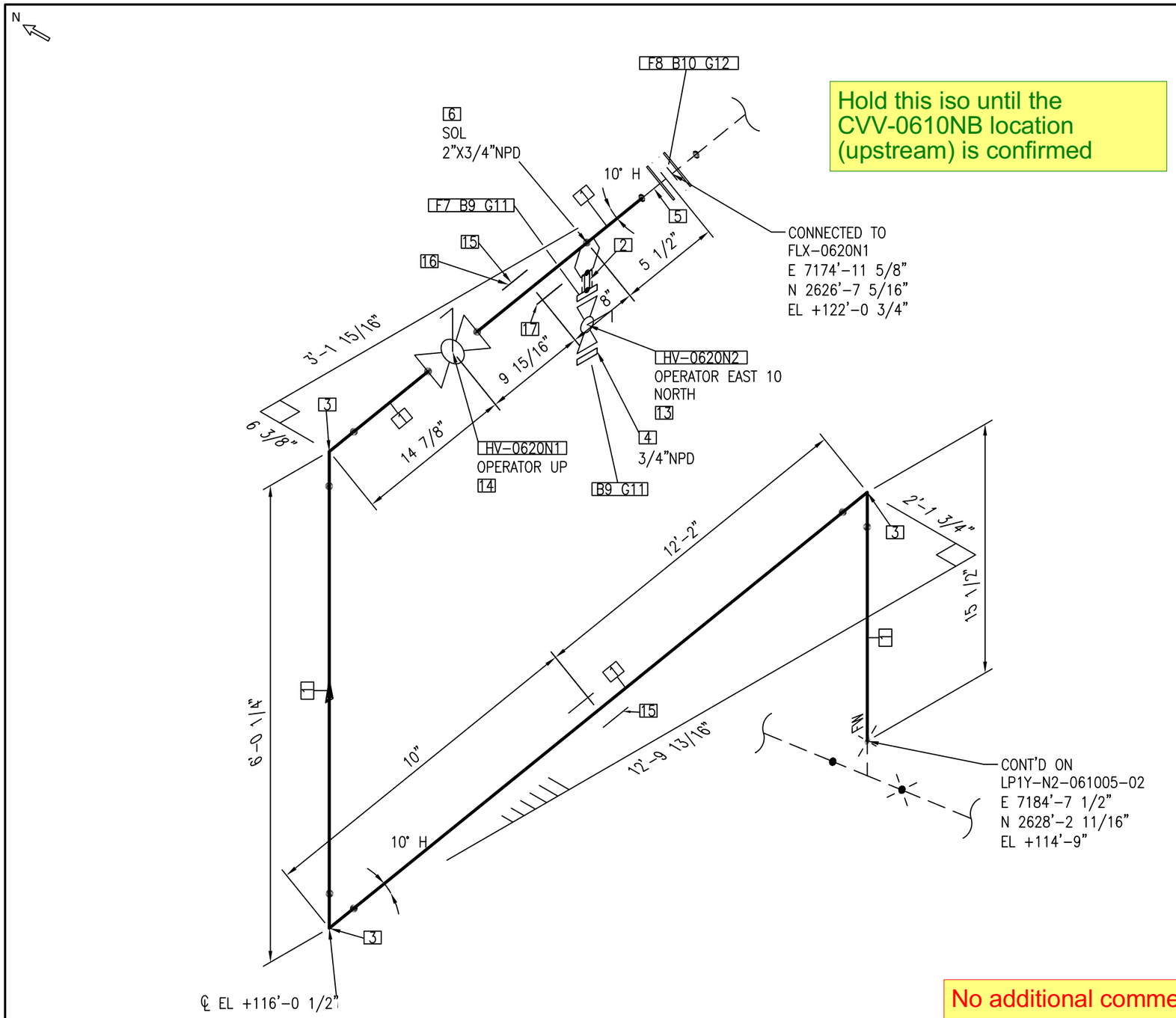
Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2 (315) E 061035

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-N2-061035-03	0

NOTICE: THIS DRAWING HAS NOT BEEN PUBLISHED AND IS THE SOLE PROPERTY OF FLUOR. AND IS LENT TO THE BORROWER FOR THEIR CONFIDENTIAL USE ONLY, AND IN CONSIDERATION OF THE LOAN OF THIS DRAWING, THE BORROWER PROMISES AND AGREES TO RETURN IT UPON REQUEST AND AGREES THAT IT WILL NOT BE REPRODUCED, COPIED, LENT, OR OTHERWISE DISPOSED OF DIRECTLY OR INDIRECTLY, NOR USED FOR ANY PURPOSE OTHER THAN THAT FOR WHICH IT IS FURNISHED.



Hold this iso until the CVV-0610NB location (upstream) is confirmed

No additional comment

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	21'-4"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	1	3/4"	NIPPLE SCH 10S 316/316L SS PBE 3" LONG	5540491
3	3	2"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
4	1	3/4"	FLG BLIND 150# RF 316/316L SS	5616520
5	1	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
6	1	2"X3/4"	PIPET 10S X 3000# SW 316/316L SS	5537081
7	1	3/4"	FLG SW 150# RF 316/316L SS 10S BORE	5608148
8	1	2"	FLG LAP JOINT 150# 316 SS	5599578L
9	8	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
10	4	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
11	2	3/4"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
12	1	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
13	1	3/4"	BALL 150# RF 316SS TFE FP HNDL OP	5352193L1
14	1	2"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
15	2	2"	5CI (ISOLATION CRADLE)	5CI-02
16	1	2"	5UG (U-BOLT GUIDE)	5UG-02
17	1	2"	5ABS2 (ANGLE BASE SUPPORT)	5ABS2

C1 - DATA INCOMPLETE - WITH COMMENTS - HOLD WORK - RESUBMIT

Authorization to proceed does not relieve Contractor/Supplier of its responsibility or liability under the Contract and or Purchase Order.

By: Ricardo DaCosta dac54131 April 21, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	13-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2 (315) E 061036

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-N2-061036-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 13-MAR-2026

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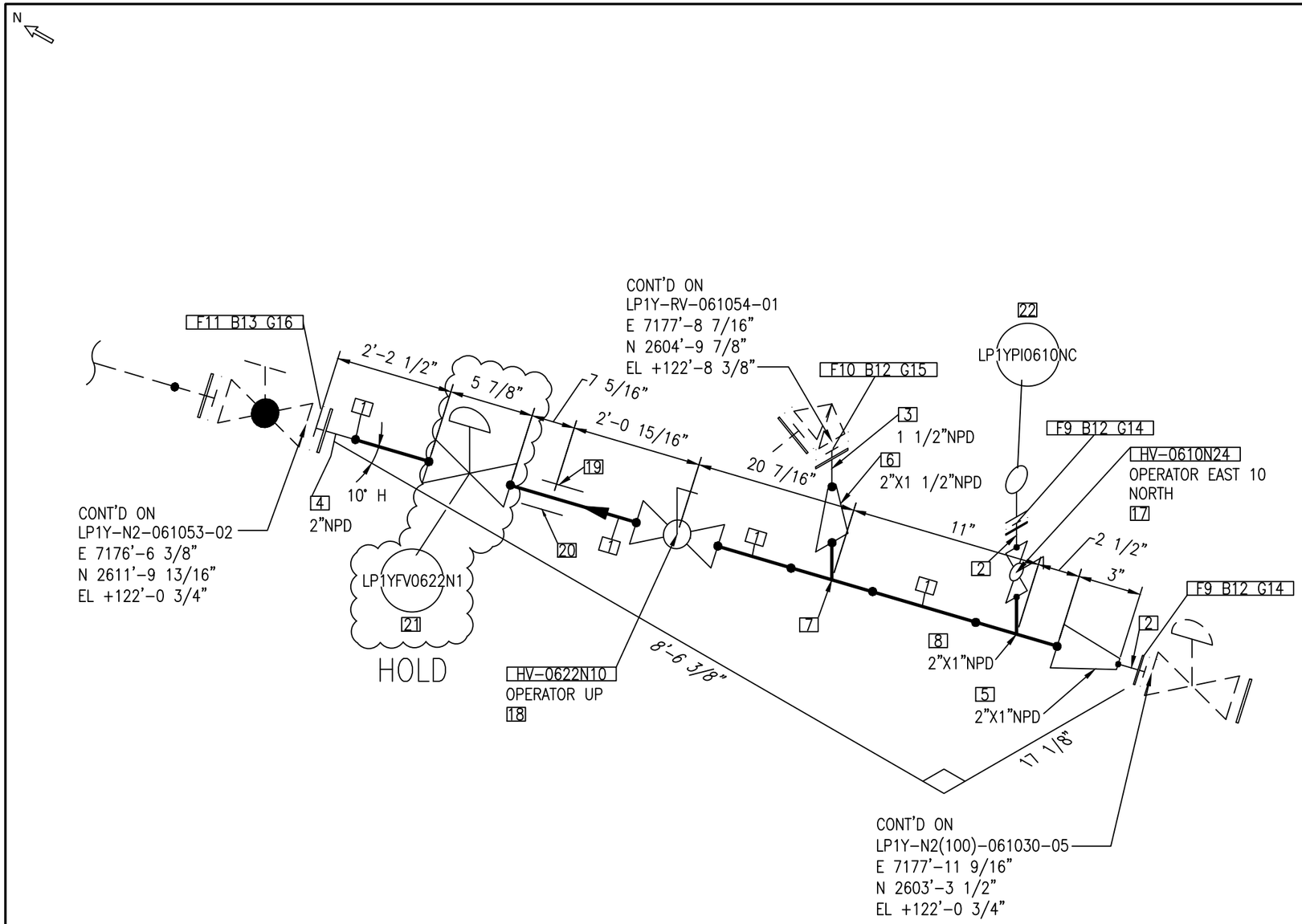

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	6'-3"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	2	1"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
3	1	1 1/2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
4	1	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
5	1	2"X1"	REDUCER CONC SCH 10S 316/316L SS A403-W	5450717
6	1	2"X1 1/2"	REDUCER CONC SCH 10S 316/316L SS A403-W	5450717
7	1	2"	TEE SCH 10S 316/316L SS A403-W	5450715
8	1	2"X1"	TEE RED SCH 10S 316/316L SS A403-W	5450714
9	2	1"	FLG LAP JOINT 150# 316 SS	5599578L
10	1	1 1/2"	FLG LAP JOINT 150# 316 SS	5599578L
11	1	2"	FLG LAP JOINT 150# 316 SS	5599578L
12	12	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
13	4	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
14	2	1"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
15	1	1 1/2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
16	1	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
17	1	1"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
18	1	2"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
19	1	2"	5CI (ISOLATION CRADLE)	5CI-02
20	1	2"	5FSS5 (FIELD BASE SUPPORT)	5FSS5-L-1
21	1	2"	SERIES 59 BALL VALVE BW	LP1YFV0622N1
22	1	1"	PRESSURE GAUGE	LP1YPI0610NC

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

GENERAL NOTE:
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2. UNLESS OTHERWISE NOTED, ALL WELDS ARE INDICATED AS SHOP WELDS. CONTRACTOR TO DETERMINE FIELD WELDS LOCATIONS AND DOCUMENT ACCORDINGLY ON AS-BUILT RECORDS.
3. WELDS AND/OR COUPLINGS DEFINING RANDOM PIPE SPOOL LENGTHS ARE FOR REFERENCE ONLY. FINAL WELD OR COUPLING LOCATION IS TO BE DETERMINED BY THE CONTRACTOR.
4. THE MECHANICAL CONTRACTOR TO FIELD VERIFY INTERFACE POINTS BETWEEN EQUIPMENT & PLUMBING FIXTURES IN 3rd FIX (3F SERIES CWP) AND YS YARD STICK BUILT (YS SERIES CWP EQUIPMENT CONNECTIONS) ISOMETRICS PRIOR TO FABRICATION.
5. THE MECHANICAL CONTRACTOR IS TO VERIFY ALL DIMENSIONAL DATA WITH ACTUAL FIELD CONDITIONS PRIOR TO INSTALLATION OF THE SYSTEM.
6. PIPE SUPPORTS SHALL BE FURNISHED AND INSTALLED IN ACCORDANCE WITH THIS DRAWING AND SPECIFICATION LP100-250-EMS-15110. MECHANICAL CONTRACTOR SHALL PROVIDE TEMPORARY SUPPORTS AS REQUIRED TO AID PRESSURE TESTING, INSTALLATION ETC.
7. PIPE SUPPORTS INDICATED ON THE DRAWING MAY BE PART OF AN ASSEMBLY ON SEVERAL ISOMETRICS. REFER TO THE 3D MODEL FOR THE COMPLETE PIPE SUPPORT ASSEMBLY.
8. MECHANICAL CONTRACTOR SHALL NOTE ALL DESIGN CHANGES ON THIS DRAWING AND RETURN IT TO THE CONSTRUCTION WPO FOR AS-BUILT RECORDS.
9. THE BILL OF MATERIAL IS FOR REFERENCE ONLY.
10. SLOPE EQUIVALENTS: 1:27 = 7/16" PER FT, 1:96 = 1/8" PER FT, 1:120 = 1/10" PER FT, 1:192 = 1/16" PER FT, 1:480 = 1/40" PER FT.

NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	13-MAR-2026	DRW	SBR	N/A	JHN / N/A	MDW

Lilly

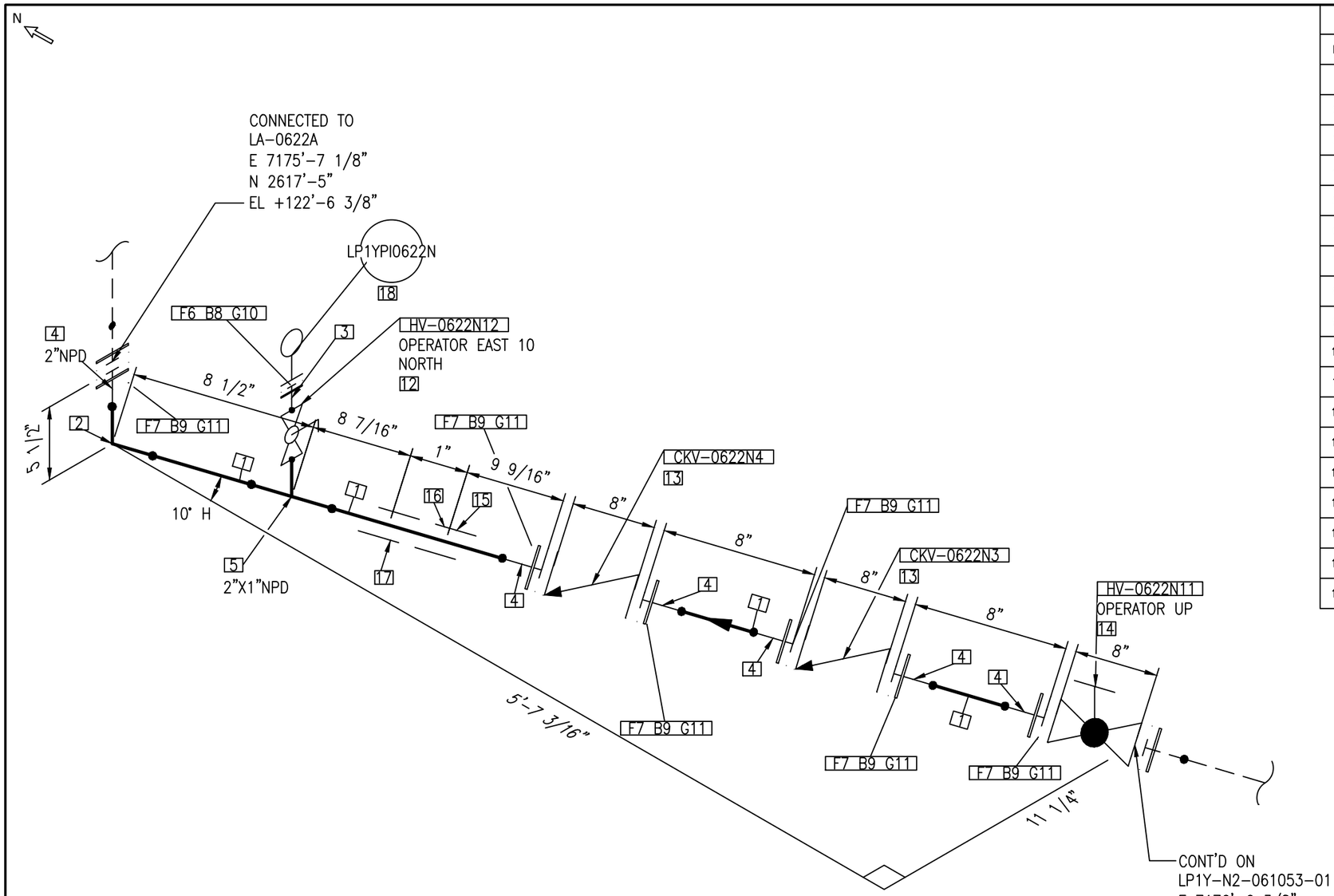
HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2 (315) E 061053

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-N2-061053-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 13-MAR-2026

NOTICE: THIS DRAWING HAS NOT BEEN PUBLISHED AND IS THE SOLE PROPERTY OF FLUOR AND IS LENT TO THE BORROWER FOR THEIR CONFIDENTIAL USE ONLY, AND IN CONSIDERATION OF THE LOAN OF THIS DRAWING, THE BORROWER PROMISES AND AGREES TO RETURN IT UPON REQUEST AND AGREES THAT IT WILL NOT BE REPRODUCED, COPIED, LENT, OR OTHERWISE DISPOSED OF DIRECTLY OR INDIRECTLY, NOR USED FOR ANY PURPOSE OTHER THAN THAT FOR WHICH IT IS FURNISHED.



BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	1'-11"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	1	2"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
3	1	1"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
4	6	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
5	1	2"X1"	TEE RED SCH 10S 316/316L SS A403-W	5450714
6	1	1"	FLG LAP JOINT 150# 316 SS	5599578L
7	6	2"	FLG LAP JOINT 150# 316 SS	5599578L
8	4	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
9	24	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
10	1	1"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
11	6	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
12	1	1"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
13	2	2"	CHECK 150# RF 316 SS SWING	5257711L
14	1	2"	GLOBE 150# RF 316 SS	5254829
15	1	2"	5CI (ISOLATION CRADLE)	5CI-02
16	1	2"	5FSS5 (FIELD BASE SUPPORT)	5FSS5-L-1
17	1	2"	5UG (U-BOLT GUIDE)	5UG-02
18	1	1"	PRESSURE GAUGE	LP1YPI0622N

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	13-MAR-2026	DRW	SBR	N/A	JHN / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2 (315) E 061053

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-N2-061053-02	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 13-MAR-2026

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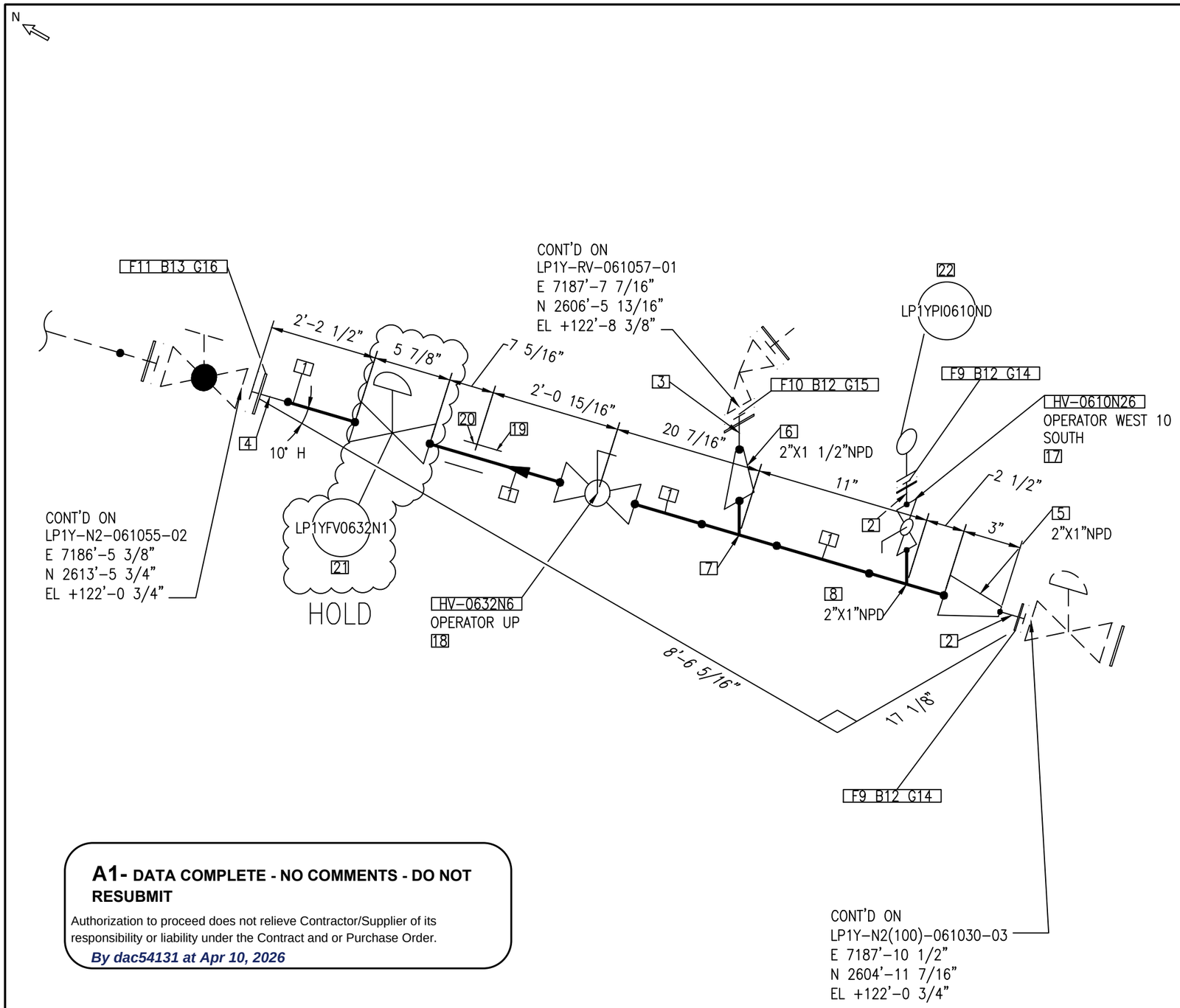

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	6'-3"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	2	1"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
3	1	1 1/2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
4	1	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
5	1	2"X1"	REDUCER CONC SCH 10S 316/316L SS A403-W	5450717
6	1	2"X1 1/2"	REDUCER CONC SCH 10S 316/316L SS A403-W	5450717
7	1	2"	TEE SCH 10S 316/316L SS A403-W	5450715
8	1	2"X1"	TEE RED SCH 10S 316/316L SS A403-W	5450714
9	2	1"	FLG LAP JOINT 150# 316 SS	5599578L
10	1	1 1/2"	FLG LAP JOINT 150# 316 SS	5599578L
11	1	2"	FLG LAP JOINT 150# 316 SS	5599578L
12	12	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
13	4	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
14	2	1"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
15	1	1 1/2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
16	1	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
17	1	1"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
18	1	2"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
19	1	2"	5CI (ISOLATION CRADLE)	5CI-02
20	1	2"	5FSS5 (FIELD BASE SUPPORT)	5FSS5-L-1
21	1	2"	SERIES 59 BALL VALVE BW	LP1YFV0632N1
22	1	1"	PRESSURE GAUGE	LP1YPI0610ND

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	13-MAR-2026	DRW	SBR	N/A	JHN / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2 (315) E 061055

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-N2-061055-01	0

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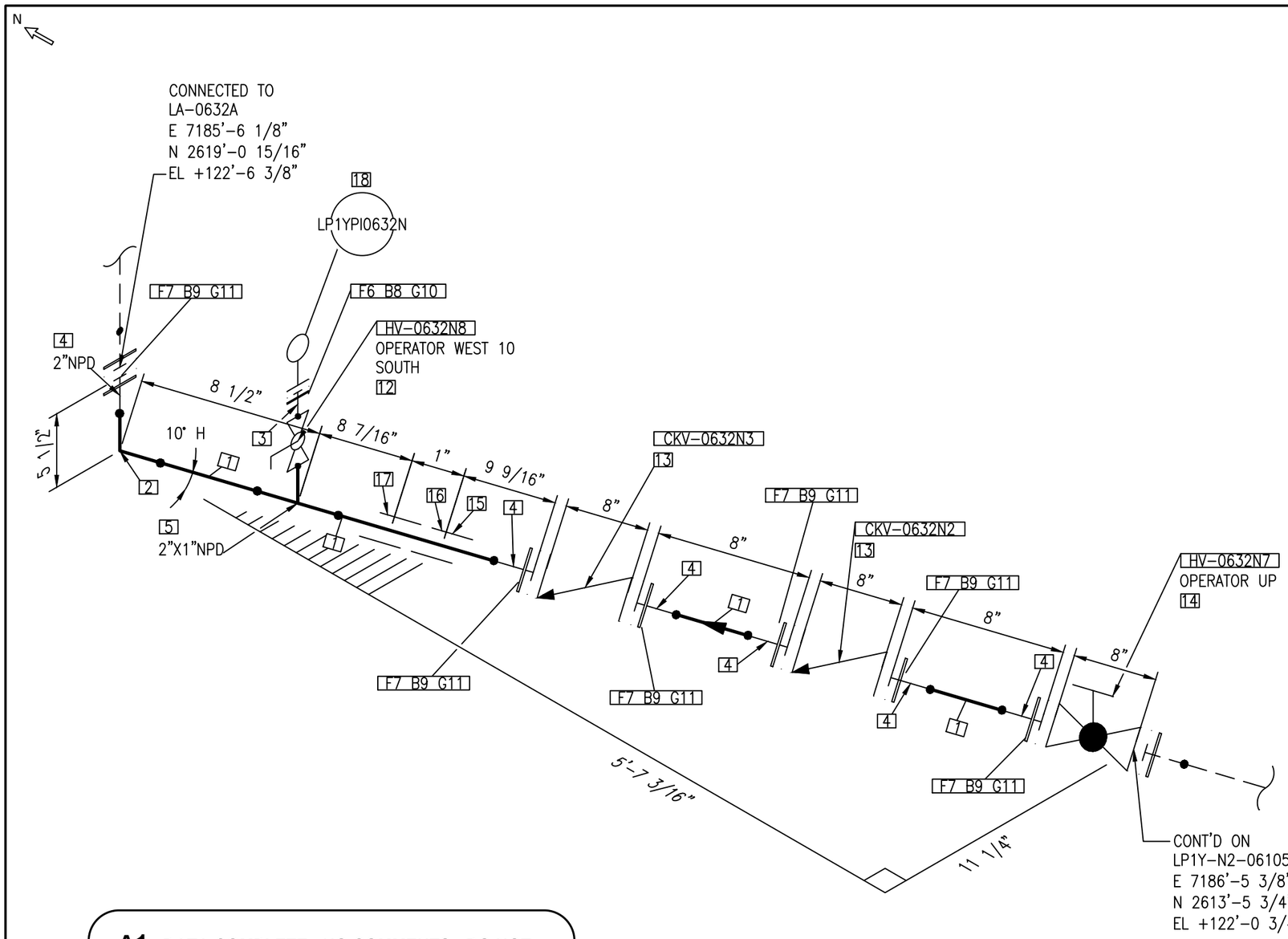

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	2'	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	1	2"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
3	1	1"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
4	6	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
5	1	2"X1"	TEE RED SCH 10S 316/316L SS A403-W	5450714
6	1	1"	FLG LAP JOINT 150# 316 SS	5599578L
7	6	2"	FLG LAP JOINT 150# 316 SS	5599578L
8	4	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
9	24	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
10	1	1"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
11	6	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
12	1	1"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
13	2	2"	CHECK 150# RF 316 SS SWING	5257711L
14	1	2"	GLOBE 150# RF 316 SS	5254829
15	1	2"	5CI (ISOLATION CRADLE)	5CI-02
16	1	2"	5FSS5 (FIELD BASE SUPPORT)	5FSS5-L-1
17	1	2"	5UG (U-BOLT GUIDE)	5UG-02
18	1	1"	PRESSURE GAUGE	LP1YPI0632N

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	13-MAR-2026	DRW	SBR	N/A	JHN / N/A	MDW

GENERAL NOTE:
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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2 (315) E 061055

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-N2-061055-02	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 13-MAR-2026

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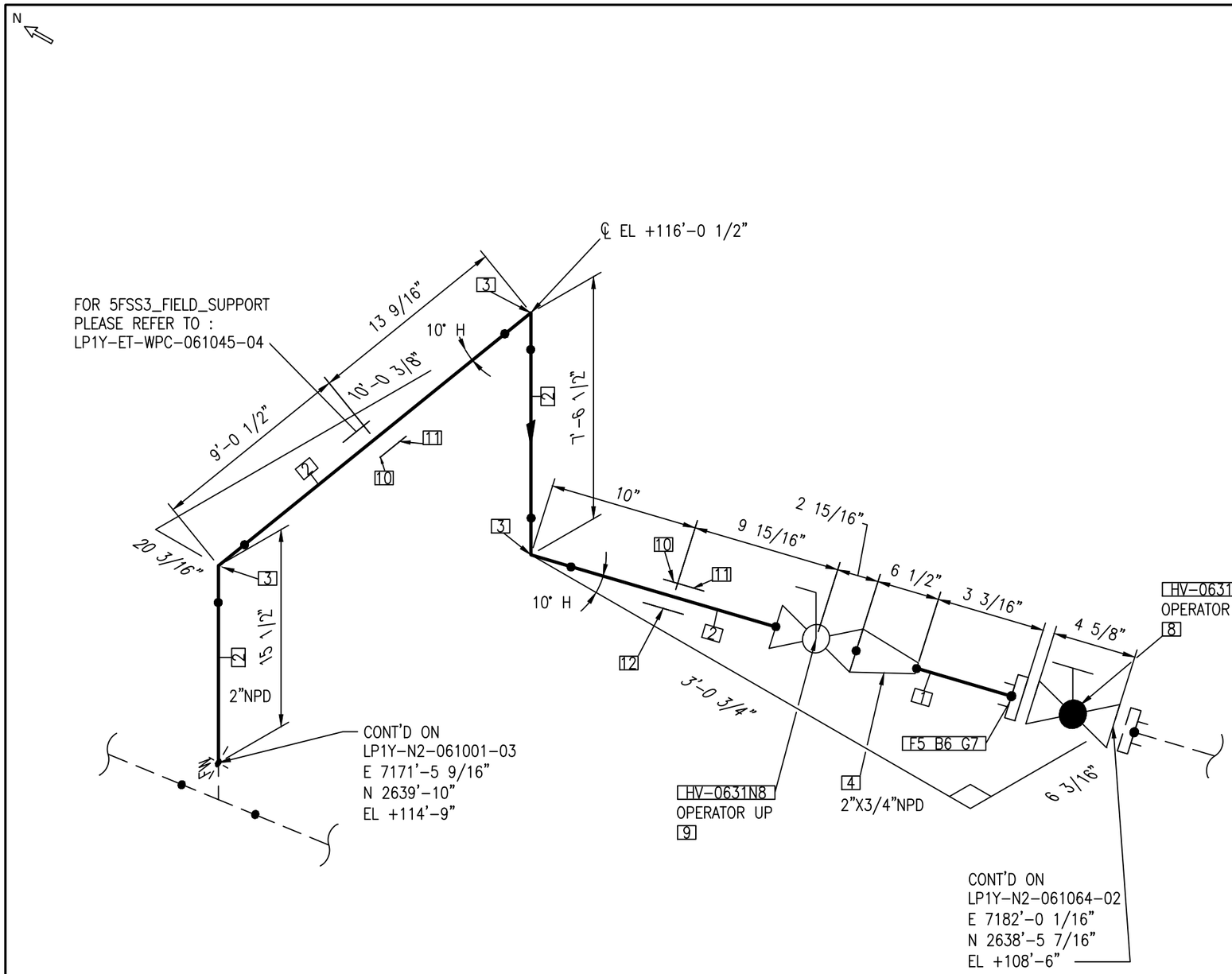

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	3"	3/4"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	19'	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
3	3	2"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
4	1	2"X3/4"	SWAGE CONC SCH 10S 316/316L SS BBE	5546561
5	1	3/4"	FLG SW 150# RF 316/316L SS 10S BORE	5608148
6	4	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
7	1	3/4"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
8	1	3/4"	GLOBE 150# RF 316 SS	5254829
9	1	2"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
10	2	2"	5CI (ISOLATION CRADLE)	5CI-02
11	2	2"	5UG (U-BOLT GUIDE)	5UG-02
12	1	2"	5ABS2 (ANGLE BASE SUPPORT)	5ABS2

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	17-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS		
BUILDING NO:	Y		
CWP NO:	N/A		
MODULE NO:	N/A		
SERVICE CODE:	N2		
PIPE SCHEDULE:	315		
INSULATION CODE:	-		
TRACING TYPE:	NT		
ELASTOMER:	E		
HYGIENIC:	NO	IMPACT:	YES

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2 (315) E 061064

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-N2-061064-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 17-MAR-2026

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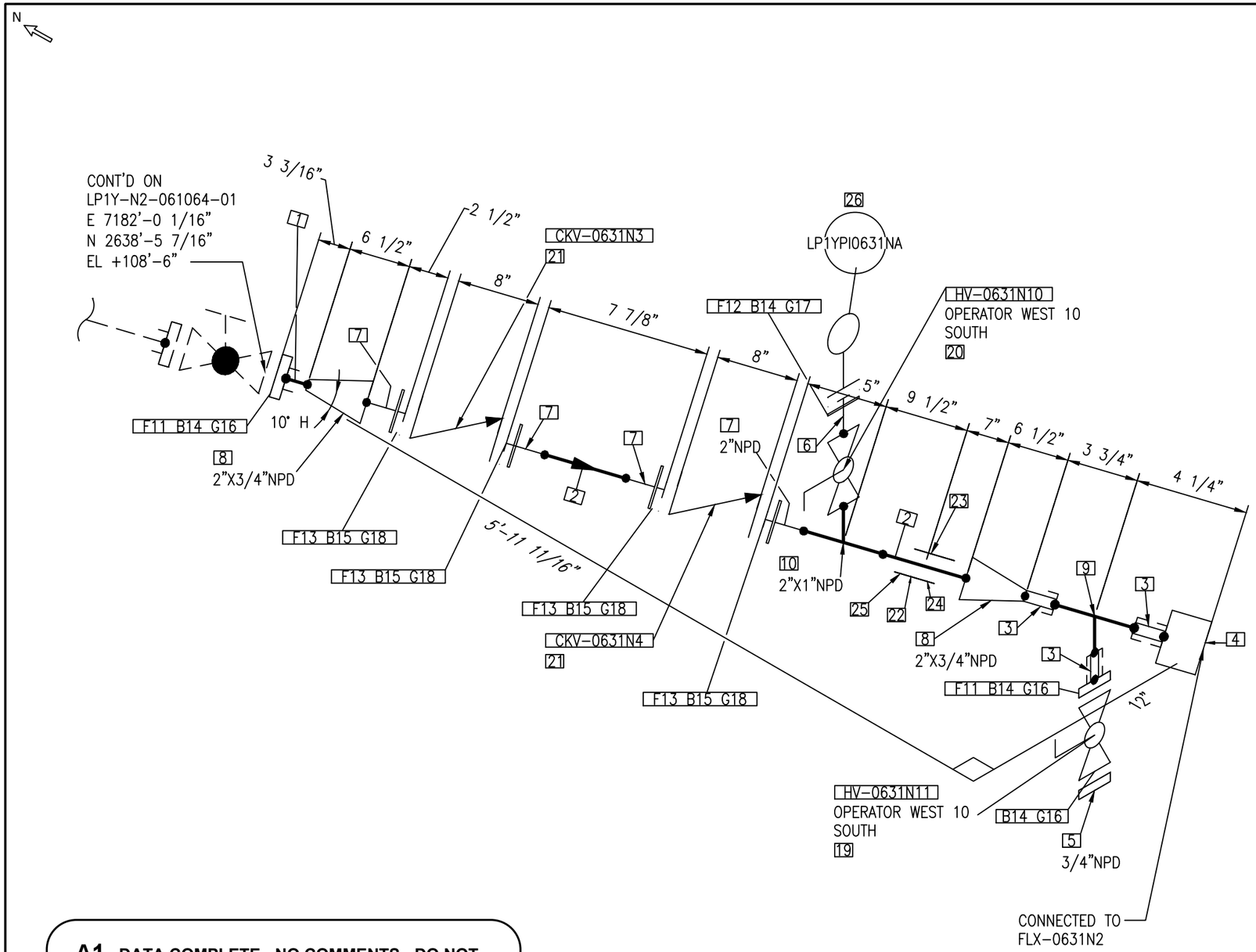

BILL OF MATERIALS

ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	4"	3/4"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	1'-5"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
3	3	3/4"	NIPPLE SCH 10S 316/316L SS PBE 3" LONG	5540491
4	1	3/4"	SOCKETWELD CAMLOCK CONNECTION UNIT	LP1YCCU0631N4
5	1	3/4"	FLG BLIND 150# RF 316/316L SS	5616520
6	1	1"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
7	4	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
8	2	2"X3/4"	SWAGE CONC SCH 10S 316/316L SS BBE	5546561
9	1	3/4"	TEE 3000# SW 316/316L SS	5480140
10	1	2"X1"	TEE RED SCH 10S 316/316L SS A403-W	5450714
11	2	3/4"	FLG SW 150# RF 316/316L SS 10S BORE	5608148
12	1	1"	FLG LAP JOINT 150# 316 SS	5599578L
13	4	2"	FLG LAP JOINT 150# 316 SS	5599578L
14	16	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
15	16	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
16	3	3/4"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
17	1	1"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
18	4	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
19	1	3/4"	BALL 150# RF 316SS TFE FP HNDL OP	5352193L1
20	1	1"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
21	2	2"	CHECK 150# RF 316 SS SWING	5257711L
22	1	2"	5CI (ISOLATION CRADLE)	5CI-02
23	1	2"	5UG (U-BOLT GUIDE)	5UG-02
24	1	2"	5DA1 (DIRECTIONAL ANCHOR)	5DA1-M
25	1	2"	5ABS2 (ANGLE BASE SUPPORT)	5ABS2
26	1	1"	PRESSURE GAUGE	LP1YPI0631NA

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	17-MAR-2026	DRW	SBR	N/A	JHN / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2 (315) E 061064

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-N2-061064-02	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 17-MAR-2026

Tim Trice

Planned Controlled Document

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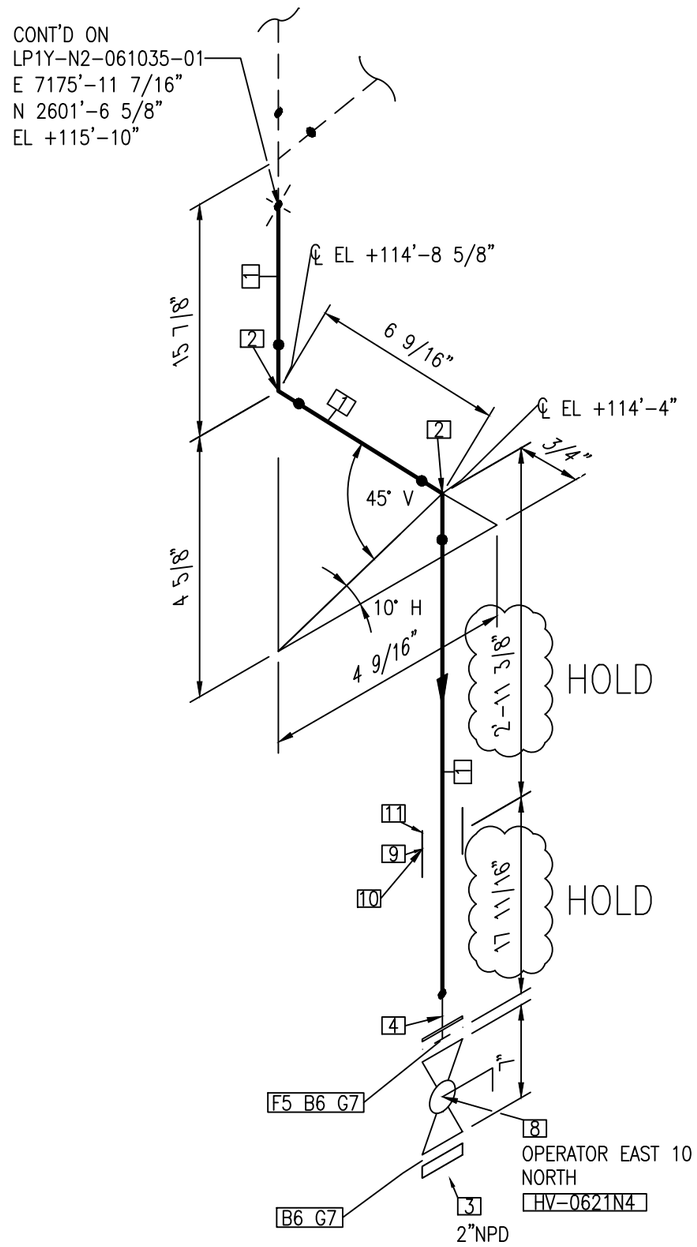

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	5'-5"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	2	2"	ELL 45 SCH 10S 316/316L SS A403-W	5450712
3	1	2"	FLG BLIND 150# RF 316/316L SS	5616520
4	1	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
5	1	2"	FLG LAP JOINT 150# 316 SS	5599578L
6	8	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
7	2	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
8	1	2"	BALL 150# RF 316SS TFE FP HNDL OP	5352194L1
9	1	2"	5CI (ISOLATION CRADLE)	5CI-02
10	1	2"	5UG (U-BOLT GUIDE)	5UG-02
11	1	2"	5FS2 (VERTICAL PIPE SUPPORT)	5FS2S1

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	10-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2 (315) E 062104

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000062	LP1Y-N2-062104-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 10-MAR-2026

Tim Trice

Planned Controlled Document

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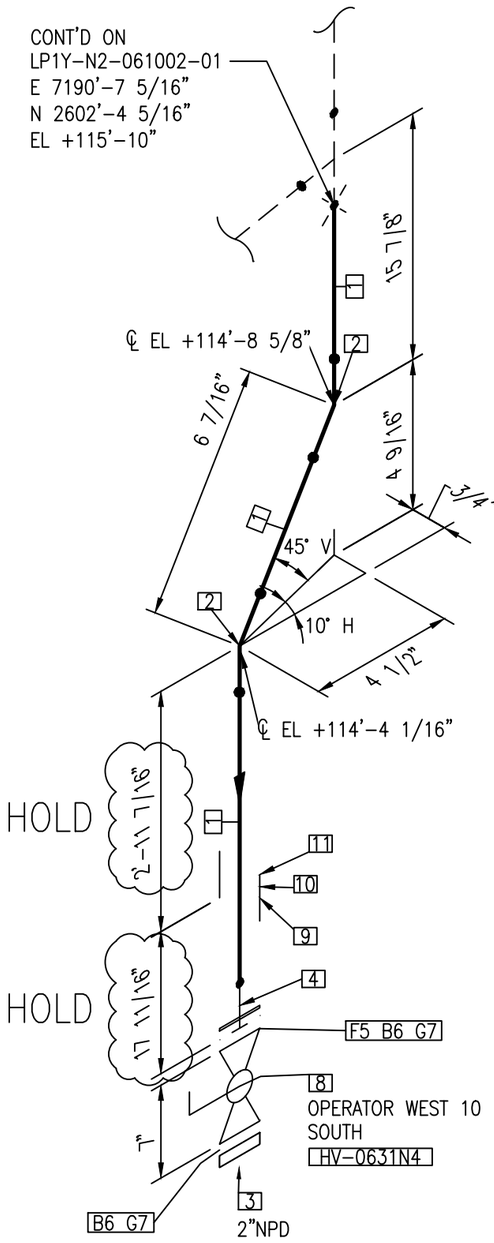

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4	1	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
5	1	2"	FLG LAP JOINT 150# 316 SS	5599578L
6	8	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
7	2	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
8	1	2"	BALL 150# RF 316SS TFE FP HNDL OP	5352194L1
9	1	2"	5CI (ISOLATION CRADLE)	5CI-02
10	1	2"	5UG (U-BOLT GUIDE)	5UG-02
11	1	2"	5FS2 (VERTICAL PIPE SUPPORT)	5FS2S1

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	10-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2 (315) E 063104

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000063	LP1Y-N2-063104-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 10-MAR-2026

Tim Trice

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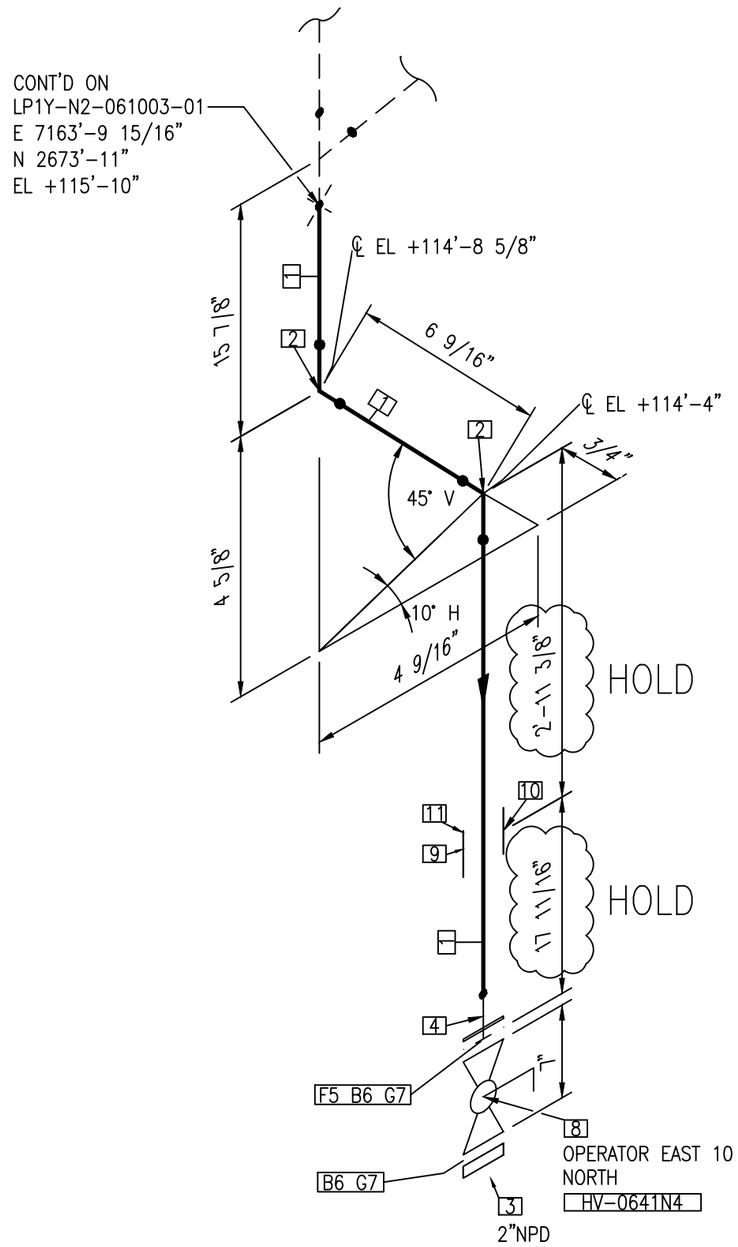

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5	1	2"	FLG LAP JOINT 150# 316 SS	5599578L
6	8	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
7	2	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
8	1	2"	BALL 150# RF 316SS TFE FP HNDL OP	5352194L1
9	1	2"	5CI (ISOLATION CRADLE)	5CI-02
10	1	2"	5UG (U-BOLT GUIDE)	5UG-02
11	1	2"	5FS2 (VERTICAL PIPE SUPPORT)	5FS2S1

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	10-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2 (315) E 064104

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000064	LP1Y-N2-064104-01	0

Tim Trice

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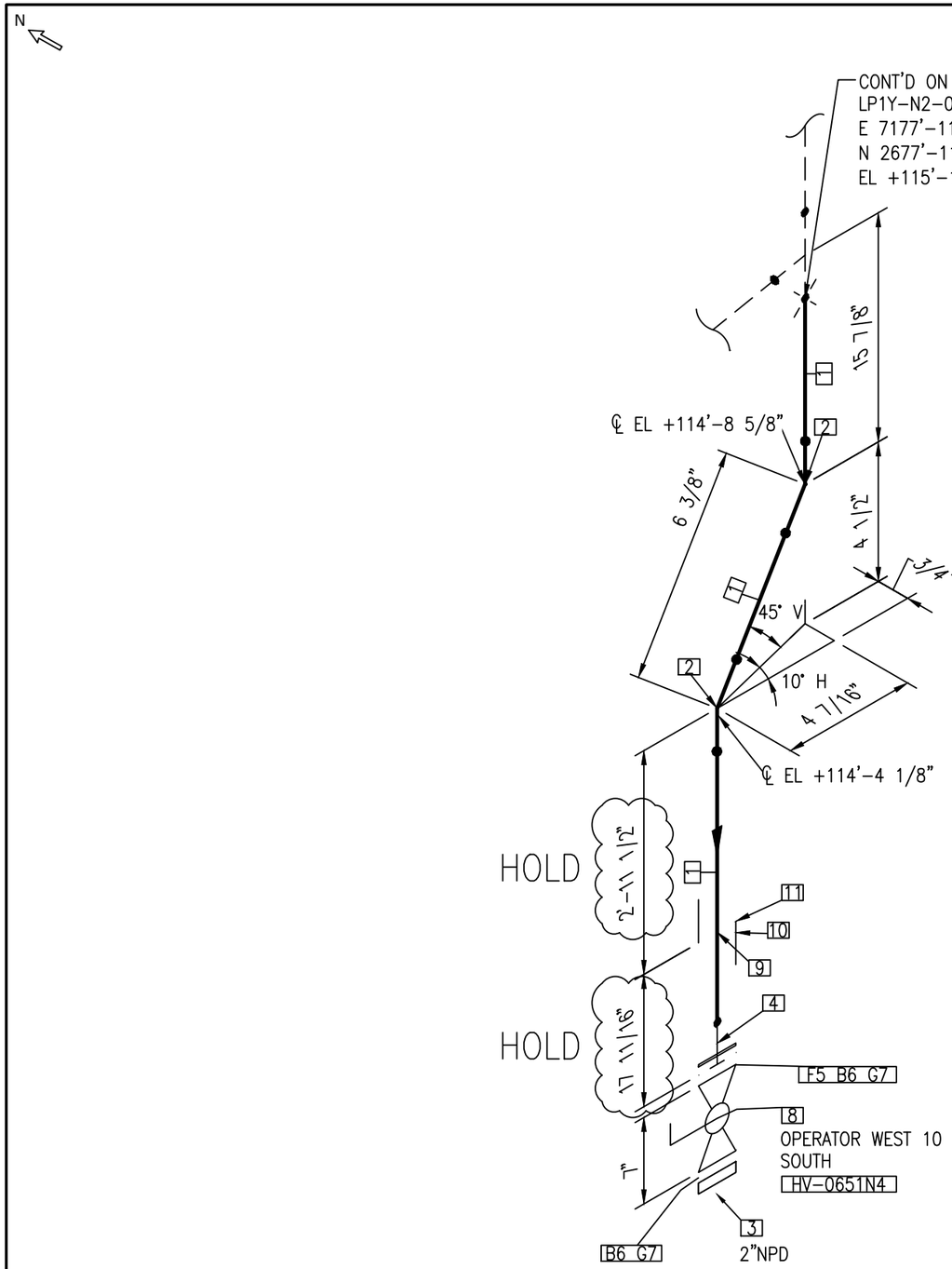

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	N2
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	E
HYGIENIC: NO	IMPACT: YES

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS N2 (315) E 065104

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000065	LP1Y-N2-065104-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 10-MAR-2026

Tim Trice

Planned Controlled Document

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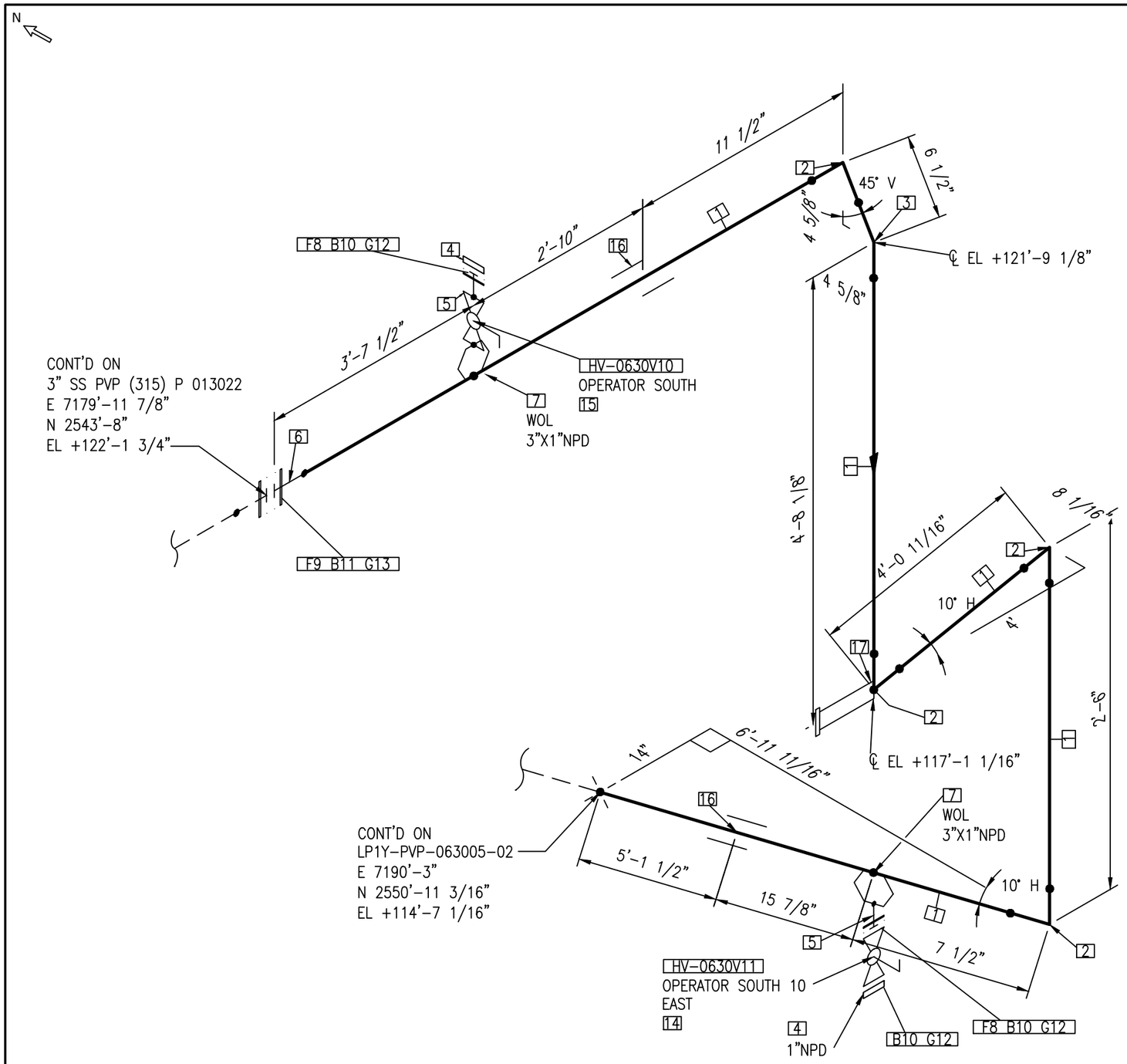

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	22'-9"	3"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	4	3"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
3	1	3"	ELL 45 SCH 10S 316/316L SS A403-W	5450712
4	2	1"	FLG BLIND 150# RF 316/316L SS	5616520
5	2	1"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
6	1	3"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
7	2	3"X1"	PIPET SCH 10S 316/316L SS	5510132
8	2	1"	FLG LAP JOINT 150# 316 SS	5599578L
9	1	3"	FLG LAP JOINT 150# 316 SS	5599578L
10	12	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
11	4	5/8"X2 3/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
12	3	1"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
13	1	3"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
14	1	1"	BALL 150# RF 316SS TFE FP HNDL OP	5352193L1
15	1	1"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
16	2	3"	5CI (ISOLATION CRADLE)	5CI-03
17	1	3"	5DS1 (DUMMY SUPPORT) L=11"	5DS1-03

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	PVP
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	P
HYGIENIC: NO	IMPACT: YES

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	05-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3" SS PVP (315) P 063005

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000063	LP1Y-PVP-063005-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 05-MAR-2026

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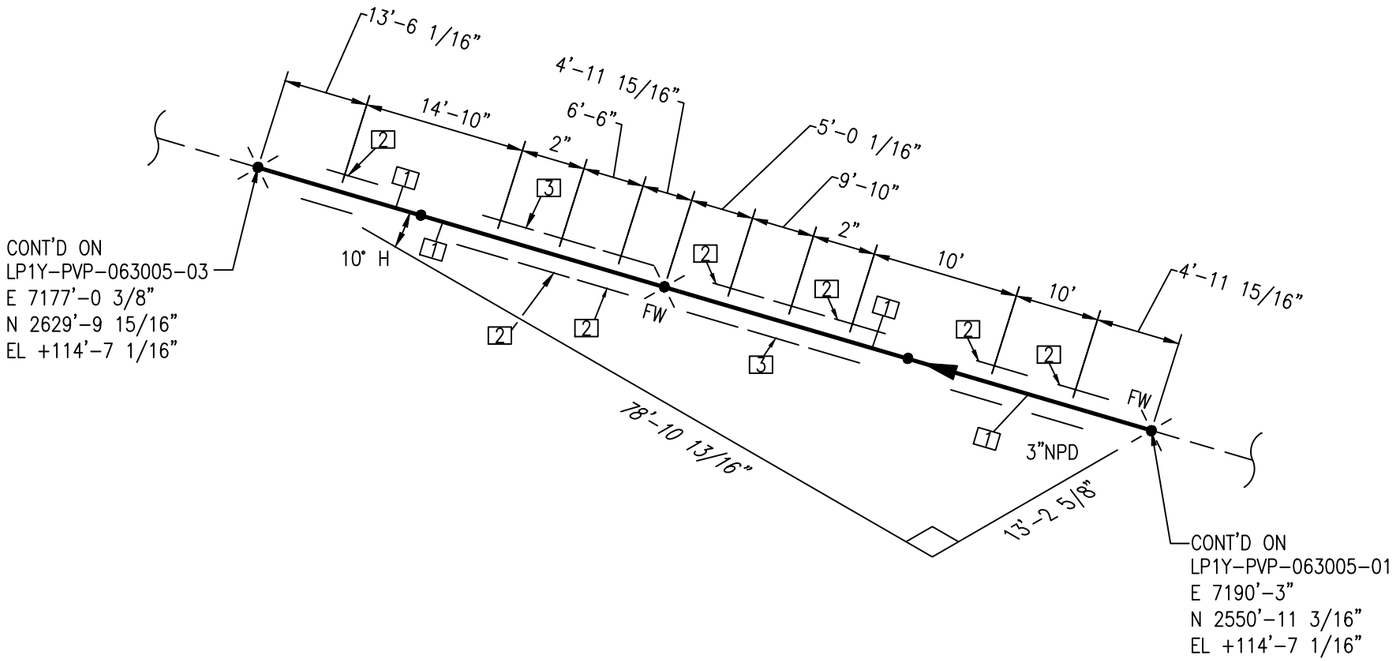
N

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	80'	3"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	7	3"	5CI (ISOLATION CRADLE)	5CI-03
3	2	3"	5UG (U-BOLT GUIDE)	5UG-03



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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	05-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	PVP
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	P
HYGIENIC: NO	IMPACT: YES

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3" SS PVP (315) P 063005

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000063	LP1Y-PVP-063005-02	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 05-MAR-2026

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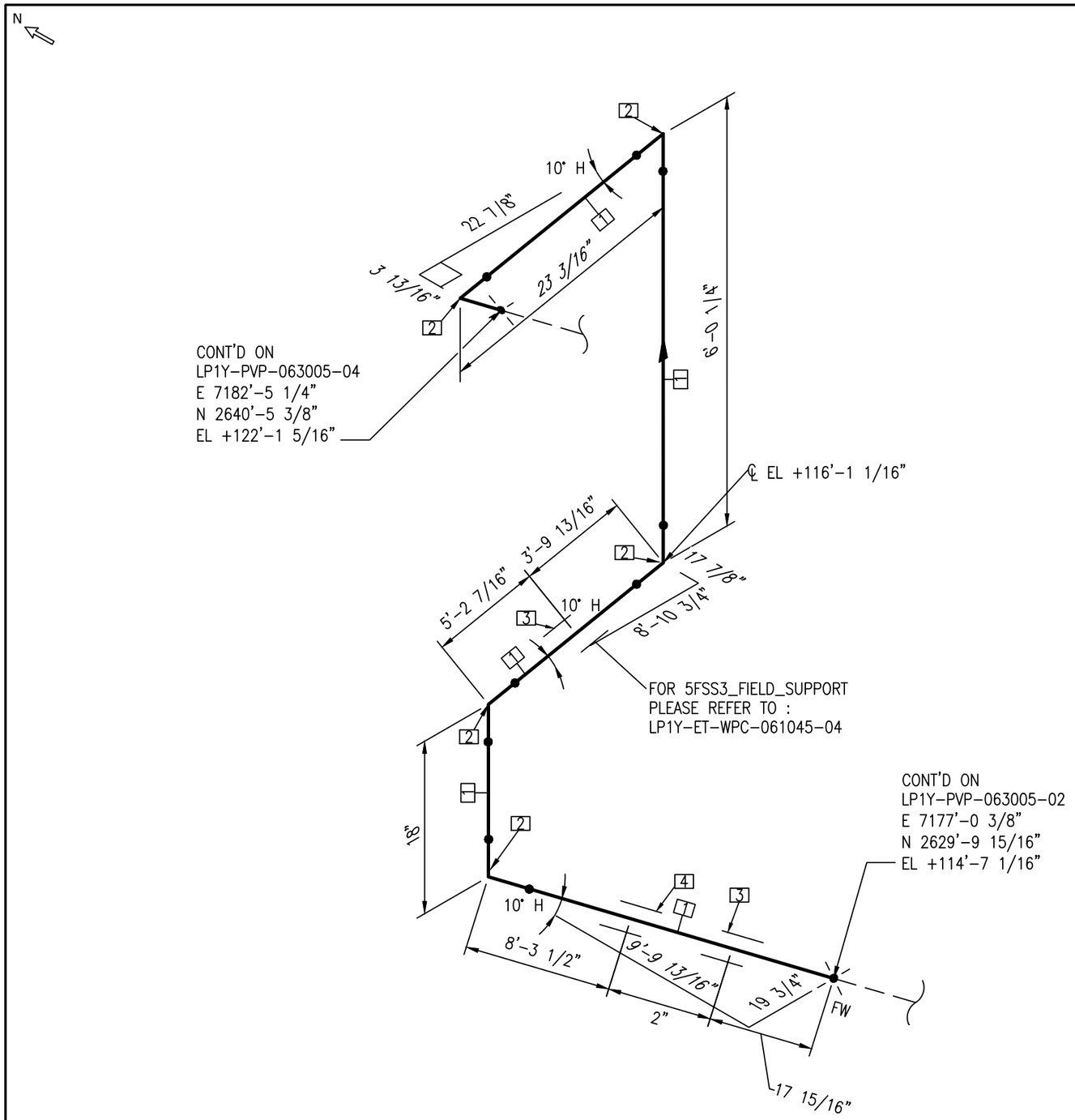

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	25'-1"	3"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	5	3"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
3	2	3"	5CI (ISOLATION CRADLE)	5CI-03
4	1	3"	5UG (U-BOLT GUIDE)	5UG-03

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	PVP
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	P
HYGIENIC: NO	IMPACT: YES

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	05-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3" SS PVP (315) P 063005

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000063	LP1Y-PVP-063005-03	0

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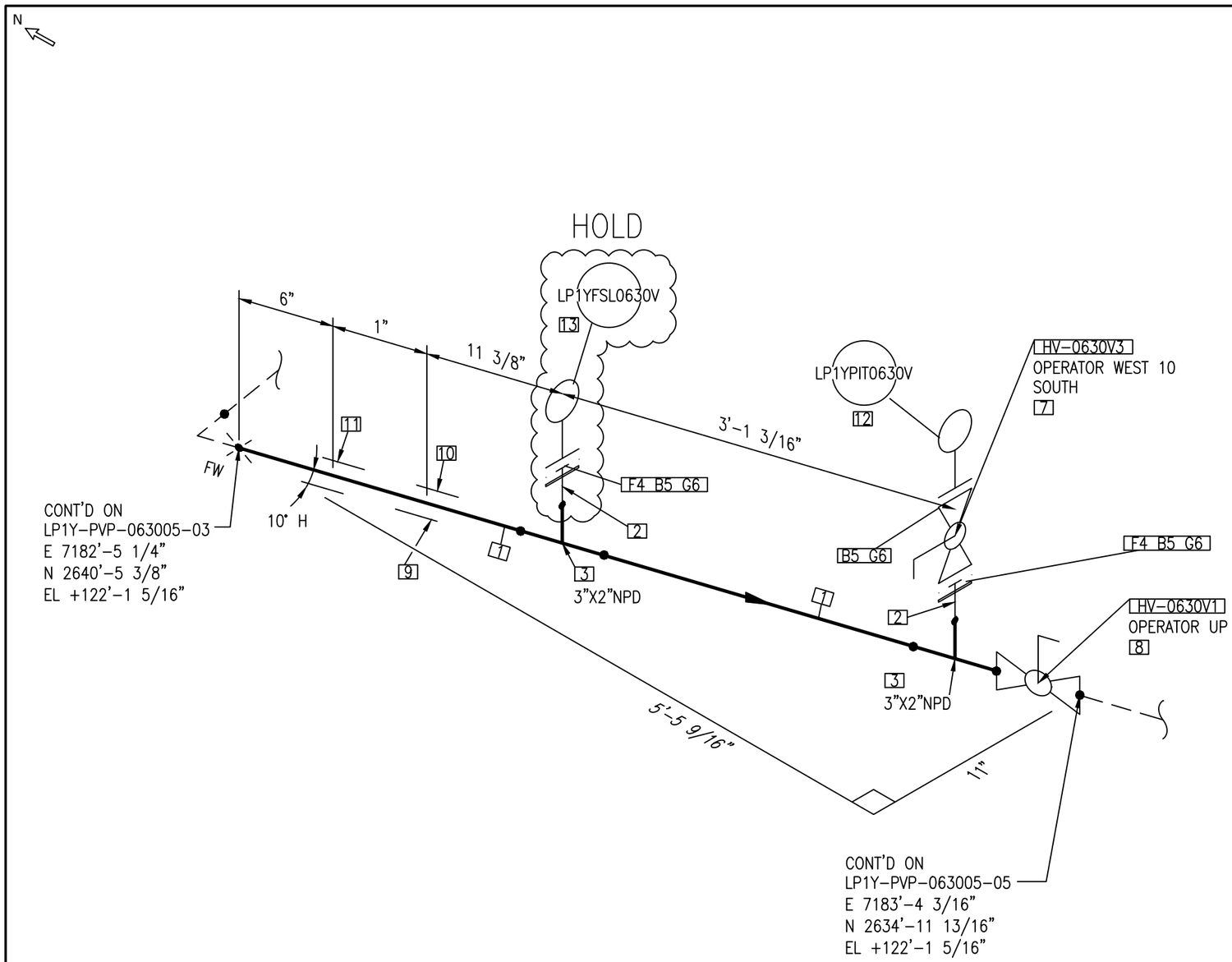

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	3'-10"	3"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	2	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
3	2	3"X2"	TEE RED SCH 10S 316/316L SS A403-W	5450714
4	2	2"	FLG LAP JOINT 150# 316 SS	5599578L
5	12	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
6	3	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
7	1	2"	BALL 150# RF 316SS TFE FP HNDL OP	5352194L1
8	1	3"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
9	1	3"	5CI (ISOLATION CRADLE)	5CI-03
10	1	3"	5FSS5 (FIELD BASE SUPPORT)	5FSS5-L-1
11	1	3"	5UG (U-BOLT GUIDE)	5UG-03
12	1	2"	PRESSURE TRANSMITTER	LP1YPIT0630V
13	1	2"	FSL	LP1YFSL0630V

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
						/	
						/	
0	ISSUE FOR CONSTRUCTION	05-MAR-2026	DRW	SBR	N/A	JHN / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	PVP
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	P
HYGIENIC: NO	IMPACT: YES

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3" SS PVP (315) P 063005

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000063	LP1Y-PVP-063005-04	0

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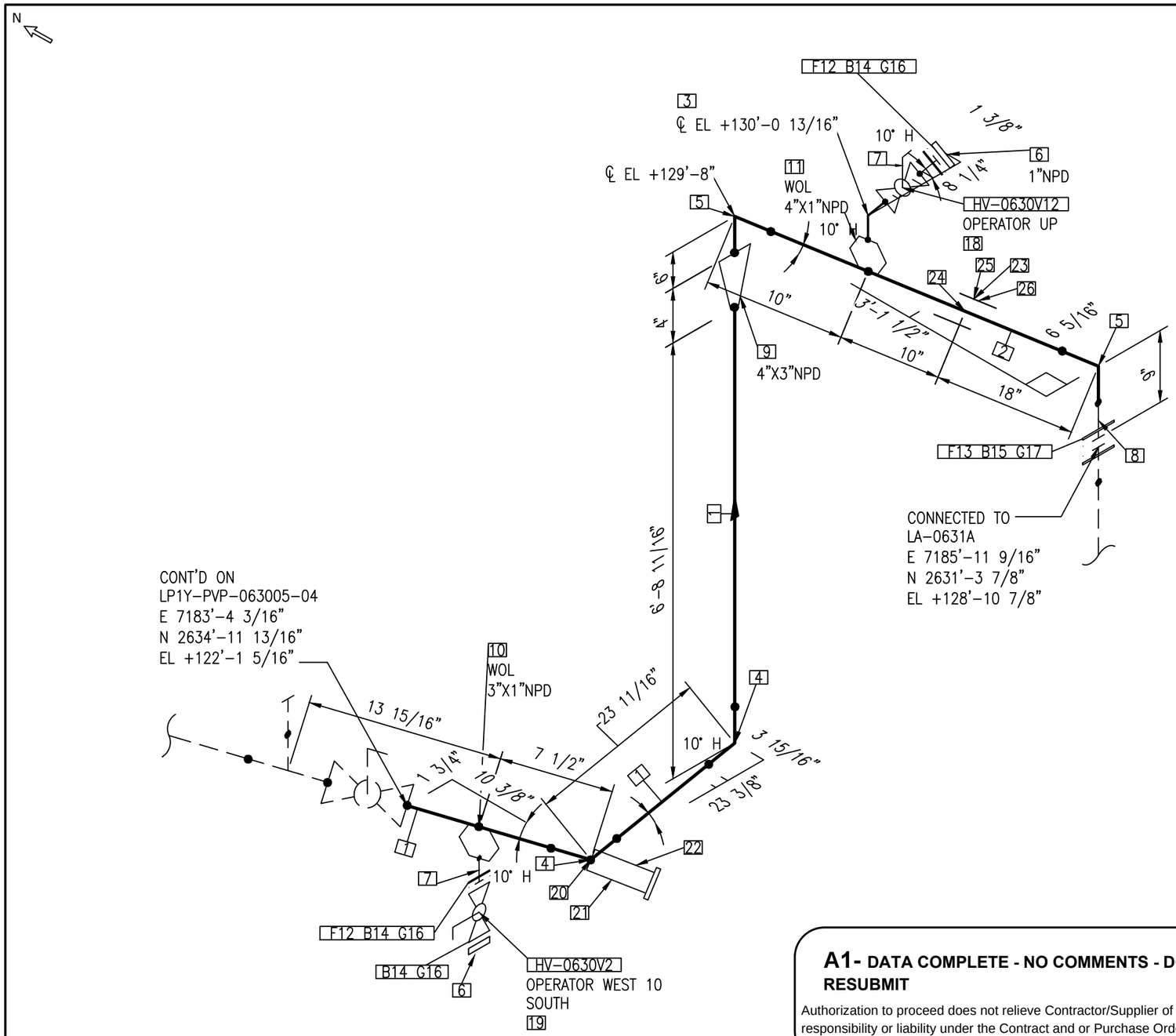

BILL OF MATERIALS

ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	8'-1"	3"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	2'-3"	4"	PIPE SCH 10S ERW 316/316L SS A312	5368751
3	1	1"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
4	2	3"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
5	2	4"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
6	2	1"	FLG BLIND 150# RF 316/316L SS	5616520
7	2	1"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
8	1	4"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
9	1	4"X3"	REDUCER CONC SCH 10S 316/316L SS A403-W	5450717
10	1	3"X1"	PIPET SCH 10S 316/316L SS	5510132
11	1	4"X1"	PIPET SCH 10S 316/316L SS	5510132
12	2	1"	FLG LAP JOINT 150# 316 SS	5599578L
13	1	4"	FLG LAP JOINT 150# 316 SS	5599578L
14	12	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
15	8	5/8"X2 3/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
16	3	1"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
17	1	4"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
18	1	1"	BALL 1000# CWP BW 316SS TFE S10S FP HNDL OP	5352191L
19	1	1"	BALL 150# RF 316SS TFE FP HNDL OP	5352193L1
20	1	3"	5DS1 (DUMMY SUPPORT) L=1'-8"	5DS1-03
21	1	3"	5CI (ISOLATION CRADLE)	5CI-03
22	1	3"	5FSS5 (FIELD BASE SUPPORT)	5FSS5-L-1
23	1	4"	5CI (ISOLATION CRADLE)	5CI-04
24	1	4"	5UG (U-BOLT GUIDE)	5UG-04
25	1	4"	5DA1 (DIRECTIONAL ANCHOR)	5DA1-M
26	1	4"	5FS1 (FIELD SUPPORT)	5FS1S2

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	PVP
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	P
HYGIENIC: NO	IMPACT: YES

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	05-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3" SS PVP (315) P 063005

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000063	LP1Y-PVP-063005-05	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 05-MAR-2026

Tim Trice

Planned Controlled Document

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Planned Controlled Document

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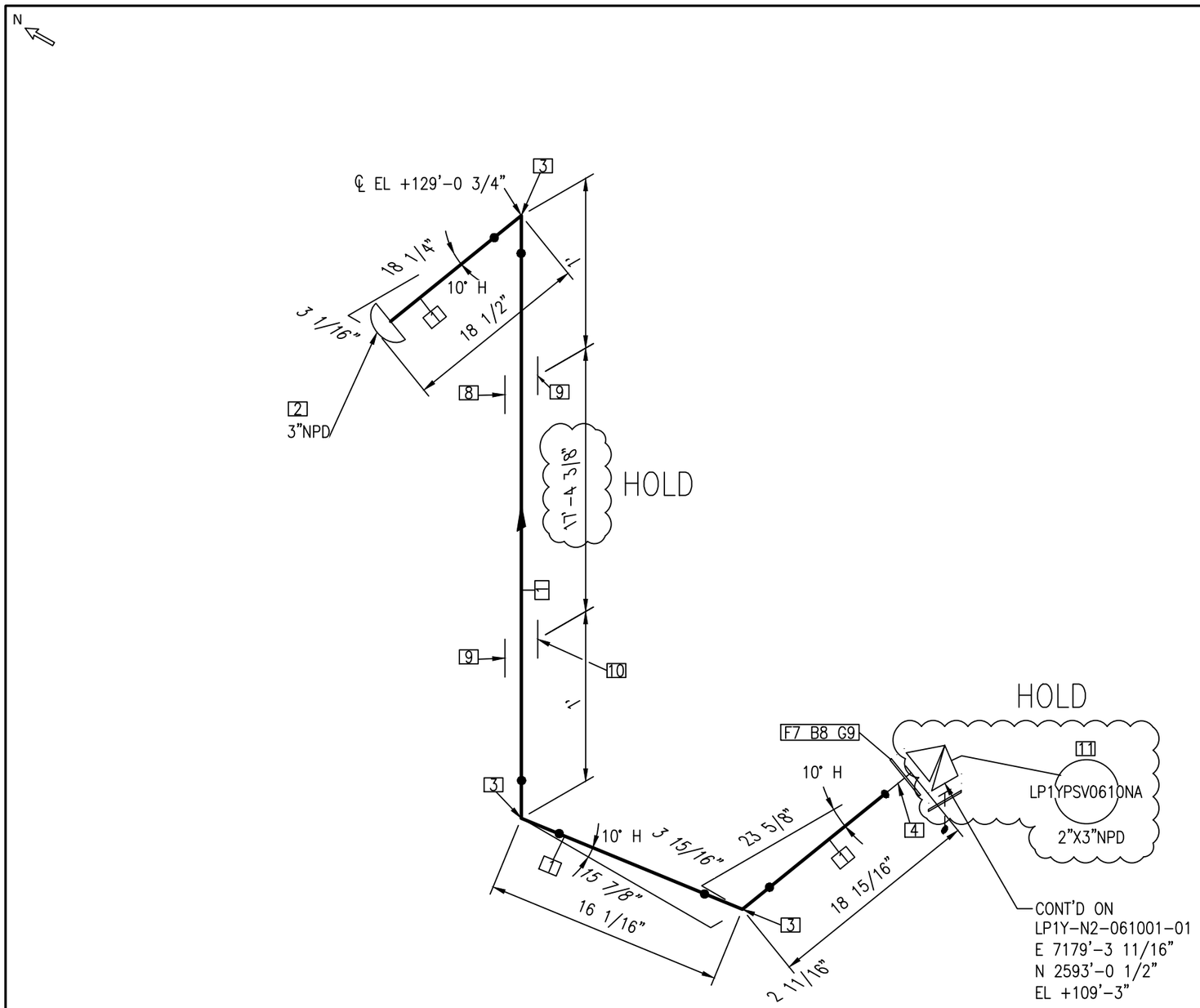

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	21'-3"	3"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	1	3"	SL&C PIPE COVER WSL SERIES	LP1YPC0610V1
3	3	3"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
4	1	3"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
5	1	3"	FLG LAP JOINT 150# 316 SS	5599578L
6	4	5/8"X2 3/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
7	1	3"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
8	1	3"	5US (U-BOLT SUPPORT)	5US-03
9	2	3"	5FS1 (FIELD SUPPORT)	5FS1S2
10	1	3"	5UG (U-BOLT GUIDE)	5UG-03
11	1	2"X3"	PRESSURE SAFETY VALVE	LP1YPSV0610NA

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	10-MAR-2026	DRW	SBR	N/A	JHN / N/A	MDW

GENERAL NOTE:
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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	RV
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3" SS RV (315) 061041

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-RV-061041-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 10-MAR-2026

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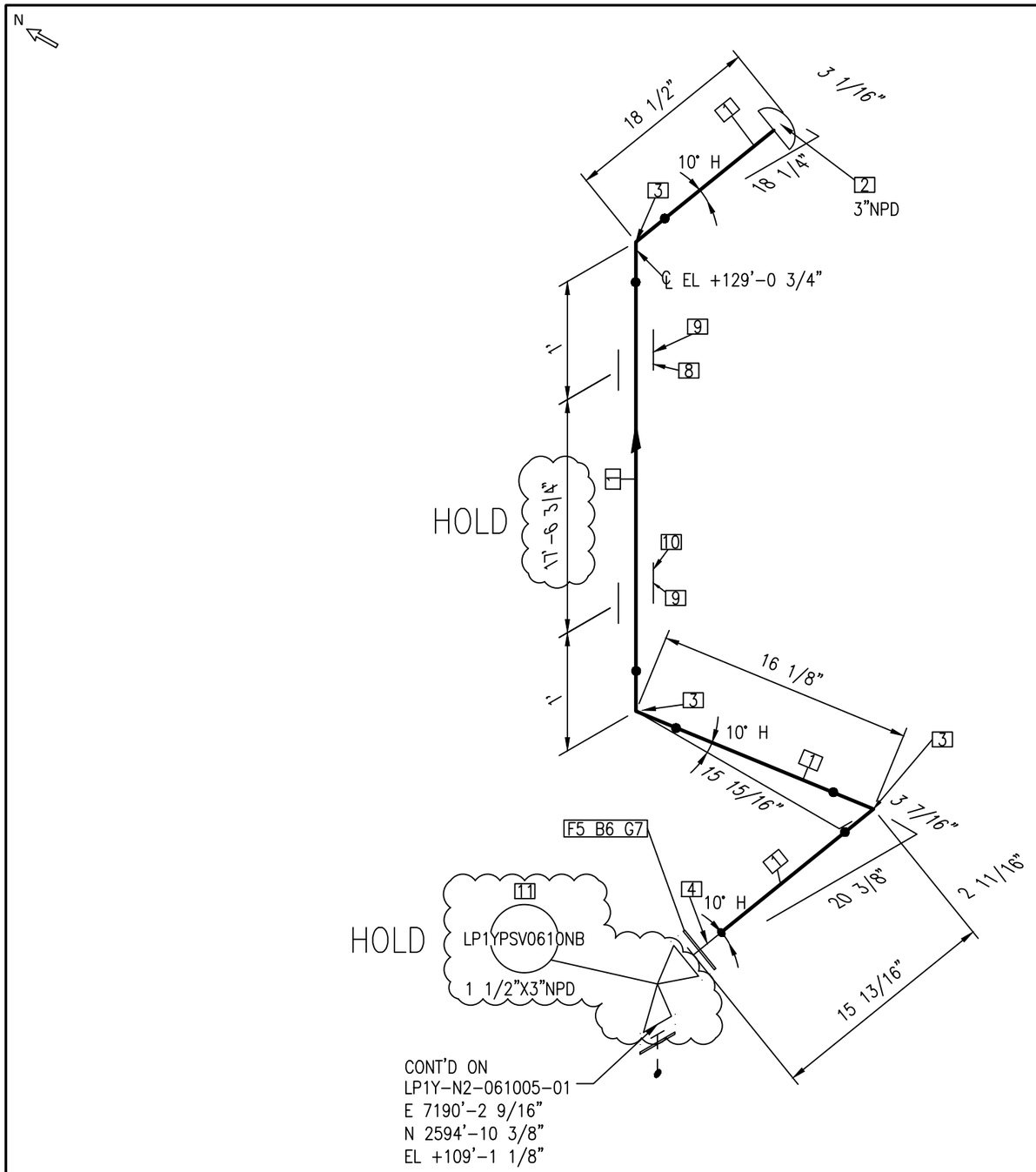

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	21'-2"	3"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	1	3"	SL&C PIPE COVER WSL SERIES	LP1YPC0610V2
3	3	3"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
4	1	3"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
5	1	3"	FLG LAP JOINT 150# 316 SS	5599578L
6	4	5/8"X2 3/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
7	1	3"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
8	1	3"	5US (U-BOLT SUPPORT)	5US-03
9	2	3"	5FS1 (FIELD SUPPORT)	5FS1S2
10	1	3"	5UG (U-BOLT GUIDE)	5UG-03
11	1	1 1/2"X3"	PRESSURE SAFETY VALVE	LP1YPSV0610NB

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	10-MAR-2026	DRW	SBR	N/A	JHN / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	RV
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3" SS RV (315) 061042

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-RV-061042-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 10-MAR-2026

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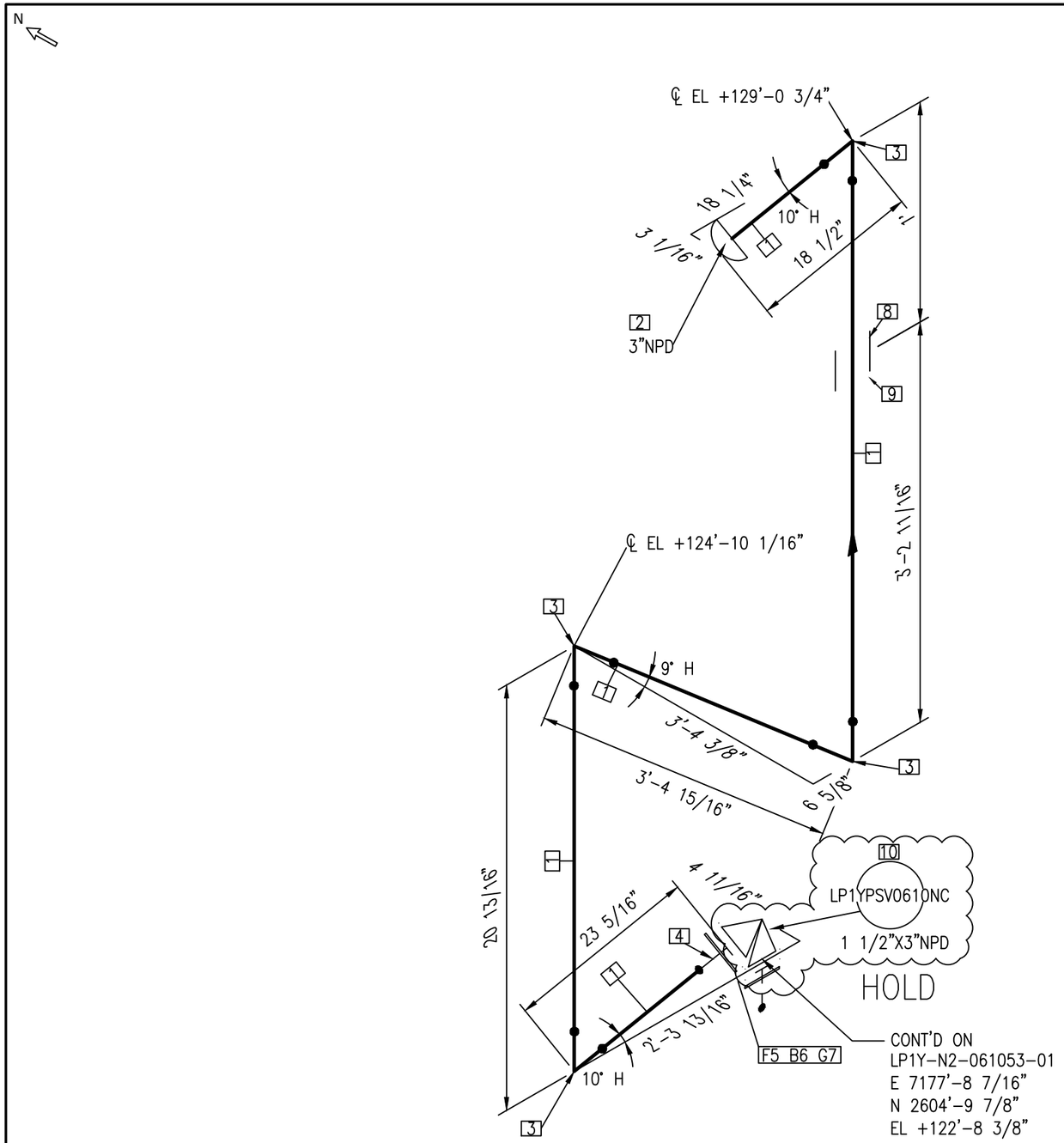

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	9'-6"	3"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	1	3"	SL&C PIPE COVER WSL SERIES	LP1YPC0610V3
3	4	3"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
4	1	3"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
5	1	3"	FLG LAP JOINT 150# 316 SS	5599578L
6	4	5/8"X2 3/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
7	1	3"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
8	1	3"	5US (U-BOLT SUPPORT)	5US-03
9	1	3"	5FS1 (FIELD SUPPORT)	5FS1S2
10	1	1 1/2"X3"	PRESSURE SAFETY VALVE	LP1YPSV0610NC

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By dac54131 at Apr 10, 2026

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0	ISSUE FOR CONSTRUCTION	10-MAR-2026	DRW	SBR	N/A	JHN / N/A	MDW
REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC

NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	RV
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3" SS RV (315) 061054

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-RV-061054-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 10-MAR-2026

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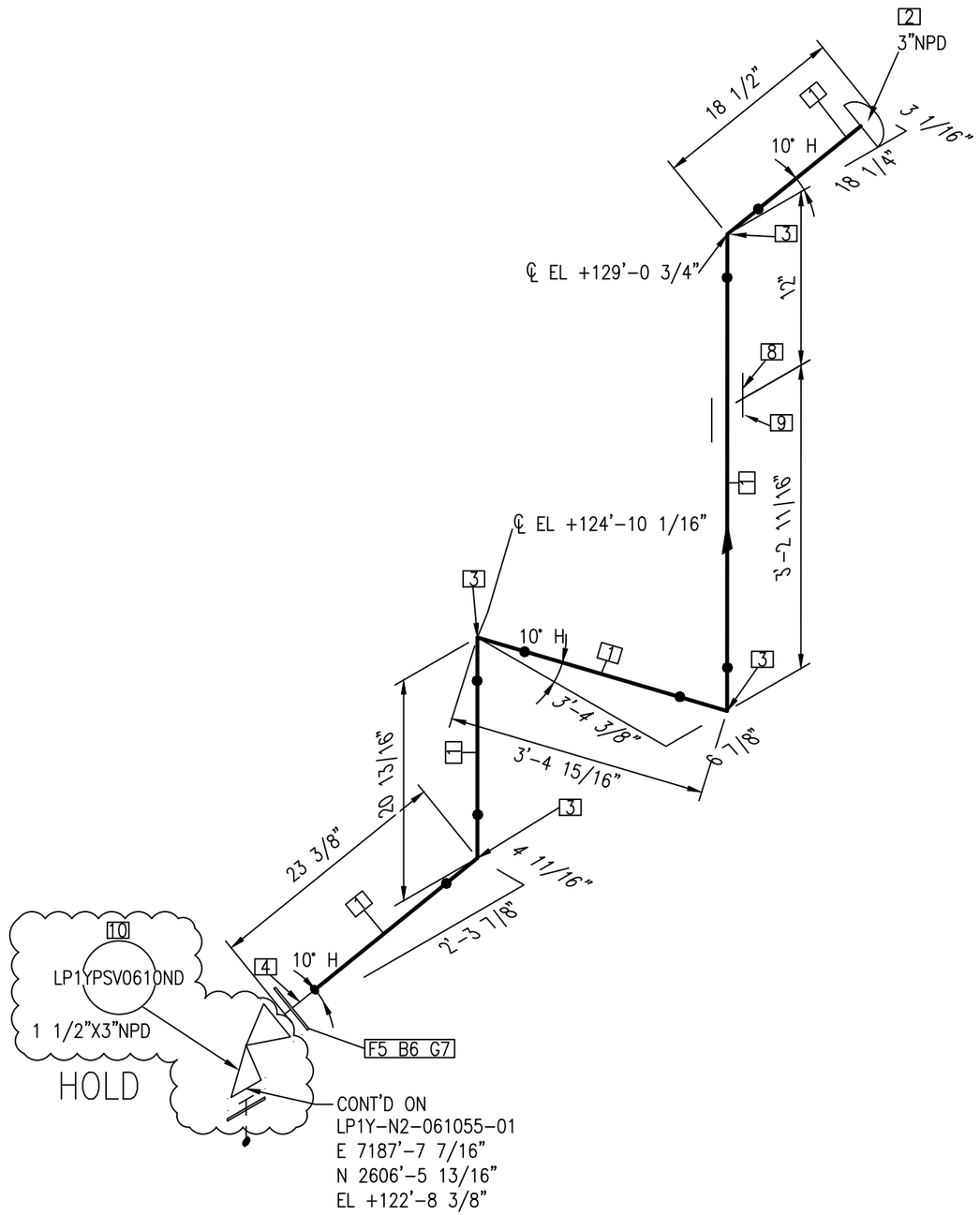

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	9'-6"	3"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	1	3"	SL&C PIPE COVER WSL SERIES	LP1YPC0610V4
3	4	3"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
4	1	3"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
5	1	3"	FLG LAP JOINT 150# 316 SS	5599578L
6	4	5/8"X2 3/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
7	1	3"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
8	1	3"	5US (U-BOLT SUPPORT)	5US-03
9	1	3"	5FS1 (FIELD SUPPORT)	5FS1S2
10	1	1 1/2"X3"	PRESSURE SAFETY VALVE	LP1YPSV0610ND

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	10-MAR-2026	DRW	SBR	N/A	JHN / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	RV
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3" SS RV (315) 061057

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-RV-061057-01	0

Tim Trice

Planned Controlled Document

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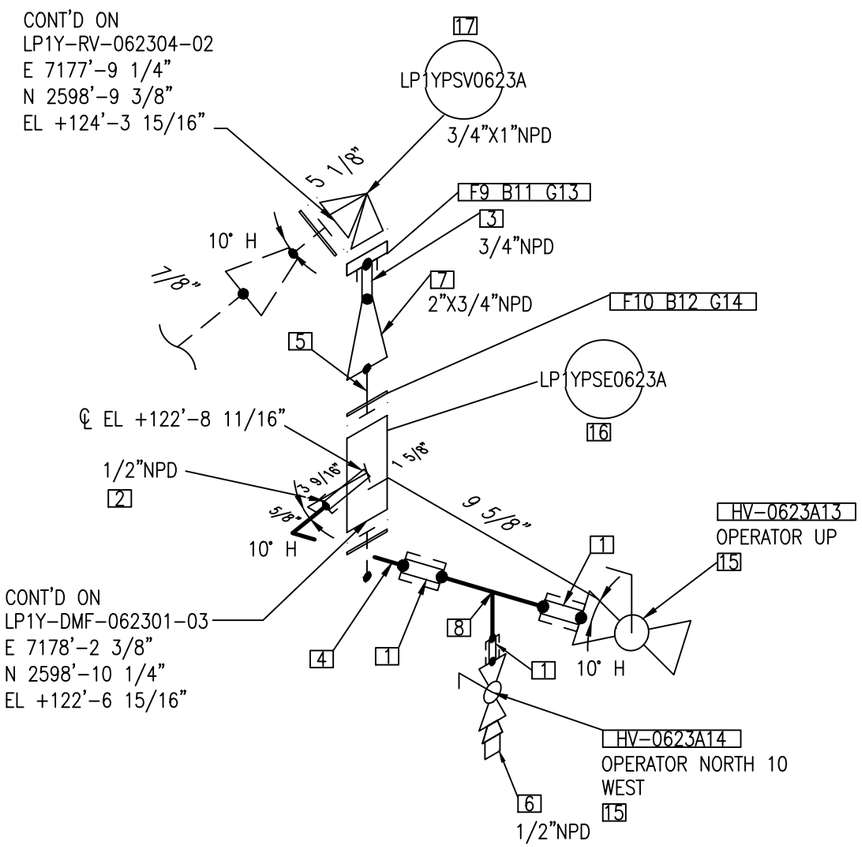

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	3	1/2"	NIPPLE SCH 10S 316/316L SS PBE 3" LONG	5540491
2	1	1/2"	NIPPLE SCH 80S 316/316L SS TOE 3" LONG	5540561
3	1	3/4"	NIPPLE SCH 10S 316/316L SS PBE 3" LONG	5540491
4	1	1/2"	ELL 90 DEG 3000# SW 316/316L SS	5480138
5	1	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
6	1	1/2"	PLUG ROUND HEAD SCRD 316/316L SS	5498383
7	1	2"X3/4"	SWAGE CONC SCH 10S 316/316L SS BBE	5546561
8	1	1/2"	TEE 3000# SW 316/316L SS	5480140
9	1	3/4"	FLG SW 150# RF 316/316L SS 10S BORE	5608148
10	1	2"	FLG LAP JOINT 150# 316 SS	5599578L
11	4	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
12	4	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
13	1	3/4"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
14	1	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
15	2	1/2"	BALL 1000# CWP SW X FNPT 316SS TFE FP HNDL OP	5352192L6
16	1	2"	RUPTURE DISK AND HOLDER ASSEMBLY	LP1YPSE0623A
17	1	3/4"X1"	PRESSURE SAFETY VALVE	LP1YPSV0623A

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	09-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	RV
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS RV (315) 062304

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000062	LP1Y-RV-062304-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 09-MAR-2026

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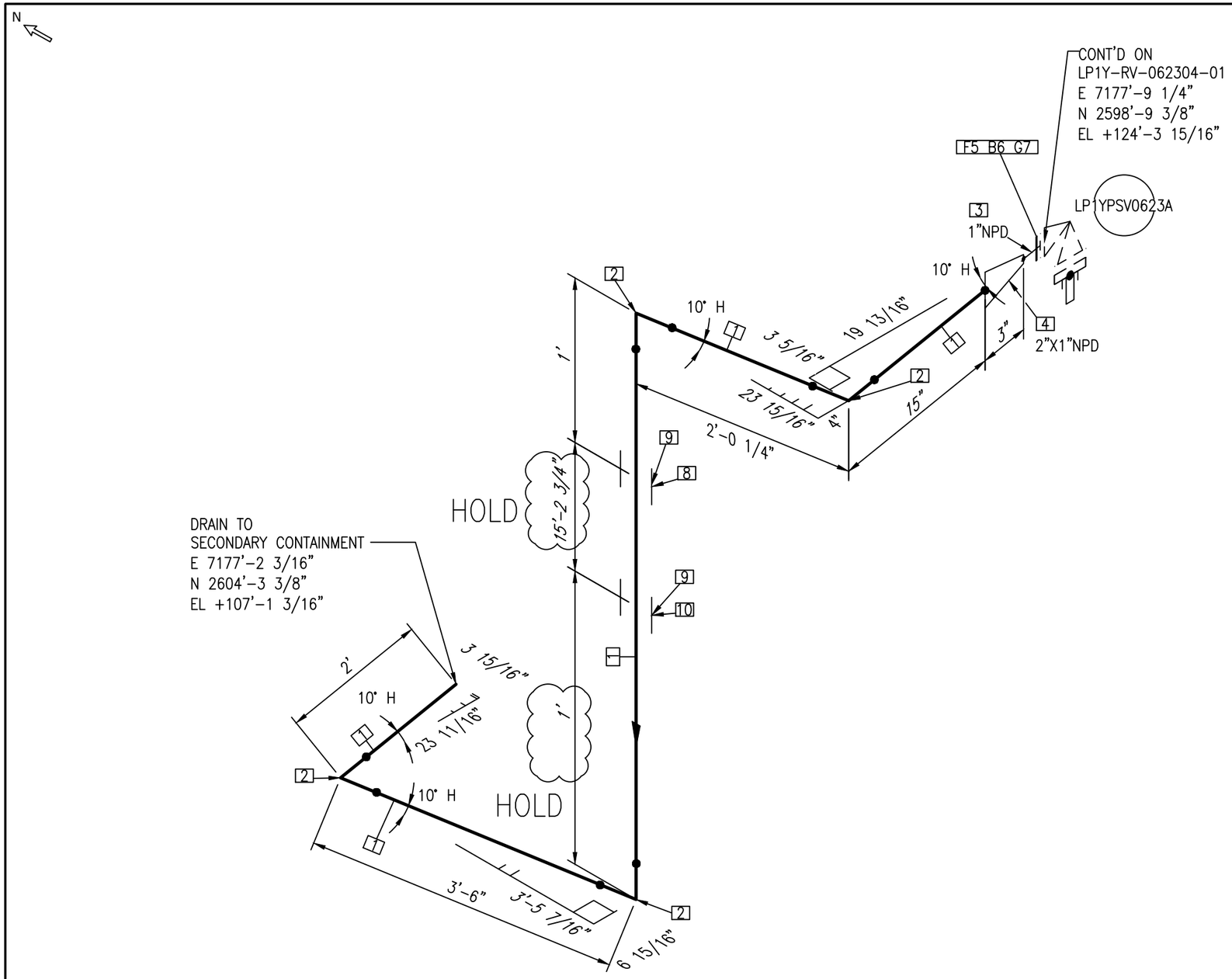

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	24'	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	4	2"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
3	1	1"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
4	1	2"X1"	REDUCER CONC SCH 10S 316/316L SS A403-W	5450717
5	1	1"	FLG LAP JOINT 150# 316 SS	5599578L
6	4	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
7	1	1"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
8	1	2"	5US (U-BOLT SUPPORT)	5US-02
9	2	2"	5FS1 (FIELD SUPPORT)	5FS1S1
10	1	2"	5UG (U-BOLT GUIDE)	5UG-02

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	09-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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7. PIPE SUPPORTS INDICATED ON THE DRAWING MAY BE PART OF AN ASSEMBLY ON SEVERAL ISOMETRICS. REFER TO THE 3D MODEL FOR THE COMPLETE PIPE SUPPORT ASSEMBLY.
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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	RV
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS RV (315) 062304

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000062	LP1Y-RV-062304-02	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 09-MAR-2026

Tim Trice

Planned Controlled Document

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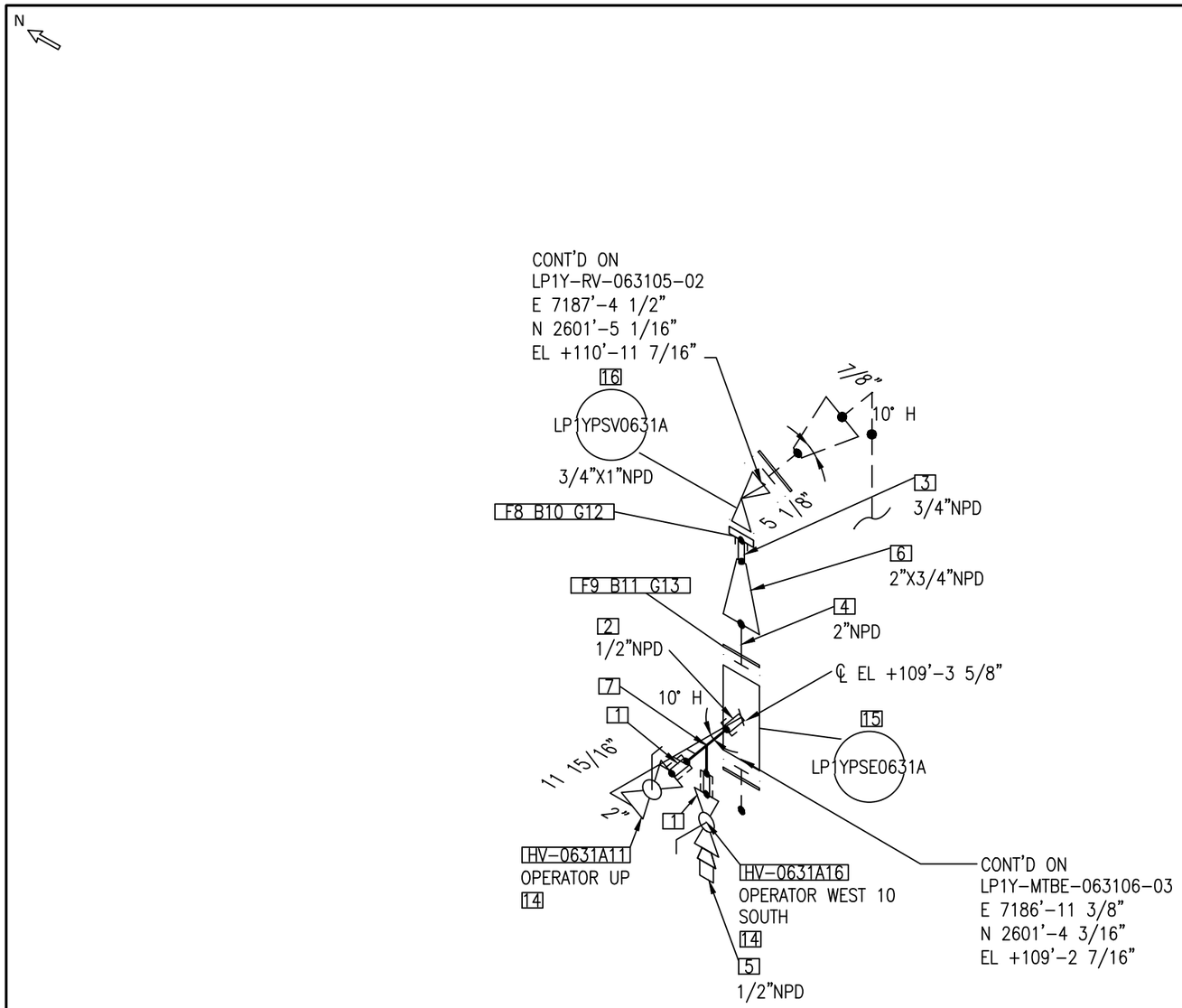

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	2	1/2"	NIPPLE SCH 10S 316/316L SS PBE 3" LONG	5540491
2	1	1/2"	NIPPLE SCH 80S 316/316L SS TOE 6" LONG	5540563
3	1	3/4"	NIPPLE SCH 10S 316/316L SS PBE 3" LONG	5540491
4	1	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
5	1	1/2"	PLUG ROUND HEAD SCRD 316/316L SS	5498383
6	1	2"X3/4"	SWAGE CONC SCH 10S 316/316L SS BBE	5546561
7	1	1/2"	TEE 3000# SW 316/316L SS	5480140
8	1	3/4"	FLG SW 150# RF 316/316L SS 10S BORE	5608148
9	1	2"	FLG LAP JOINT 150# 316 SS	5599578L
10	4	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
11	4	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
12	1	3/4"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
13	1	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
14	2	1/2"	BALL 1000# CWP SW X FNPT 316SS TFE FP HNDL OP	5352192L6
15	1	2"	RUPTURE DISK AND HOLDER ASSEMBLY	LP1YPSE0631A
16	1	3/4"X1"	PRESSURE SAFETY VALVE	LP1YPSV0631A

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	10-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	RV
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS RV (315) 063105

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000063	LP1Y-RV-063105-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 10-MAR-2026

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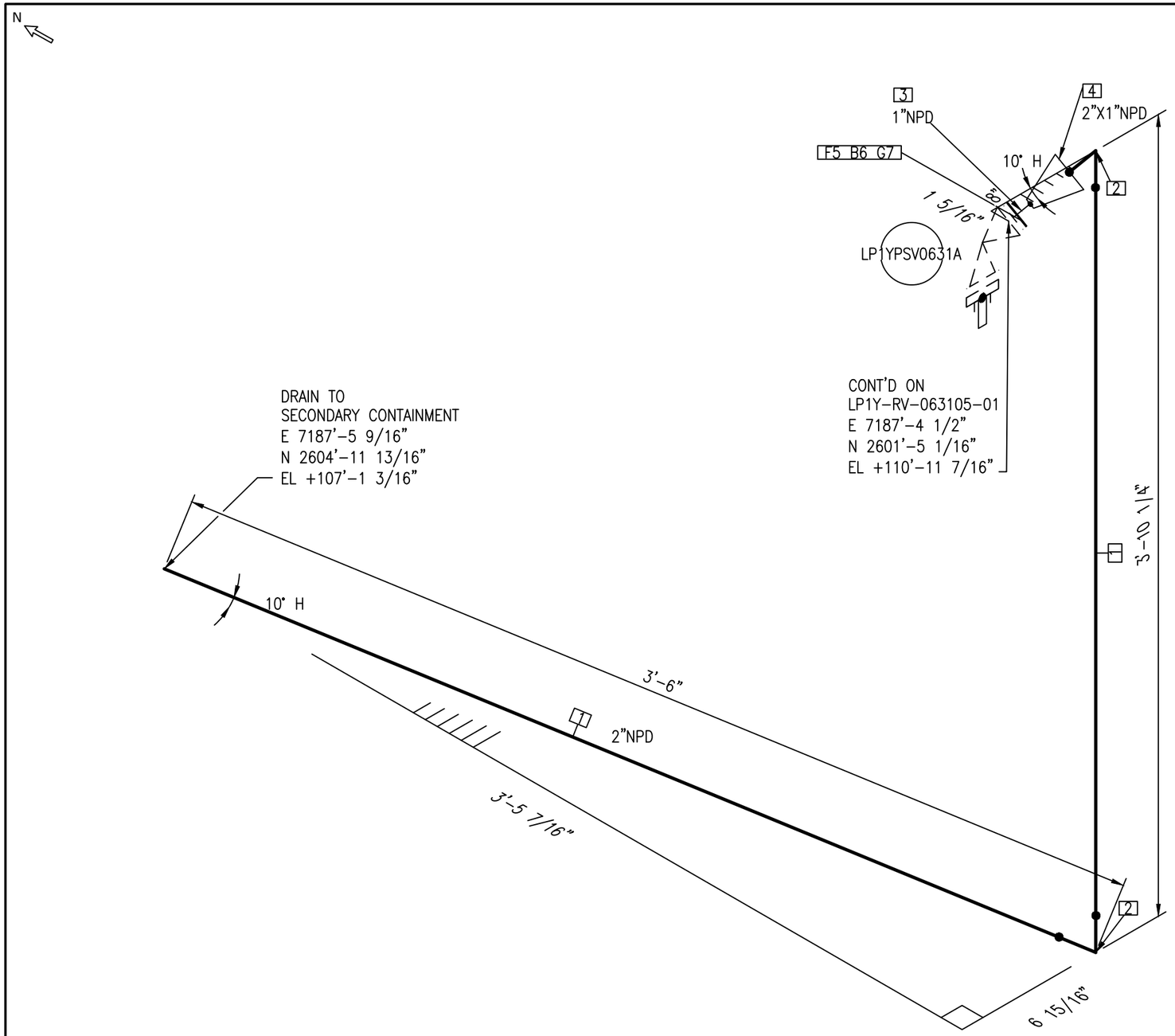

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	6'-8"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	2	2"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
3	1	1"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
4	1	2"X1"	REDUCER CONC SCH 10S 316/316L SS A403-W	5450717
5	1	1"	FLG LAP JOINT 150# 316 SS	5599578L
6	4	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
7	1	1"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	10-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	RV
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS RV (315) 063105

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000063	LP1Y-RV-063105-02	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 10-MAR-2026

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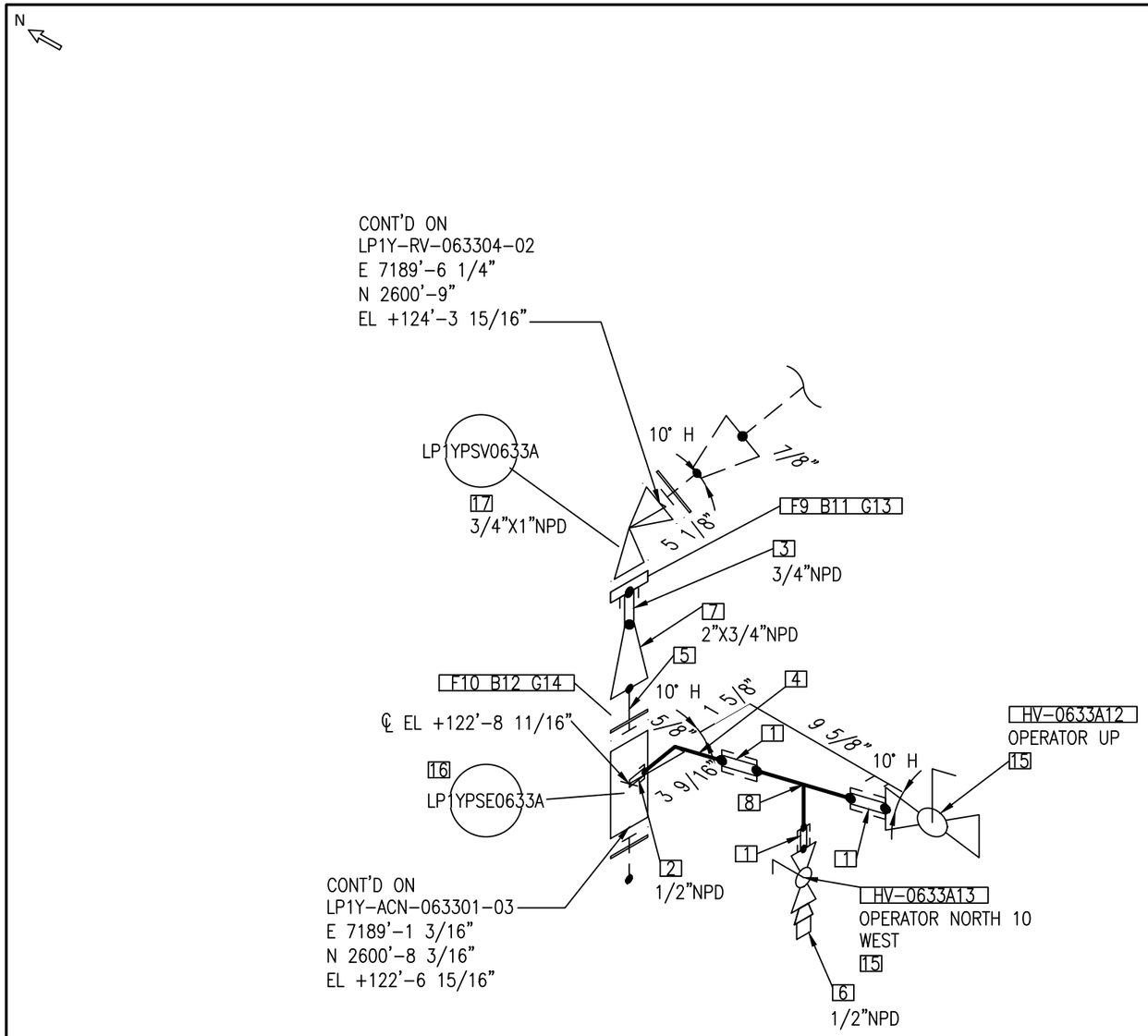

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	3	1/2"	NIPPLE SCH 10S 316/316L SS PBE 3" LONG	5540491
2	1	1/2"	NIPPLE SCH 80S 316/316L SS TOE 3" LONG	5540561
3	1	3/4"	NIPPLE SCH 10S 316/316L SS PBE 3" LONG	5540491
4	1	1/2"	ELL 90 DEG 3000# SW 316/316L SS	5480138
5	1	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
6	1	1/2"	PLUG ROUND HEAD SCRD 316/316L SS	5498383
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9	1	3/4"	FLG SW 150# RF 316/316L SS 10S BORE	5608148
10	1	2"	FLG LAP JOINT 150# 316 SS	5599578L
11	4	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
12	4	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
13	1	3/4"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
14	1	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
15	2	1/2"	BALL 1000# CWP SW X FNPT 316SS TFE FP HNDL OP	5352192L6
16	1	2"	RUPTURE DISK AND HOLDER ASSEMBLY	LP1YPSE0633A
17	1	3/4"X1"	PRESSURE SAFETY VALVE	LP1YPSV0633A

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	RV
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	09-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS RV (315) 063304

REFERENCE P&ID NUMBER
LP1Y-DI-00000063

ISOMETRIC DRAWING NUMBER
LP1Y-RV-063304-01

REV
0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 09-MAR-2026

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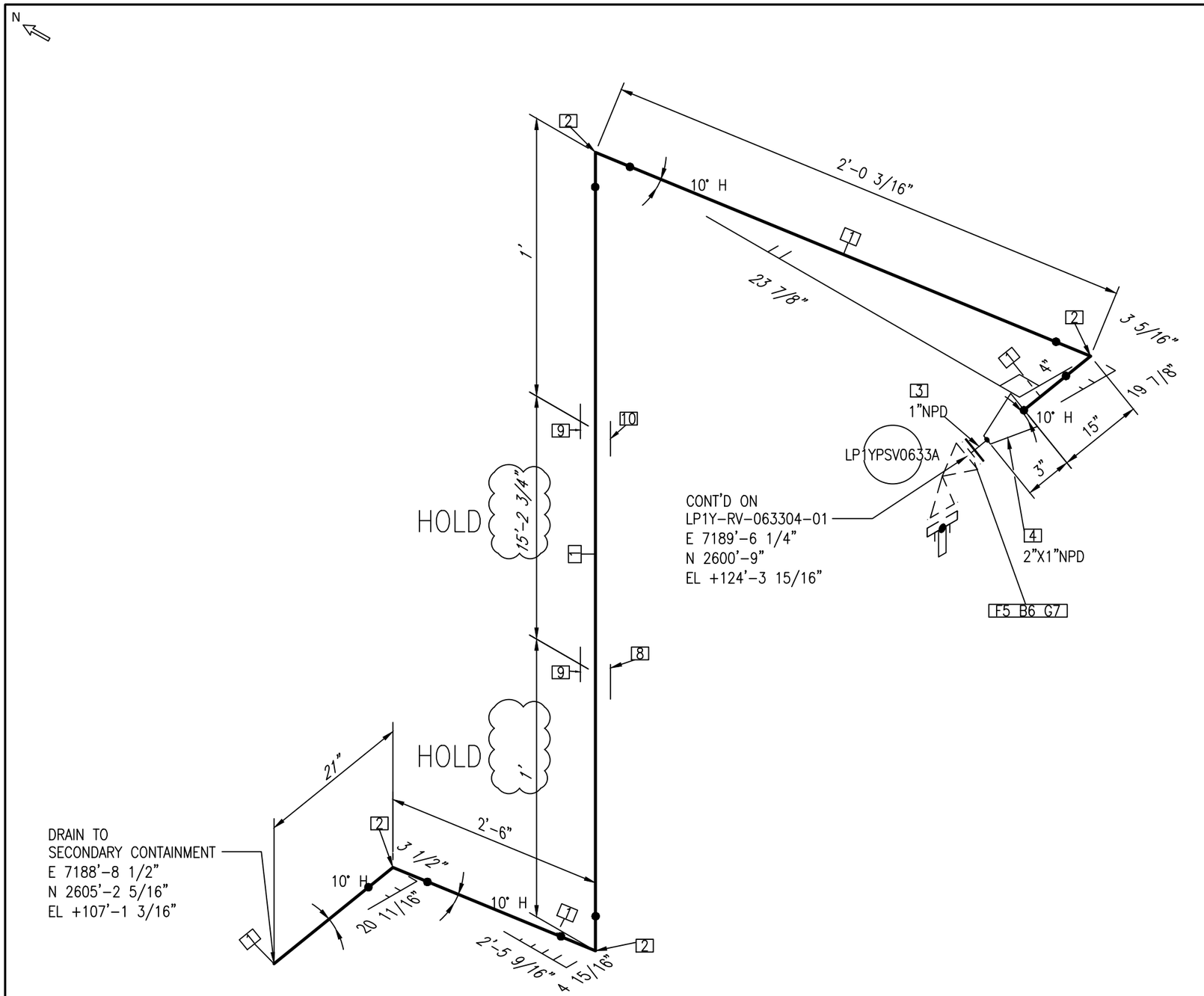

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	22'-9"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	4	2"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
3	1	1"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
4	1	2"X1"	REDUCER CONC SCH 10S 316/316L SS A403-W	5450717
5	1	1"	FLG LAP JOINT 150# 316 SS	5599578L
6	4	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
7	1	1"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
8	1	2"	5UG (U-BOLT GUIDE)	5UG-02
9	2	2"	5FS1 (FIELD SUPPORT)	5FS1S1
10	1	2"	5US (U-BOLT SUPPORT)	5US-02

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By dac54131 at Apr 10, 2026

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0	ISSUE FOR CONSTRUCTION	09-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	RV
PIPE SCHEDULE:	315
INSULATION CODE:	-
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS RV (315) 063304

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000063	LP1Y-RV-063304-02	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 09-MAR-2026

Tim Trice

Planned Controlled Document

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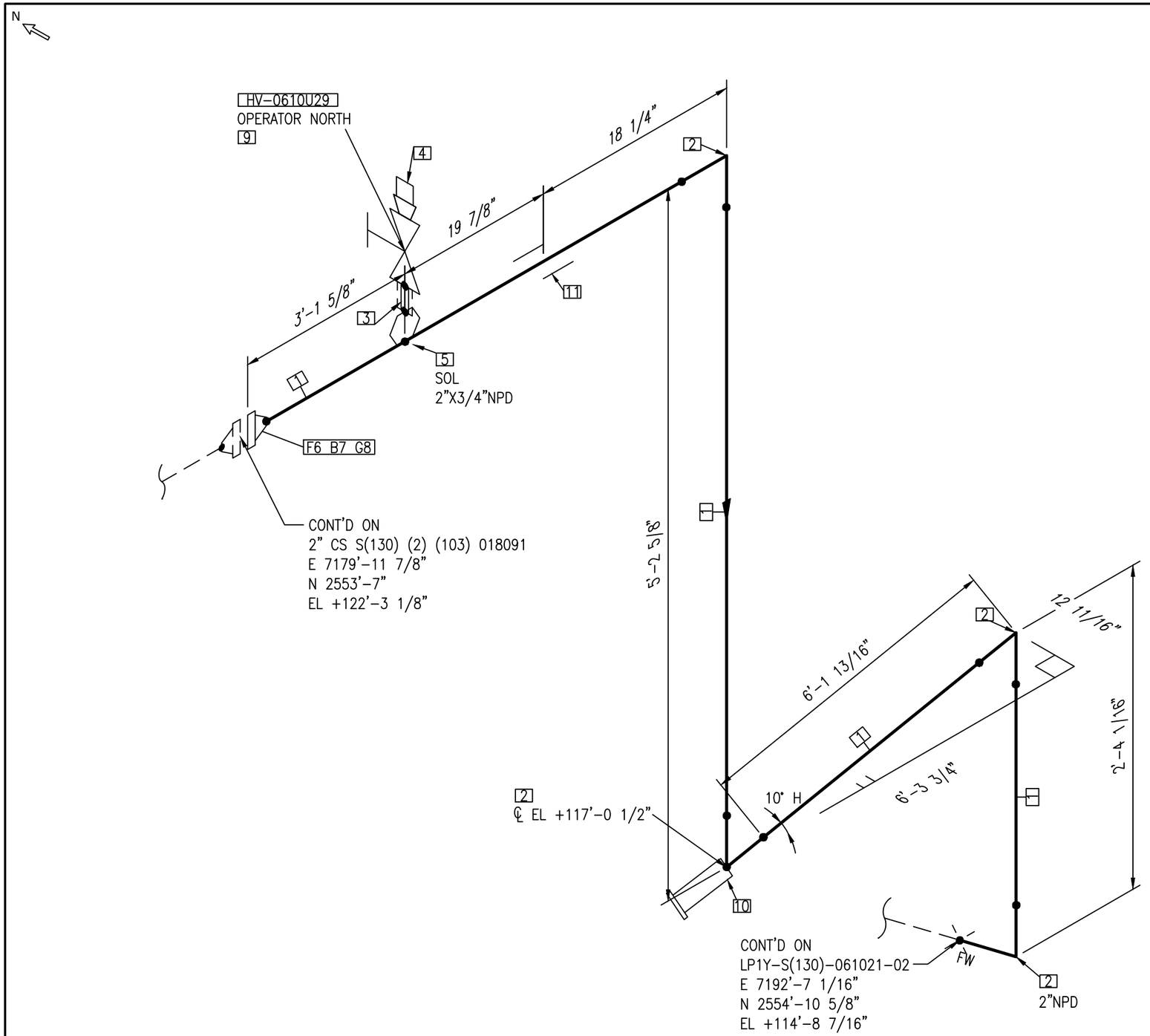

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	18'-4"	2"	PIPE SCH 40 ERW STL A53-B	5356648
2	4	2"	ELL 90 DEG LR SCH 40 STL A234 WPB	5374060
3	1	3/4"	NIPPLE SCH 80 STL A53-B PBE 6" LONG	5539043L1
4	1	3/4"	PLUG ROUND HEAD SCRD STL A105	5490665
5	1	2"X3/4"	SOCKOLET 3000# STL A105	5532671
6	1	2"	FLG WN 150# RF STL 40 BORE A105	5554401
7	4	5/8"X2 1/2"	STUD-BOLT A193 GR B7 W/A194 GR 2H NUTS	5675648
8	1	2"	GASKET 150# 304SS SP WND GRAPHITE IR	5672120L
9	1	3/4"	GATE 800# SW/SCRD STL, F6/STEL SEATS	5115273
10	1	2"	5DS1 (DUMMY SUPPORT) L=1'-6"	5DS1-02
11	1	2"	5CH (HOT SERVICE INSULATION CRADLE)	5CH-02-15

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

GENERAL NOTE:
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3. WELDS AND/OR COUPLINGS DEFINING RANDOM PIPE SPOOL LENGTHS ARE FOR REFERENCE ONLY. FINAL WELD OR COUPLING LOCATION IS TO BE DETERMINED BY THE CONTRACTOR.
4. THE MECHANICAL CONTRACTOR TO FIELD VERIFY INTERFACE POINTS BETWEEN EQUIPMENT & PLUMBING FIXTURES IN 3rd FIX (3F SERIES CWP) AND YS YARD STICK BUILT (YS SERIES CWP EQUIPMENT CONNECTIONS) ISOMETRICS PRIOR TO FABRICATION.
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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	S(130)
PIPE SCHEDULE:	2
INSULATION CODE:	103
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	05-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" CS S(130) (2) (103) 061021

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-S(130)-061021-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 05-MAR-2026

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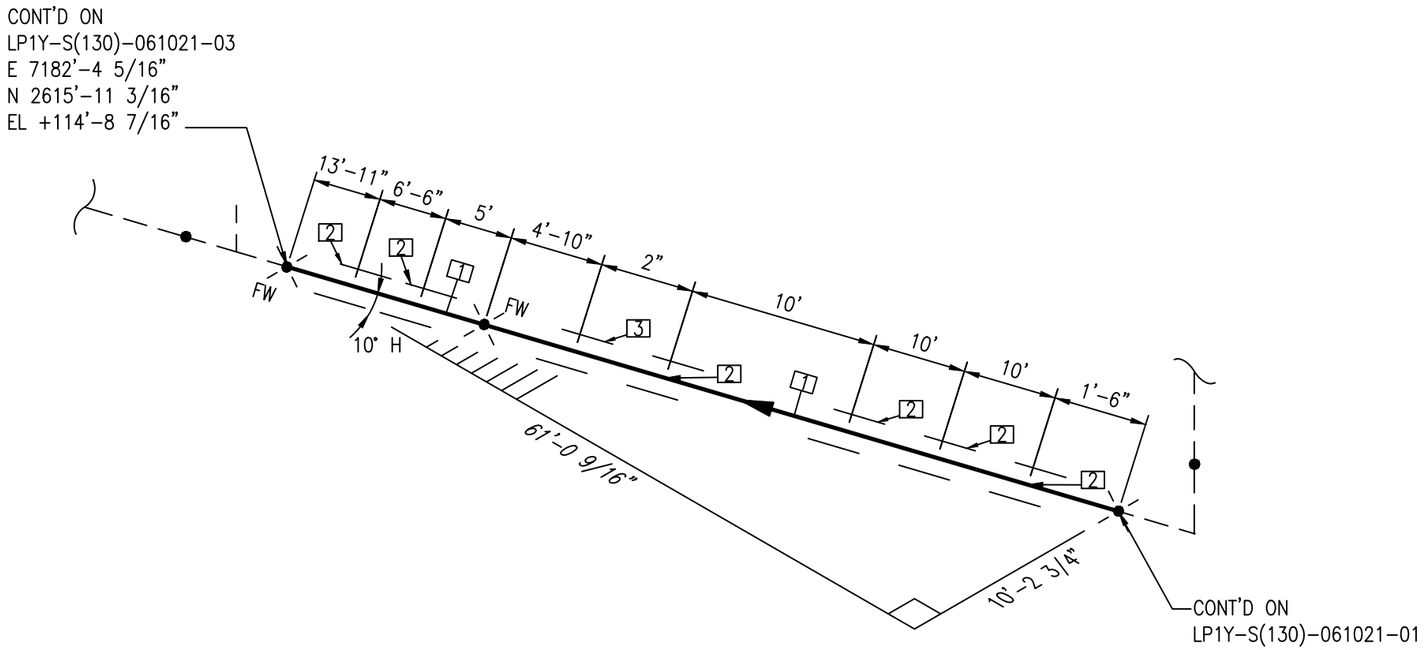
N

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	61'-11"	2"	PIPE SCH 40 ERW STL A53-B	5356648
2	6	2"	5CH (HOT SERVICE INSULATION CRADLE)	5CH-02-15
3	1	2"	5GUSP (INSULATED U-BOLT GUIDE)	5UGSP-02-15



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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
						/	
						/	
0	ISSUE FOR CONSTRUCTION	05-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	S(130)
PIPE SCHEDULE:	2
INSULATION CODE:	103
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

EFDEE Engineering PC
PROJECT NO. B1LG

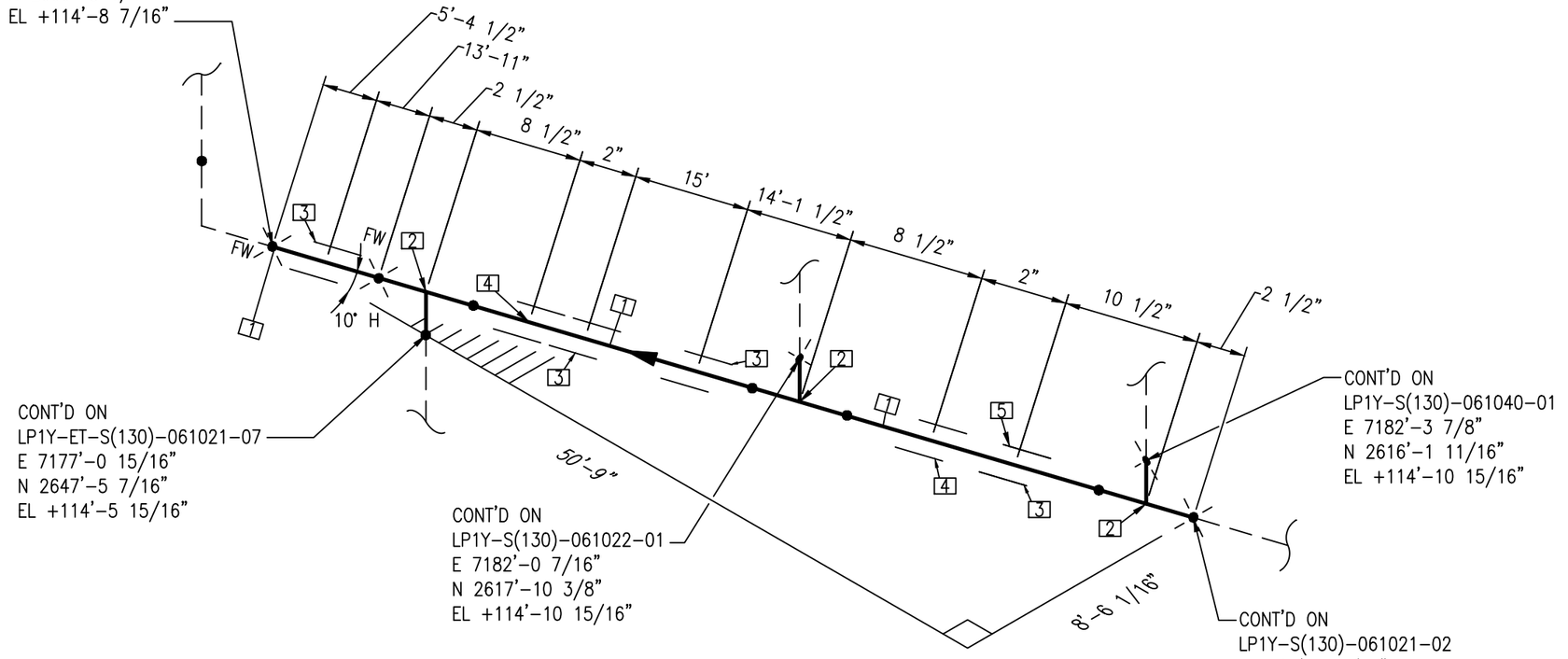
LINE NUMBER
LP1Y 2" CS S(130) (2) (103) 061021

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-S(130)-061021-02	0

NOTICE: THIS DRAWING HAS NOT BEEN PUBLISHED AND IS THE SOLE PROPERTY OF FLUOR. AND IS LENT TO THE BORROWER FOR THEIR CONFIDENTIAL USE ONLY, AND IN CONSIDERATION OF THE LOAN OF THIS DRAWING, THE BORROWER PROMISES AND AGREES TO RETURN IT UPON REQUEST AND AGREES THAT IT WILL NOT BE REPRODUCED, COPIED, LENT, OR OTHERWISE DISPOSED OF DIRECTLY OR INDIRECTLY, NOR USED FOR ANY PURPOSE OTHER THAN THAT FOR WHICH IT IS FURNISHED.

N

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	50'-3"	2"	PIPE SCH 40 ERW STL A53-B	5356648
2	3	2"	TEE SCH 40 STL A234 WPB	5374069
3	4	2"	5CH (HOT SERVICE INSULATION CRADLE)	5CH-02-15
4	2	2"	5GUSP (INSULATED U-BOLT GUIDE)	5UGSP-02-15
5	1	2"	5DA1 (DIRECTIONAL ANCHOR)	5DA1-C



CONT'D ON
LP1Y-S(130)-061021-04
E 7173'-10 1/4"
N 2666'-8 1/4"
EL +114'-8 7/16"

CONT'D ON
LP1Y-ET-S(130)-061021-07
E 7177'-0 15/16"
N 2647'-5 7/16"
EL +114'-5 15/16"

CONT'D ON
LP1Y-S(130)-061022-01
E 7182'-0 7/16"
N 2617'-10 3/8"
EL +114'-10 15/16"

CONT'D ON
LP1Y-S(130)-061040-01
E 7182'-3 7/8"
N 2616'-1 11/16"
EL +114'-10 15/16"

CONT'D ON
LP1Y-S(130)-061021-02
E 7182'-4 5/16"
N 2615'-11 3/16"
EL +114'-8 7/16"

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	05-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	S(130)
PIPE SCHEDULE:	2
INSULATION CODE:	103
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" CS S(130) (2) (103) 061021

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-S(130)-061021-03	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 05-MAR-2026

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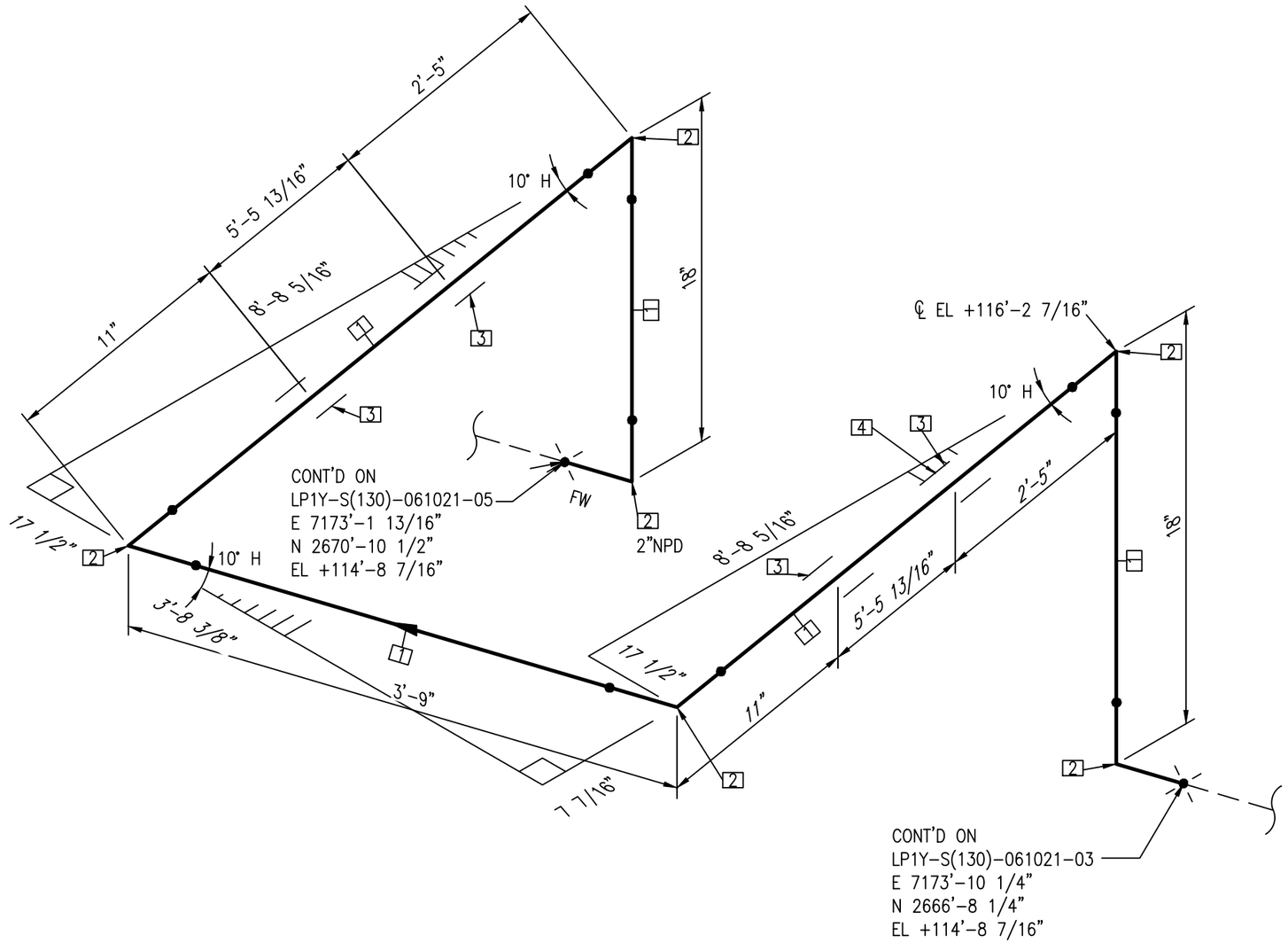
N



BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	21'-11"	2"	PIPE SCH 40 ERW STL A53-B	5356648
2	6	2"	ELL 90 DEG LR SCH 40 STL A234 WPB	5374060
3	4	2"	5CH (HOT SERVICE INSULATION CRADLE)	5CH-02-15
4	1	2"	5FSS3 (TRAPEZE FIELD SUPPORT)	5FSS3-L-4-3

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	05-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	S(130)
PIPE SCHEDULE:	2
INSULATION CODE:	103
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" CS S(130) (2) (103) 061021

REFERENCE P&ID NUMBER
LP1Y-DI-00000061

ISOMETRIC DRAWING NUMBER
LP1Y-S(130)-061021-04

REV
0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 05-MAR-2026

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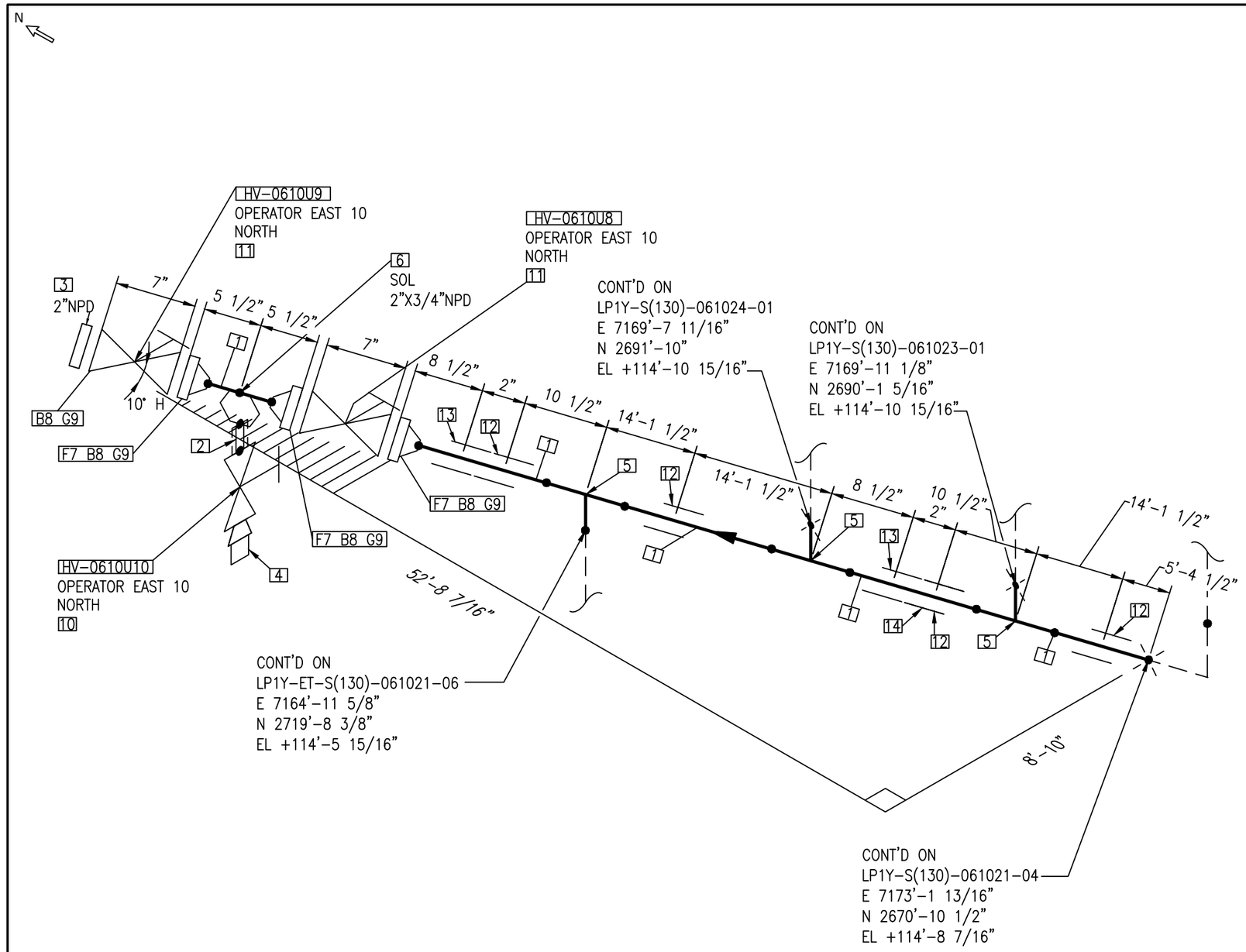

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	50'-4"	2"	PIPE SCH 40 ERW STL A53-B	5356648
2	1	3/4"	NIPPLE SCH 80 STL A53-B PBE 6" LONG	5539043L1
3	1	2"	FLG BLIND 150# RF STL A105	5615476
4	1	3/4"	PLUG ROUND HEAD SCRD STL A105	5490665
5	3	2"	TEE SCH 40 STL A234 WPB	5374069
6	1	2"X3/4"	SOCKOLET 3000# STL A105	5532671
7	3	2"	FLG WN 150# RF STL 40 BORE A105	5554401
8	16	5/8"X2 1/2"	STUD-BOLT A193 GR B7 W/A194 GR 2H NUTS	5675648
9	4	2"	GASKET 150# 304SS SP WND GRAPHITE IR	5672120L
10	1	3/4"	GATE 800# SW/SCRD STL, F6/STEL SEATS	5115273
11	2	2"	GATE 150# RF STL, F6/STEL SEATS	5156140
12	4	2"	5CH (HOT SERVICE INSULATION CRADLE)	5CH-02-15
13	2	2"	5GUSP (INSULATED U-BOLT GUIDE)	5UGSP-02-15
14	1	2"	5DA1 (DIRECTIONAL ANCHOR)	5DA1-C

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	05-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	S(130)
PIPE SCHEDULE:	2
INSULATION CODE:	103
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" CS S(130) (2) (103) 061021

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-S(130)-061021-05	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 05-MAR-2026

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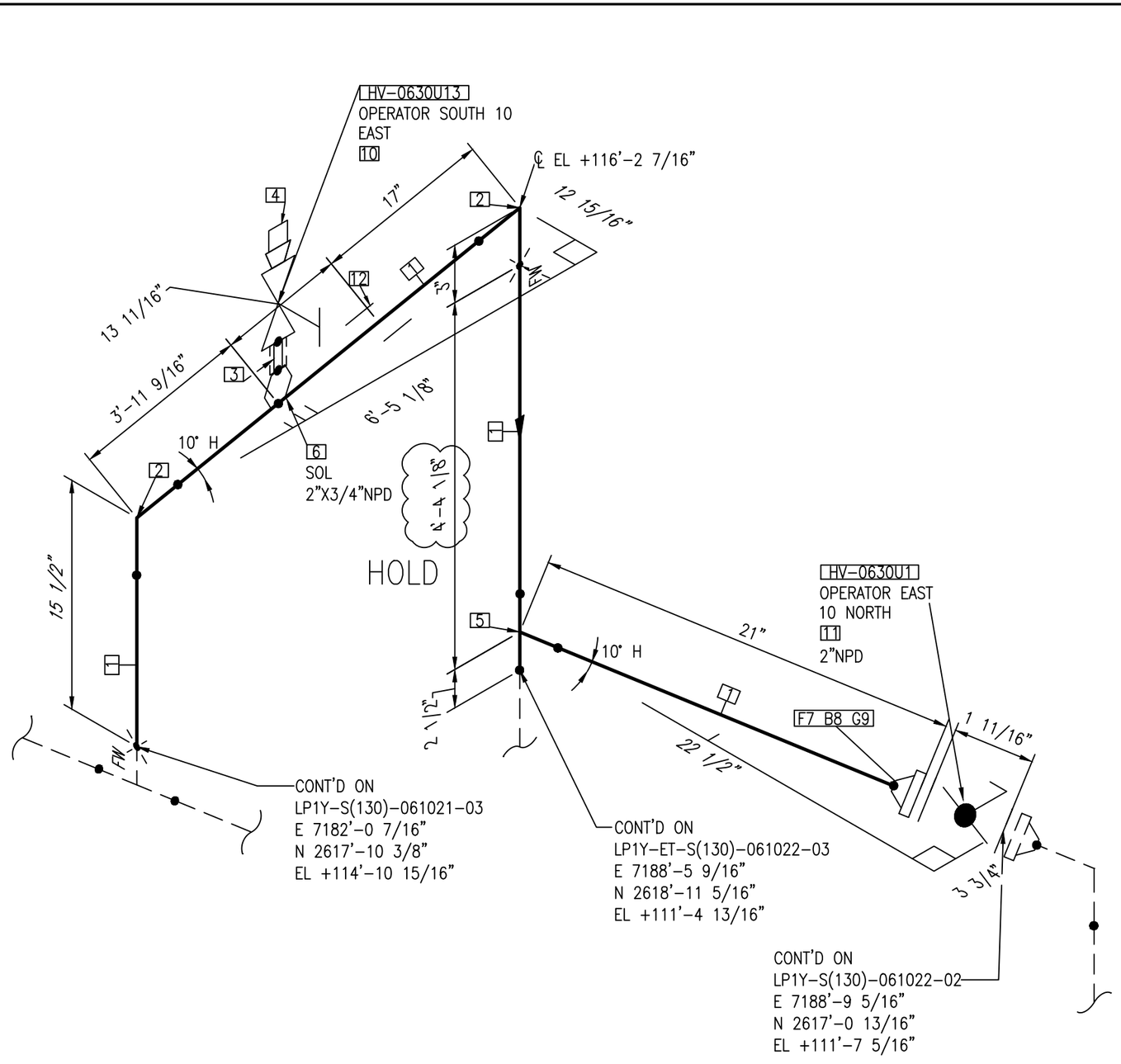

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	12'-7"	2"	PIPE SCH 40 ERW STL A53-B	5356648
2	2	2"	ELL 90 DEG LR SCH 40 STL A234 WPB	5374060
3	1	3/4"	NIPPLE SCH 80 STL A53-B PBE 6" LONG	5539043L1
4	1	3/4"	PLUG ROUND HEAD SCRD STL A105	5490665
5	1	2"	TEE SCH 40 STL A234 WPB	5374069
6	1	2"X3/4"	SOCKOLET 3000# STL A105	5532671
7	1	2"	FLG WN 150# RF STL 40 BORE A105	5554401
8	4	5/8"X2 1/2"	STUD-BOLT A193 GR B7 W/A194 GR 2H NUTS	5675648
9	1	2"	GASKET 150# 304SS SP WND GRAPHITE IR	5672120L
10	1	3/4"	GATE 800# SW/SCRD STL, F6/STEL SEATS	5115273
11	1	2"	BUTTERFLY TRIPLE 150# LUG STL GO ANSI	5308832L1
12	1	2"	5CH (HOT SERVICE INSULATION CRADLE)	5CH-02-15

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	05-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" CS S(130) (2) (103) 061022

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000063	LP1Y-S(130)-061022-01	0

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	S(130)
PIPE SCHEDULE:	2
INSULATION CODE:	103
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 05-MAR-2026

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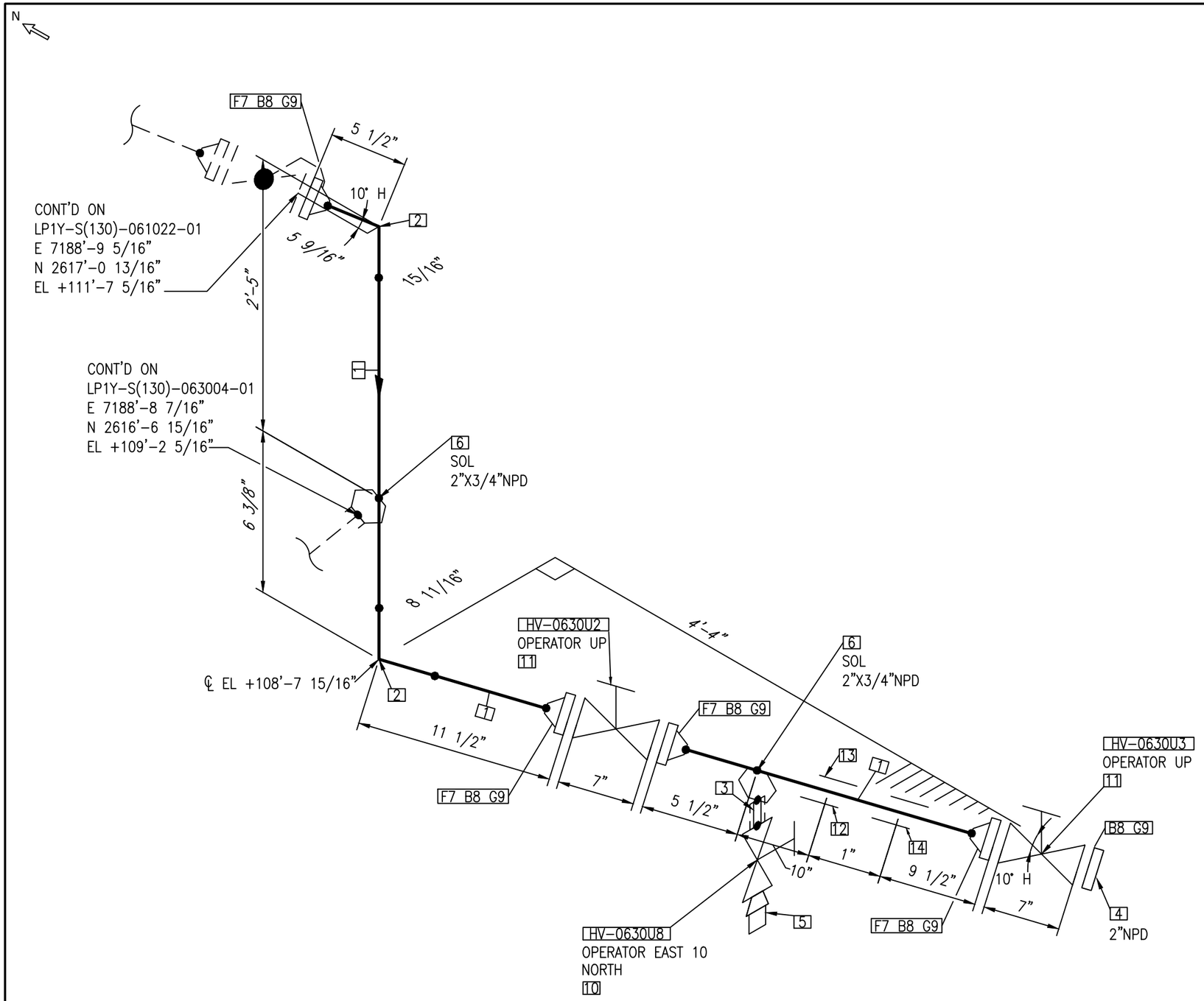

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	4'-9"	2"	PIPE SCH 40 ERW STL A53-B	5356648
2	2	2"	ELL 90 DEG LR SCH 40 STL A234 WPB	5374060
3	1	3/4"	NIPPLE SCH 80 STL A53-B PBE 6" LONG	5539043L1
4	1	2"	FLG BLIND 150# RF STL A105	5615476
5	1	3/4"	PLUG ROUND HEAD SCRD STL A105	5490665
6	2	2"X3/4"	SOCKOLET 3000# STL A105	5532671
7	4	2"	FLG WN 150# RF STL 40 BORE A105	5554401
8	20	5/8"X2 1/2"	STUD-BOLT A193 GR B7 W/A194 GR 2H NUTS	5675648
9	5	2"	GASKET 150# 304SS SP WND GRAPHITE IR	5672120L
10	1	3/4"	GATE 800# SW/SCRD STL, F6/STEL SEATS	5115273
11	2	2"	GATE 150# RF STL, F6/STEL SEATS	5156140
12	1	2"	5CH (HOT SERVICE INSULATION CRADLE)	5CH-02-15
13	1	2"	5FSS5 (FIELD BASE SUPPORT)	5FSS5-L-1
14	1	2"	5GUSP (INSULATED U-BOLT GUIDE)	5UGSP-02-15

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	S(130)
PIPE SCHEDULE:	2
INSULATION CODE:	103
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	05-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" CS S(130) (2) (103) 061022

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000063	LP1Y-S(130)-061022-02	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 05-MAR-2026

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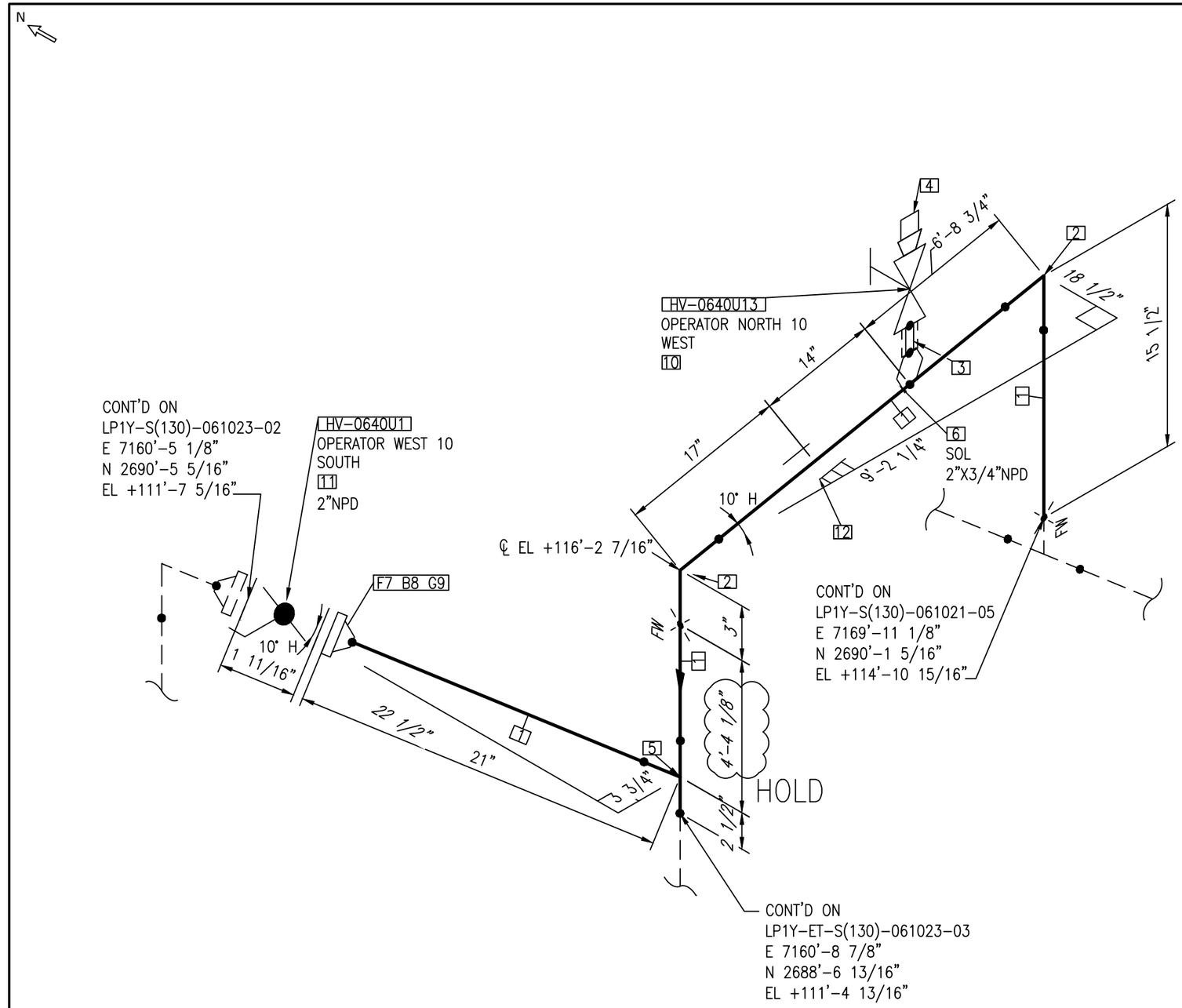

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	15'-4"	2"	PIPE SCH 40 ERW STL A53-B	5356648
2	2	2"	ELL 90 DEG LR SCH 40 STL A234 WPB	5374060
3	1	3/4"	NIPPLE SCH 80 STL A53-B PBE 6" LONG	5539043L1
4	1	3/4"	PLUG ROUND HEAD SCRD STL A105	5490665
5	1	2"	TEE SCH 40 STL A234 WPB	5374069
6	1	2"X3/4"	SOCKOLET 3000# STL A105	5532671
7	1	2"	FLG WN 150# RF STL 40 BORE A105	5554401
8	4	5/8"X2 1/2"	STUD-BOLT A193 GR B7 W/A194 GR 2H NUTS	5675648
9	1	2"	GASKET 150# 304SS SP WND GRAPHITE IR	5672120L
10	1	3/4"	GATE 800# SW/SCRD STL, F6/STEL SEATS	5115273
11	1	2"	BUTTERFLY TRIPLE 150# LUG STL GO ANSI	5308832L1
12	1	2"	5CH (HOT SERVICE INSULATION CRADLE)	5CH-02-15

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	05-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	S(130)
PIPE SCHEDULE:	2
INSULATION CODE:	103
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" CS S(130) (2) (103) 061023

REFERENCE P&ID NUMBER
LP1Y-DI-00000064

ISOMETRIC DRAWING NUMBER
LP1Y-S(130)-061023-01

REV
0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 05-MAR-2026

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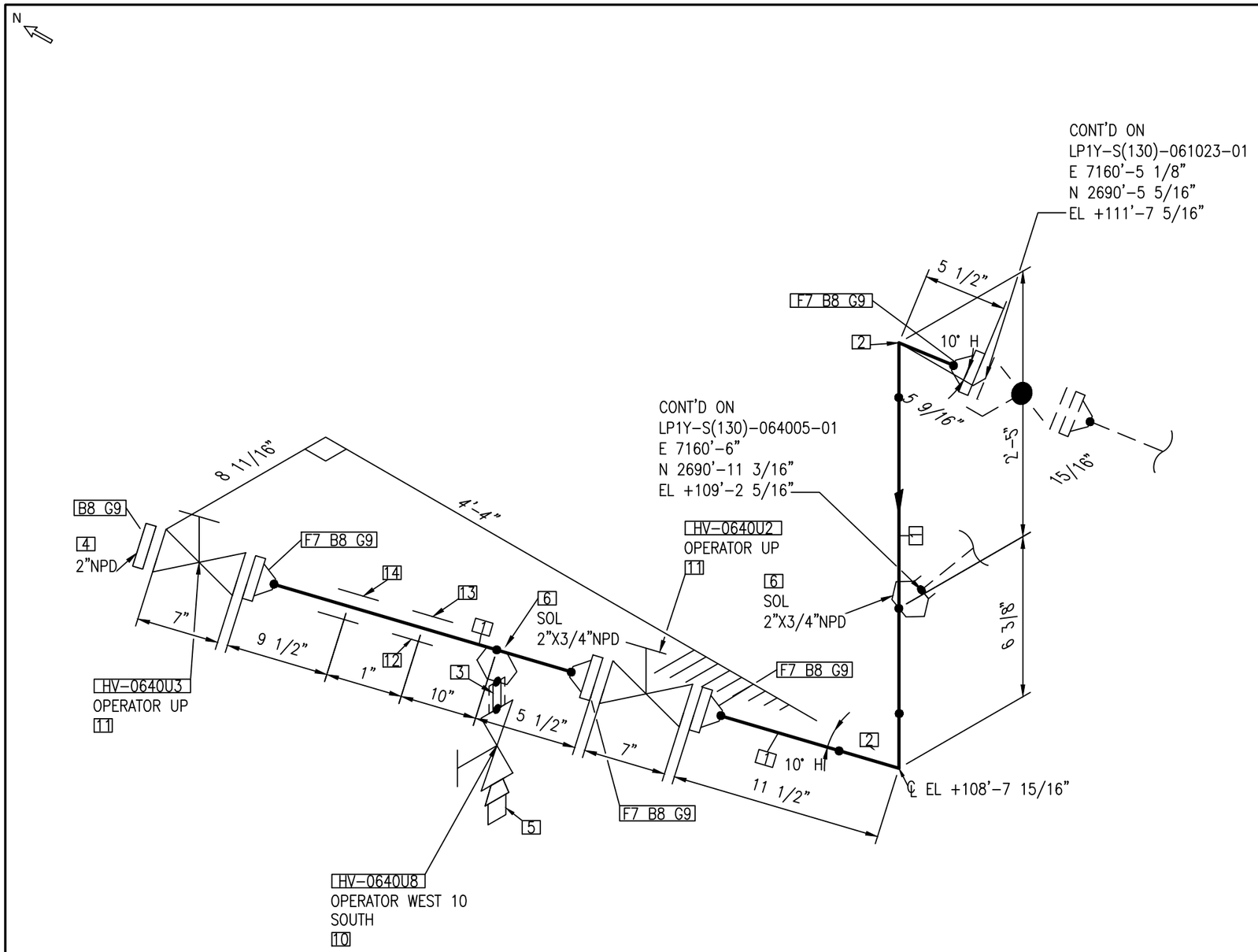

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ID	QTY	ND	DESCRIPTION	CMDTY CODE
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9	5	2"	GASKET 150# 304SS SP WND GRAPHITE IR	5672120L
10	1	3/4"	GATE 800# SW/SCRD STL, F6/STEL SEATS	5115273
11	2	2"	GATE 150# RF STL, F6/STEL SEATS	5156140
12	1	2"	5CH (HOT SERVICE INSULATION CRADLE)	5CH-02-15
13	1	2"	5FSS5 (FIELD BASE SUPPORT)	5FSS5-L-1
14	1	2"	5GUSP (INSULATED U-BOLT GUIDE)	5UGSP-02-15

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	S(130)
PIPE SCHEDULE:	2
INSULATION CODE:	103
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	05-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" CS S(130) (2) (103) 061023

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000064	LP1Y-S(130)-061023-02	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 05-MAR-2026

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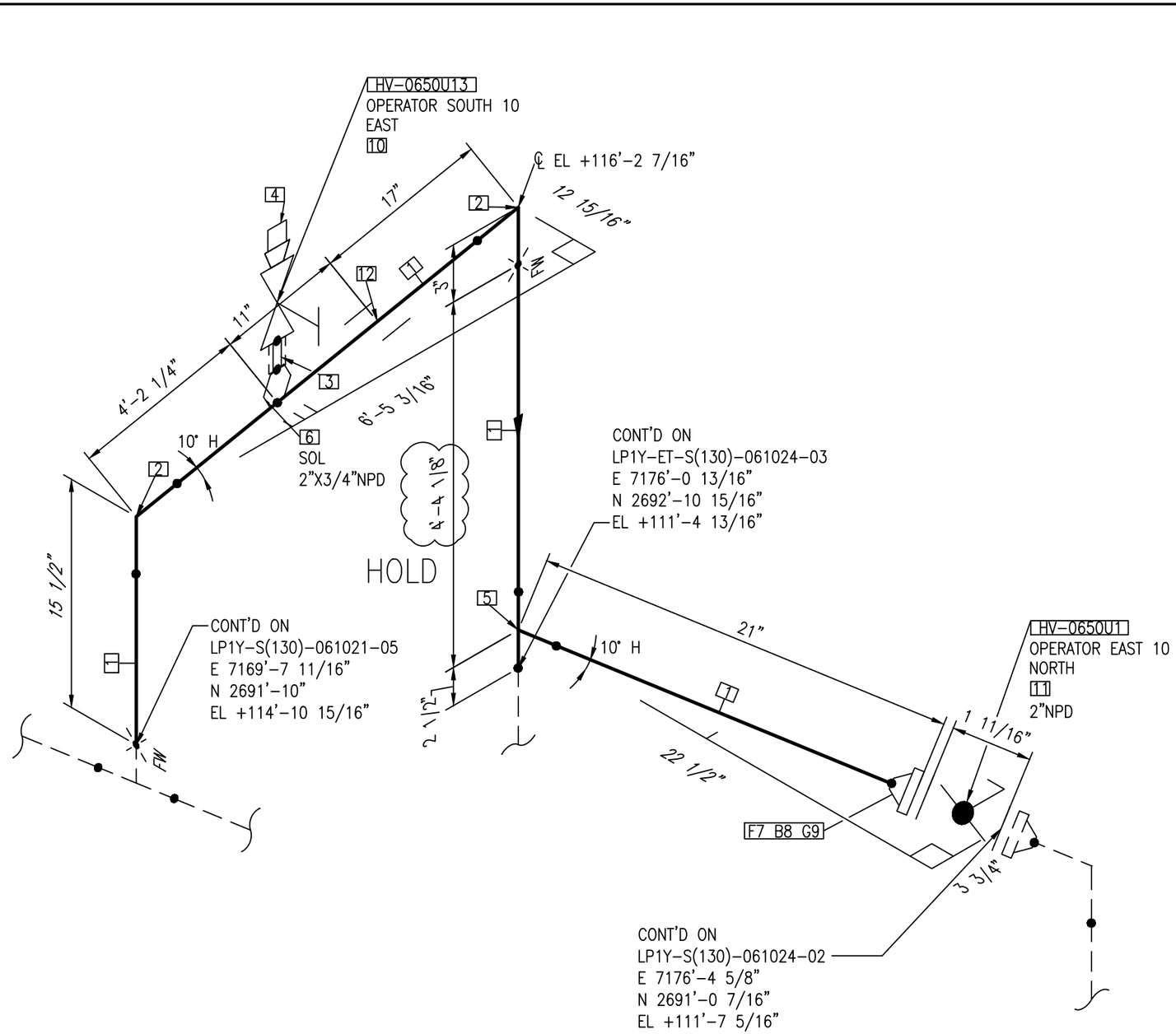

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
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3	1	3/4"	NIPPLE SCH 80 STL A53-B PBE 6" LONG	5539043L1
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11	1	2"	BUTTERFLY TRIPLE 150# LUG STL GO ANSI	5308832L1
12	1	2"	5CH (HOT SERVICE INSULATION CRADLE)	5CH-02-15

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	S(130)
PIPE SCHEDULE:	2
INSULATION CODE:	103
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	05-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" CS S(130) (2) (103) 061024

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000065	LP1Y-S(130)-061024-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 05-MAR-2026

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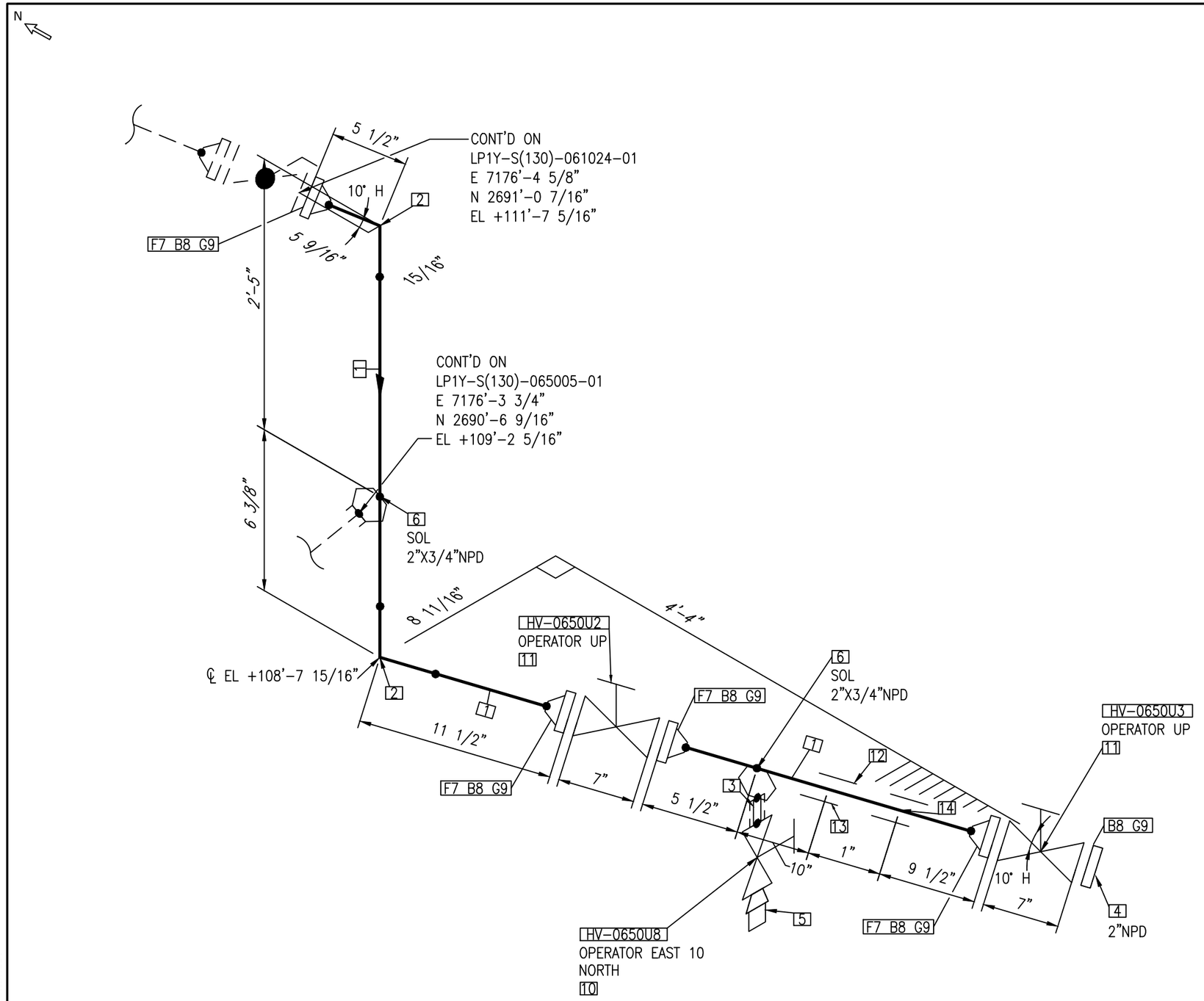

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12	1	2"	5CH (HOT SERVICE INSULATION CRADLE)	5CH-02-15
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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	S(130)
PIPE SCHEDULE:	2
INSULATION CODE:	103
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	05-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" CS S(130) (2) (103) 061024

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000065	LP1Y-S(130)-061024-02	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 05-MAR-2026

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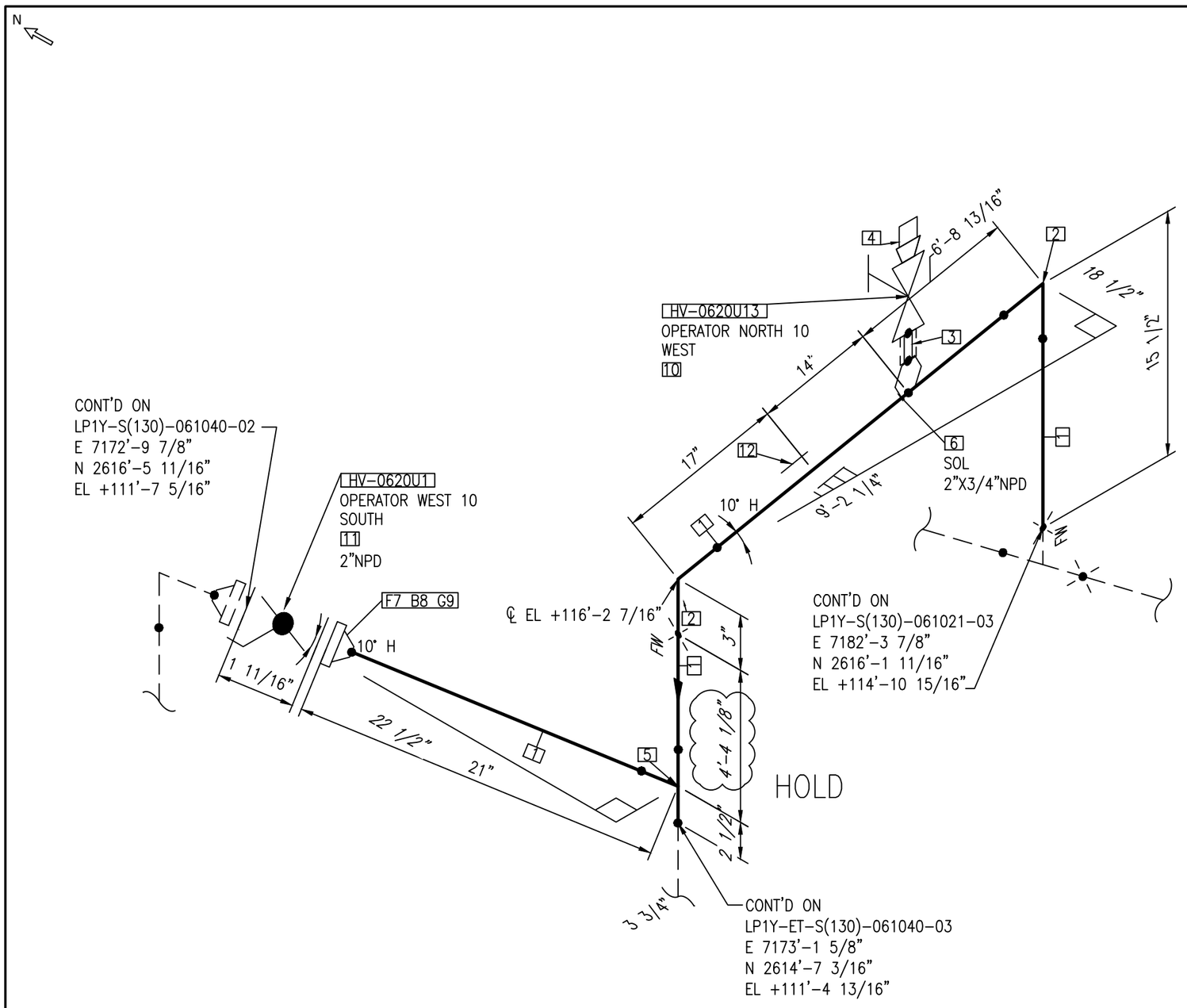

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	15'-4"	2"	PIPE SCH 40 ERW STL A53-B	5356648
2	2	2"	ELL 90 DEG LR SCH 40 STL A234 WPB	5374060
3	1	3/4"	NIPPLE SCH 80 STL A53-B PBE 6" LONG	5539043L1
4	1	3/4"	PLUG ROUND HEAD SCRD STL A105	5490665
5	1	2"	TEE SCH 40 STL A234 WPB	5374069
6	1	2"X3/4"	SOCKOLET 3000# STL A105	5532671
7	1	2"	FLG WN 150# RF STL 40 BORE A105	5554401
8	4	5/8"X2 1/2"	STUD-BOLT A193 GR B7 W/A194 GR 2H NUTS	5675648
9	1	2"	GASKET 150# 304SS SP WND GRAPHITE IR	5672120L
10	1	3/4"	GATE 800# SW/SCRD STL, F6/STEL SEATS	5115273
11	1	2"	BUTTERFLY TRIPLE 150# LUG STL GO ANSI	5308832L1
12	1	2"	5CH (HOT SERVICE INSULATION CRADLE)	5CH-02-15

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	05-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	S(130)
PIPE SCHEDULE:	2
INSULATION CODE:	103
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" CS S(130) (2) (103) 061040

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000062	LP1Y-S(130)-061040-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 05-MAR-2026

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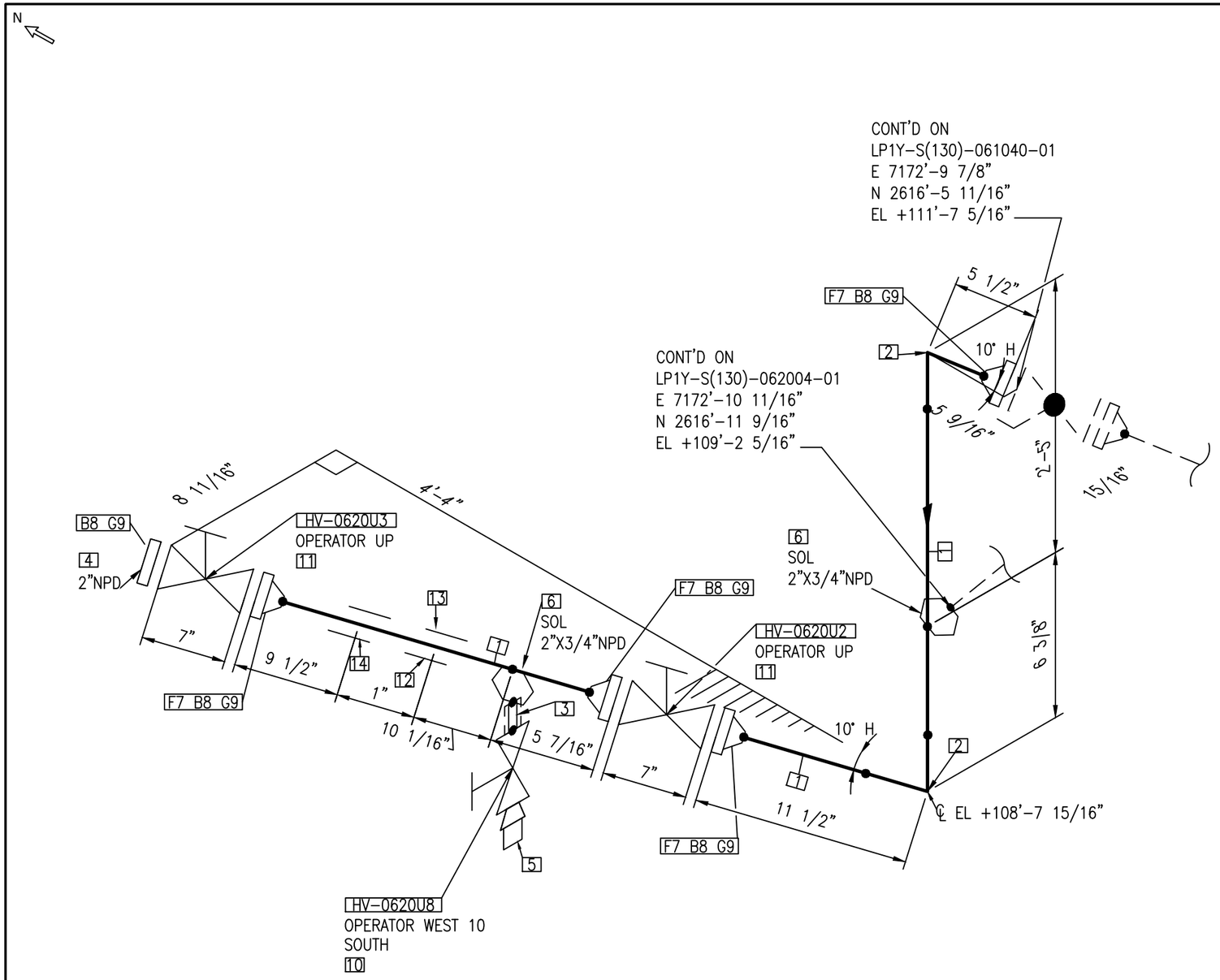

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
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3	1	3/4"	NIPPLE SCH 80 STL A53-B PBE 6" LONG	5539043L1
4	1	2"	FLG BLIND 150# RF STL A105	5615476
5	1	3/4"	PLUG ROUND HEAD SCRD STL A105	5490665
6	2	2"X3/4"	SOCKOLET 3000# STL A105	5532671
7	4	2"	FLG WN 150# RF STL 40 BORE A105	5554401
8	20	5/8"X2 1/2"	STUD-BOLT A193 GR B7 W/A194 GR 2H NUTS	5675648
9	5	2"	GASKET 150# 304SS SP WND GRAPHITE IR	5672120L
10	1	3/4"	GATE 800# SW/SCRD STL, F6/STEL SEATS	5115273
11	2	2"	GATE 150# RF STL, F6/STEL SEATS	5156140
12	1	2"	5CH (HOT SERVICE INSULATION CRADLE)	5CH-02-15
13	1	2"	5FSS5 (FIELD BASE SUPPORT)	5FSS5-L-1
14	1	2"	5GUSP (INSULATED U-BOLT GUIDE)	5UGSP-02-15

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By dac54131 at Apr 10, 2026

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0	ISSUE FOR CONSTRUCTION	05-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW
REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC

NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	S(130)
PIPE SCHEDULE:	2
INSULATION CODE:	103
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" CS S(130) (2) (103) 061040

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000062	LP1Y-S(130)-061040-02	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 05-MAR-2026

Tim Trice

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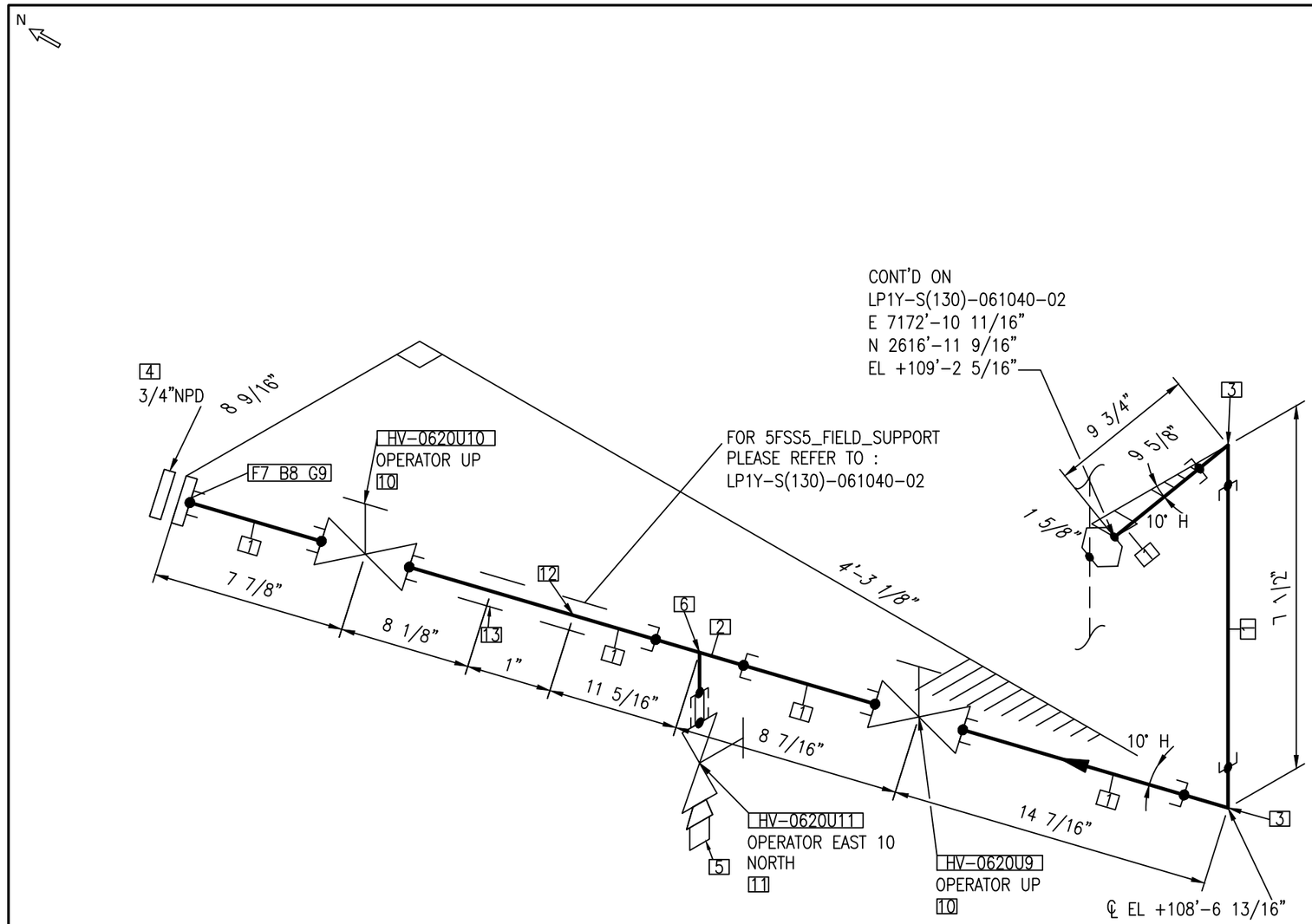

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	4'-10"	3/4"	PIPE SCH 80 ERW STL A53-B	5355898L
2	1	3/4"	NIPPLE SCH 80 STL A53-B PBE 6" LONG	5539043L1
3	2	3/4"	ELL 90 DEG 3000# SW STL A105	5478105
4	1	3/4"	FLG BLIND 150# RF STL A105	5615476
5	1	3/4"	PLUG ROUND HEAD SCRD STL A105	5490665
6	1	3/4"	TEE 3000# SW STL A105	5478107
7	1	3/4"	FLG SW 150# RF STL 80 BORE A105	5601737
8	4	1/2"X2 1/4"	STUD-BOLT A193 GR B7 W/A194 GR 2H NUTS	5675648
9	1	3/4"	GASKET 150# 304SS SP WND GRAPHITE IR	5672120L
10	2	3/4"	GATE 800# SW STL, F6/STEL SEATS	5061780L0
11	1	3/4"	GATE 800# SW/SCRD STL, F6/STEL SEATS	5115273
12	1	3/4"	5C (INSULATION CRADLE)	5C-75-15
13	1	3/4"	5GUSP (INSULATED U-BOLT GUIDE)	5UGSP-75-15

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	05-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	S(130)
PIPE SCHEDULE:	2
INSULATION CODE:	103
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3/4" CS S(130) (2) (103) 062004

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000062	LP1Y-S(130)-062004-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 05-MAR-2026

Tim Trice

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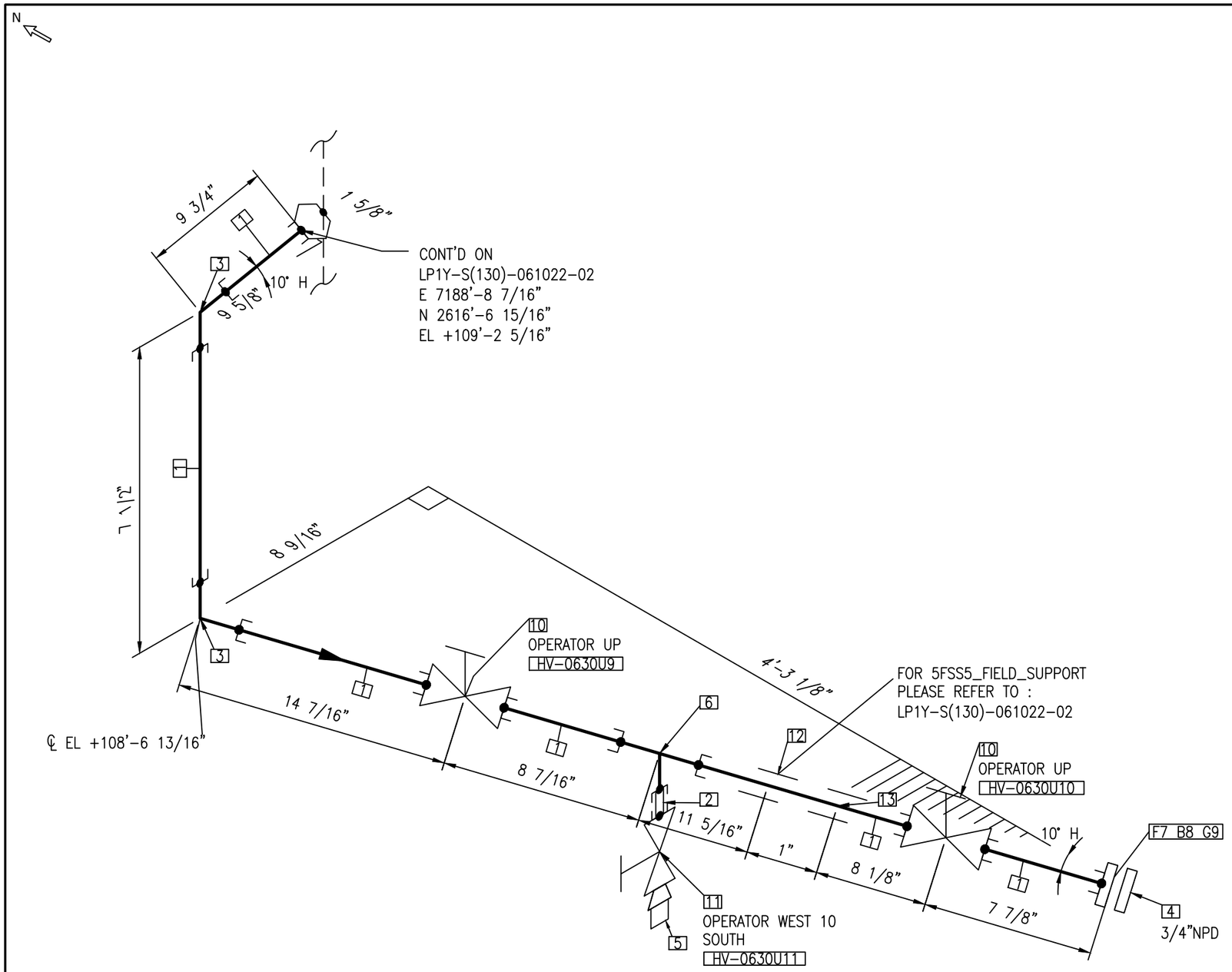

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	4'-9"	3/4"	PIPE SCH 80 ERW STL A53-B	5355898L
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3	2	3/4"	ELL 90 DEG 3000# SW STL A105	5478105
4	1	3/4"	FLG BLIND 150# RF STL A105	5615476
5	1	3/4"	PLUG ROUND HEAD SCRD STL A105	5490665
6	1	3/4"	TEE 3000# SW STL A105	5478107
7	1	3/4"	FLG SW 150# RF STL 80 BORE A105	5601737
8	4	1/2"X2 1/4"	STUD-BOLT A193 GR B7 W/A194 GR 2H NUTS	5675648
9	1	3/4"	GASKET 150# 304SS SP WND GRAPHITE IR	5672120L
10	2	3/4"	GATE 800# SW STL, F6/STEL SEATS	5061780L0
11	1	3/4"	GATE 800# SW/SCRD STL, F6/STEL SEATS	5115273
12	1	3/4"	5C (INSULATION CRADLE)	5C-75-15
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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	05-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	S(130)
PIPE SCHEDULE:	2
INSULATION CODE:	103
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3/4" CS S(130) (2) (103) 063004

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000063	LP1Y-S(130)-063004-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 05-MAR-2026

Tim Trice

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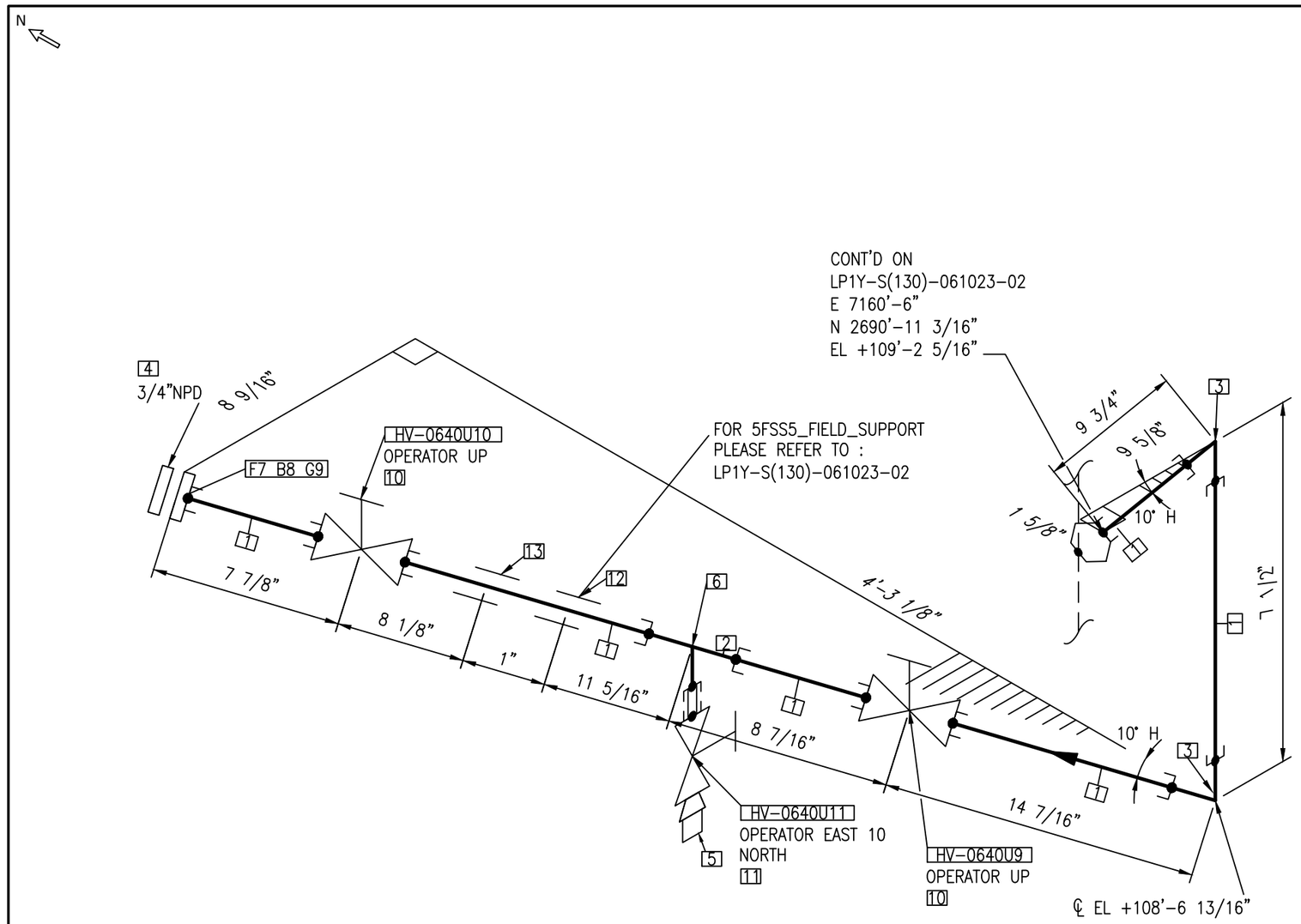

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
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2	1	3/4"	NIPPLE SCH 80 STL A53-B PBE 6" LONG	5539043L1
3	2	3/4"	ELL 90 DEG 3000# SW STL A105	5478105
4	1	3/4"	FLG BLIND 150# RF STL A105	5615476
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6	1	3/4"	TEE 3000# SW STL A105	5478107
7	1	3/4"	FLG SW 150# RF STL 80 BORE A105	5601737
8	4	1/2"X2 1/4"	STUD-BOLT A193 GR B7 W/A194 GR 2H NUTS	5675648
9	1	3/4"	GASKET 150# 304SS SP WND GRAPHITE IR	5672120L
10	2	3/4"	GATE 800# SW STL, F6/STEL SEATS	5061780L0
11	1	3/4"	GATE 800# SW/SCRD STL, F6/STEL SEATS	5115273
12	1	3/4"	5C (INSULATION CRADLE)	5C-75-15
13	1	3/4"	5GUSP (INSULATED U-BOLT GUIDE)	5UGSP-75-15

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
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9. THE BILL OF MATERIAL IS FOR REFERENCE ONLY.
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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	S(130)
PIPE SCHEDULE:	2
INSULATION CODE:	103
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3/4" CS S(130) (2) (103) 064005

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000064	LP1Y-S(130)-064005-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 05-MAR-2026

Tim Trice

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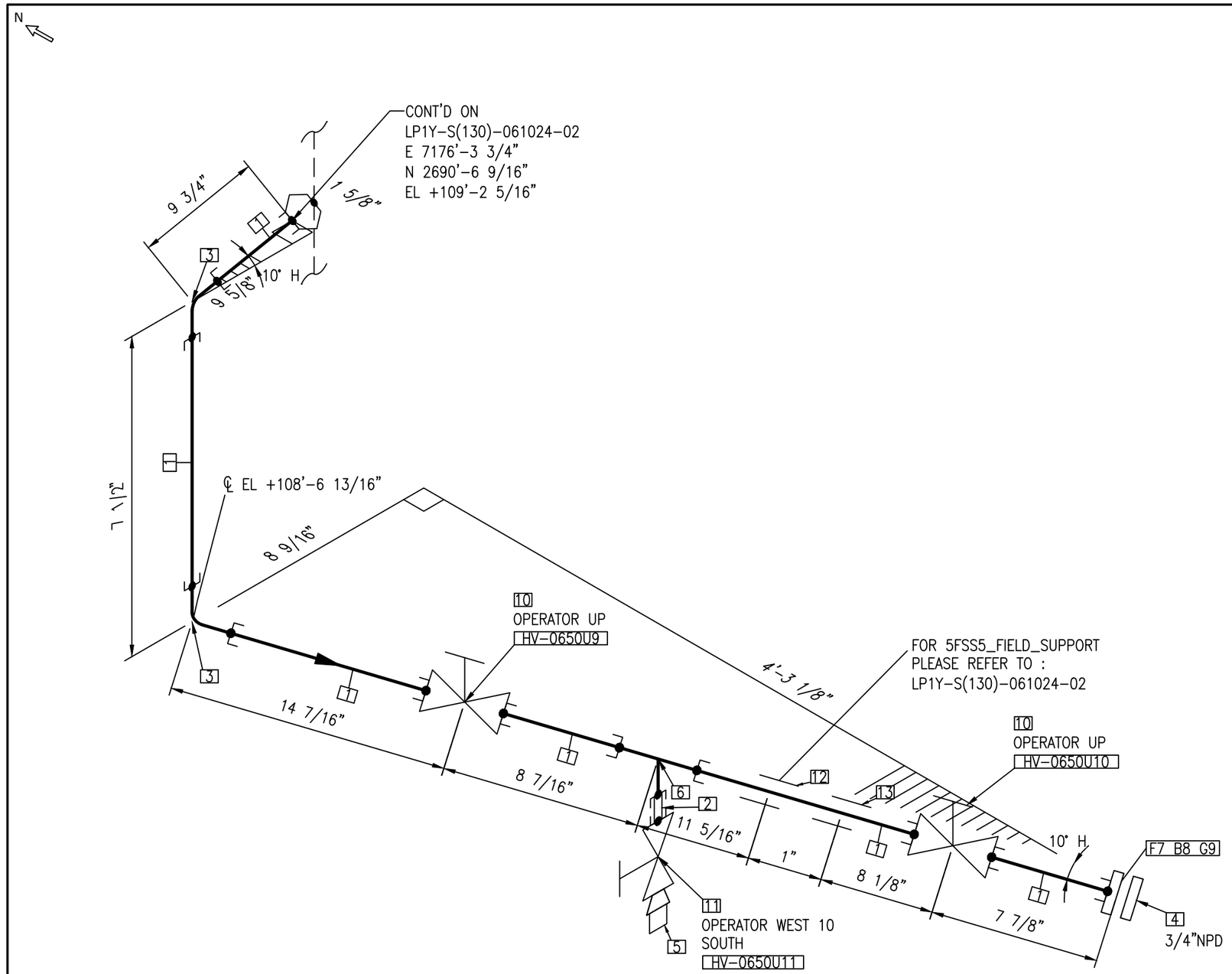

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	4'-10"	3/4"	PIPE SCH 80 ERW STL A53-B	5355898L
2	1	3/4"	NIPPLE SCH 80 STL A53-B PBE 6" LONG	5539043L1
3	2	3/4"	ELL 90 DEG 3000# SW STL A105	5478105
4	1	3/4"	FLG BLIND 150# RF STL A105	5615476
5	1	3/4"	PLUG ROUND HEAD SCRD STL A105	5490665
6	1	3/4"	TEE 3000# SW STL A105	5478107
7	1	3/4"	FLG SW 150# RF STL 80 BORE A105	5601737
8	4	1/2"X2 1/4"	STUD-BOLT A193 GR B7 W/A194 GR 2H NUTS	5675648
9	1	3/4"	GASKET 150# 304SS SP WND GRAPHITE IR	5672120L
10	2	3/4"	GATE 800# SW STL, F6/STEL SEATS	5061780L0
11	1	3/4"	GATE 800# SW/SCRD STL, F6/STEL SEATS	5115273
12	1	3/4"	5C (INSULATION CRADLE)	5C-75-15
13	1	3/4"	5GUSP (INSULATED U-BOLT GUIDE)	5UGSP-75-15

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

GENERAL NOTE:
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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	S(130)
PIPE SCHEDULE:	2
INSULATION CODE:	103
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	05-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3/4" CS S(130) (2) (103) 065005

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000065	LP1Y-S(130)-065005-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 05-MAR-2026

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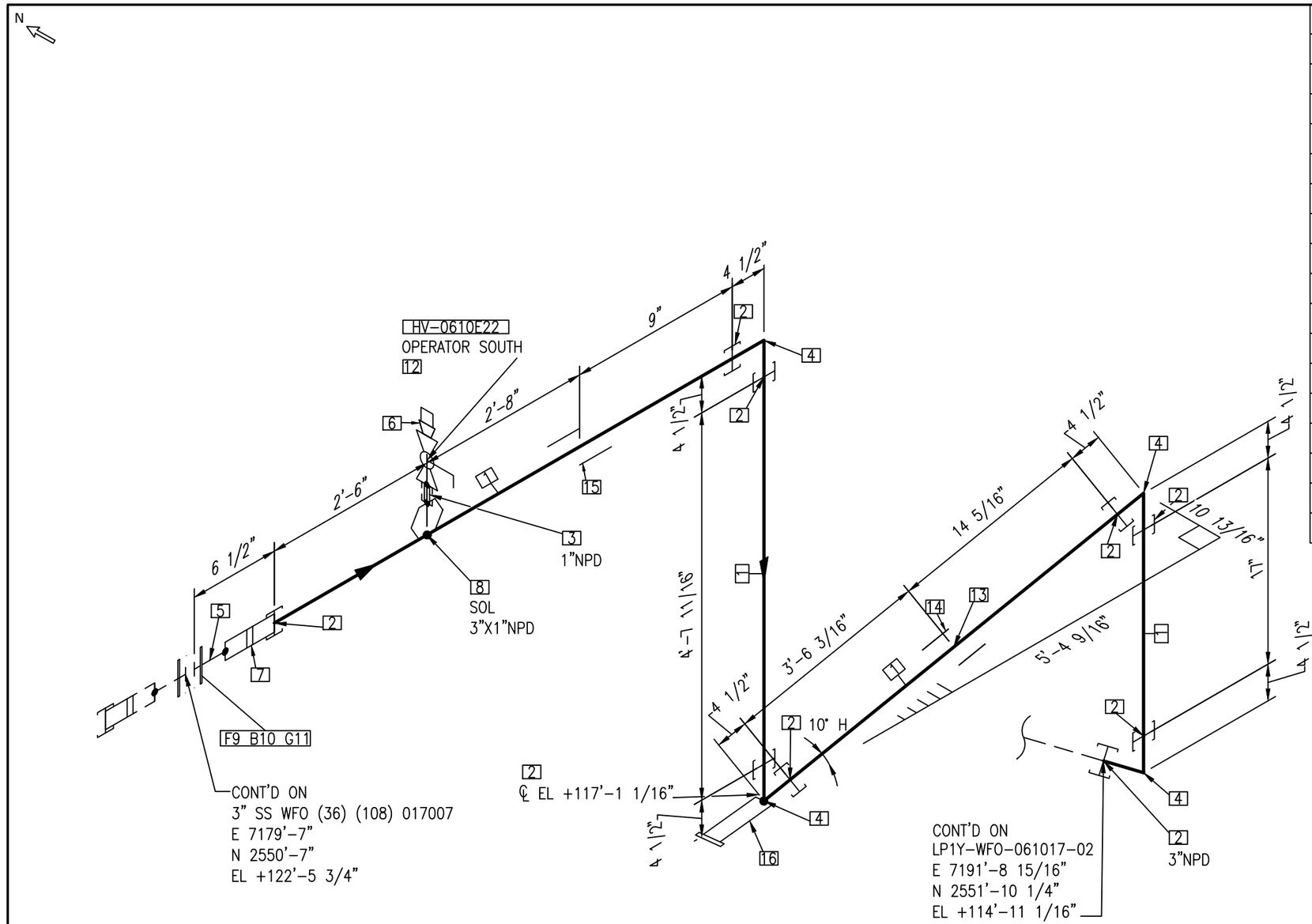

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	16'-9"	3"	PIPE SCH 10S ERW 316/316L SS A312 NSF	5365189L
2	9	3"	COUPLING RIGID GRV DI GRADE P GASKET PRE-AS NSF	5714906L0
3	1	1"	NIPPLE SCH 10S 316/316L SS PBE 6" LONG	5540493
4	4	3"	ELL 90 DEG GRV 316L SCH 10S NSF	5484951S1
5	1	3"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
6	1	1"	PLUG ROUND HEAD SCRD 316/316L SS	5498383
7	1	3"	ADAPTER NIPPLE GRV X BE 316L SCH 10S NSF	5540181
8	1	3"X1"	PIPET 10S X 3000# SW 316/316L SS	5537081
9	1	3"	FLG LAP JOINT 150# 316 SS	5599578L
10	4	5/8"X2 3/4"	STUD-BOLT A193 GR B7 W/A194 GR 2H NUTS	5675648
11	1	3"	GASKET 150# SOFT PTFE RING 1/16" THK NSF	5669345
12	1	1"	BALL 1440# CWP SW X FNPT 316 SS, 316 TRIM RP EXTD STEM, HNDL OP	5300985L8
13	1	3"	5CC (COLD SERVICE INSULATION CRADLE)	5CC-03-01
14	1	3"	5HR4 (HANGER ROD)	5HR4-06
15	1	3"	5ISC (COLD SERVICE INSULATION CRADLE AND SHOE)	5ISC-03-01
16	1	3"	5DS1 (DUMMY SUPPORT) L=1'-2"	5DS1-03

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	02-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	WFO
PIPE SCHEDULE:	36
INSULATION CODE:	108
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3" SS WFO (36) (108) 061017

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-WFO-061017-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 02-MAR-2026

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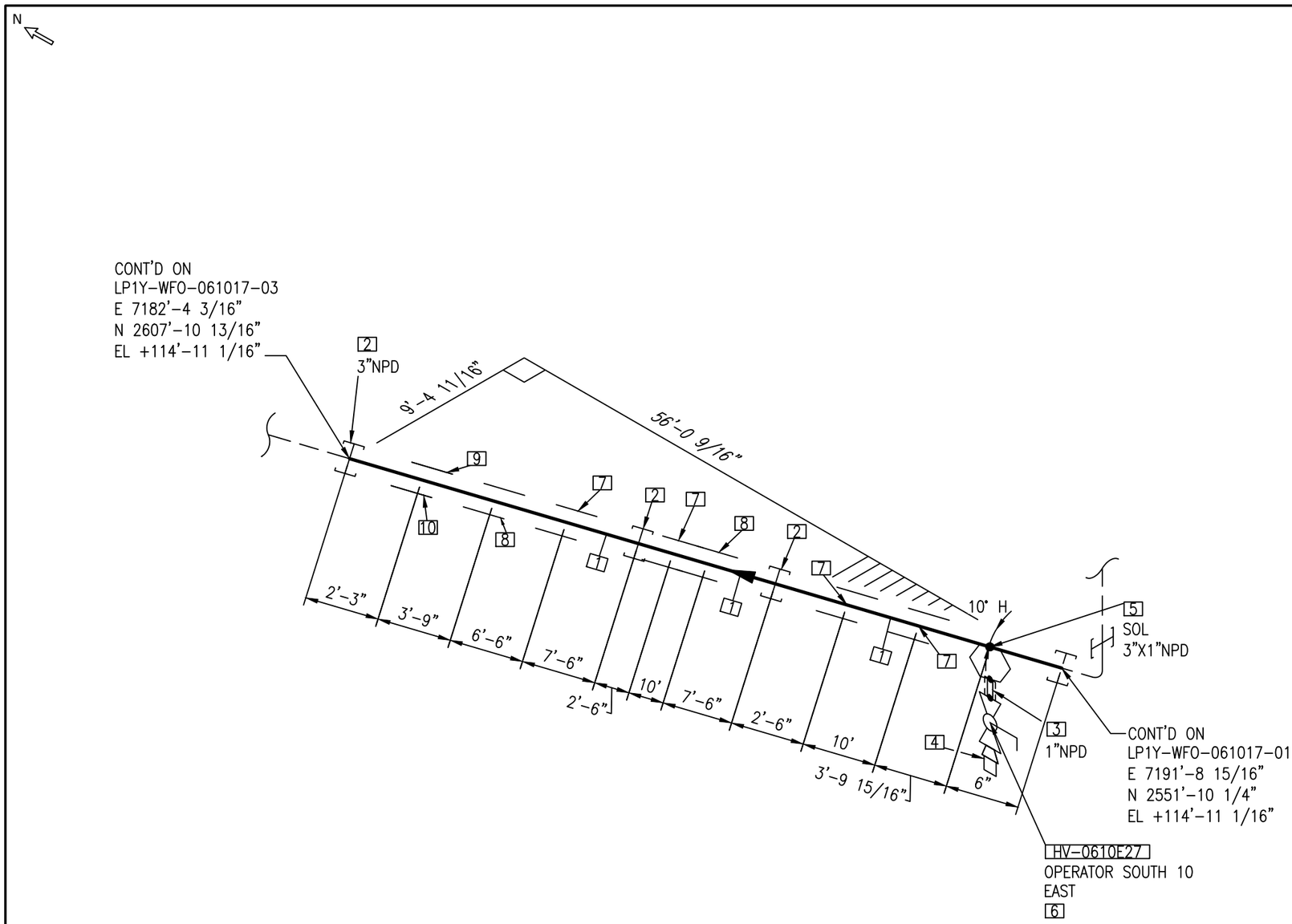

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	56'-10"	3"	PIPE SCH 10S ERW 316/316L SS A312 NSF	5365189L
2	3	3"	COUPLING RIGID GRV DI GRADE P GASKET PRE-AS NSF	5714906L0
3	1	1"	NIPPLE SCH 10S 316/316L SS PBE 6" LONG	5540493
4	1	1"	PLUG ROUND HEAD SCRD 316/316L SS	5498383
5	1	3"X1"	PIPET 10S X 3000# SW 316/316L SS	5537081
6	1	1"	BALL 1440# CWP SW X FNPT 316 SS, 316 TRIM RP EXTD STEM, HNDL OP	5300985L8
7	4	3"	5ISC (COLD SERVICE INSULATION CRADLE AND SHOE)	5ISC-03-01
8	2	3"	5IGC (COLD SERVICE INSULATION CRADLE WITH SHOE AND GUIDE)	5IGC-03-01
9	1	3"	5CC (COLD SERVICE INSULATION CRADLE)	5CC-03-01
10	1	3"	5HR4 (HANGER ROD)	5HR4-06

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	WFO
PIPE SCHEDULE:	36
INSULATION CODE:	108
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	02-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3" SS WFO (36) (108) 061017

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-WFO-061017-02	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 02-MAR-2026

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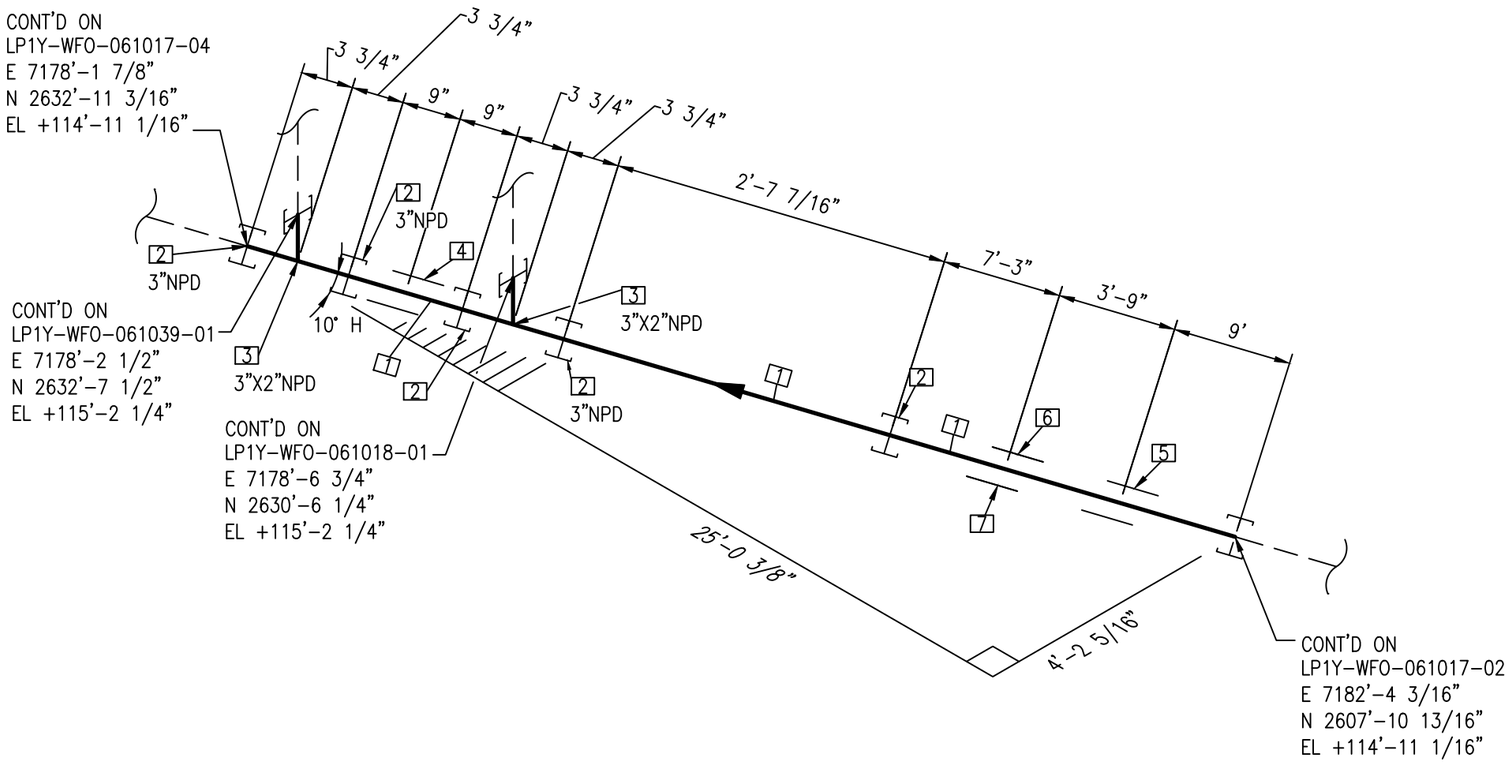
N

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	24'-2"	3"	PIPE SCH 10S ERW 316/316L SS A312 NSF	5365189L
2	5	3"	COUPLING RIGID GRV DI GRADE P GASKET PRE-AS NSF	5714906L0
3	2	3"X2"	TEE RED GRV 316L SCH 10S NSF	5484959S1
4	1	3"	5IGC (COLD SERVICE INSULATION CRADLE WITH SHOE AND GUIDE)	5IGC-03-01
5	1	3"	5ISC (COLD SERVICE INSULATION CRADLE AND SHOE)	5ISC-03-01
6	1	3"	5CC (COLD SERVICE INSULATION CRADLE)	5CC-03-01
7	1	3"	5HR4 (HANGER ROD)	5HR4-06



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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	02-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	WFO
PIPE SCHEDULE:	36
INSULATION CODE:	108
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3" SS WFO (36) (108) 061017

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-WFO-061017-03	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 02-MAR-2026

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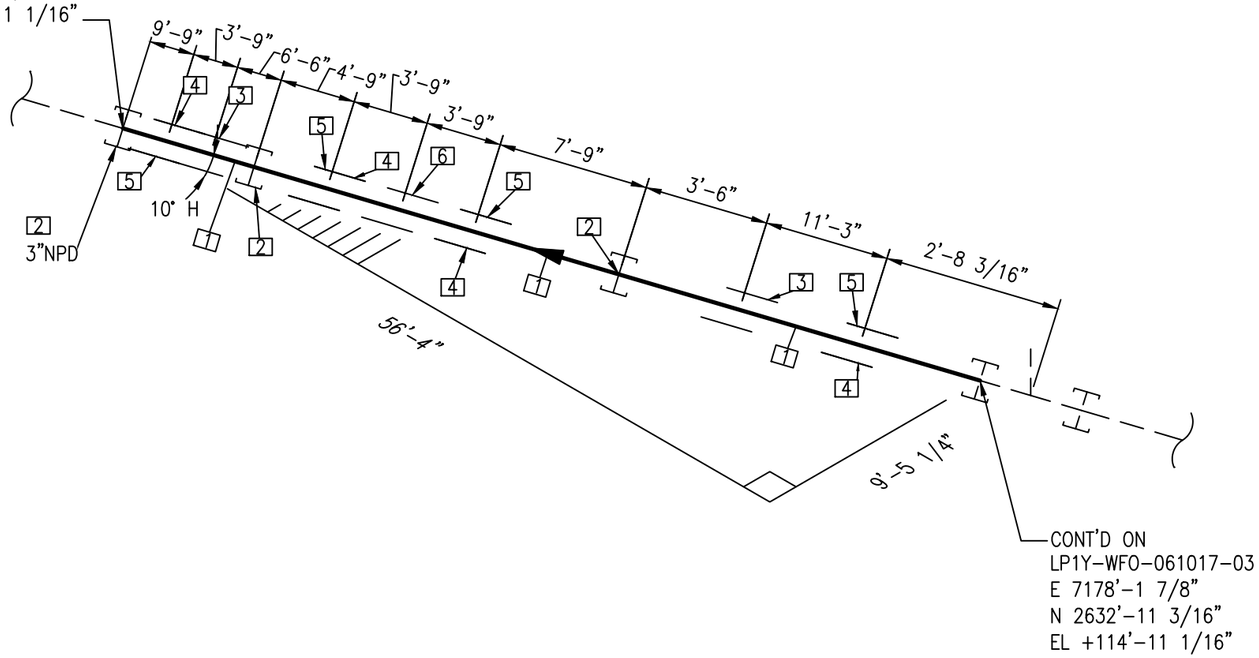
N

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	57'-2"	3"	PIPE SCH 10S ERW 316/316L SS A312 NSF	5365189L
2	3	3"	COUPLING RIGID GRV DI GRADE P GASKET PRE-AS NSF	5714906L0
3	2	3"	5ISC (COLD SERVICE INSULATION CRADLE AND SHOE)	5ISC-03-01
4	4	3"	5CC (COLD SERVICE INSULATION CRADLE)	5CC-03-01
5	4	3"	5HR4 (HANGER ROD)	5HR4-06
6	1	3"	5IGC (COLD SERVICE INSULATION CRADLE WITH SHOE AND GUIDE)	5IGC-03-01

CONT'D ON
LP1Y-WFO-061017-05
E 7168'-8 5/8"
N 2689'-3 3/16"
EL +114'-11 1/16"



CONT'D ON
LP1Y-WFO-061017-03
E 7178'-1 7/8"
N 2632'-11 3/16"
EL +114'-11 1/16"

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	02-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	WFO
PIPE SCHEDULE:	36
INSULATION CODE:	108
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3" SS WFO (36) (108) 061017

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-WFO-061017-04	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 02-MAR-2026

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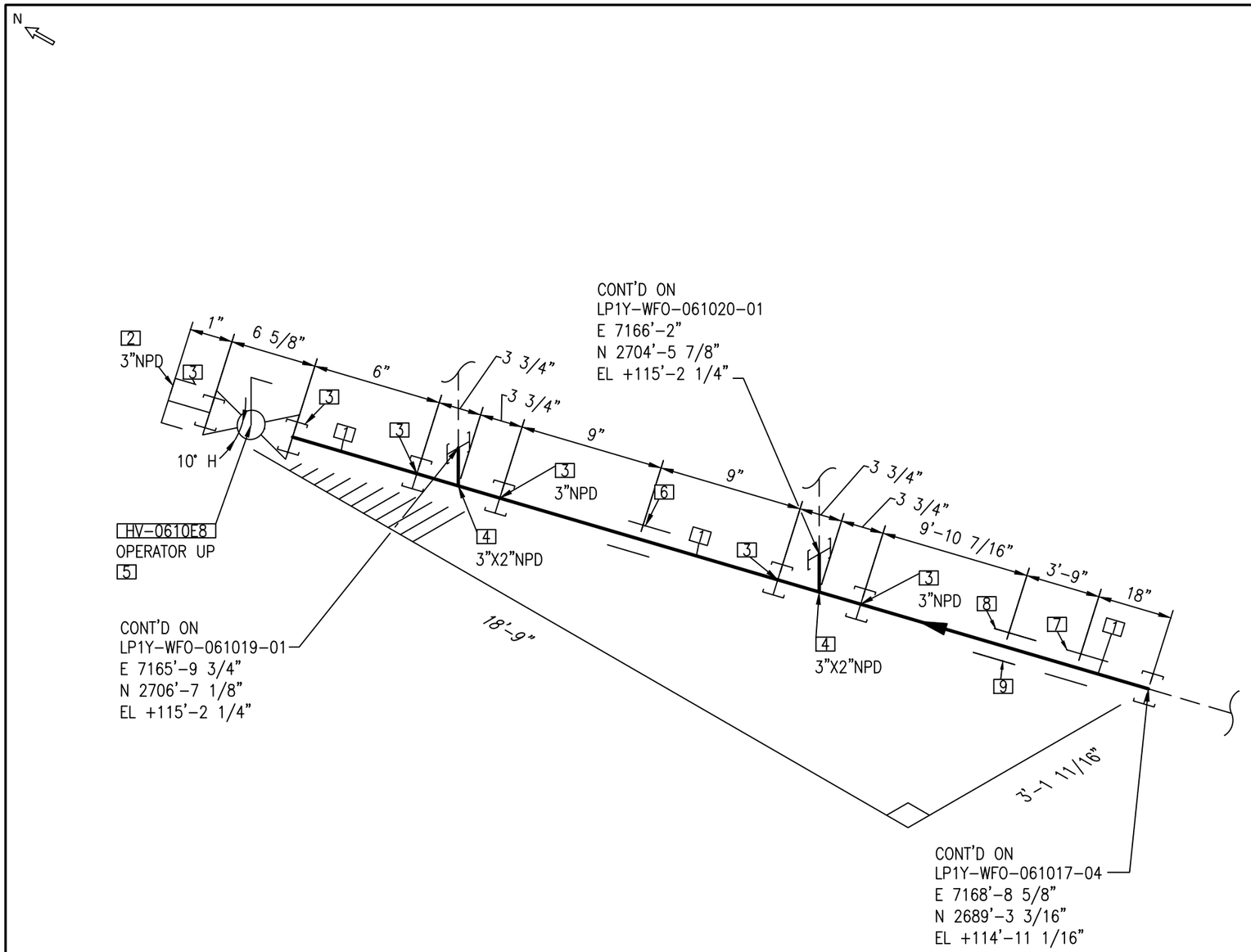

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	17'-2"	3"	PIPE SCH 10S ERW 316/316L SS A312 NSF	5365189L
2	1	3"	CAP GRV 316L SCH 10S NSF	5484957S1
3	6	3"	COUPLING RIGID GRV DI GRADE P GASKET PRE-AS NSF	5714906L0
4	2	3"X2"	TEE RED GRV 316L SCH 10S NSF	5484959S1
5	1	3"	BALL 1000# CWP GRV SS, RTFE 2-PC RP HNDL OP NSF	511453L3
6	1	3"	5ISC (COLD SERVICE INSULATION CRADLE AND SHOE)	5ISC-03-01
7	1	3"	5IGC (COLD SERVICE INSULATION CRADLE WITH SHOE AND GUIDE)	5IGC-03-01
8	1	3"	5CC (COLD SERVICE INSULATION CRADLE)	5CC-03-01
9	1	3"	5HR4 (HANGER ROD)	5HR4-06

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	02-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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10. SLOPE EQUIVALENTS: 1:27 = 7/16" PER FT, 1:96 = 1/8" PER FT, 1:120 = 1/10" PER FT, 1:192 = 1/16" PER FT, 1:480 = 1/40" PER FT.

NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	WFO
PIPE SCHEDULE:	36
INSULATION CODE:	108
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3" SS WFO (36) (108) 061017

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-WFO-061017-05	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 02-MAR-2026

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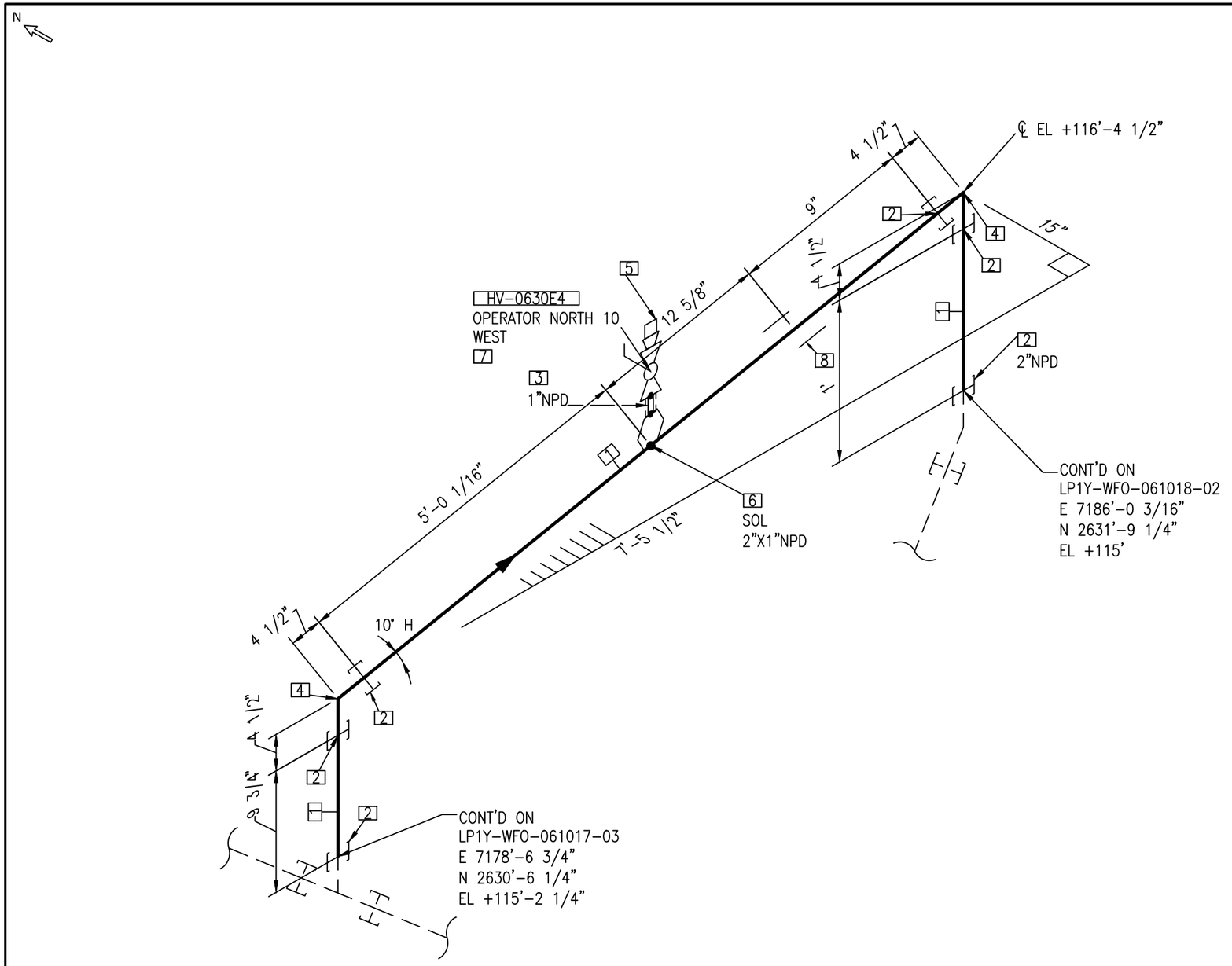

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	8'-8"	2"	PIPE SCH 10S ERW 316/316L SS A312 NSF	5365189L
2	6	2"	COUPLING RIGID GRV DI GRADE P GASKET PRE-AS NSF	5714906L0
3	1	1"	NIPPLE SCH 10S 316/316L SS PBE 6" LONG	5540493
4	2	2"	ELL 90 DEG GRV 316L SCH 10S NSF	5484951S1
5	1	1"	PLUG ROUND HEAD SCRD 316/316L SS	5498383
6	1	2"X1"	PIPET 10S X 3000# SW 316/316L SS	5537081
7	1	1"	BALL 1440# CWP SW X FNPT 316 SS, 316 TRIM RP EXTD STEM, HNDL OP	5300985L8
8	1	2"	5ISC (COLD SERVICE INSULATION CRADLE AND SHOE)	5ISC-02-01

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	02-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	WFO
PIPE SCHEDULE:	36
INSULATION CODE:	108
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS WFO (36) (108) 061018

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-WFO-061018-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 02-MAR-2026

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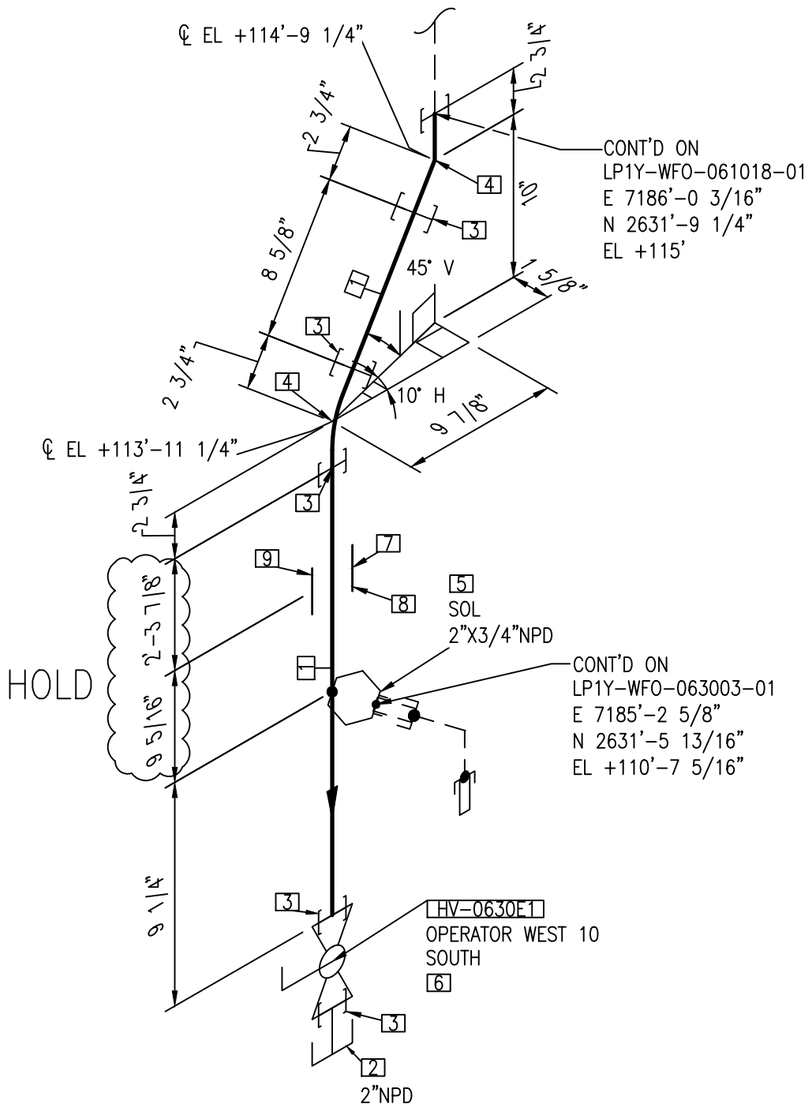

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	4'-8"	2"	PIPE SCH 10S ERW 316/316L SS A312 NSF	5365189L
2	1	2"	CAP GRV 316L SCH 10S NSF	5484957S1
3	5	2"	COUPLING RIGID GRV DI GRADE P GASKET PRE-AS NSF	5714906L0
4	2	2"	ELL 45 DEG GRV 316L SCH 10S NSF	5484952S1
5	1	2"X3/4"	PIPET 10S X 3000# SW 316/316L SS	5537081
6	1	2"	BALL 1000# CWP GRV SS, RTFE 2-PC RP HNDL OP NSF	511453L3
7	1	2"	5CC (COLD SERVICE INSULATION CRADLE)	5CC-02-01
8	1	2"	5GUSP (INSULATED U-BOLT GUIDE)	5UGSP-02-01
9	1	2"	5FS2 (VERTICAL PIPE SUPPORT)	5FS2S1

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	02-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	WFO
PIPE SCHEDULE:	36
INSULATION CODE:	108
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS WFO (36) (108) 061018

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-WFO-061018-02	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 02-MAR-2026

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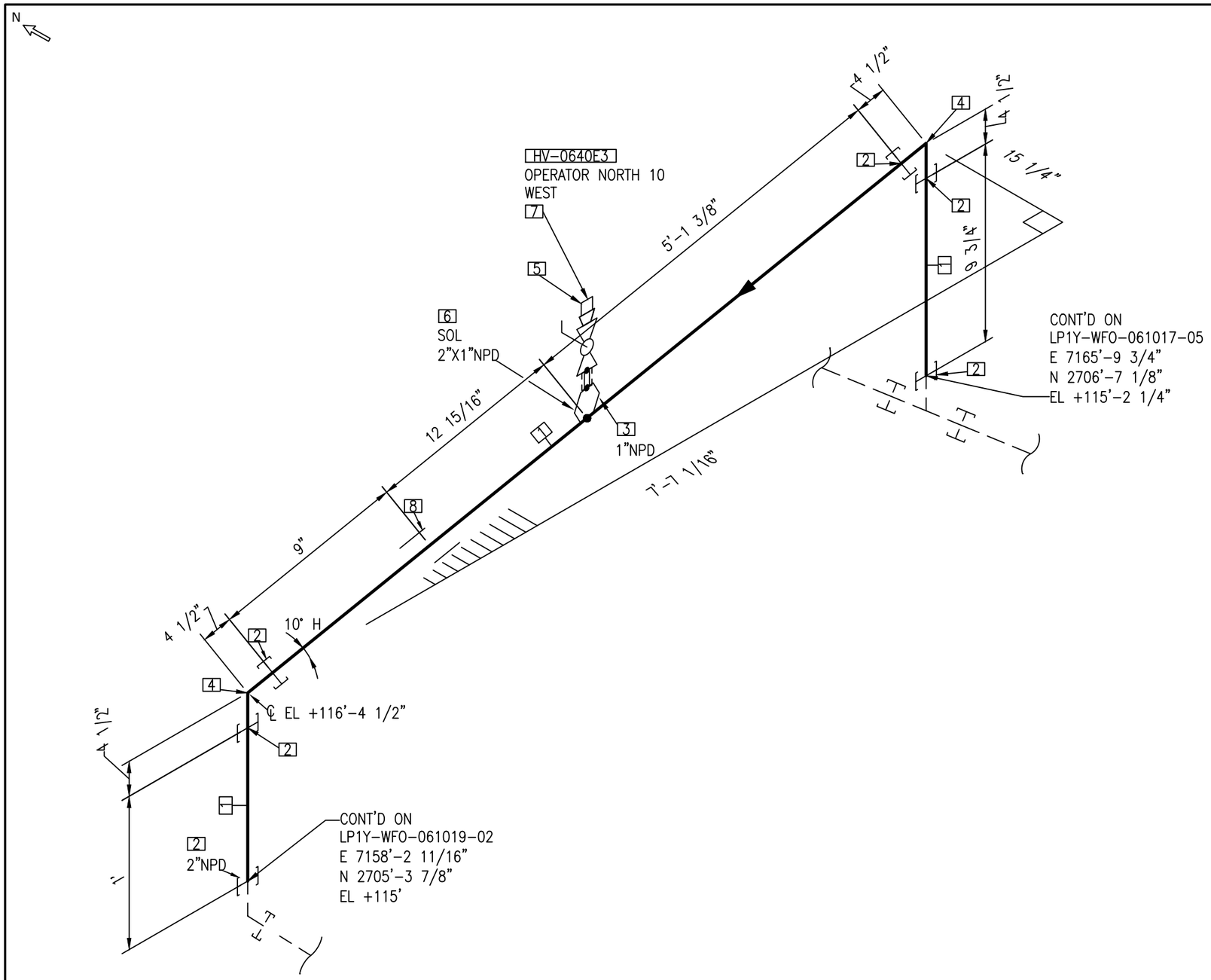

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	8'-10"	2"	PIPE SCH 10S ERW 316/316L SS A312 NSF	5365189L
2	6	2"	COUPLING RIGID GRV DI GRADE P GASKET PRE-AS NSF	5714906L0
3	1	1"	NIPPLE SCH 10S 316/316L SS PBE 6" LONG	5540493
4	2	2"	ELL 90 DEG GRV 316L SCH 10S NSF	5484951S1
5	1	1"	PLUG ROUND HEAD SCRD 316/316L SS	5498383
6	1	2"X1"	PIPET 10S X 3000# SW 316/316L SS	5537081
7	1	1"	BALL 1440# CWP SW X FNPT 316 SS, 316 TRIM RP EXTD STEM, HNDL OP	5300985L8
8	1	2"	5ISC (COLD SERVICE INSULATION CRADLE AND SHOE)	5ISC-02-01

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	WFO
PIPE SCHEDULE:	36
INSULATION CODE:	108
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	02-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS WFO (36) (108) 061019

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-WFO-061019-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 02-MAR-2026

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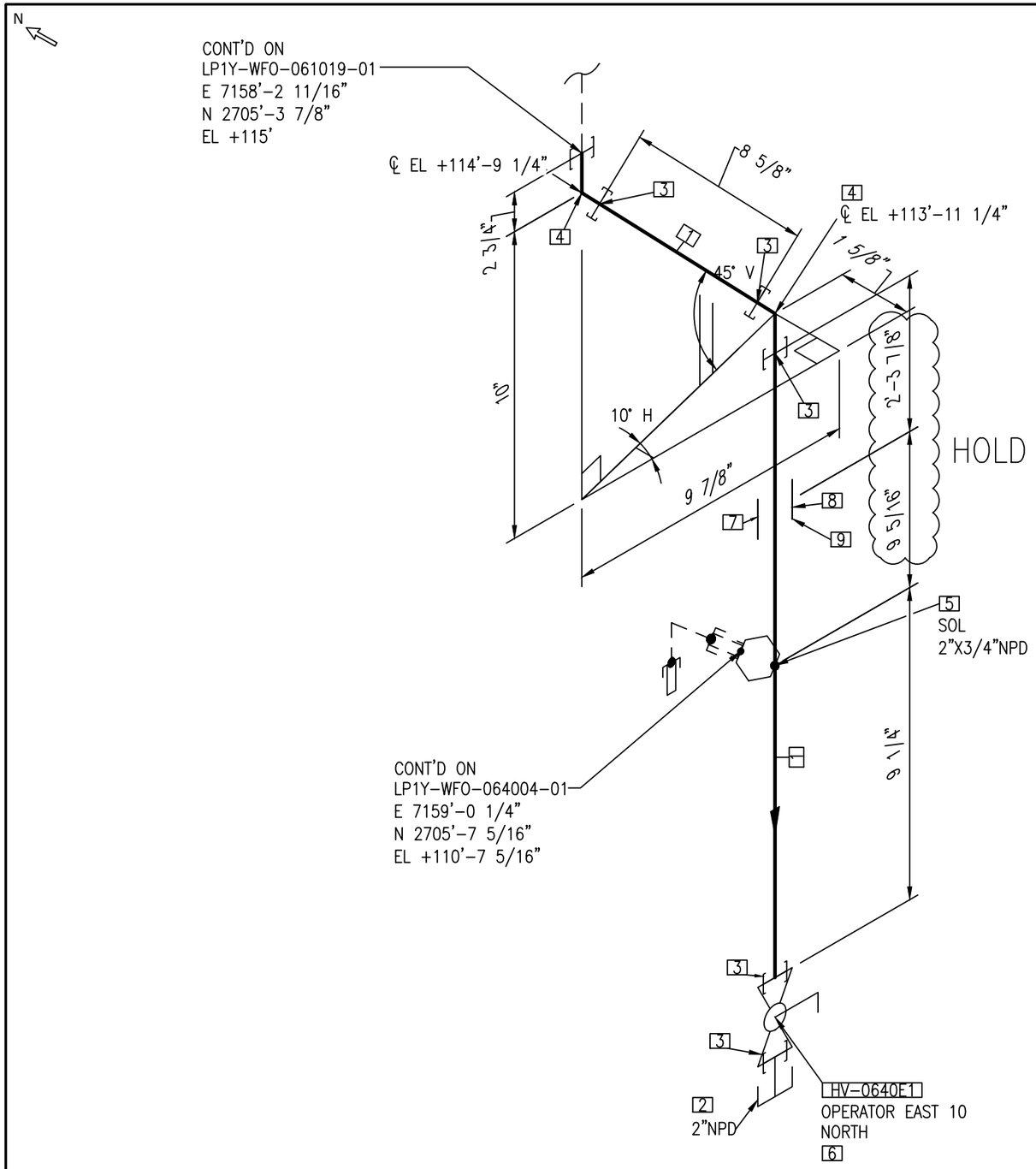

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	4'-8"	2"	PIPE SCH 10S ERW 316/316L SS A312 NSF	5365189L
2	1	2"	CAP GRV 316L SCH 10S NSF	5484957S1
3	5	2"	COUPLING RIGID GRV DI GRADE P GASKET PRE-AS NSF	5714906L0
4	2	2"	ELL 45 DEG GRV 316L SCH 10S NSF	5484952S1
5	1	2"X3/4"	PIPET 10S X 3000# SW 316/316L SS	5537081
6	1	2"	BALL 1000# CWP GRV SS, RTFE 2-PC RP HNDL OP NSF	511453L3
7	1	2"	5CC (COLD SERVICE INSULATION CRADLE)	5CC-02-01
8	1	2"	5GUSP (INSULATED U-BOLT GUIDE)	5UGSP-02-01
9	1	2"	5FS2 (VERTICAL PIPE SUPPORT)	5FS2S1

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	WFO
PIPE SCHEDULE:	36
INSULATION CODE:	108
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	02-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS WFO (36) (108) 061019

REFERENCE P&ID NUMBER
LP1Y-DI-00000061

ISOMETRIC DRAWING NUMBER
LP1Y-WFO-061019-02

REV
0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 02-MAR-2026

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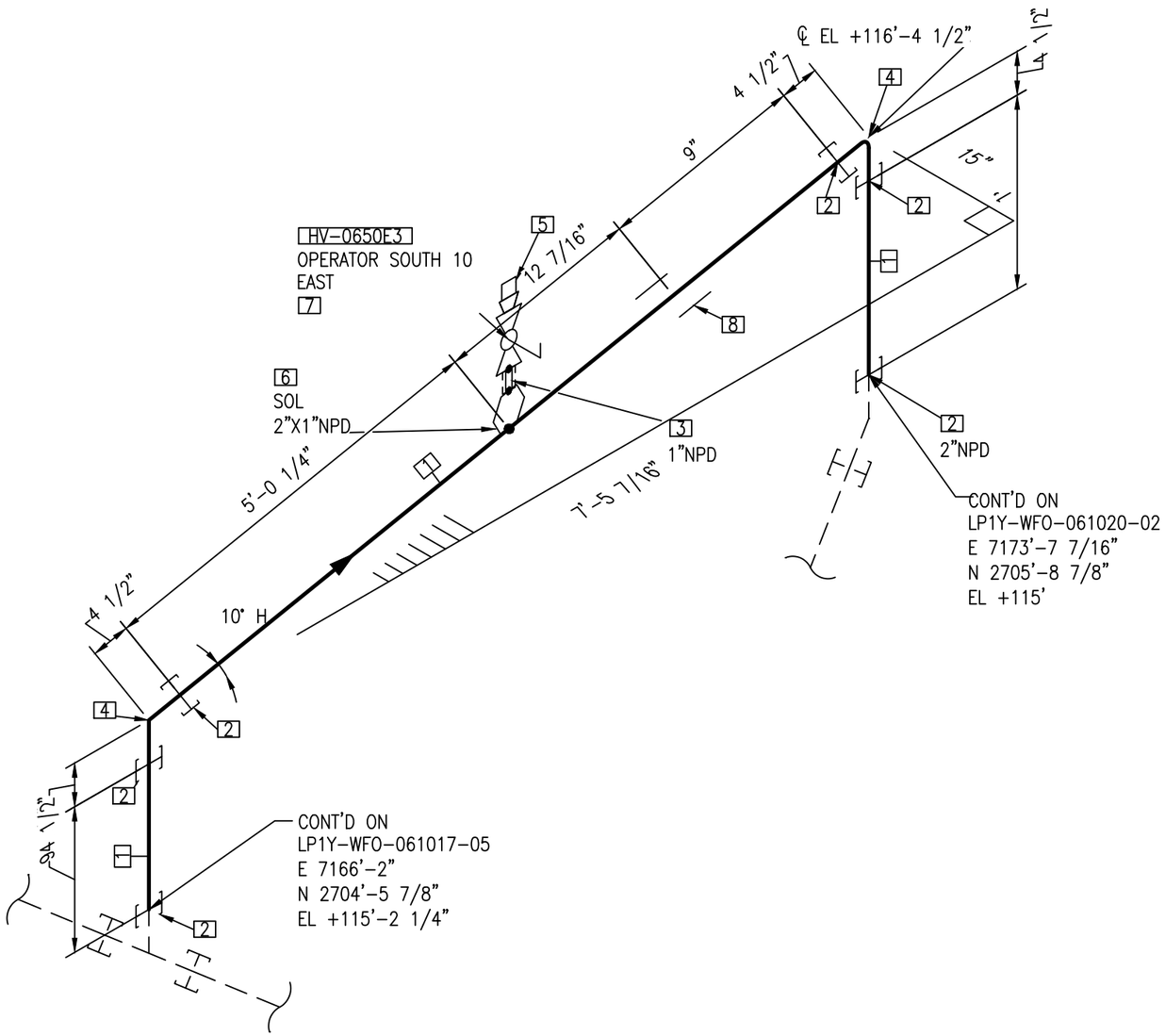

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
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8	1	2"	5ISC (COLD SERVICE INSULATION CRADLE AND SHOE)	5ISC-02-01

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	02-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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7. PIPE SUPPORTS INDICATED ON THE DRAWING MAY BE PART OF AN ASSEMBLY ON SEVERAL ISOMETRICS. REFER TO THE 3D MODEL FOR THE COMPLETE PIPE SUPPORT ASSEMBLY.
8. MECHANICAL CONTRACTOR SHALL NOTE ALL DESIGN CHANGES ON THIS DRAWING AND RETURN IT TO THE CONSTRUCTION WPO FOR AS-BUILT RECORDS.
9. THE BILL OF MATERIAL IS FOR REFERENCE ONLY.
10. SLOPE EQUIVALENTS: 1:27 = 7/16" PER FT, 1:96 = 1/8" PER FT, 1:120 = 1/10" PER FT, 1:192 = 1/16" PER FT, 1:480 = 1/40" PER FT.

NODE:	TFS		
BUILDING NO:	Y		
CWP NO:	N/A		
MODULE NO:	N/A		
SERVICE CODE:	WFO		
PIPE SCHEDULE:	36		
INSULATION CODE:	108		
TRACING TYPE:	NT		
ELASTOMER:	-		
HYGIENIC:	NO	IMPACT:	NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS WFO (36) (108) 061020

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000065	LP1Y-WFO-061020-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 02-MAR-2026

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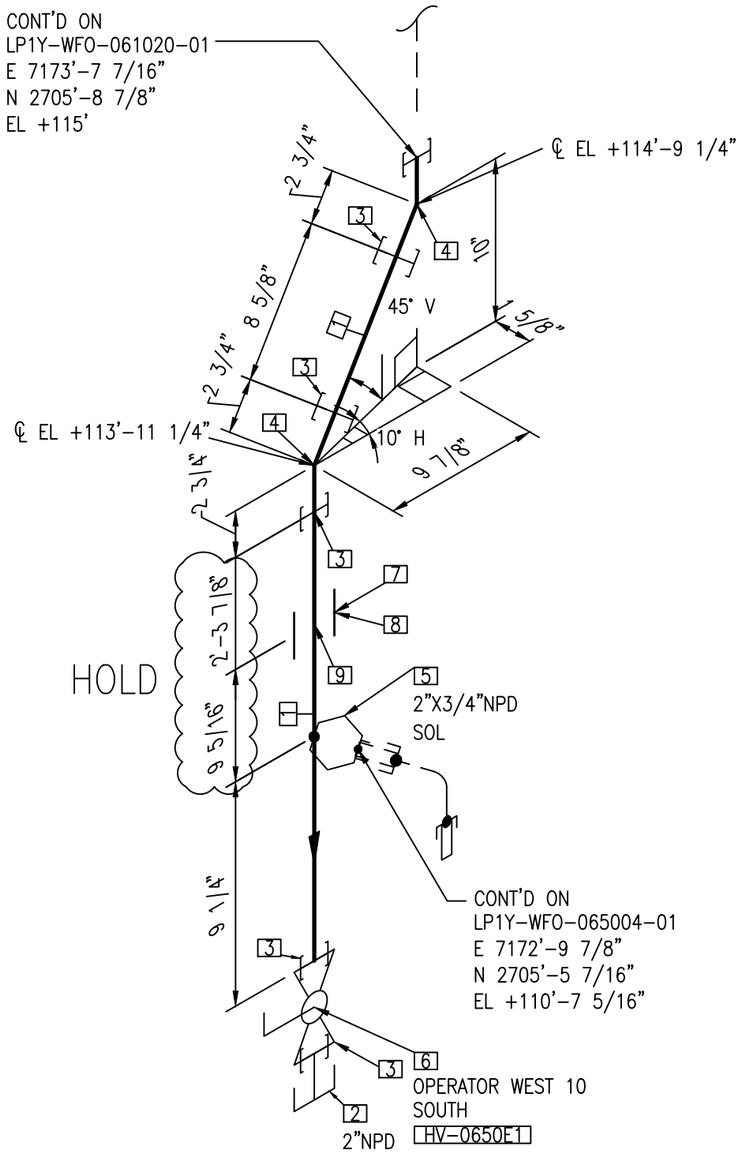
N



BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	4'-8"	2"	PIPE SCH 10S ERW 316/316L SS A312 NSF	5365189L
2	1	2"	CAP GRV 316L SCH 10S NSF	5484957S1
3	5	2"	COUPLING RIGID GRV DI GRADE P GASKET PRE-AS NSF	5714906L0
4	2	2"	ELL 45 DEG GRV 316L SCH 10S NSF	5484952S1
5	1	2"X3/4"	PIPET 10S X 3000# SW 316/316L SS	5537081
6	1	2"	BALL 1000# CWP GRV SS, RTFE 2-PC RP HNDL OP NSF	511453L3
7	1	2"	5CC (COLD SERVICE INSULATION CRADLE)	5CC-02-01
8	1	2"	5GUSP (INSULATED U-BOLT GUIDE)	5UGSP-02-01
9	1	2"	5FS2 (VERTICAL PIPE SUPPORT)	5FS2S1

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	02-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	WFO
PIPE SCHEDULE:	36
INSULATION CODE:	108
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS WFO (36) (108) 061020

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000065	LP1Y-WFO-061020-02	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 02-MAR-2026

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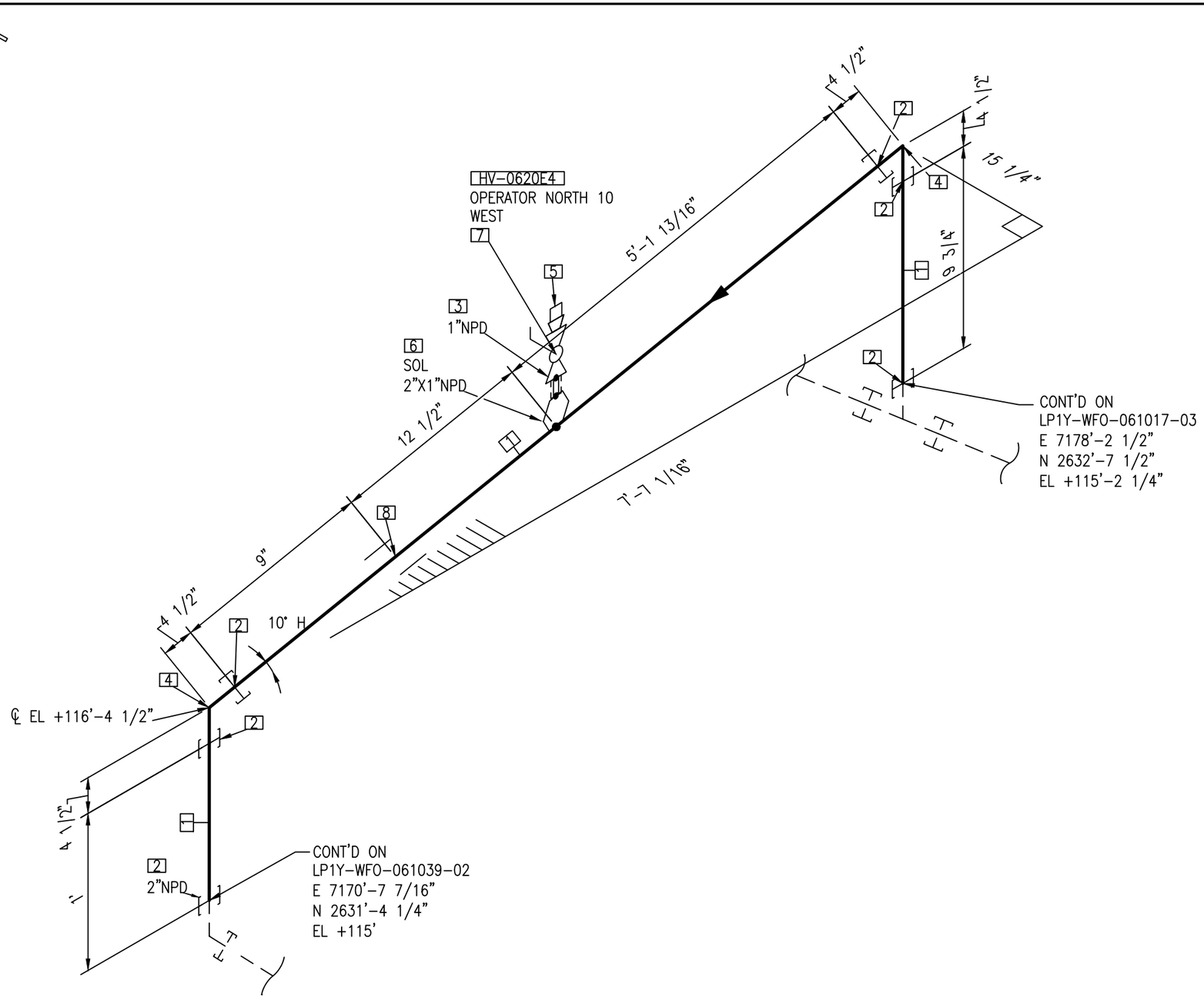

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	8'-10"	2"	PIPE SCH 10S ERW 316/316L SS A312 NSF	5365189L
2	6	2"	COUPLING RIGID GRV DI GRADE P GASKET PRE-AS NSF	5714906L0
3	1	1"	NIPPLE SCH 10S 316/316L SS PBE 6" LONG	5540493
4	2	2"	ELL 90 DEG GRV 316L SCH 10S NSF	5484951S1
5	1	1"	PLUG ROUND HEAD SCRD 316/316L SS	5498383
6	1	2"X1"	PIPET 10S X 3000# SW 316/316L SS	5537081
7	1	1"	BALL 1440# CWP SW X FNPT 316 SS, 316 TRIM RP EXTD STEM, HNDL OP	5300985L8
8	1	2"	5ISC (COLD SERVICE INSULATION CRADLE AND SHOE)	5ISC-02-01

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	02-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS		
BUILDING NO:	Y		
CWP NO:	N/A		
MODULE NO:	N/A		
SERVICE CODE:	WFO		
PIPE SCHEDULE:	36		
INSULATION CODE:	108		
TRACING TYPE:	NT		
ELASTOMER:	-		
HYGIENIC:	NO	IMPACT:	NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS WFO (36) (108) 061039

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-WFO-061039-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 02-MAR-2026

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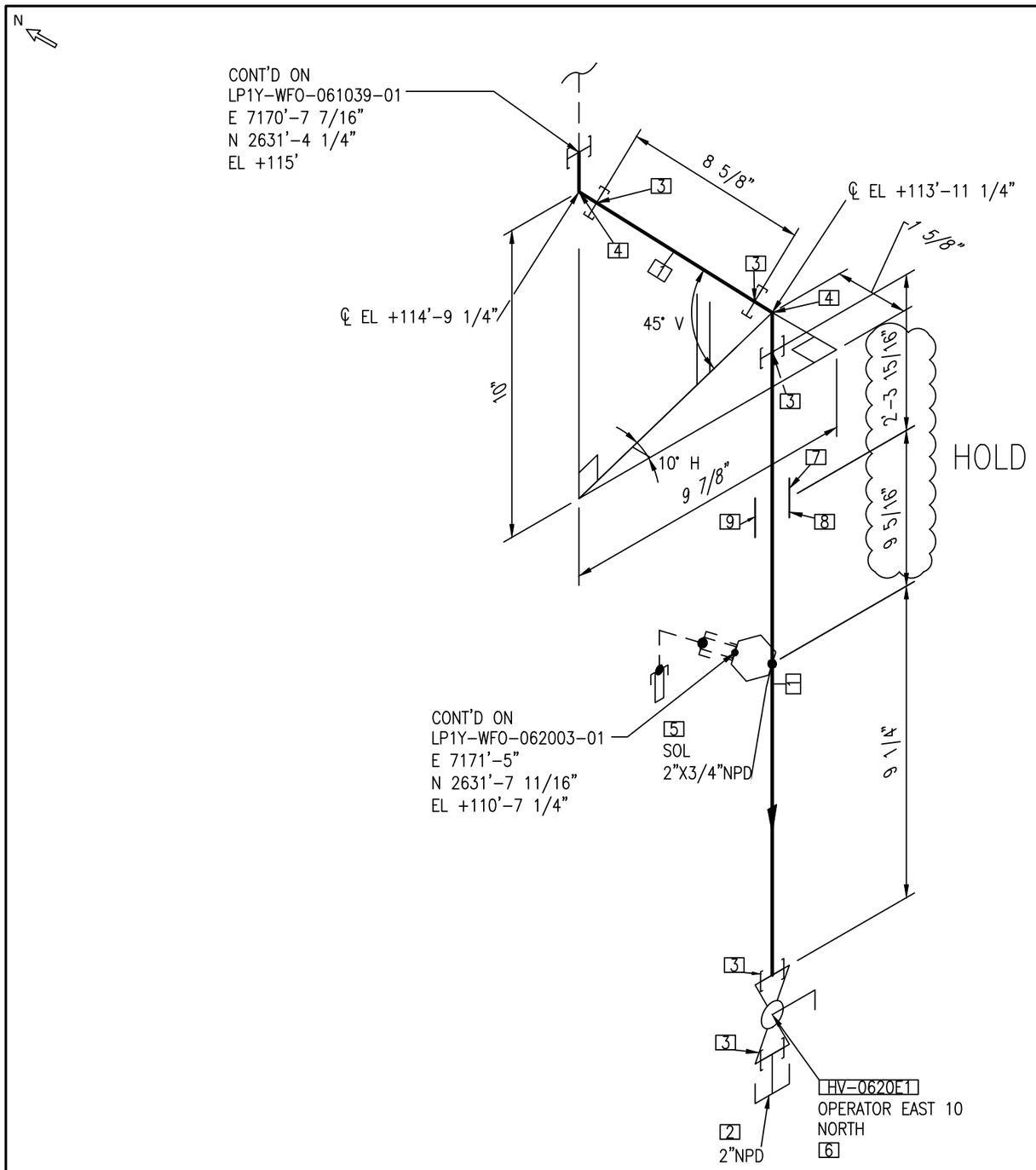

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
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3	5	2"	COUPLING RIGID GRV DI GRADE P GASKET PRE-AS NSF	5714906L0
4	2	2"	ELL 45 DEG GRV 316L SCH 10S NSF	5484952S1
5	1	2"X3/4"	PIPET 10S X 3000# SW 316/316L SS	5537081
6	1	2"	BALL 1000# CWP GRV SS, RTFE 2-PC RP HNDL OP NSF	511453L3
7	1	2"	5CC (COLD SERVICE INSULATION CRADLE)	5CC-02-01
8	1	2"	5GUSP (INSULATED U-BOLT GUIDE)	5UGSP-02-01
9	1	2"	5FS2 (VERTICAL PIPE SUPPORT)	5FS2S1

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	WFO
PIPE SCHEDULE:	36
INSULATION CODE:	108
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
						/	
						/	
0	ISSUE FOR CONSTRUCTION	02-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS WFO (36) (108) 061039

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-WFO-061039-02	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 02-MAR-2026

Tim Trice

Planned Controlled Document

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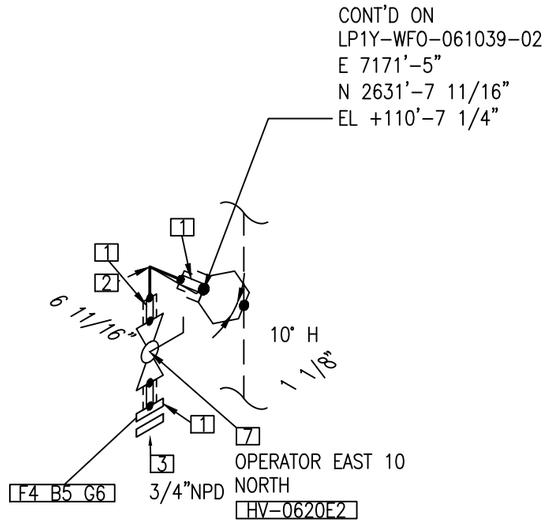
N

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	3	3/4"	NIPPLE SCH 10S 316/316L SS PBE 6" LONG	5540493
2	1	3/4"	ELL 90 DEG 3000# SW 316/316L SS	5480138
3	1	3/4"	FLG BLIND 150# RF 316/316L SS	5616520
4	1	3/4"	FLG SW 150# RF 316/316L SS 10S BORE	5608148
5	4	1/2"X2 1/4"	STUD-BOLT A193 GR B7 W/A194 GR 2H NUTS	5675648
6	1	3/4"	GASKET 150# SOFT PTFE RING 1/16" THK NSF	5669345
7	1	3/4"	BALL 1440# CWP SW 316 SS, 316 TRIM RP EXTD STEM, HNDL OP	5300985L7



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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	WFO
PIPE SCHEDULE:	36
INSULATION CODE:	108
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3/4" SS WFO (36) (108) 062003

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000062	LP1Y-WFO-062003-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 02-MAR-2026

Tim Trice

Planned Controlled Document

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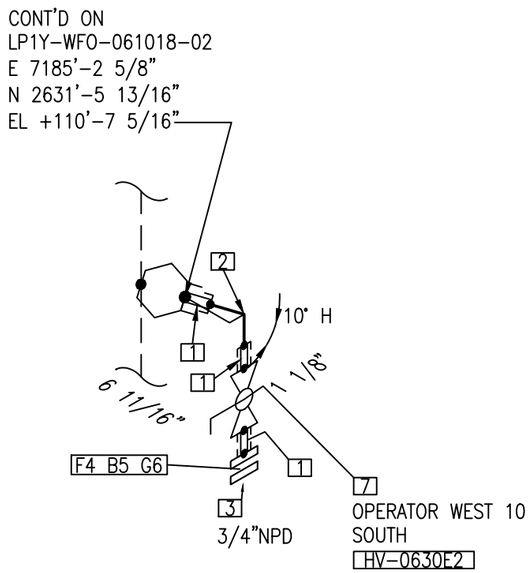

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By dac54131 at Apr 10, 2026

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NODE:	TFS
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CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	WFO
PIPE SCHEDULE:	36
INSULATION CODE:	108
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3/4" SS WFO (36) (108) 063003

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000063	LP1Y-WFO-063003-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 02-MAR-2026

Tim Trice

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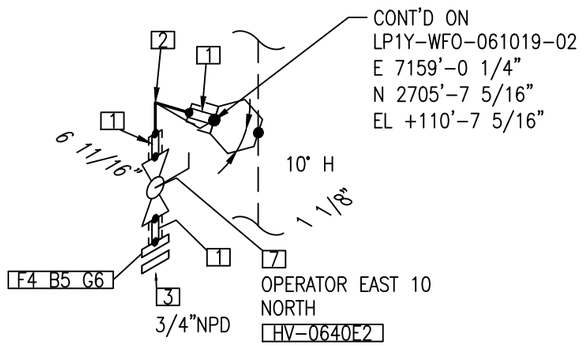
N

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	3	3/4"	NIPPLE SCH 10S 316/316L SS PBE 6" LONG	5540493
2	1	3/4"	ELL 90 DEG 3000# SW 316/316L SS	5480138
3	1	3/4"	FLG BLIND 150# RF 316/316L SS	5616520
4	1	3/4"	FLG SW 150# RF 316/316L SS 10S BORE	5608148
5	4	1/2"X2 1/4"	STUD-BOLT A193 GR B7 W/A194 GR 2H NUTS	5675648
6	1	3/4"	GASKET 150# SOFT PTFE RING 1/16" THK NSF	5669345
7	1	3/4"	BALL 1440# CWP SW 316 SS, 316 TRIM RP EXTD STEM, HNDL OP	5300985L7



A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

Authorization to proceed does not relieve Contractor/Supplier of its responsibility or liability under the Contract and or Purchase Order.

By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	02-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

GENERAL NOTE:
1. REFER TO THE PIPING LINE LIST FOR PROCESS DATA/CONDITIONS AND OTHER FABRICATION DETAILS SUCH AS TESTING, CLEANING, AND PAINTING REQUIREMENTS.
2. UNLESS OTHERWISE NOTED, ALL WELDS ARE INDICATED AS SHOP WELDS. CONTRACTOR TO DETERMINE FIELD WELDS LOCATIONS AND DOCUMENT ACCORDINGLY ON AS-BUILT RECORDS.
3. WELDS AND/OR COUPLINGS DEFINING RANDOM PIPE SPOOL LENGTHS ARE FOR REFERENCE ONLY. FINAL WELD OR COUPLING LOCATION IS TO BE DETERMINED BY THE CONTRACTOR.
4. THE MECHANICAL CONTRACTOR TO FIELD VERIFY INTERFACE POINTS BETWEEN EQUIPMENT & PLUMBING FIXTURES IN 3rd FIX (3F SERIES CWP) AND YS YARD STICK BUILT (YS SERIES CWP EQUIPMENT CONNECTIONS) ISOMETRICS PRIOR TO FABRICATION.
5. THE MECHANICAL CONTRACTOR IS TO VERIFY ALL DIMENSIONAL DATA WITH ACTUAL FIELD CONDITIONS PRIOR TO INSTALLATION OF THE SYSTEM.
6. PIPE SUPPORTS SHALL BE FURNISHED AND INSTALLED IN ACCORDANCE WITH THIS DRAWING AND SPECIFICATION LP100-250-EMS-15110. MECHANICAL CONTRACTOR SHALL PROVIDE TEMPORARY SUPPORTS AS REQUIRED TO AID PRESSURE TESTING, INSTALLATION ETC.
7. PIPE SUPPORTS INDICATED ON THE DRAWING MAY BE PART OF AN ASSEMBLY ON SEVERAL ISOMETRICS. REFER TO THE 3D MODEL FOR THE COMPLETE PIPE SUPPORT ASSEMBLY.
8. MECHANICAL CONTRACTOR SHALL NOTE ALL DESIGN CHANGES ON THIS DRAWING AND RETURN IT TO THE CONSTRUCTION WPO FOR AS-BUILT RECORDS.
9. THE BILL OF MATERIAL IS FOR REFERENCE ONLY.
10. SLOPE EQUIVALENTS: 1:27 = 7/16" PER FT, 1:96 = 1/8" PER FT, 1:120 = 1/10" PER FT, 1:192 = 1/16" PER FT, 1:480 = 1/40" PER FT.

NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	WFO
PIPE SCHEDULE:	36
INSULATION CODE:	108
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3/4" SS WFO (36) (108) 064004

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000064	LP1Y-WFO-064004-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 02-MAR-2026

Tim Trice

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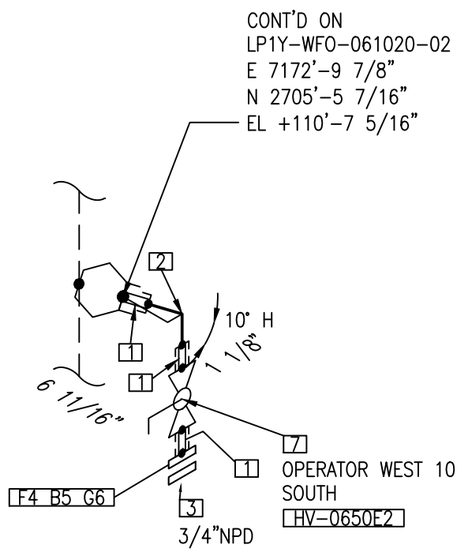

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	02-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	WFO
PIPE SCHEDULE:	36
INSULATION CODE:	108
TRACING TYPE:	NT
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3/4" SS WFO (36) (108) 065004

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000065	LP1Y-WFO-065004-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 02-MAR-2026

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